

BOUNDARY LAWS AND RIGIDITY OF PERIODIC DISPERSING BILLIARDS

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ABSTRACT. We prove relative boundary laws near nongrazing periodic dispersing billiard orbits. Exact periodic Jacobi normalization and a two-ended trace-class comparison retain distinct future and past half-line actions on a flight-independent collar. The relative interaction identifies their nonlinear transmission coordinates. For alternating channels, fixed positive-offset laws determine the nonlinear reflecting geometry. A finite stationary-envelope argument then identifies the action filtration for actual smooth boundaries; its signed contact inverse has determinant-one blocks at every order, with order-uniform bounds under geometric admissibility. The gap and two same-type conditional endpoint laws determine all finite contact jets without supplied curvatures or flux amplitudes. For analytic contacts, the complete action map has a local analytic inverse, and a local analytic prior yields conditional Hölder stability for complete contact germs on a smaller disc from real density error. For an exactly calibrated channel, finite histograms from charged independent full-phase preparations recover the complete local analytic pair with an explicit confidence rate; estimated calibration has a separate error budget. For noncircular analytic periodic tables, selected marked channels determine finitely many complete realizations, including the unknown marked lattice; proper asymmetry gives uniqueness. Reconstruction remains valid on visible interior windows with unknown transverse origins and separate positive recording factors. The differential has precisely the common Euclidean kernel, and immersed finite-dimensional models admit finite observable coordinates. Realized comparisons separate nonlinear information from leading endpoint data, finite matching ambiguity, and continuous symmetry. A persistent position pilot yields a further rate in the complete preparation budget, including calibration and every rejected attempt.

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1. INTRODUCTION

A long alternating billiard bridge has two different geometric scales. Its middle approaches a hyperbolic two-periodic orbit, whereas its transverse endpoints remain in a fixed contact collar. Does conditioning on this increasingly rare event retain the nonlinear geometry of the reflecting boundaries, or only their contact Hessians? We prove that it retains the two signed half-line actions and that these actions determine every finite contact jet. For analytic periodic tables, a finite selected set of marked channels then determines complete obstacle images and the unknown marked lattice, with finite ambiguity in the noncircular class and uniqueness in the properly asymmetric class.

The offset and the flight number play different roles. We let the number of flights tend to infinity at a *fixed positive excess time*; we do not replace that positive-offset law by its leading small-offset term. The unconditioned mass decays exponentially. The normalized law, however, retains a nonlinear endpoint interaction on a collar independent of flight number. Thus a long-flight limit is not a quadratic truncation. The separation is realized by actual periodic tables: an analytic family has the same gap, free area, contact curvatures, and leading endpoint metrics and count amplitudes at every flight number, but different nonlinear limiting laws. Its construction and its nonzero response are proved in Theorem A.4.2.

There are two analytical steps. First, one must control the physical mixed derivative relative to an exponentially small reference twist. An absolute estimate for the stationary action cannot do this. An exact cofactor identity followed by a two-ended trace-class comparison produces the relative limit before conditioning and before offset differentiation. Second, one must show that the action jet is a function of the actual boundary jet, independent of arbitrary smooth remainders. A finite stationary-envelope identity, with its terminal term retained until it vanishes, proves this factorization. The resulting signed inverse has a determinant-one block at every order, without reflection symmetry or equal contact curvatures. These are the two uses of the boundary-layer construction on which the reconstruction rests.

1.1. Relative transmission along an arbitrary periodic itinerary. The forward mechanism extends to every selected nongrazing periodic orbit with clear flights and strictly dispersing contacts. Its proof uses the same two-ended normalization, now with periodic Jacobi coefficients

and distinct future and past boundary actions. We give the geometric verification rather than assuming a hyperbolic normal form. A three-obstacle orbit and its nonsymmetric perturbations provide realizations outside the normal alternating setting (Proposition 8.6).

Here is the normalization which makes the extension precise. Let P be an even oriented period, L_i the flight lengths, κ_i the curvatures, and ν_i the outgoing normal cosines. Set $k_i = L_i^{-1}$, $m_i = 2\kappa_i/\nu_i$ and

$$Q_i = \begin{pmatrix} 1 + m_i/k_i & 1/k_i \\ m_i & 1 \end{pmatrix}, \quad \mathcal{M}_b = Q_{b+P-1} \cdots Q_b, \quad \text{tr } \mathcal{M}_b = 2 \cosh \chi.$$

Positive curvature gives $\chi > 0$. For $N = nP$ flights starting at phase b , the exact reference twist is

$$D_{N,b} = \frac{\sinh \chi}{(\mathcal{M}_b)_{12} \sinh(n\chi)}.$$

In scaled signed arclength coordinates, let p_b be the reference endpoint momentum and $L = \sum L_i$. Theorem 8.3 proves, on a collar independent of n ,

$$\begin{aligned} W_{N,b} - nL + p_b u - p_b v &= S_b^-(u) + S_b^+(v) + O_{C^k}(\tau^N), \\ -W_{N,b,uv}/D_{N,b} &= B_b^-(u)B_b^+(v) + O_{C^k}(\tau^N). \end{aligned} \quad (1.1)$$

Each differentiation order is fixed separately. The future and past actions need not agree. The physical window law contains $d - S_b^-(u) - S_b^+(v) + p_b u - p_b v$; the endpoint momentum terms cannot be discarded as though the reference collisions were normal. Theorem 8.5 derives this law from normalized phase volume and retains its exact exponentially small acceptance mass.

The same estimate also identifies the nonlinear coefficient of the weak interaction between the ends. If $C_N = W_N(u, v) - W_N(u, 0) - W_N(0, v) + W_N(0, 0)$ and $\zeta_b^\pm(x) = \int_0^x B_b^\pm(t) dt$, then

$$-C_N/D_{N,b} = \zeta_b^-(u)\zeta_b^+(v) + O_{C^k}(\tau^N).$$

The error before normalization is $O(D_{N,b}\tau^N|uv|)$. Moreover ζ_b^- linearizes the future stable period map, while ζ_b^+ linearizes the reversed unstable period map, both with multiplier $e^{-\chi}$ (Corollary 8.4). Scalar linearization and the final double integration are consequences of the relative theorem; they are not substitutes for its geometric determinant estimate.

This extension explains a structural feature of the mechanism: it requires dispersing coercivity, a clear nongrazing itinerary and summable endpoint layers, rather than equal flight lengths or a normal two-orbit. The inversion below exploits the additional geometry of an alternating pair. All of its complete-jet, analytic-germ, global and finite-observation conclusions are retained. No two-contact inverse is silently applied to the distinct actions of a general periodic itinerary.

1.2. The local relative and smooth inverse mechanism. Fix a clear closest channel between two smooth strictly convex reflecting arcs, with gap $g > 0$, positive curvatures κ_0, κ_1 , and a common signed transverse coordinate. In contact frames the arcs are $(-\psi_0(u), u)$ and $(g + \psi_1(v), v)$, with $\psi_b(0) = \psi_b'(0) = 0$ and $\psi_b''(0) = \kappa_b$. The stationary length of j alternating flights is $W_j(u, v)$. Put $c_b = 1 + g\kappa_b$, $c = \sqrt{c_0 c_1}$, $\gamma = \text{arcosh } c$, and $p = j \bmod 2$. Geometric parameter derivatives, when taken, keep the excess time $d = t - jg$ fixed in these moving intrinsic coordinates.

Theorem 1.1 (Relative boundary law and actual smooth contact inverse). *For a selected clear channel in a smooth periodic dispersing table, there are a flight-independent contact collar, smooth strictly convex actions S_b and positive smooth amplitudes B_b , normalized by $S_b(0) = S_b'(0) = 0$ and $B_b(0) = 1$, with the following properties.*

- (1) Write $a_b = S_b''(0) = c \sinh \gamma / (gc_{1-b})$ and $d_j^0 = \sqrt{a_0 a_p} / \sinh(j\gamma)$. At every separately fixed finite differentiation order, on a sufficiently small common collar,

$$\|W_j - jg - (S_0 \oplus S_p)\|_{C^k} + \left\| \frac{-W_{j,uv}}{d_j^0} - B_0 \otimes B_p \right\|_{C^k} \leq C_k \tau^j, \quad 0 < \tau < 1. \quad (1.2)$$

The estimates include the stipulated fixed-order family derivatives; the constants may depend on higher smooth norms. No estimate uniform in the differentiation order is asserted.

- (2) At fixed sufficiently small positive excess d , the limiting same-type conditional endpoint law obtained from the full phase ensemble has positive interior density

$$f_b(u, v) = Z_b^{-1} B_b(u) B_b(v) \{d - S_b(u) - S_b(v)\}. \quad (1.3)$$

The gap and one such law of each type determine the two signed action germs and every finite labelled contact jet, without supplied amplitudes, curvatures or normal collision coordinates. The extraction is smooth at each fixed finite order on the positive, nondegenerate domains specified in the density inverse below.

- (3) For $n \geq 3$, after lower jets are fixed, write $s_n = (S_0^{(n)}(0), S_1^{(n)}(0))^t$ and $q_n = (\psi_0^{(n)}(0), \psi_1^{(n)}(0))^t$. Then

$$s_n = M_n q_n + R_n, \quad M_n = \begin{pmatrix} \coth(n\gamma) & \mathfrak{r}_0^n \operatorname{csch}(n\gamma) \\ \mathfrak{r}_1^n \operatorname{csch}(n\gamma) & \coth(n\gamma) \end{pmatrix}, \quad \mathfrak{r}_b = \sqrt{c_b/c_{1-b}}, \quad \det M_n = 1. \quad (1.4)$$

Here R_n depends only on the gap and lower graph jets. The triangular inverse holds for actual smooth representatives, including odd contact orders. More precisely, let $M \geq 2$ and let two anchored graph pairs have the same gap and equal graph jets through order M at their respective anchored contacts. Under common functional smooth bounds on a sufficiently small collar, their actions satisfy

$$|S_b^{[1]}(u) - S_b^{[0]}(u)| \leq \frac{C_M}{1 - \rho^{M+1}} |u|^{M+1}, \quad 0 < \rho < 1. \quad (1.5)$$

The highest-degree blocks have the additional bounds

$$\begin{aligned} \max\{\|M_n\|_\infty, \|M_n^{-1}\|_\infty\} &\leq \frac{3 + t^6}{1 - t^6}, \\ \max\{\|M_n - I\|_\infty, \|M_n^{-1} - I\|_\infty\} &\leq \frac{4\vartheta^n}{1 - t^6}, \end{aligned} \quad t = e^{-\gamma}, \quad \vartheta = \frac{1 + t^2}{2} < 1. \quad (1.6)$$

Here the norm is the maximum absolute row sum. These bounds are uniform in $n \geq 3$, and uniformly over geometries with $\gamma \geq \gamma_* > 0$. They do not assert an order-uniform bound for the lower-jet remainders or the complete nonlinear inverse.

The local forward and finite-jet conclusions do not require analytic boundaries, equal curvatures, or reflection symmetry.

Proof. Lemma 5.2 constructs the finite stationary bridges. Lemma 7.1 constructs their half-line limits. The two-ended normalized determinant comparison in Theorem 7.2 proves (1.2). The full phase formula (6.1), the common-domain integration in Lemma 6.1, and Theorem 7.3 give (1.3). Theorem 14.1 and Proposition 14.2 extract the actions and leading curvatures on the positive interior domain. Lemma 13.4 proves (1.5) before the last-jet calculation is made; Proposition 13.6 gives (1.4) and its inverse. Proposition 13.9 uses the positive-curvature constraint to prove (1.6). All of these supporting proofs are printed in the present article. The block bounds are order-uniform; the nonlinear smooth estimates and reconstruction remain at each separately prescribed finite order. $\square$

1.3. Why the relative estimate is the relevant limit. The estimates have different roles. To prove the relative estimate one uses, before division by d_j^0 , the exact identity

$$\log b_j = \sum_{i=0}^{j-1} \log \{g[-\ell_{i \bmod 2, uv}(y_i, y_{i+1})]\} - \log \det(I + G_j \Delta H_j), \quad b_j = -W_{j, uv}/d_j^0. \quad (1.7)$$

The perturbation ΔH_j is localized near two ends; its trace norm is estimated by a summable weight, not by the number of interior sites. The finite Green compressions are then compared to their half-line limits. Fixed powers introduced by differentiation are absorbed by strict exponential margins. For the inverse, one interpolates actual smooth graphs and differentiates finite stationary action sums. Interior variations cancel by stationarity; the terminal term is retained until its decay has been proved. Only then is the finite Taylor recursion used. Thus neither an absolute-error estimate nor a formal power series is substituted for the required nonlinear argument.

The distinction can be stated directly at the level of the observation. On a fixed positive interior square, integrating out the residual time and normalizing gives

$$f_{j,b,d}(u, v) = \frac{b_{j,b}(u, v)\{d - E_{j,b}(u, v)\}}{\int (d - E_{j,b}) + b_{j,b}}.$$

The exponentially small reference twist cancels from this quotient. That cancellation is useful only after $b_{j,b}$, not merely $E_{j,b}$, has been controlled. The relative estimate supplies the amplitude and normalization bounds needed for the positive-offset limit. In particular, Corollary 14.3 gives, at each fixed order M , the geometric error $C_M(\tau^j + \epsilon + \epsilon_g)$ from a finite-flight interior C^M density error ϵ and gap error ϵ_g . This connects the relative limit to its inverse without an assumption of order-uniform real differentiation or a bound on the number of preparations required to estimate the density.

1.4. The action as a nonlinear boundary coordinate. The geometric parameter restriction also identifies where an all-order conditioning issue can occur. Positivity gives $\tau_b e^{-\gamma} < (1 + e^{-2\gamma})/2 < 1$. Thus the highest-degree response approaches the identity rather than becoming singular with order. Proposition 13.9 places this statement on coefficient spaces and for actual smooth pairs with common lower jets. The remaining lower-degree couplings, smooth remainder bounds and analytic continuation are distinct parts of the full inverse; no estimate for them follows merely from the diagonal bounds.

The analytic germ problem admits a further conclusion for the complete nonlinear map, not just its diagonal blocks. In Theorem 13.12, the half-line actions are local analytic coordinates for anchored analytic contact pairs in a bounded-holomorphic norm on one small fixed disc. The derivative is

$$(D\mathcal{S}(\psi)\eta)_b(u) = \alpha_b(u)\eta_b(u) + \sum_{i \geq 1} v_{b,i}(u)\eta_{b+i}(x_{b,i}(u)), \quad |x_{b,i}(u)| \leq \rho^i |u|, \quad \inf |\alpha_b| > 0.$$

This identity follows from the actual stationary envelope, including its vanishing terminal term. The first term is an invertible multiplier; the remaining evaluations contract high-order vanishing functions. One finite low-jet system and a norm-convergent tail inverse therefore control all lower-degree couplings. A local contraction gives the nonlinear inverse. In the induced analytic quotient norms, its complete linearized finite-jet inverses have bounds independent of the truncation order. The analytic norm is an additional hypothesis: it is not inferred from finitely many density derivatives, total variation, or a finite sample. No continuation to an entire obstacle or uniform acquisition rate is part of this local analytic estimate.

There is also a conditional conclusion in the real observation topology. Theorem 13.15 fixes a local bounded analytic neighborhood and recovers the complete contact germs on a strictly

smaller complex disc from uniform real endpoint-density error ϵ , with a bound $C(\epsilon_g + \epsilon^\vartheta)$, $0 < \vartheta < 1$. Curvatures may vary and are not separately supplied. With a uniform first-order real density bound, total variation or histogram errors give exponent $\vartheta/3$. The proof combines amplitude-free extraction on a real square, a displayed interpolation estimate, and the complete analytic inverse on the inner disc. It does not infer same-disc analytic smallness from real data: the outer analytic prior, radius loss, density and anchor margins, and local choice of inverse branch are explicit. No global registration, unknown-origin calibration, or preparation budget is a consequence of this local estimate. Proposition 13.13 isolates the elementary functional-analytic criterion; the protected half-lines and stationary envelope remain its billiard-specific inputs.

The analytic inverse is another consequence of the stationary envelope, not a consequence of the determinant of each finite block alone. The scalar restriction estimate and the interior interpolation used in the real-observation theorem have a different role: they transfer that inverse to a weaker observation topology under stated priors. They are not additional mechanisms for reconstructing a billiard boundary. In particular, the all-order conditional estimate and the fixed-order finite-flight estimate above answer different questions and retain their different hypotheses.

The density algebra has a separate, simpler role. On a contact-centered positive square,

$$1 - \frac{f(u, v)f(0, 0)}{f(u, 0)f(0, v)} = t(u)t(v), \quad t(u) = \frac{S(u)}{d - S(u)}.$$

A nonzero scalar anchor fixes the positive normalization of t , and $S = dt/(1 + t)$ recovers the signed action. This eliminates unknown separate factors; it is not a new general association invariant. The signed finite-jet inverse is what turns that invariant into boundary geometry. Equality of all smooth jets alone is not identified with equality of arbitrary smooth germs. Analyticity enters in the next step, to recover complete obstacle images.

For fixed contact conventions, the sequence of recoveries is therefore

$$(g, f_0, f_1) \mapsto (g, S_0, S_1) \mapsto (j_0^M \psi_0, j_0^M \psi_1), \quad M \geq 2.$$

The first arrow is an explicit operation on the interaction of the law; the second is the inverse of a well-defined map on actual smooth jets. On an analytic inverse neighborhood the second arrow recovers the entire contact pair. This formulation does not identify arbitrary smooth germs with their jets, nor does it assert that every pair of observed densities is realized by a billiard. The theorems reconstruct within the specified physical class.

1.5. The observation and the main theorem. A table T consists of N labelled strictly convex analytic obstacles $K_1, \dots, K_N$ and their translates by $L\mathbb{Z}^2$, where $L \in \text{GL}^+(2, \mathbb{R})$ is unknown. Distinct lifted closures are disjoint and all curvatures are positive. The basis of the abstract deck group $\mathbb{Z}^2$ is marked; its Euclidean realization is not. A selected clear channel $e = (a, b, \ell)$ joins the closest pair of K_a and $K_b + L\ell$. Each channel has its own oriented contact frame and signed transverse coordinate, with no inter-channel registration. The full geometric conventions are stated in Section 2 and Section 17.

For each channel retain its gap g_e and, for each starting contact type $b \in \{0, 1\}$, the limiting conditional joint law $P_{e,b}$ of the two signed transverse endpoints of an even bridge at one known positive excess time d_e . The intrinsic datum is

$$\mathcal{D}_E(T) = (g_e, P_{e,0}, P_{e,1})_{e \in E}. \quad (1.8)$$

Obstacle, channel and deck marks and the signed conventions are design information. No longitudinal position, residual time, waiting count, curvature, flux amplitude, Euclidean deck vector or inter-channel pose is supplied. A law is a function-valued datum, not one scalar sample. The gap is retained separately from the conditional laws.

Let $\mathcal{C}_N$ be the class in which every obstacle is noncircular, and let $\mathcal{A}_N \subset \mathcal{C}_N$ be the class in which every obstacle has trivial proper Euclidean symmetry. The latter is open and dense in the relative C^2 analytic-support topology (Proposition 20.6). A *common-strip analytic-support family* is C^1 into the Banach space of bounded holomorphic support functions on one fixed complex strip, with C^1 lattice columns; see (22.1). In particular the parameter derivative is itself analytic on a smaller strip. The differential statements below do not merely assume that each parameter slice is analytic.

Theorem A (Single-offset periodic rigidity). *For each $T_0 \in \mathcal{C}_N$ there is a persistent selected design E of $N + 1$ channels, consisting of an obstacle spanning tree and two independent marked cycle gains, and a common sufficiently small positive offset, with the following properties.*

- (1) *The datum $\mathcal{D}_E(T_0)$ determines a finite set of complete labelled periodic tables, including their marked lattices, modulo one common proper Euclidean motion. This set is obtained by finite choices of incidence congruences followed by explicit lattice reconstruction and geometric compatibility tests. If $T_0 \in \mathcal{A}_N$, it consists of one table.*
- (2) *For a common-strip analytic-support variation through T_0 , if all gap derivatives and all endpoint-density derivatives on contact-centered positive interior squares vanish, the table variation is one common infinitesimal proper Euclidean motion. Conversely such a motion has zero intrinsic derivative. In a fixed Euclidean gauge the data derivative is injective.*
- (3) *On every immersed p -dimensional C^1 common-strip analytic-support model in that gauge, $p \geq 1$, one can select p scalar functions, each a gap or an expectation $\int \phi dP_{e,b}$ with $\phi \in C_c^\infty$ supported in a positive interior square, which form local C^1 coordinates at T_0 . Their inverse is locally Lipschitz, with local control of every fixed real support norm and of the lattice matrix.*

Proof. Theorem 20.3 supplies the persistent design. The signed law inverse gives complete analytic channel-frame images. Theorem 21.3 enumerates the finite matching alternatives and reconstructs their lattices; the asymmetric uniqueness conclusion is also Theorem 20.7. Theorems 22.5 and 22.6 prove the second and third assertions. Their local registration uses Lemma 21.2, not a choice of one globally preferred symmetry branch. $\square$

The finite bound in the first assertion is a product of orders of obstacle rotation groups along a spanning tree of channel incidences ((21.3)); it is not a sharp degree assertion. The $N + 1$ count is minimal for the stipulated connected two-cycle architecture, not an information lower bound over arbitrary experiments. The selected design and its marks are not discovered by an unmarked trajectory algorithm. The scalar coordinates in the third assertion may depend on the base table and model; finite dimensionality is not a restriction on the first two assertions.

1.6. Unknown origins and separate recording efficiencies. The contact-centered formulation isolates the geometric inverse. Its origins can themselves be recovered from the law, even when its two endpoint amplitudes are independently distorted. Suppose the recorded coordinates are $x = u + \xi$ and $y = v + \eta$, with unknown constants ξ, η , and successful records are retained with a positive factor $e_1(x)e_2(y)$. The physical signed units and directions are unchanged; neither an unknown clock nor an unknown coordinate scale is included in these constants. Section 15 proves the following extension using the interaction of the two endpoints.

Theorem B (Rigidity with unknown transverse origins and weights). *In the geometric setting of Theorem A, retain the same marks, gaps and known positive offsets, but replace every ideal same-type law by its translated law with unknown positive smooth separate endpoint recording factors. Assume that the complete positive cap is recorded and that the numerical coordinates*

have the specified signed transverse units and directions. Then the laws determine their two transverse origins and the signed action germs. The compatible table fiber is exactly the one in Theorem A(1), including the unknown marked lattices. In a differentiable table-and-recording family of the stated smooth class, no table variation outside the common Euclidean tangent space can be masked by varying those origins or separate recording factors.

Proof. Lemma 15.1 extracts the origins from the zero fibers of $\partial_x \partial_y \log \tilde{f}$. Theorem 15.2 then recovers the action by its scalar-anchored four-density ratio. Proposition 15.3 justifies the finite-order differentiable extraction, and Theorem 15.4 proves the exact-fiber and joint derivative kernel assertions through the smooth contact inverse and periodic cochain. $\square$

The cancellation underlying this extension is a classical association invariant [14, 15]. What the boundary-law mechanism supplies is its specific form

$$\partial_x \partial_y \log \tilde{f} = -\frac{S'(x - \xi)S'(y - \eta)}{[d - S(x - \xi) - S(y - \eta)]^2}.$$

Strict convexity locates the two origins, and the signed contact inverse turns the recovered action into geometry. No equality of the two recording profiles is required. This is an exact-law reconstruction with a local finite smooth stability estimate, not a claim that a fixed finite vector of noisy observations or the old preparation budget suffices for arbitrary unknown recording efficiencies. Theorem A(3) remains a separate statement on its stipulated finite-dimensional ideal-law model.

For the complete-cap observation in this statement, the origins can also be recovered without a density derivative: the vertical and horizontal fiber widths of the positive support have their unique maxima at the two origins (Lemma 15.5). Thus centering a complete cap is not, by itself, a new geometric rigidity mechanism. The benefit of the interaction construction is its dependence on a fixed positive interior rectangle. Theorem 15.6 makes this locality explicit: a known rectangular recording window containing both unknown origins, compactly inside the positive cap, suffices even when the cap boundary and discarded mass are not observed. Its positive support is the same window for every admitted law. The signed action is nevertheless recovered from the interior density, and Corollary 15.7 gives exactly the same analytic table fiber and projected derivative kernel. The visibility of the two origins remains a hypothesis; there is no unmarked search for that window.

The separate-factor cancellation is already the calculation of Theorem 14.1, with no equality of the two factors needed. The interior extension uses this algebra rather than inventing another association invariant. Its regularity is the local C^{m+3} -to- C^m estimate of Proposition 15.3, now on a window not reaching the cap boundary. A complete support can center the law but need not determine its action; the functional comparison in Proposition 14.13 isolates that distinction. None of these statements provides a total-variation estimate or a uniform preparation bound for an unconstrained recording class.

1.7. The recovered object and the supplied information. The observation in Theorems A and B is local in each channel but function-valued. It supplies the marked channel design, the signed physical transverse units, gaps and positive offsets. It does not supply a lattice Gram matrix, a common pose for the channel frames, normal collision coordinates, or the separate flux amplitudes. No claim that $N + 1$ laws constitute $N + 1$ numerical samples is needed for either theorem. For interior windows, both critical lines must remain visible; the full cap boundary and discarded mass are not supplied.

This information permits a precise hierarchy of conclusions. The relative and actual-smooth mechanism is local and requires only the stated smooth bounds. Analyticity propagates the recovered contact germs into complete obstacle images. Noncircularity makes global congruence

choices finite, whereas proper asymmetry makes them unique. One nonzero support harmonic determines angular velocity along an actual local alignment branch, so finite symmetry produces no extra infinitesimal kernel. The lattice is then recovered from two independent marked cycle translations using their actual integer gain matrix. The finite-dimensional scalar-coordinate conclusion is a consequence of this geometric kernel theorem on a stipulated immersed model.

The nonlinear content is not encoded by the rarity of preparation alone. The analytic family in Theorem A.4.2 fixes the gap, free area, contact curvatures and every leading endpoint covariance and count amplitude, while its fourth contact jet is $3 - 24s$. The signed block inverse and Theorem 14.1 therefore separate its conditional laws at a fixed sufficiently small positive offset. Thus the full law distinguishes geometries that its leading quadratic record does not. The relative estimate supplies that nonlinear record at long flight number; the smooth remainder factorization licenses its geometric inversion. These two analytic steps, rather than the subsequent finite congruence bookkeeping or finite-dimensional coordinate selection, are the mechanism of the inverse.

For the leading-data comparison the geometric distinction is explicit. The area-preserving support family in Theorem A.4.2 has $g = 1$, $A = 12 - \pi$ and $\kappa_0 = \kappa_1 = 1$, while its fourth graph derivative is $q_s = 3 - 24s$. The normalized onset response satisfies

$$\left. \frac{d}{ds} \partial_d \mathcal{F}^{(s)}(0) \right|_{s=0} = \frac{\sqrt{3}}{2}.$$

The full proof computes both the action and determinant contributions; it does not infer this coefficient from an uncontrolled absolute error. Independently, the signed action inverse separates the fixed-positive-offset endpoint laws when the fourth contact jets differ. The example establishes strict additional geometric information beyond the leading record. It does not compare the information of endpoint laws with that of a marked length spectrum, and its symmetric realization is not an assumption in the general signed inverse.

The three most direct comparison cases are proved in the article. Theorem A.4.2 exhibits an analytic family with the same leading endpoint metrics and count amplitudes but different nonlinear laws. Proposition 21.5 gives a noncircular datum with exactly two complete periodic realizations and different marked Gram forms; Proposition 21.6 separates those two branches with one additional marked gap. The circular comparison in Proposition 21.4 instead exhibits a continuous lattice ambiguity. These examples distinguish nonlinear information, finite exact ambiguity, and an infinitesimal symmetry; they are not three counterexamples to the structural theorem.

The periodic conclusion addresses a further question after local reconstruction. The recovered obstacle images lie in separate channel frames, not in one supplied Euclidean frame. Noncircularity makes the possible incidence congruences finite; the marked cycle gains then turn the recovered displacement cochain into the unknown lattice. Thus the lattice is an output, not hidden metric information in the channel marks. The congruence and cochain arguments are consequences of the local reconstruction, rather than independent substitutes for its analytical content. Their complete proofs also show why finite rotational symmetry and a continuous infinitesimal ambiguity must be distinguished.

1.8. Relation to earlier inverse problems. Separate-factor cancellation belongs to the association framework of Holland and Wang [14] and Osius [15]. The present dynamical input is the limiting density (1.3) with its actual nonlinear actions, and the geometric input is the signed smooth inverse. These are different assertions from the algebraic invariance of an odds ratio.

Finamore and Leguil [4] determine finite-horizon Sinai billiards from an enriched marked length spectrum. That enriched datum includes the auxiliary closed-geodesic lengths in their definition

and is not identified here with ordinary periodic-orbit marked lengths. De Simoi, Kaloshin and Leguil [2] treat analytic open billiards under non-eclipse, symmetry and genericity hypotheses. No reduction between either marked-length observation and the present endpoint-law observation is used. The fifth version of Florio and Leguil [3] removes an earlier geometric spectral-rigidity assertion affected by an error while retaining dynamical conclusions; that removed assertion is not a premise here. Our comparison concerns what is observed and what is recovered, not an assertion that the present theorem removes hypotheses from those results.

Zelditch's inverse spectral theorem provides a useful orbit-local comparison [20]. Theorem 1.1 of that paper determines a simply connected analytic plane domain in the class $\mathcal{D}_{1,L}$ from its Dirichlet or Neumann spectrum. This class includes an involution reversing a nondegenerate bouncing-ball orbit, multiplicity one for every iterated length, and the additional nonresonance and endpoint conditions specified there. The proof recovers boundary Taylor coefficients from localized wave-trace invariants. The symmetry relates the two boundary graphs. Thus neither localization near a periodic orbit nor all-order boundary recovery, by itself, distinguishes the problem considered here.

The distinction lies in the observation map and in the passage from a long bridge to its nonlinear endpoint action. Our datum is a controlled conditional law obtained from phase-volume preparations, not a Laplace spectrum or a sequence of wave invariants. The selected bridge becomes exponentially rare, but its endpoints remain on a fixed collar at a fixed positive offset. Theorem 7.2 compares the mixed endpoint derivative with its exponentially small reference value in *relative* size. An absolute error bound would not suffice for the physical conditional normalization. The resulting law retains the nonlinear half-line actions; separate-factor cancellation then extracts those actions from the density. It is the relative limit, rather than the cancellation identity alone, that connects an observable law to the geometric inverse.

The signed contact inverse treats both anchored graphs without imposing a symmetry between them. Its finite-order content is a statement about actual smooth graphs: Lemma 13.4 controls functional remainders by finite-truncation stationarity before passage to any jet coordinates. Analyticity enters the subsequent determination of a complete germ and its continuation, not that smooth factorization. The function-space inverse controls the full coupled action map, including its lower-order terms. With an outer analytic prior, its real-observation consequence estimates the complete germ on a smaller disc. The finite preparation theorems further distinguish the exactly calibrated rate from the rate that charges the position pilot. Their minimum-distance construction is an existence result for a measurable estimator; it does not supply a finite running-time bound for computational reconstruction.

There is also an internal comparison for which a geometric separation is proved, rather than inferred from different descriptions of the data. The family in Theorem A.4.2 has the same gap, area, quadratic contact data and leading count amplitudes, while its positive-offset boundary law detects a varying fourth-order contact term. This shows that the present law retains information absent from those specified leading data. It is not a separation theorem between the law and a full Laplace spectrum, nor between the law and either marked-length datum discussed above. No such comparison is needed for the relative boundary limit or the contact inverse. These distinctions locate the mathematical contribution in the nonlinear relative-action mechanism and its geometric consequences, without treating different observations as interchangeable.

1.9. Proof organization and the complete technical manuscript. Part I contains the full geometric proof: the phase ensemble and finite bridge, the relative boundary limit, the signed contact inverse, interior interaction, analytic continuation, and periodic registration. The differential theorem follows this same chain. The appendices supply the graph-to-support

conversion, marked source identity and the complete local comparison inverses invoked in the discussion. The principal proof does not use the later stopped-record probability reductions.

The accompanying *complete technical manuscript* retains, in full, the additional local experiments, the finite-resolution inverse, physical calibration, and every historical theorem and auxiliary proof. References prefixed F. point to its independently numbered statements. They are explicit references to that companion, not missing internal numbers. One has a further logical role: Corollary 20.9 is an application of Theorem F.48.3 of the complete technical manuscript. It uses that theorem's acquisition and risk estimates in addition to the geometric results proved here. Its proof verifies the persistent-design hypotheses; it does not reprove the physical pilot or capped estimation theorem. This application is not a premise of Theorem 1.1 or Theorems A and B. The companion's quantified acquisition theory uses additional analytic and sensor bounds; it is not a statistical conclusion under the hypotheses of Theorem A. Its successful-mark, waiting-count and capped-bit experiments are different records of a forward measure, not additional principal rigidity theorems. Their complete proofs remain available without placing those likelihood calculations on the proof path of Theorem 1.1 and Theorems A and B.

1.10. Finite experiments and the complete local analytic target. The relative law and the analytic inverse together give a finite-experiment consequence without truncating the contact geometry. On the local bounded analytic class of Section 14.4, two actual finite-bridge endpoint densities at the fixed offset satisfy

$$\|\psi - \tilde{\psi}\|_r \leq C\{|g - \tilde{g}| + (\|f_j - \tilde{f}_k\|_{C^0(Q)} + e^{-\omega^j} + e^{-\omega^k})^\vartheta\}.$$

The argument compares their realizable limiting laws; it does not regard a finite density as exactly factorized. In an exactly calibrated selected channel experiment, $2N$ independent full-phase preparations, including all failures, yield a measurable estimator of the entire contact pair with confidence error

$$C \left(\frac{\log(C_* N / \alpha)}{N} \right)^{\vartheta \omega / (4\omega + \Gamma)}.$$

Here $e^{-\omega^j}$ is a uniform relative error, Γ bounds the acceptance exponent, and the positive offset and analytic prior are fixed. The flight number is even and proportional to $\log N$. This is an upper bound for complete germs on a smaller disc, not a claim of statistical optimality or a finite-dimensional parametrization. The proof estimates cell densities, fits a realizable limiting law, and then invokes the complete analytic inverse. It charges the exponential rarity rather than replacing raw preparations by free conditional samples.

Estimated calibration adds the separately visible terms $jv_g + v_t$ and r_c/h , where h is the transverse cell size, and its preliminary preparation cap is added to the total charge (Corollary 14.8). This application substantively uses the existing position-sensor calibration theorem; the exactly calibrated power bound is not asserted for its total preliminary cost. All original smooth, global matching, unknown-origin, differential and acquisition results retain their hypotheses and proofs.

With the persistent position pilot and its compact marked-class hypotheses, Corollary 14.9 further gives error $C\{[\log(CB/\alpha)]^2/B\}^{\vartheta \omega / (12\omega + \Gamma)}$ with confidence $1 - \alpha$ and at most B preparations in total. Its slower power includes the entire onset scan and coordinate calibration. This remains a local analytic-germ estimate, not a quantitative assertion for unmarked discovery or global continuation.

Part 1. The relative law and the complete geometric inverse

2. GEOMETRIC SETTING AND THE PRINCIPAL RELATIVE THEOREM

Fix a full-rank lattice $\Lambda \subset \mathbb{R}^2$ and finitely many bounded C^∞ strictly convex obstacles $D_1, \dots, D_s$ with positive curvature. Distinct lattice translates have disjoint closures. Let $A > 0$ be the free area per lattice cell and prepare the unit-speed billiard with normalized phase volume $dq d\vartheta/(2\pi A)$ on the torus complement. We observe the collision count and, when specified, the first and last collision positions. Finite horizon, congruence and symmetry are not assumed for the general forward or boundary-law results.

Consider a fixed configuration and a smooth finite-dimensional family $D_{a,\xi}$ through it, with ξ in a compact set K_0 contained in a sufficiently small parameter neighborhood. At every fixed derivative order, constants may depend on the corresponding higher norms of this family, on positive separation and curvature bounds, and on the lattice, but not on the collision order.

Let $\mathcal{E}$ be the finite set, modulo common lattice translation, of oriented closest pairs attaining the minimum gap at the reference configuration. An index $e = (a, b, \ell)$ represents a segment from $D_{a,\xi}$ to $D_{b,\xi} + \ell$; reversing it gives $\bar{e} = (b, a, -\ell)$. Write

$$\begin{aligned} g_* &= \min_{e \in \mathcal{E}} g_e, & c_{e,b} &= 1 + g_e \kappa_{e,b} \quad (b = 0, 1), \\ c_e &= \sqrt{c_{e,0} c_{e,1}}, & \gamma_e &= \operatorname{arcosh} c_e, & d_{j,e} &= t - j g_e. \end{aligned} \quad (2.1)$$

Here $\kappa_{e,0}$ and $\kappa_{e,1}$ are the positive curvatures at the two contact points and need not be equal. A channel event $E_{j,e}(d)$ consists of the $j+1$ impacts alternating between the specified facing patches in the time window $t = j g_e + d$. These patches are fixed around the reference contacts before observation and define a measurable part of the physical record. The localization theorem below proves that the relevant pairs continue smoothly and that a common onset collar is available.

Parameter derivatives are multi-index derivatives in a fixed Euclidean parameter chart. Unless a statement explicitly says otherwise, every geometric derivative is taken in the *channel-centered* coordinates (ξ, d) , with

$$t = j g_e(\xi) + d, \quad (2.2)$$

so the offset d is held fixed while the physical time moves with the channel onset. In particular the mixed geometric derivatives in Theorem 2.1 are not derivatives at fixed physical time. This convention also applies to the parameter derivatives of $S_b, B_b, \mathcal{F}_p$ and to the uniform constants below. Matrix norms are Euclidean operator norms unless a trace norm is specified.

2.1. Local multiplier convention. Suppress the channel index for the local hyperbolic calculations and set

$$\begin{aligned} \varrho &= e^{-\gamma}, & \lambda &= \varrho^2 = e^{-2\gamma}, \\ \mathfrak{r}_b &= \sqrt{c_b/c_{1-b}}, & r_b &= \mathfrak{r}_b \varrho \quad (b = 0, 1). \end{aligned} \quad (2.3)$$

The number $\mathfrak{r}_b$ is a curvature ratio, whereas r_b is the derivative at zero of the physical first-hit map φ_b . Thus $\mathfrak{r}_0 \mathfrak{r}_1 = 1$, $r_0 r_1 = \lambda$, and $R_b = \varphi_{1-b} \circ \varphi_b$ has multiplier λ . The integer j counts flights; n iterations of R_b mean $j = 2n$. The symbol ρ in exponential estimates denotes a freely chosen strict decay margin, not the exact factor ϱ .

2.2. The nonlinear relative law. Fix an oriented channel e and abbreviate $g = g_e$, $\gamma = \gamma_e$. Use transverse endpoint coordinates u, v at its two contacts. Let $W_j(u, v)$ be the stationary length of its j alternating flights, $E_j = W_j - j g$, $d_j^0 = -W_{j,uv}(0, 0)$, and $b_j = -W_{j,uv}/d_j^0$. For $p = j \bmod 2$ put

$$a_b = \frac{\sqrt{c_0 c_1} \sinh \gamma}{g c_{1-b}}, \quad q_p = \sqrt{a_0 a_p}, \quad d_j^0 = \frac{q_p}{\sinh(j\gamma)}.$$

Theorem 2.1 (Relative boundary law). *There are positive smooth functions B_b and strictly convex smooth functions S_b , $b = 0, 1$, with*

$$S_b(0) = S'_b(0) = 0, \quad S''_b(0) = a_b, \quad B_b(0) = 1,$$

a common endpoint box, a number $d_ > 0$, and $0 < \tau < 1$ such that, for every fixed derivative order k and every $j \geq 1$,*

$$\|E_j - S_0 \oplus S_p\|_{C^k} + \|b_j - B_0 \otimes B_p\|_{C^k} \leq C_k \tau^j, \quad p = j \bmod 2. \quad (2.4)$$

The action difference vanishes to second order at the origin. Define

$$\mathcal{F}_p(d) = \frac{q_p}{\pi d^2} \int (d - S_0(u) - S_p(v))_+ B_0(u) B_p(v) du dv. \quad (2.5)$$

Then $\mathcal{F}_p$ extends smoothly to $[0, d_]$, is positive there, and has value one at zero. The exact selected-channel probabilities obey*

$$\left\| \frac{2A \sinh(j\gamma)}{d^2} \Pr(E_{j,e}(d)) - \mathcal{F}_p(d) \right\|_{C^k([0, d_*])} \leq C_k \tau^j. \quad (2.6)$$

The norm includes every fixed mixed geometric and offset derivative in the centered convention (2.2). The limiting conditional endpoint-residual law at $d > 0$ has density proportional to

$$B_0(u) B_p(v) \mathbf{1}_{\{r > 0, S_0(u) + S_p(v) + r < d\}}.$$

All assertions hold uniformly on the compact local families specified above. For analytic families the endpoint constructions are analytic.

Proof and dependencies. Lemma 5.2 constructs the finite bridge and its relative determinant. Lemma 7.1 constructs S_b, B_b with the normalizations above. Theorem 7.2 proves (2.4) by comparing two separated perturbation blocks in trace norm, including mixed derivatives. Theorem 7.3 proves (2.6) using the common Morse domain of Lemma 6.1 and identifies the conditional density. The full proofs occur in Part I. $\square$

3. UNIFORM THRESHOLDS AND CONDITIONAL CONTACT METRICS

Theorem 3.1 (Uniform thresholds for periodic dispersing configurations). *There is $\varepsilon_0 > 0$, independent of every integer $j \geq 1$, such that $\varepsilon_0 < \min_{K_0} g_*/4$ and, for $0 \leq t - jg_* < \varepsilon_0$,*

$$\Pr_\xi(N_t \geq j+1) = \Pr_\xi(N_t = j+1) = \sum_{e \in \mathcal{E}} \frac{d_{j,e,+}^2}{2A(\xi) \sinh(j\gamma_e(\xi))} [1 + d_{j,e,+} \mathcal{R}_{j,e}(\xi, d_{j,e,+})]. \quad (3.1)$$

Every fixed mixed derivative of $\mathcal{R}_{j,e}$ is bounded uniformly in j ; it has a smooth right extension at offset zero. For $t \leq jg_$ the probability is zero. At the selected channel's own window $t = jg_e + d$, $0 < d < \varepsilon_0$, the same individual summand is exactly $\Pr_\xi(E_{j,e}(d))$, whether or not e minimizes the gap.*

The exponent two is independent of j , but its coefficient is exponentially small. The essential estimate is relative to that coefficient. The proof uses a weighted Dirichlet construction and a trace-norm estimate for its relative determinant, rather than a long initial-value iteration. The local hypotheses for this construction are verified at every reference configuration above. In the triangular one-obstacle near-circle class, $\mathcal{E}$ consists of the six unit lattice directions, so (3.1) is its specialization, with all competing onsets.

For the next theorem, lift the first and last collision positions to the fixed contact patches in $\mathbb{R}^2$, and denote them by Q_0, Q_j . Define the two scalar conditional variances

$$V_{j,b}(d) = \frac{1}{d} \left\{ \mathbb{E}(|Q_b|^2 \mid E_{j,e}(d)) - |\mathbb{E}(Q_b \mid E_{j,e}(d))|^2 \right\}, \quad b = 0, j. \quad (3.2)$$

These are translation- and rotation-invariant within the chosen lifts. Their definition requires neither the unknown contact point nor its tangent or curvature. The gap g_e is identified by its channel onset; quantitative timing calibration is a separate experiment below.

Theorem 3.2 (Uniform two-variance curvature recovery). *For every odd $j \geq 1$ the right limits $v_{j,0}, v_{j,1}$ of $V_{j,0}(d), V_{j,j}(d)$ exist, and*

$$(v_{j,0}, v_{j,1}) = \frac{g_e \coth(j\gamma_e)}{3 \sinh \gamma_e} \left(\sqrt{\frac{c_{e,1}}{c_{e,0}}}, \sqrt{\frac{c_{e,0}}{c_{e,1}}} \right). \quad (3.3)$$

Each difference $V_{j,b}(d) - v_{j,b}$ is d times a smooth function with uniform fixed-order derivative bounds. At known g_e , these two limits determine the ordered pair $(\kappa_{e,0}, \kappa_{e,1})$ without knowing A . The inverse is given by

$$\begin{aligned} \frac{\coth(j\gamma)}{\sinh \gamma} &= \frac{3}{g} \sqrt{v_0 v_1}, & r &= \sqrt{v_1/v_0}, \\ \kappa_0 &= \frac{r \cosh \gamma - 1}{g}, & \kappa_1 &= \frac{r^{-1} \cosh \gamma - 1}{g}. \end{aligned} \quad (3.4)$$

On every compact positive gap-curvature box this observation map and its inverse are Lipschitz, with constants independent of odd j . They remain uniformly conditioned at $\kappa_0 = \kappa_1$. If both measured variances have error at most δ , a reconstruction constrained to such a box has curvature error at most $C(d + \delta)$.

Thus the full conditional metric retains information lost by the scalar flux coefficient. Theorem A.3.3 realizes the scalar ambiguity by actual smooth periodic obstacles with the same gap, area and every leading count amplitude, while their endpoint curvatures vary. The two variances distinguish that family. The statement is local contact-curvature recovery, not global shape rigidity. Conditioning on an exponentially rare event is not free: Proposition A.3.2 records the number of independent physical preparations required.

4. SHORTEST CHANNELS IN ARBITRARY PERIODIC CONFIGURATIONS

The relevant localization is a consequence of separated strict convexity, not proximity to a particular model shape.

Lemma 4.1 (A geometric criterion for the complete event). *At any reference configuration of Section 2, the minimum distance between distinct lifted obstacles is positive and is attained by finitely many oriented pairs modulo common translation. Every such pair has a unique closest segment, positive-definite distance Hessian and positive clearance from the remaining obstacles. Its contact points, length and curvatures continue smoothly under small smooth deformations. There are fixed contact and direction neighborhoods and $\varepsilon_0 > 0$ such that every physical sequence with j complete flights and total length at most $jg_* + \varepsilon_0$ alternates in one of these neighborhoods. The constants are independent of j .*

Proof. Choose bounded representatives of the finitely many obstacles. The distance between D_a and $D_b + \lambda$ tends to infinity with $|\lambda|$, uniformly in a, b . Only finitely many lattice translates can have distance below any fixed bound. Each of those finitely many positive distances is attained; disjoint closures make their minimum positive. Among pairs not attaining this minimum, the next distance is separated from it by a positive amount. Pairs outside the finite list already exceed a larger fixed bound. These statements persist under a small Hausdorff perturbation, with a strict gap to all pairs outside $\mathcal{E}$.

The closest displacement vector between two disjoint convex bodies is unique: if two different displacement vectors both minimized its norm, their midpoint would have smaller norm and

would still be a displacement. For that vector, each endpoint is a support point in its normal direction. Strict convexity makes those support points unique. Rotate the segment to $[0, g]$ and use a common transverse coordinate. The facing graphs are $(-\psi_0(u), u)$ and $(g + \psi_1(v), v)$, with $\psi_b(0) = \psi'_b(0) = 0$ and $\psi''_b(0) = \kappa_b > 0$. The distance Hessian at its minimum is

$$\frac{1}{g} \begin{pmatrix} 1 + g\kappa_0 & -1 \\ -1 & 1 + g\kappa_1 \end{pmatrix}. \quad (4.1)$$

Its determinant and its diagonal entries are positive. The implicit function theorem gives smooth contact points and distance. Compactness keeps both eigenvalues uniformly positive and all chosen charts uniform.

A reference minimum segment cannot meet a third obstacle in its open interior: the initial part would give a smaller distance from its first obstacle to that third obstacle. The closed segment also stays a positive distance from every other obstacle, since its endpoints belong only to its own two disjoint bodies and only finitely many other bodies are nearby. Small contact patches and small geometric perturbations preserve this clearance. Strict convexity makes each connecting chord leave the first obstacle and enter the second transversely, with no earlier hit. Thus these local chords are physical first hits.

At a fixed outgoing obstacle, distinct minimizing channels give distinct normal outgoing collision states. If two shared a contact and outgoing normal, their other contacts at the same distance would coincide, contradicting disjointness. The finite collection of states is therefore separated. Choose their neighborhoods so small that normal reflection of a flight arriving along e can belong to a short outgoing channel only in the neighborhood of $\bar{e}$. This property is open and uniform.

Every physical free flight joins distinct lifted obstacles: a straight ray leaving a convex body cannot return to it before another hit. Hence its length is at least g_* . If a sum of j such lengths is at most $jg_* + \varepsilon_0$, every summand is at most $g_* + \varepsilon_0$. The gap to the other pairs restricts that summand to $\mathcal{E}$. Away from the unique contact patches the distance to that pair's minimum is bounded below; decrease ε_0 to exclude this complement. Each short flight is then nearly normal. The separated outgoing states and specular reflection force every following short flight to be the reverse channel. Induction uses the same neighborhoods at each impact and introduces no length-dependent constant. Grazing states and their regular preimages have phase measure zero. $\square$

The lemma also applies to a specified nonminimal closest chord with positive clearance, for its selected local itinerary. It need not then exhaust the whole count event. All original six near-circle channels satisfy the criterion. For example, the criterion also applies to arbitrarily eccentric ellipses and to noncongruent obstacles on any fixed lattice, provided the indicated disjointness and positive curvature conditions hold; there is no small eccentricity assumption.

For one channel the exact one-flight actions are

$$\ell_b(u, v) = \sqrt{(g + \psi_b(u) + \psi_{1-b}(v))^2 + (v - u)^2}, \quad b = 0, 1. \quad (4.2)$$

Uniformly on a small fixed box,

$$\ell_b(u, v) = g + \frac{c_b u^2 + c_{1-b} v^2 - 2uv}{2g} + O((|u| + |v|)^3), \quad (4.3)$$

$$\ell_b(u, v) - g \geq \psi_b(u) + \psi_{1-b}(v) \geq c_{\text{loc}}(u^2 + v^2). \quad (4.4)$$

The full broken-action Hessian has strictly positive diagonal-dominance margins: κ_b in endpoint rows and $2\kappa_b$ in interior rows at zero. Every row perturbation involves only neighboring coordinates. The same small box therefore gives strict convexity at every length.

5. A PARAMETER-DEPENDENT BOUNDARY-VALUE AND RELATIVE-FLUX LEMMA

Let $W_{j,e}(\xi, u, v)$ be the stationary value of the sum of (4.2), with $y_0 = u, y_j = v$. We construct that value on one common neighborhood. Endpoint Hessians are denoted by H ; scalar relative remainders are denoted by $\mathcal{R}$.

Lemma 5.1 (Alternating Jacobi reduction). *Put $c = \sqrt{c_0 c_1}$, $\gamma = \operatorname{arcosh} c$, $\sigma_i = \sqrt{c_{1-(i \bmod 2)}}$, and $D_j = \operatorname{diag}(\sigma_0, \sigma_j)$. The quadratic endpoint Hessian is*

$$H_{j,e} = \frac{c \sinh \gamma}{g} D_j^{-1} \begin{pmatrix} \coth(j\gamma) & -\operatorname{csch}(j\gamma) \\ -\operatorname{csch}(j\gamma) & \coth(j\gamma) \end{pmatrix} D_j^{-1}. \quad (5.1)$$

Consequently

$$d_{j,e}^0 := -W_{uv}(\xi, 0, 0) = \frac{c \sinh \gamma}{g \sigma_0 \sigma_j \sinh(j\gamma)}, \quad \sqrt{\det H_{j,e}} = \frac{c \sinh \gamma}{g \sigma_0 \sigma_j}, \quad \frac{d_{j,e}^0}{\sqrt{\det H_{j,e}}} = \operatorname{csch}(j\gamma). \quad (5.2)$$

The endpoint eigenvalues have positive uniform upper and lower bounds. The period-two return multiplier is $e^{2\gamma}$.

Proof. The interior equation is $y_{i+1} - 2c_{i \bmod 2} y_i + y_{i-1} = 0$. Set $y_i = \sigma_i z_i$. Since $c_{i \bmod 2} \sigma_i / \sigma_{i+1} = c$, the recurrence becomes $z_{i+1} - 2c z_i + z_{i-1} = 0$ and the whole quadratic action becomes

$$\frac{c}{2g} \sum_{i=0}^{j-1} [c(z_i^2 + z_{i+1}^2) - 2z_i z_{i+1}].$$

Its endpoint solution is $z_i = \sinh((j-i)\gamma) z_0 / \sinh(j\gamma) + \sinh(i\gamma) z_j / \sinh(j\gamma)$. The first endpoint derivative is $c(c z_0 - z_1)/g$. The identity $c \sinh(j\gamma) - \sinh((j-1)\gamma) = \sinh \gamma \cosh(j\gamma)$, and its reversed version, give (5.1). Now use $\coth^2 - \operatorname{csch}^2 = 1$. The middle matrix has eigenvalues $\tanh(j\gamma/2)$ and $\coth(j\gamma/2)$. Since $\gamma \geq \gamma_* > 0$ and the σ_i have uniform positive bounds, this proves the eigenvalue bounds. The two-periodic scaling conjugates the two-step recurrence to the square of $\begin{pmatrix} 2c & -1 \\ 1 & 0 \end{pmatrix}$. The nonzero one-flight twist $1/g$ identifies position-pair and collision position-momentum coordinates, so its unstable eigenvalue is the stated physical return multiplier. In particular the formula at $j = 1$ is exactly (4.1). $\square$

Lemma 5.2 (Uniform nonlinear bridge and relative determinant). *Suppose g and the two graph functions in (4.2) range over a compact smooth parameter family, with g, κ_0, κ_1 bounded above and bounded away from zero. There is a common endpoint neighborhood on which the stationary segment exists uniquely and is smooth. For some fixed $\rho < 1$, with $a = |u| + |v|$ and $w_i = \rho^i + \rho^{j-i}$,*

$$y_i^{\text{lin}} = \sigma_i \left[\frac{\sinh((j-i)\gamma)}{\sinh(j\gamma)} \frac{u}{\sigma_0} + \frac{\sinh(i\gamma)}{\sinh(j\gamma)} \frac{v}{\sigma_j} \right], \quad (5.3)$$

$$|y_i| \leq C a w_i, \quad |y_i - y_i^{\text{lin}}| \leq C a^2 w_i,$$

$$W_{j,e} - jg = \frac{1}{2}(u, v) H_{j,e}(u, v)^t + O(a^3),$$

$$-W_{uv}(\xi, u, v) = d_{j,e}^0(\xi) b_{j,e}(\xi, u, v), \quad b_{j,e}(\xi, 0, 0) = 1, \quad b_{j,e} = 1 + O(a) > 0. \quad (5.4)$$

Every fixed mixed derivative of y_i and $y_i - y_i^{\text{lin}}$ is bounded by $C_k w_i$ times respectively $a^{\max(1-|\beta|, 0)}$ and $a^{\max(2-|\beta|, 0)}$, where β counts endpoint derivatives. The action remainder and $b_{j,e}$ have uniform fixed-order bounds. If both graphs are even, the bridge correction is $O(a^3 w_i)$, the action remainder is $O(a^4)$, and $b_{j,e} - 1 = O(a^2)$ is even. In an analytic family these bounds hold on a common complex neighborhood and are analytic bounds.

Proof. Let H_{int}^0 be the quadratic interior Hessian. Its inverse, for $1 \leq i \leq k < j$, is

$$G_{ik} = \frac{g\sigma_i\sigma_k}{c} \frac{\sinh(i\gamma) \sinh((j-k)\gamma)}{\sinh \gamma \sinh(j\gamma)}, \quad G_{ki} = G_{ik}. \quad (5.5)$$

Substitution verifies the homogeneous recurrence away from $i = k$, zero endpoint values and the unit jump at $i = k$. In particular $|G_{ik}| \leq Ce^{-\gamma^*|i-k|}$. Choose $e^{-\gamma^*} < \rho_0 < \rho < 1$. A geometric-series sum yields

$$\sum_k \rho_0^{|i-k|} (\rho^k + \rho^{j-k}) \leq C_\rho w_i. \quad (5.6)$$

For its ρ^k part, split at $k = i$ and sum the ratios ρ_0/ρ and $\rho_0\rho$; the other part follows by reversal. Thus G is bounded independently of j on the norm $\|z\|_w = \max_i |z_i|/w_i$.

Parameter differentiation of (5.5) has the same property. To see this without a hidden growing dimension, write the hyperbolic quotient as a bounded factor times $e^{-\gamma(k-i)}$, using factors $1 - e^{-2r\gamma}$ for its distances to the endpoints. Derivatives of those factors produce $r^m e^{-2r\gamma}$, uniformly bounded for each fixed m ; derivatives of the first exponential produce $|i-k|^m e^{-\gamma|i-k|} \leq C_m \rho_0^{|i-k|}$. All derivatives of the positive σ_i are bounded. The same argument for the endpoint solution gives $|\partial_\xi^\alpha y_i^{\text{lin}}| \leq C_\alpha a w_i$. Consequently every fixed parameter derivative of G is bounded in the weighted norm, with no j factor.

The interior gradient is $H_{\text{int}}^0(y - y^{\text{lin}}) + \mathcal{N}_\xi(y)$, including the fixed endpoint coordinates in the local remainder. Its i th component depends on only three neighboring coordinates and vanishes quadratically there. Adjacent weights have bounded ratios. On $\|y\|_w \leq C_0 a$, its weighted norm is $O(a^2)$ and its derivative has norm $O(a)$. The map $y \mapsto y^{\text{lin}} - G\mathcal{N}_\xi(y)$ preserves this ball and has Lipschitz constant at most $1/2$ when a is uniformly small. This proves existence and the first estimates. The diagonal-dominance bound after (4.4) gives uniqueness throughout the contact box, not just inside the weighted ball.

For clarity, the derivative induction takes place in these same weighted spaces. One derivative of the fixed-point equation gives

$$(I + GD_y \mathcal{N}_\xi) Dy = Dy^{\text{lin}} - (DG)\mathcal{N}_\xi - GD_{\text{explicit}} \mathcal{N}_\xi.$$

Its left inverse has norm at most two. At order k , the only term containing an order- k derivative of y is the same left operator; every other term contains derivatives of G , explicit parameter or endpoint derivatives of the local remainder, and lower derivatives of y . The local chain rule and bounded neighboring-weight ratios bound their products in $\|\cdot\|_w$. Induction proves the estimates at order k . Parameter derivatives preserve vanishing at zero; Taylor's integral formula in (u, v) therefore gives precisely the powers of a stated in the lemma, including the correction of order two.

For every $p > 0$, $\sum_i w_i^p \leq C_p$. Sum the cubic action remainders using this inequality. The linear bridge is stationary for the quadratic action with endpoints fixed, so its change on substituting the nonlinear bridge is $O(a^4)$. This proves the action estimate. The same summability and the derivative induction prove all its fixed-order bounds.

For the relative estimate put $b_i = -\ell_{i \bmod 2, uv}(y_i, y_{i+1})$ and $H_{\text{int}} = H_{\text{int}}^0 + E$. Schur complementation gives

$$-W_{uv} = \frac{\prod_{i=0}^{j-1} b_i}{\det H_{\text{int}}}. \quad (5.7)$$

Indeed, the endpoint-to-interior entries are $-b_0, -b_{j-1}$ and the corner entry of the inverse tridiagonal interior matrix is $b_1 \cdots b_{j-2} / \det H_{\text{int}}$. At $j = 1$ the determinant is empty and equals

one. The local expansions and endpoint weights give

$$\sum_i |\log(gb_i)| \leq Ca, \quad \|E\|_1 \leq \sum_{i,k} |E_{ik}| \leq Ca, \quad \|(H_{\text{int}}^0)^{-1}\| \leq C.$$

The trace-norm bound uses tridiagonality; each matrix unit has trace norm one. Take the endpoint neighborhood so small that $\|(H_{\text{int}}^0)^{-1}E\| < 1/2$. Then

$$\log \frac{\det H_{\text{int}}}{\det H_{\text{int}}^0} = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} \text{tr}[(H_{\text{int}}^0)^{-1}E]^m = O(a). \quad (5.8)$$

The absolute trace is bounded by one trace norm times $m-1$ operator norms. Summing therefore costs $C\|E\|_1$, not the matrix dimension. Divide (5.7) by its value at zero and subtract logarithms to obtain (5.4).

Every differentiated E has uniformly bounded trace norm by the already proved coordinate bounds. The quadratic Hessian and each of its parameter derivatives are uniformly banded with bounded entries; hence derivatives of its inverse have bounded operator norm, by $\partial H^{-1} = -H^{-1}(\partial H)H^{-1}$ and induction. Termwise differentiation of (5.8) is justified by geometric convergence even after its fixed polynomial factors in m . Every term has at least one trace-class factor. The differentiated sum of $\log(gb_i)$ is controlled by the same endpoint weights. This proves all derivative bounds for the logarithmic ratio and its exponential. Positive twists and a positive interior determinant give positivity.

If the graphs are even, the action expansion is even, its nonlinear gradient is cubic, and its Hessian and twist errors are quadratic. The preceding estimates then give the improved orders. Uniqueness implies simultaneous reversal of every coordinate, proving evenness. For a jointly analytic family all local functions extend to a common small complex neighborhood of the fixed compact real parameter set. Shrink it to retain the strict exponential and inverse-norm margins. The same contraction is uniform there. The determinant series remains convergent, and Cauchy estimates on a smaller neighborhood prove the analytic assertion. No assertion about analytic extension is used for a merely smooth family. $\square$

6. FULL-PHASE INTEGRATION AND THE THRESHOLD LAW

Let $P(\xi)$ be the sum of obstacle perimeters in one cell. The collision section measure is $ds dp/(2P)$, the mean roof $\pi A/P$, and the stationary collision intensity $P/(\pi A)$. These identities follow from a flight-tube change of variables: irrational torus lines hit an open obstacle, rational directions form a null angular set, and the positive minimal gap excludes finite-time accumulation. In particular, in first-impact endpoint coordinates the full phase measure is

$$\frac{-W_{uv}}{2\pi A(\xi)} du dv dr. \quad (6.1)$$

Indeed, first variation gives $p = -W_u/(ds/du)$; the arclength factor cancels in $|ds dp| = |W_{uv}| du dv$. The positive relative twist makes the endpoint map injective at fixed u . The localized stationary path is physical by Lemma 4.1. Thus (6.1) is not a substituted transverse ensemble. The total angle-boundary flux is $2P$, and its flight-tube integral is $2\pi A$; this also proves the normalization without an upper roof bound.

Lemma 6.1 (Uniform smooth radial integration). *There are smooth Morse inverses $\Phi_{j,e}(\xi, w)$ on a common ball, with all fixed derivatives bounded uniformly in j , such that*

$$W_{j,e}(\xi, \Phi_{j,e}(\xi, w)) - jg_e = |w|^2/2, \quad D_w \Phi_{j,e}(\xi, 0) = H_{j,e}^{-1/2}.$$

The phase density (6.1) in (w, r) is

$$\frac{a_{j,e}(\xi, w)}{2\pi A(\xi) \sinh(j\gamma_e)} dw dr, \quad a_{j,e}(\xi, 0) = 1, \quad a_{j,e} = 1 + O(|w|) > 0, \quad (6.2)$$

with the same derivative bounds. If a is any such uniformly smooth amplitude, then

$$\frac{2}{\pi} \int_{|z|<1} (1 - |z|^2) a(\xi, \sqrt{2d} z) dz = a(\xi, 0) + dH(\xi, d),$$

where H is smooth for $d \geq 0$ and has uniform fixed-order bounds. No evenness of a or of the scatterer is required.

Proof. For $E = W - jg$, Taylor's integral formula writes $E(y) = \frac{1}{2}y^t M(y)y$, where $M(y) = 2 \int_0^1 (1-s) D^2 E(sy) ds$ is uniformly positive near zero. The map $y \mapsto M(y)^{1/2}y$ has uniformly invertible differential at zero and small nonlinear differential on a common ball. Its inverse is obtained by contraction, then differentiated to each finite order. This construction uses the uniform endpoint Hessian, not the exponentially small twist as an inverse bound. Inserting the Jacobian and (5.2) into (6.1) gives (6.2).

For the last assertion replace $a(w)$ by $(a(w) + a(-w))/2$ in the integral. Every odd Taylor term vanishes. Taylor expansion to any prescribed order, with a sufficiently high differentiated remainder, then yields an expansion in integer powers of d . One can also set $r = \sqrt{d}$ first: the integral is a smooth even function of r with uniform bounds. Successive division of its even Taylor remainder by r^2 shows that it is a smooth function of r^2 on the right. Using two more endpoint derivatives for each division proves the uniform bounds for every fixed derivative of H . $\square$

Proof of Theorem 3.1. In a channel's window $jg_e + d$, each of its j complete flights is at least g_e . Hence the allowed first residual time satisfies $0 < r < d - (W - jg_e) < d < g_*$. The preceding roof cannot truncate this interval, and the next roof cannot fit after the last hit. The first-impact phase formula (6.1) therefore applies to this selected channel at its own threshold: integrating the residual variable gives exactly that interval length. The coercivity (4.4) implies $|y_0|^2 + |y_j|^2 + 2 \sum_{0 < i < j} |y_i|^2 \leq Cd$ on its sublevel. It places the entire active set inside the common Morse coordinates, so no chart-boundary term occurs. Its exact probability is

$$\frac{1}{2\pi A \sinh(j\gamma_e)} \int_{|w|^2 < 2d} (d - |w|^2/2) a_{j,e}(\xi, w) dw.$$

Since the quadratic integral is πd^2 , Lemma 6.1 proves the channel assertion, including its derivative bounds.

For the complete event let $t = jg_* + \varepsilon$ with $0 < \varepsilon < \varepsilon_0$. No $j+2$ hits are possible, since $t < (j+1)g_*$. Lemma 4.1 partitions every contributing j -flight segment into exactly one of the finitely many oriented channels. Its available excess there is $d_{j,e} = \varepsilon - j(g_e - g_*) \leq \varepsilon$. Channels with nonpositive excess contribute zero. Summing the exact channel formulas gives (3.1). The minimum-roof bound proves the negative-side assertion. $\square$

Proof of Theorem A.2.1. Write the mark on a stationary bridge as $w_{j,e} + V_{j,e}$, subtracting the normal value separately at each impact. Chord directions and normal reflections are smooth maps of neighboring transverse coordinates, with uniformly positive chord denominators. Lemma 5.2 therefore gives $|V_{j,e}| \leq C(|u| + |v|)$ and uniform fixed-order bounds for all its derivatives. The normal states now depend on ξ , but each summand's deviation vanishes at $(u, v) = 0$ for every ξ ; parameter differentiation preserves its endpoint localization. The finite chain rule and $\sum_i (\rho^i + \rho^{j-i}) \leq C$ prove the bounds. Multiplying the Morse amplitude by $e^{qV_{j,e}}$ and applying Lemma 6.1 proves (A.2.1).

For the convergence assertion, exact parity counting gives

$$w_{j,e} = (j+1)\bar{f}_e + \mathbf{1}_{\{j \text{ even}\}}(a_{e,0} - a_{e,1})/2. \quad (6.3)$$

For fixed real q the parity factor is bounded above and below by positive constants, as is the normalized integral of $a_{j,e}e^{qV}$ on a sufficiently small collar. Also $\text{csch}(j\gamma_e) \asymp 2e^{-j\gamma_e}$, uniformly in j . Thus a channel's normalized term is comparable to $e^{j(q\bar{f}_e - \gamma_e)}$. Positivity rules out cancellation: equality in (A.2.2) gives a nondecaying term, and a positive exponent gives a growing term. This proves necessity as well as sufficiency, including the right limit at zero. On a compact subset of the strict domain there is a common margin $\delta > 0$. Parameter and source derivatives of the leading factor cost only a polynomial in j ; offset derivatives act on the uniform smooth remainder. Every fixed derivative is bounded by $C(1+j)^m e^{-\delta j}$ and is summable. The vector-source assertion is the same argument with scalar products. $\square$

Corollary 6.2 (Moving threshold interfaces). *Within the collar of (3.1), the probability is a C^1 function of the smooth physical variables (ξ, t) , with smooth pieces separated by $t = jg_e(\xi)$. For an orientation e , along a smooth scalar path $s \mapsto (\xi(s), t(s))$ entering $d_{j,e} > 0$ transversely, its contribution has second-derivative jump from the inactive to the active side*

$$\frac{(t'(s) - j \partial_s g_e)^2}{A(\xi) \sinh(j\gamma_e)}. \quad (6.4)$$

Coincident contributions, including the reversed orientation, add with their respective activation signs. The marked formula has the additional factor $e^{qw_{j,e}}$. One-sided and distributional higher derivatives follow by differentiating the positive-part expression, with polynomial-in- j coefficient bounds.

Proof. Each summand is $d_+^2[C(\xi) + d_+C(\xi)H(\xi, d_+)]$. It and its first derivatives match at zero. Only $2C(\partial_s d)^2$ survives in the second-derivative jump. All other terms contain a factor d . Finite addition proves the claim at intersections. For higher derivatives the ordinary product and chain rules on each side, and the derivatives of d_+^2 in the distributional sense, give the stated formulas; no differentiation of g_* is used. $\square$

In the triangular one-obstacle specialization, fix $R_* \in (0, 1/2)$. For a concrete change of minimizing mechanism, take $h_s(\theta) = R_* + s \cos(2\theta)$. Differentiating the closest distance at the circular normal chord gives $g'_e(0) = -2 \cos(2 \arg e)$: the endpoint variation is the normal support variation, since the unperturbed endpoint gradient is zero. The horizontal pair has derivative -2 and the other two pairs derivative 1 . Reflection symmetry keeps those two lengths equal. Consequently the horizontal pair minimizes for small positive s , while the other two minimize for small negative s . Formula (3.1) continues through $s = 0$. A nonminimal pair contributes exactly while $j(g_e - g_*) < \varepsilon$; the parameter transition scale is therefore ε/j at a transverse length crossing. This assertion is about activation of physical channels, not a new family of symbolic itineraries.

For the conditional record, put $w = \sqrt{2d}z$ and $r = d\eta$. The fixed latent domain and its probability measure are

$$\mathcal{U} = \{(z, \eta) : |z| < 1, 0 < \eta < 1 - |z|^2\}, \quad \text{dm} = \frac{2}{\pi} dz d\eta. \quad (6.5)$$

This notation will also be used for the general record response.

7. TWO NONLINEAR BOUNDARY LAYERS

Fix one oriented channel and suppress its index and the smooth geometric parameter. All bounds below include each fixed number of parameter derivatives on the compact families

specified in Section 2. Shrinking the common contact box is allowed once, independently of the flight number. The functions $E_j = W_j - jg$ and $b_j = -W_{j,uv}/d_j^0$ refer to the actual nonlinear stationary action and its normalized twist, not their quadratic models.

7.1. The half-line construction. For a starting type $b \in \{0, 1\}$ put $\sigma_i^{(b)} = \sqrt{c_{1-((b+i) \bmod 2)}}$. The quadratic half-line interior Hessian acts on sequences indexed by $i \geq 1$, with fixed zero boundary at $i = 0$. Its diagonal entries are $2c_{(b+i) \bmod 2}/g$ and its adjacent entries are $-1/g$. Its inverse is

$$G_{ik}^{(b)} = \frac{g\sigma_i^{(b)}\sigma_k^{(b)}}{2c \sinh \gamma} \left(e^{-\gamma|i-k|} - e^{-\gamma(i+k)} \right), \quad i, k \geq 1. \quad (7.1)$$

The formula follows either from the recurrence and its unit jump or from (5.5) by sending the right endpoint to infinity. It is a bounded positive operator on $\ell^2(\mathbb{N})$, and is bounded on $\sup_i \rho^{-i}|x_i|$ whenever $e^{-\gamma} < \rho < 1$. Every fixed parameter derivative obeys the same assertion after leaving a strict exponential margin.

Lemma 7.1 (Semi-infinite stationary segments). *For $|u| < u_*$ there is a unique small stationary half-line segment $x^{(b)}(u) = (x_i^{(b)}(u))_{i \geq 0}$ with $x_0^{(b)} = u$ and $x_i^{(b)} \rightarrow 0$. It is smooth, and on a common neighborhood*

$$\begin{aligned} x_i^{(b)}(u) &= \frac{\sigma_i^{(b)}}{\sigma_0^{(b)}} e^{-i\gamma} u + O(u^2 \rho^i), \\ |\partial_\xi^\alpha \partial_u^r x_i^{(b)}(u)| &\leq C_{\alpha,r} \rho^i |u|^{\max(1-r,0)}. \end{aligned} \quad (7.2)$$

The boundary action

$$S_b(u) = \sum_{i \geq 0} \left[\ell_{(b+i) \bmod 2}(x_i^{(b)}(u), x_{i+1}^{(b)}(u)) - g \right] \quad (7.3)$$

is smooth and strictly convex near zero. It satisfies

$$S_b(0) = S'_b(0) = 0, \quad a_b := S''_b(0) = \frac{c \sinh \gamma}{g c_{1-b}} > 0. \quad (7.4)$$

These assertions are analytic on a common smaller neighborhood for jointly analytic graph families.

Proof. Use the half-line Green operator in the weighted contraction from Lemma 5.2. The linear solution is the first term of (7.2). The nonlinear gradient is local, vanishes quadratically, and its derivative has norm $O(|u|)$ on the weighted ball of radius $C|u|$. Thus the map has contraction constant at most one half on a common ball. Differentiating its fixed-point equation leaves the same invertible operator on the highest derivative; every other term is a product of a local derivative and weighted lower ones. The convolution estimate used in (5.6) proves (7.2) by induction.

Uniqueness also holds among all sufficiently small stationary sequences tending to zero, not just among elements of that weighted ball. Their difference solves a homogeneous Jacobi equation whose averaged Hessian is strictly diagonally dominant. Its inverse on bounded sequences is bounded by the reciprocal dominance margin: write the Hessian as its positive diagonal times $I - T$, with $\|T\|_\infty < 1$. With zero left boundary the difference is zero. This argument applies to every fixed small contact box.

Every summand of (7.3) is $O(u^2 \rho^{2i})$. Its differentiated sums converge by the same weighted bounds. Interior stationarity cancels the first-variation terms, leaving only the left endpoint derivative; the boundary term at a truncation tends to zero. At quadratic order this derivative is $(c_b u - x_1^{(b), \text{lin}})/g$. Since $c_b \sigma_0^{(b)}/\sigma_1^{(b)} = c$, it equals $a_b u$. This proves (7.4), hence strict convexity.

The uniform complex contraction and Cauchy estimates, as in Lemma 5.2, prove the analytic assertion. $\square$

Let $H_b(u)$ be the half-line interior Hessian evaluated on this segment, and write $\Delta H_b(u) = H_b(u) - H_b(0)$. It is tridiagonal. Locality and (7.2) imply

$$\|\Delta H_b(u)\|_1 \leq C|u|, \quad \|\partial_\xi^\alpha \partial_u^r \Delta H_b(u)\|_1 \leq C_{\alpha,r}. \quad (7.5)$$

Here and below $\|\cdot\|_1$ on operators is the trace norm. Indeed, the sum of absolute entries bounds that norm, and the entries are bounded by sums of neighboring endpoint weights. Define

$$U_b(u) = \sum_{i \geq 0} \log \left(g[-\ell_{(b+i) \bmod 2, uv}(x_i^{(b)}, x_{i+1}^{(b)})] \right), \quad (7.6)$$

$$B_b(u) = \exp \left\{ U_b(u) - \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} \operatorname{tr}[(G^{(b)} \Delta H_b(u))^m] \right\}. \quad (7.7)$$

This also defines the Fredholm determinant in the denominator of B_b : no additional determinant normalization is implicit. The series converges absolutely after requiring $\|G^{(b)} \Delta H_b\| < 1/2$. Each trace is bounded by one trace norm times $m-1$ operator norms. The edge sum converges by endpoint localization. Consequently $B_b > 0$, $B_b(0) = 1$, and every fixed mixed derivative is uniformly bounded. Differentiation of the trace series retains a trace-class factor, so its bound does not involve the number of sites of a truncation.

7.2. Relative factorization.

Theorem 7.2 (Nonlinear two-boundary factorization). *There are a common endpoint box and $\tau \in (0, 1)$ such that for every fixed k ,*

$$\|E_j - S_0 \oplus S_p\|_{C^k} + \|b_j - B_0 \otimes B_p\|_{C^k} \leq C_k \tau^j, \quad p = j \bmod 2. \quad (7.8)$$

The norms include mixed geometric-parameter and endpoint derivatives; $S_0 \oplus S_p$ means $S_0(u) + S_p(v)$ and $B_0 \otimes B_p$ means $B_0(u)B_p(v)$. The action difference has zero value and first endpoint differential at zero, and therefore is $O(\tau^j(|u| + |v|)^2)$. The amplitude difference vanishes at zero. The fixed-order constants, but not j , may depend on higher boundary norms. Both limits are analytic for analytic families.

Proof. We give the gluing and determinant estimates separately. Reverse a half-line segment starting at type p at the right endpoint. Its i th coordinate in a length- j indexing is $x_{j-i}^{(p)}(v)$; the reversal is consistent because the flight action is symmetric under reversal together with interchange of endpoint types. For $0 < i < j$ put

$$z_i = x_i^{(0)}(u) + x_{j-i}^{(p)}(v), \quad z_0 = u, \quad z_j = v.$$

The endpoint overwrites cost $O(\rho^j)$ in every fixed differentiated norm. In each interior stationarity equation the linear terms cancel. The remaining interaction is a sum of products of left and right coordinates or their neighboring values. Its size is bounded by $C\rho^i \rho^{j-i} = C\rho^j$. Thus, for the interior residual $r(z)$,

$$\|\partial^\alpha r(z)\|_{\ell^1} \leq C_\alpha(1+j)\rho^j. \quad (7.9)$$

The assertion for derivatives follows from the local chain rule; parameter derivatives of the separated linear exponentials introduce only fixed powers of indices, absorbed by a strict exponential margin.

Let y be the finite stationary segment already constructed in Lemma 5.2. The identity

$$r(z) - r(y) = \left[\int_0^1 H_{\text{int}}(y + t(z - y)) \, dt \right] (z - y)$$

has a symmetric strictly diagonally dominant matrix with a uniform positive margin. Its inverse has bounded ℓ^∞ norm by diagonal splitting and a geometric series; symmetry gives the same ℓ^1 bound. It follows that $\|y - z\|_{\ell^1} \leq C(1 + j)\rho^j$. Differentiating this identity leaves the same uniformly invertible operator on the highest difference derivative. All other terms contain lower difference derivatives and uniformly bounded local coefficients. Induction proves the corresponding ℓ^1 estimate at every fixed order, with a polynomial in j multiplying an exponential. Fixing a slower exponential absorbs every such fixed polynomial.

For the action, subtract the two half-line sums. The missing tails are exponentially small. On overlapping edges the difference between the value at the sum and the sum of the two values minus g is a local interaction bounded by $C\rho^j$. The replacement of z by y is controlled by its ℓ^1 error and bounded local derivatives. The same argument after differentiation proves the first term of (7.8). Both finite and limiting actions have zero value and gradient at zero; Taylor's integral remainder yields the additional quadratic vanishing.

For the normalized twist, the finite cofactor formula gives

$$\log b_j = \sum_{i=0}^{j-1} \log(g[-\ell_{i \bmod 2, uv}(y_i, y_{i+1})]) - \log \det(I + G_j \Delta H_j), \quad (7.10)$$

where $G_j = (H_{\text{int}}^0)^{-1}$ and $\Delta H_j = H_{\text{int}} - H_{\text{int}}^0$. The edge sum converges exponentially to $U_0(u) + U_p(v)$ by exactly the preceding local interaction and tail estimates. It remains to control the determinant relatively.

Here are the needed finite-dimensional-to-half-line details. For $j \geq 6$ choose $r = \lfloor j/3 \rfloor$ and retain the first and last r interior sites. Set to zero all perturbation entries outside their two diagonal blocks, including crossing entries. The trace-norm error is bounded by

$$C_k(1 + j)^{M_k} \rho^r \quad (7.11)$$

after k fixed derivatives. This uses tridiagonality and the decay of the two endpoint layers, not a dimension times operator-norm estimate. Within each retained block, replacing the finite segment by its own half-line segment has an exponentially small entrywise sum, by (7.9) and the opposite tail.

For a perturbation supported in these blocks, the identity $\det(I + AB) = \det(I + BA)$ reduces the determinant to that support. The left compression of G_j differs from the compression of $G^{(0)}$ by an exponentially small operator norm. The reversed right compression differs in the same way from $G^{(p)}$. To check the claim, use (5.5): after expanding hyperbolic sines into exponentials, the reflection from the remote boundary has a factor $e^{-\gamma(2j-i-k)}$ for $i, k \leq r$. The two off-diagonal block kernels have a factor $e^{-\gamma(j-2r)}$. Summing at most r^2 entries and differentiating any fixed number of times only adds a polynomial factor. Consequently the compressed Green matrix differs exponentially from the direct sum of the two compressed half-line operators. The same estimate holds after multiplication by the bounded trace-norm perturbation.

For completeness, if T, T' are the resulting trace-class operators with norms at most $q < 1$, telescoping powers gives

$$|\text{tr}(T^m - (T')^m)| \leq mq^{m-1} \|T - T'\|_1.$$

Summation with coefficient $1/m$ proves local Lipschitz continuity of the logarithmic determinant in trace norm. Its differentiated series has the same property with a fixed polynomial factor in m , still summable. This justifies each truncation and comparison above, for all fixed mixed derivatives. Sending the block size to infinity at either end costs $O(\rho^r)$ again. The direct-sum logarithm is the sum of the half-line logarithms in (7.7).

We have proved exponential convergence of (7.10) to $\log B_0(u) + \log B_p(v)$. Exponentiation preserves the bounds, since all logarithms range over one bounded interval. Choose $\tau < 1$ slower

than the finitely many exponential rates in the gluing, truncation, and off-diagonal estimates. Polynomial factors at each fixed derivative order are absorbed in C_k . The finitely many $j < 6$ are absorbed by enlarging those constants. At zero the finite and limiting normalized twists are one, which proves the last vanishing assertion. Uniform complex versions of the same estimates prove analyticity. In particular the argument never divides an absolute action error by the exponentially small reference twist. $\square$

7.3. Physical limiting laws. For $p \in \{0, 1\}$ let $q_p = \sqrt{a_0 a_p}$ and define, for $d > 0$,

$$\mathcal{F}_p(d) = \frac{q_p}{\pi d^2} \int_{\mathbb{R}^2} (d - S_0(u) - S_p(v))_+ B_0(u) B_p(v) \, du \, dv. \quad (7.12)$$

The integral is restricted to the common contact box; for the collars below its support lies strictly inside that box, so the displayed extension by zero is unambiguous.

Theorem 7.3 (Fixed-offset long-bridge law). *There is $d_* > 0$ independent of j such that $\mathcal{F}_p$ has a smooth right extension to $[0, d_*]$, with $\mathcal{F}_p(0) = 1$ and $\mathcal{F}_p > 0$. On this interval,*

$$\left\| \frac{2A \sinh(j\gamma)}{d^2} \Pr(E_{j,e}(d)) - \mathcal{F}_p(d) \right\|_{C^k} \leq C_k \tau^j, \quad p = j \bmod 2, \quad (7.13)$$

for every fixed mixed parameter and offset derivative order. The conditional endpoint and first-residual-time law converges to

$$\frac{B_0(u) B_p(v)}{I_p(d)} \mathbf{1}_{\{r > 0, S_0(u) + S_p(v) + r < d\}} \, du \, dv \, dr, \quad I_p(d) = \int (d - S_0 - S_p)_+ B_0 B_p \, du \, dv. \quad (7.14)$$

For the scaled variables $(u/\sqrt{d}, v/\sqrt{d}, r/d)$ the convergence has bounded-Lipschitz error at most $C\tau^j$, uniformly for $0 < d \leq d_*$. Any fixed number of collisions at the two ends may be included, with their limiting coordinates given by the corresponding half-line segments. The two ends are coupled by the common residual energy constraint; (7.14) is not a product probability.

Proof. At parity p , (5.2) and (7.4) give $d_j^0 = q_p / \sinh(j\gamma)$. Full-phase integration therefore gives exactly

$$F_j(d) = \frac{q_p}{\pi d^2} \int (d - E_j)_+ b_j \, du \, dv.$$

To compare this integral down to $d = 0$, use the constructive Morse maps of Lemma 6.1 for both E_j and $S_0 \oplus S_p$. Their quadratic matrices are uniformly positive. The matrix $2 \int_0^1 (1-s) D^2 E(sy) \, ds$, its positive square root and the inverse coordinate map depend locally Lipschitz-continuously in each fixed C^k norm on E , at the cost of finitely many additional derivatives. This follows by differentiating the defining inverse equation and using the uniform inverse differential; the positive-square-root map has bounded derivatives on the common compact positive cone. Thus their inverse maps and transformed amplitudes differ by $C_k \tau^j$. The inverse maps both vanish at zero, so their difference at w is also $O(\tau^j |w|)$.

Put $w = \sqrt{2d} z$. The normalized integrals are over the same unit disk with weight $1 - |z|^2$. Their odd Taylor terms vanish. Taylor's integral remainder, to two additional endpoint orders for each requested offset derivative, bounds the difference of the smooth functions of d by $C_k \tau^j$. This proves (7.13) including its derivatives at zero; it is not differentiation of a moving sharp boundary without justification. The quadratic integral at zero is $\pi d^2 / q_p$, proving $\mathcal{F}_p(0) = 1$. Positivity and a smaller common collar give a uniform positive lower bound.

For conditional laws retain $r = d\eta$ and the fixed domain $\mathcal{U}$ in (6.5). The normalized densities there differ in L^1 by $C\tau^j$, since the denominators are uniformly positive. The two endpoint maps, after division by $\sqrt{d}$, differ uniformly by $C\tau^j$. Couple the common part of the latent densities and use these maps; the remaining mass is $O(\tau^j)$ on a bounded scaled domain. This

proves the bounded-Lipschitz estimate. A fixed number of end coordinates follows from the differentiated gluing estimate and its vanishing at zero. First variations give directions from neighboring coordinates when desired. The explicit density (7.14) follows by returning to (u, v, r) coordinates. $\square$

Corollary 7.4 (Nonlinear weights at a tied onset). *Suppose a fixed finite set of selected ground channels has the same gap. Let $\gamma_{\min}$ be the least of their exponents. At a fixed $0 < d \leq d_*$, along a parity subsequence their conditional weights in the complete maximal-count event converge to*

$$\frac{\mathbf{1}_{\{\gamma_e = \gamma_{\min}\}} \mathcal{F}_{e,p}(d)}{\sum_{e': \gamma_{e'} = \gamma_{\min}} \mathcal{F}_{e',p}(d)}.$$

The convergence is exponential. At $d = 0$ the minimizing leading weights are equal, whereas at positive offset the boundary laws supply their corrections.

Proof. Divide the finite sum in (3.1) by $d^2 e^{-j\gamma_{\min}}/A$. Theorem 7.3 and $\text{csch}(j\gamma) = 2e^{-j\gamma}/(1 - e^{-2j\gamma})$ give the limit. All limiting denominators are positive. A finite positive gap between distinct exponents and the factorization rate give an exponential error; if all exponents tie only the latter rate is needed. $\square$

Corollary 7.5 (The first poles of the onset series). *For a fixed geometry put $\mathcal{G}(z, d) = \sum_{j \geq 1} z^j \Pr(E_{j,e}(d))/d^2$. There is $\nu > 0$ such that, with $x = ze^{-\gamma}$,*

$$\mathcal{G}(z, d) = \frac{\mathcal{F}_1(d)x + \mathcal{F}_0(d)x^2}{A(1 - x^2)} + \mathcal{H}(z, d), \quad (7.15)$$

where $\mathcal{H}$ is holomorphic for $|z| < e^{\gamma+\nu}$ and smooth in $d \in [0, d_]$. Every fixed geometric and offset derivative has the same assertion on strictly smaller common domains. The positive singularity is a simple pole at $z = e^\gamma$, with residue $-e^\gamma[\mathcal{F}_0(d) + \mathcal{F}_1(d)]/(2A)$. The possible pole at $-e^\gamma$ is removable exactly when $\mathcal{F}_0(d) = \mathcal{F}_1(d)$.*

Proof. The coefficient in question is

$$\frac{e^{-j\gamma}}{A(1 - e^{-2j\gamma})} F_j(d) = \frac{e^{-j\gamma}}{A} \mathcal{F}_{j \bmod 2}(d) + O(e^{-(\gamma+\nu)j}),$$

for any positive ν strictly smaller than $-\log \tau$ and 2γ . Fixed parameter derivatives add only fixed powers of j , absorbed by a smaller strict margin. The error series is normally convergent, as are its stated derivatives. Summing its even and odd leading terms gives (7.15). Positivity gives the nonzero positive residue. At the negative point the numerator is $\mathcal{F}_0 - \mathcal{F}_1$, proving the removability criterion. $\square$

The series in (7.15) uses a different physical window $jd + d$ at each coefficient. It is not the generating function of all itineraries in one observation window. Its continuation is a consequence of a nonlinear relative-flux limit for these specified events.

7.4. Differentiated operator bounds and moving poles. For use in differentiated forms of the relative factorization, the operator estimates can be recorded in a single common norm convention. Let G denote either a finite reference Green matrix or a half-line Green operator, and let ΔH be the corresponding tridiagonal Hessian perturbation on its stationary segment. For every fixed k ,

$$\max_{|\alpha| \leq k} \|\partial_\xi^\alpha G\|_{\text{op}} \leq C_k, \quad \max_{|\alpha|+|\beta| \leq k} \|\partial_\xi^\alpha \partial_Y^\beta \Delta H\|_1 \leq C_k, \quad \|G\Delta H\|_{\text{op}} \leq q < 1. \quad (7.16)$$

Here Y is the finite endpoint vector or the single half-line endpoint. There is no assertion that derivatives of $G\Delta H$ are themselves small. They are bounded in trace norm, whereas its zeroth-order operator norm is small on the fixed contact box.

For completeness, the first bound follows directly from the Green kernels: their parameter derivatives introduce fixed powers of the indices into exponentially summable kernels, absorbed by the strict exponential margin. The Schur row and column estimates give the operator bound independently of matrix size. For the second, each tridiagonal entry is a local smooth function of neighboring stationary coordinates. The endpoint localization and its differentiated bounds make the sum of its absolute entries bounded. This sum bounds the trace norm. The last inequality follows by choosing the common box once, using $\Delta H(0) = 0$. The product rule consequently gives trace-norm bounds for every fixed derivative of $T = G\Delta H$.

These estimates also explain the differentiated determinant comparison. Differentiating $\text{tr}(T^n)$ a fixed k times produces at most $C_k n^k$ products with at most k differentiated factors. One factor is bounded in trace norm and all others in operator norm. The undifferentiated factors contribute the geometric decay q^{n-k} ; the finitely many $n < k$ are harmless. In a difference of two such expressions, telescope one factor at a time. The differentiated trace-norm differences from the block comparison in the proof of Theorem 7.2 supply $C_k \tau^j$, and summation in n is uniform. Thus differentiated logarithmic determinants retain a trace-class factor and never acquire a factor equal to the number of interior sites.

In Corollary 7.5, simple poles refer to the undifferentiated series at a fixed geometry. If a parameter moves γ , differentiation of its displayed rational principal part can increase the pole order. After k parameter derivatives it is at most $k + 1$. For example, $\partial_\xi(1 - ze^{-\gamma})^{-1} = -ze^{-\gamma}(\partial_\xi \gamma)/(1 - ze^{-\gamma})^2$. The differentiated remainder remains holomorphic on the stated strictly smaller common domains. This separates the regularity of the remainder from the order of the moving principal poles.

8. RELATIVE BOUNDARY LAWS ALONG A PERIODIC ITINERARY

The relative normalization is not specific to a normal two-periodic orbit. We prove it for a selected nongrazing periodic orbit of a smooth planar dispersing billiard. The two ends now carry different half-line actions. There is also a linear endpoint contribution to the physical length; it must be restored before a probability law is formed. The signed two-contact inverse in the subsequent sections retains its alternating-channel hypotheses.

8.1. The geometric Jacobi operator. Let a periodic orbit have a fixed itinerary, with lifted reflection points q_i , flight lengths $L_i > 0$, curvatures $\kappa_i > 0$, and outgoing normal cosines $\nu_i > 0$. The indices are periodic modulo an even integer P : an odd primitive period is doubled so that the signed coordinate frames also return. A lifted period is allowed to end at a lattice translate of its initial point. Put $L = \sum_{i=0}^{P-1} L_i$. Every open flight is clear of unintended obstacles. We work with a compact smooth family of such marked orbits, with fixed P and positive lower bounds for clearance, L_i , κ_i and ν_i , and finite upper bounds for the flight lengths and each separately used smooth norm. Clearance here refers to portions outside small fixed endpoint neighborhoods; nongrazing incidence controls those neighborhoods. Family derivatives keep the scaled endpoint coordinates and the excess $d = t - nL$ fixed; the physical observation time moves with the reference period length.

Choose unit-speed signed boundary charts $r_i(s)$ centered at q_i . Their tangent orientations alternate relative to the positively oriented obstacle tangents. Write

$$x_i = \nu_i s_i, \quad \ell_i(x, y) = |r_{i+1}(y/\nu_{i+1}) - r_i(x/\nu_i)|, \quad k_i = L_i^{-1}, \quad m_i = 2\kappa_i/\nu_i.$$

The endpoint arguments in r_i are signed arclengths. Reflection gives numbers p_i such that

$$\ell_{i,1}(0, 0) = -p_i, \quad \ell_{i-1,2}(0, 0) = p_i.$$

Introduce the normalized edge action

$$h_i(x, y) = \ell_i(x, y) - L_i + p_i x - p_{i+1} y. \quad (8.1)$$

Thus $h_i(0, 0) = 0$ and $Dh_i(0, 0) = 0$. The linear terms telescope; this normalization does not change an interior reflection equation or a mixed endpoint derivative.

Lemma 8.1 (Coercivity and persistence). *The Hessian of an edge at the reference orbit and the interior Jacobi operator are respectively*

$$D^2 h_i(0, 0) = \begin{pmatrix} k_i + m_i/2 & -k_i \\ -k_i & k_i + m_{i+1}/2 \end{pmatrix}, \quad (H^0 x)_i = (k_{i-1} + k_i + m_i)x_i - k_{i-1}x_{i-1} - k_i x_{i+1}. \quad (8.2)$$

They are uniformly positive. On a sufficiently small common coordinate box the interior Hessians are symmetric, tridiagonal, and strictly diagonally dominant with a uniform positive margin. The marked orbit and these properties persist under sufficiently small smooth perturbations of the reflecting arcs and of the lattice realization, with the lifted itinerary fixed.

Proof. For a chord with unit direction e and endpoint tangent t , the same-end second derivative in signed arclength is $(1 - (e \cdot t)^2)/L$ plus the curvature contribution. At an outgoing reflection $e \cdot n = \nu$, with n the outward obstacle normal and $r'' = -\kappa n$, that contribution is $\kappa\nu$; the incoming end has the same sign. The mixed derivative is $-[t_i \cdot t_{i+1} - (e \cdot t_i)(e \cdot t_{i+1})]/L$. The alternating tangent orientations make the two projections onto $e^\perp$ have the same sign, so it is $-\nu_i \nu_{i+1}/L_i$. Dividing the endpoint derivatives by the respective normal cosines proves the edge formula. Summing adjacent edge derivatives proves the second formula.

The edge quadratic form is $k_i(x - y)^2 + (m_i/2)x^2 + (m_{i+1}/2)y^2$. The finite and half-line interior forms are therefore bounded below by $m_* \sum x_i^2$, where $m_* := \min_i m_i > 0$. Continuity, finite periodicity, and compactness preserve a positive row-dominance margin on one smaller box. For persistence, apply the finite-dimensional implicit function theorem to the reflection equations with the prescribed periodic identification of the endpoints. The periodic Hessian has quadratic form $\sum k_i(x_{i+1} - x_i)^2 + \sum m_i x_i^2$, hence is invertible. The continued orbit remains in its charts; the clearance and incidence margins keep the chords physical. This argument also covers a repeated period and the prescribed deck translation. $\square$

8.2. The exact reference twist. Set

$$Q_i = \begin{pmatrix} 1 + m_i/k_i & 1/k_i \\ m_i & 1 \end{pmatrix}, \quad \mathcal{M}_b = Q_{b+P-1} \cdots Q_b, \quad \text{tr } \mathcal{M}_b = 2 \cosh \chi. \quad (8.3)$$

Each matrix Q_i has determinant one and positive entries. The products for different starting phases are conjugate. Since $(\mathcal{M}_b)_{12}(\mathcal{M}_b)_{21} > 0$ and $\det \mathcal{M}_b = 1$, their trace is greater than two. In particular $\chi > 0$, and χ has a positive lower bound on the compact class.

For $N = nP$, let $W_{N,b}(u, v)$ be the stationary length along N flights, starting at phase b , with scaled endpoint coordinates u, v . Its construction on a fixed box is given below. Define

$$D_{N,b} = -W_{N,b,uv}(0, 0) = \frac{\sinh \chi}{(\mathcal{M}_b)_{12} \sinh(n\chi)}. \quad (8.4)$$

In particular $D_{N,b} > 0$, and its decay is comparable to $e^{-n\chi}$. This formula is exact, rather than an unspecified exponential equivalent.

Lemma 8.2 (Transfer and cofactor normalizations). *Formula (8.4) holds. More generally, if y is the stationary segment and $H_N(y)$ its interior Hessian, then*

$$-W_{N,b,uv}(u, v) = \frac{\prod_{i=0}^{N-1} [-\ell_{b+i,12}(y_i, y_{i+1})]}{\det H_N(y)}. \quad (8.5)$$

Proof. For the reference recurrence write $z_i = k_{i-1}(x_i - x_{i-1})$. Then $z_{i+1} = z_i + m_i x_i$ and $x_{i+1} = x_i + z_{i+1}/k_i$, which is precisely the transfer Q_i . At the initial endpoint z_0 can be defined as $k_b(x_1 - u) - m_b u$; no exterior reflection equation is being imposed. The left boundary derivative of the quadratic stationary action is $-z_0 - m_b u/2$. The relation $v = (\mathcal{M}_b^n)_{11}u + (\mathcal{M}_b^n)_{12}z_0$ therefore gives $-W_{uv}(0, 0) = 1/(\mathcal{M}_b^n)_{12}$. Cayley–Hamilton, or diagonalization at the distinct eigenvalues e^χ and $e^{-\chi}$, gives

$$(\mathcal{M}_b^n)_{12} = (\mathcal{M}_b)_{12} \frac{\sinh(n\chi)}{\sinh \chi}.$$

The first assertion follows. Eliminating the interior variables from the tridiagonal Hessian gives the mixed Schur-complement entry. The corner cofactor is the product of the adjacent mixed entries, with the sign in (8.5). This proves the second assertion, including its normalization at zero. $\square$

8.3. A relative theorem on a fixed collar. Write

$$E_{N,b}(u, v) = W_{N,b}(u, v) - nL + p_b u - p_b v, \quad \beta_{N,b}(u, v) = \frac{-W_{N,b,uv}(u, v)}{D_{N,b}}.$$

The superscripts $-$ and $+$ below distinguish the left future tail from the right past tail. They do not indicate signs of the functions.

Theorem 8.3 (Periodic-itinerary relative boundary law). *For the marked periodic orbits above there is a coordinate interval, independent of n , on which the finite stationary segments exist uniquely within the prescribed small itinerary collars. There are smooth strictly convex functions S_b^-, S_b^+ and positive smooth functions B_b^-, B_b^+ , with*

$$S_b^\pm(0) = (S_b^\pm)'(0) = 0, \quad B_b^\pm(0) = 1,$$

such that, for one $\tau \in (0, 1)$ and each fixed k ,

$$\|E_{N,b} - S_b^- \oplus S_b^+\|_{C^k} + \|\beta_{N,b} - B_b^- \otimes B_b^+\|_{C^k} \leq C_k \tau^N, \quad N = nP. \quad (8.6)$$

The norms include each stipulated fixed number of smooth family derivatives. Constants may depend on the corresponding higher norms; no order-uniform smooth estimate is asserted. The action error has zero value and endpoint differential at zero and is $O(\tau^N(|u| + |v|)^2)$. The amplitude error vanishes at zero. For analytic families all four limits are analytic on smaller common complex collars.

Proof. We give the Green estimates, the construction, and the relative normalization explicitly. None of them uses a two-dimensional normalizing chart.

Put $a_i = k_{i-1} + k_i + m_i$, and split the reference interior operator as $H^0 = A(I - T)$, where $A = \text{diag}(a_i)$ and the two nonzero entries in row i of T are k_{i-1}/a_i and k_i/a_i . On a finite interval or a half-line,

$$\|T\|_\infty \leq q < 1, \quad (H^0)^{-1} = \sum_{r \geq 0} T^r A^{-1}, \quad |G_{ij}| \leq C q^{|i-j|}.$$

The last estimate holds because a nearest-neighbor walk from i to j has length at least $|i - j|$. Symmetry also gives uniform ℓ^1 bounds. The same series is bounded on the exponentially weighted sup norm for any weight $\rho \in (q, 1)$, by summing $q^{|i-j|} \rho^{j-i}$. For a finite segment one uses the sum of the two endpoint weights $\rho^i + \rho^{N-i}$.

These estimates include fixed derivatives in the marked family. A derivative of order r of a product representing a walk of length h introduces at most a fixed polynomial in h , times a bound $C_r q^{h-r}$. Alternatively one may differentiate the resolvent identity. A strict enlargement

of the exponential base absorbs that polynomial. The finite Green kernel and the left half-line kernel also satisfy

$$|G_{ij}^N - G_{ij}^-| \leq Cq^{2N-i-j} \quad (i, j < N), \quad |G_{ij}^N| \leq Cq^{|i-j|}. \quad (8.7)$$

Indeed a walk contributing to their difference must reach the removed right boundary and return. Its length is at least $2N - i - j$. The reflected right version and the fixed differentiated versions follow in the same way, with a slower base when necessary. The diagonal coefficients on retained sites are identical; imposing a Dirichlet boundary deletes walks, not those diagonal coefficients.

Solve the finite reflection equations as their reference linear solution minus G^N times the nonlinear gradient remainder. That remainder is local, vanishes quadratically at zero, and has derivative $O(r)$ in a box of radius r . The weighted Green bounds make this a contraction with constant at most $1/2$ on a ball of radius $C(|u| + |v|)$ when the common endpoint interval is sufficiently small. This constructs the physical segment with

$$|y_i| \leq C(|u|\rho^i + |v|\rho^{N-i}).$$

Strict convexity gives uniqueness among all segments in the small box, not just in the contraction ball. The same construction on the future and past half-lines gives $x_i^-(u)$ and $x_i^+(v)$, decaying as $C|u|\rho^i$ and $C|v|\rho^i$. Differentiating the fixed-point equation leaves the same invertible operator on its highest derivative; the other terms are local products of lower derivatives. Induction therefore gives the weighted bounds at every fixed order.

Define the half-line actions by the absolutely convergent sums

$$\begin{aligned} S_b^-(u) &= \sum_{i \geq 0} h_{b+i}(x_i^-(u), x_{i+1}^-(u)), \\ S_b^+(v) &= \sum_{i \geq 0} h_{b-i-1}(x_{i+1}^+(v), x_i^+(v)). \end{aligned} \quad (8.8)$$

Every summand is quadratic in its exponentially decaying arguments. The fixed differentiated sums converge as well. In the first variation of a finite truncation the interior terms cancel. The remaining remote boundary term tends to zero by the weighted estimates, so these sums have the stated endpoint derivatives. The positive edge forms in Lemma 8.1 bound each minimized half-line energy below by a fixed positive multiple of its boundary value squared. Their Hessians at zero are consequently positive. Shrinking the interval proves strict convexity and the stated anchoring.

For the action comparison, glue the two half-lines by $z_i = x_i^-(u) + x_{N-i}^+(v)$ and overwrite the two endpoint values by u, v . Locality and $Dh_i(0, 0) = 0$ imply that the interior residual has ℓ^1 norm at most $C(1 + N)\rho^N$: every interaction contains one factor from each end. The endpoint overwrites have the same order. The averaged Hessian between z and the finite stationary segment has uniformly bounded ℓ^1 inverse by diagonal dominance and symmetry. Hence $\|y - z\|_1 \leq C(1 + N)\rho^N$. Subtracting and differentiating these equations proves the same assertion at each fixed order, with only a polynomial loss in N . Summing edge interactions and the omitted half-line tails now proves the action part of (8.6).

For the amplitude, let $G_N = (H_N(0))^{-1}$ and $\Delta H_N = H_N(y) - H_N(0)$. The exact cofactor identity gives

$$\log \beta_{N,b} = \sum_{i=0}^{N-1} \log \frac{-\ell_{b+i,12}(y_i, y_{i+1})}{k_{b+i}} - \log \det(I + G_N \Delta H_N). \quad (8.9)$$

In particular the exponentially small $D_{N,b}$ has already been cancelled exactly. Tridiagonality and endpoint localization give $\|\Delta H_N\|_1 \leq C(|u| + |v|)$, where the norm is the trace norm: the sum of absolute matrix entries is an upper bound for that norm. All its fixed derivatives have

uniformly bounded trace norms. Take the endpoint interval small enough that $\|G_N \Delta H_N\| < 1/2$, uniformly in N .

On the future half-line define $\Delta H_b^-(u)$ by the same rule, and put

$$\begin{aligned} \log B_b^-(u) &= \sum_{i \geq 0} \log \frac{-\ell_{b+i,12}(x_i^-(u), x_{i+1}^-(u))}{k_{b+i}} \\ &\quad - \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} \operatorname{tr}[(G_b^- \Delta H_b^-(u))^r]. \end{aligned} \quad (8.10)$$

The analogous reversed formula defines B_b^+ . Each edge sum is absolutely convergent. The perturbation is trace class and the operator norm is less than $1/2$, so the trace series and its fixed derivatives converge. Its exponential is positive for real arguments, and equals one at zero. This specifies the determinant normalization completely.

Here is the two-end comparison for (8.9). For $N \geq 6$ retain the first and last $r = \lfloor N/3 \rfloor$ interior sites, and delete all other perturbation entries. The trace-norm cost is $C_k(1+N)^{a_k} \rho^r$ after k fixed derivatives. On either retained block the gluing estimate replaces the finite segment by its own half-line with exponentially small entrywise sum. The identity $\det(I+AB) = \det(I+BA)$ reduces a finitely supported perturbation to those two blocks. Equation (8.7) compares the diagonal Green compressions to their half-line compressions; the opposite-end Green blocks are $O(q^{N-2r})$, up to a polynomial factor in N . Multiplication by the bounded trace-norm perturbation preserves these bounds. Extending each retained block to its half-line costs $O(\rho^r)$ again.

For trace-class $T, \tilde{T}$ with norms at most $q_1 < 1$, the exact telescoping expansion of their powers gives

$$|\operatorname{tr}(T^h - \tilde{T}^h)| \leq h q_1^{h-1} \|T - \tilde{T}\|_1.$$

Summing with coefficient $1/h$ proves the required continuity of the logarithmic determinant. At each fixed differentiated order the series has an additional fixed polynomial in h , still summable. Thus the finite logarithm tends relatively to the sum of the two logarithms in (8.10). The edge-product comparison follows from the same gluing and tail bounds. Exponentiation proves the amplitude part of (8.6).

Choose one $\tau < 1$ slower than the Green, gluing, and block-truncation rates. The strict margins absorb every separately fixed polynomial loss; the constants, not this choice of τ , depend on the order. Finitely many shorter segments are absorbed by enlarging those constants. At zero the actions have zero value and differential, and the normalized twists and amplitudes equal one. Taylor's integral remainder gives the additional quadratic vanishing. Finally the contraction, summable perturbations, and logarithmic series extend to uniform smaller complex boxes for analytic families. Their locally uniform convergence proves the analytic assertion. $\square$

8.4. Nonlinear transmission and the stable return. Define the connected endpoint action and the two increasing coordinates by

$$\begin{aligned} C_{N,b}(u, v) &= W_{N,b}(u, v) - W_{N,b}(u, 0) - W_{N,b}(0, v) + W_{N,b}(0, 0), \\ \zeta_b^\pm(x) &= \int_0^x B_b^\pm(t) \, dt. \end{aligned}$$

Corollary 8.4 (Relative nonlinear transmission). *On a smaller fixed interval,*

$$-\frac{C_{N,b}(u, v)}{D_{N,b}} = \zeta_b^-(u) \zeta_b^+(v) + R_{N,b}(u, v), \quad \|R_{N,b}\|_{C^k} \leq C_k \tau^N, \quad (8.11)$$

with $|R_{N,b}(u, v)| \leq C\tau^N|uv|$. Let T_b^- be the next-period position on the future stable half-line, and T_b^+ the previous-period position on the past half-line. Then

$$\zeta_b^\pm(T_b^\pm(x)) = e^{-\chi}\zeta_b^\pm(x). \quad (8.12)$$

Each $\zeta_b^\pm$ is the unique differentiable local linearizing coordinate with value zero and derivative one at zero.

Proof. Twice integrating the mixed derivative, with oriented integrals on negative intervals, gives the exact identity

$$-C_{N,b}(u, v)/D_{N,b} = \int_0^u \int_0^v \beta_{N,b}(s, t) \, dt \, ds.$$

The relative theorem proves (8.11), including its endpoint vanishing and fixed mixed derivatives. Notice that the unnormalized remainder has the smaller size $O(D_{N,b}\tau^N|uv|)$; an absolute action estimate alone does not give this transmission coefficient.

For the future return concatenate a P -flight segment with an N -flight tail. The joining endpoint $w_N(u, v)$ is uniquely determined by stationarity and tends to $T_b^-(u)$ with every fixed derivative. The denominator in its implicit equation is the sum of two positive same-end Hessians. Exact stationary differentiation gives

$$-W_{N+P,b,uv}(u, v) = [-W_{N,b,12}(w_N(u, v), v)] \partial_u w_N(u, v).$$

Since $D_{N+P,b}/D_{N,b} \rightarrow e^{-\chi}$, relative factorization and cancellation of the positive right factor yield

$$B_b^-(T_b^-(u))(T_b^-)'(u) = e^{-\chi}B_b^-(u).$$

Integration gives the minus identity in (8.12). Reversing the itinerary proves the plus identity. In particular both return maps have derivative $e^{-\chi}$ at zero and are contractions on a smaller interval. If $\tilde{\zeta}$ is another normalized differentiable linearizer, then $h = \tilde{\zeta} \circ (\zeta_b^\pm)^{-1}$ satisfies $h(\lambda x) = \lambda h(x)$, $h(0) = 0$, $h'(0) = 1$, where $\lambda = e^{-\chi}$. Thus $h(x) = \lambda^{-n}h(\lambda^n x) \rightarrow x$, proving uniqueness. $\square$

8.5. The physical window law. The gauge in (8.1) is useful analytically, but it is not a change of the clock. Define the physical boundary actions

$$\Sigma_b^-(u) = S_b^-(u) - p_b u, \quad \Sigma_b^+(v) = S_b^+(v) + p_b v. \quad (8.13)$$

In general they do not have critical points at zero. Their linear parts are the oblique endpoint momenta of the reference orbit.

Theorem 8.5 (Charged phase-volume law for a repeated itinerary). *Let A be the free area of the periodic billiard cell and normalize its unit-speed phase volume by $2\pi A$. Fix a sufficiently small positive offset interval $[d_-, d_+]$. There is a smaller closed endpoint interval I , independent of $N = nP$, such that the following experiment is physical. The first collision is in the phase- b collar; there are the selected N successive flights and no further collision by time $t_N = nL + d$; the two recorded endpoints lie in I^2 . The intervening collisions are required to remain in the prescribed itinerary collars.*

Its endpoint measure, before conditioning on success, has density

$$\frac{D_{N,b}}{2\pi A} \beta_{N,b}(u, v) \{d - E_{N,b}(u, v) + p_b u - p_b v\}_+, \quad (u, v) \in I^2. \quad (8.14)$$

On this interval the expression inside braces is bounded below by $d_-/2$ and above by a constant smaller than either adjacent free-flight length. Write $Z_{N,b,d}$ for the integral of the numerator after the constant $D_{N,b}/(2\pi A)$ is removed. The success probability is exactly $D_{N,b}Z_{N,b,d}/(2\pi A)$ and is bounded above and below by positive constant multiples of $e^{-n\chi}$.

The conditional endpoint density converges, in every fixed $C^k(I^2)$ norm including the stated family derivatives, to

$$f_{b,d}(u, v) = Z_{b,d}^{-1} B_b^-(u) B_b^+(v) \{d - \Sigma_b^-(u) - \Sigma_b^+(v)\}, \quad \|f_{N,b,d} - f_{b,d}\|_{C^k} \leq C_k \tau^N. \quad (8.15)$$

All rejected independent phase preparations remain charged through the success probability. The result is a law for this selected windowed itinerary, not an identification with a maximal-collision event.

Proof. Clearance and nongrazing incidence imply that the actual preceding and following free flights of every sufficiently small stationary segment have a common positive lower bound l_* . Choose $d_+ < l_*/4$. The action has a uniform quadratic bound on the fixed box, whereas $|p_b|$ is uniformly bounded. Shrink I until $|E_{N,b} - p_b u + p_b v| \leq d_-/2$, uniformly in N . Then the available residual time $r = t_N - W_{N,b} = d - E_{N,b} + p_b u - p_b v$ lies in $[d_-/2, 3d_+/2]$, below l_* . Any split $r = s + (r - s)$, $0 \leq s \leq r$, lies strictly within the adjacent flights (apart from measure-zero collision endpoints). It gives exactly the required first and last collision times. Conversely a successful trajectory has this unique first collision, its unique small stationary bridge, and such a split. Thus no multiplicity is introduced by repeating the word.

If x is scaled boundary arclength, its canonical tangential momentum is the ordinary arclength momentum divided by ν_b . Hence $dx \wedge d\pi$ is the usual billiard flux form. The endpoint generating action gives $\pi = -W_{N,b,u}$, so that the absolute Jacobian to (u, v) is $-W_{N,b,uv} > 0$. The full phase form is this flux times ds . Integrating over $0 \leq s \leq r$ proves (8.14), with precisely the normalization $2\pi A$. In particular the endpoint gauge terms have been restored.

The residual interval has a positive floor. The finite and limiting amplitudes have positive upper and lower bounds on a common box, by (8.9) and (8.10). Consequently $Z_{N,b,d}$ and $Z_{b,d}$ have positive uniform floors and finite uniform upper bounds. The relative theorem compares their unnormalized integrands and fixed derivatives at rate $C_k \tau^N$. Integration on the fixed square and division by these floors prove (8.15). Finally (8.4) gives the stated charged probability. At no stage is an absolute action error divided by that probability. $\square$

8.6. A nonnormal orbit and the alternating specialization.

Proposition 8.6 (A three-obstacle realization). *There is a periodic dispersing table with a nongrazing three-periodic orbit to which Theorems 8.3 and 8.5 apply, and an open family of such realizations with unequal flight lengths and incidence angles.*

Proof. In a square cell of side 12 place three unit disks whose centers form an equilateral triangle of side $D = 5/2$, centered in the cell. The disks and their translates are disjoint. At each disk choose the point on its radius directed toward the triangle center. These three points form an equilateral triangle of side $\ell = D - \sqrt{3} > 0$. Its inscribed three-flight path satisfies reflection: the outward disk normal bisects the two outgoing directions, with normal cosine $\nu = \sqrt{3}/2$. The distance from the opposite disk center to a whole side line is $\sqrt{3}D/2 - 1/2 > 1$; hence no third disk blocks that side. Neighboring cells are much farther away. The flights are therefore clear and the incidence is not normal or grazing.

Take the oriented period $P = 6$. Then $k_i = 1/\ell$, $m_i = 2/\nu$, and

$$\chi = 6 \operatorname{arcosh}(1 + \ell/\nu).$$

For explicit nonsymmetric realizations, choose radii R_0, R_1, R_2 close to one and prescribe a nearby scalene contact triangle. Place center i at distance R_i behind vertex i along its interior-angle bisector. The triangle then satisfies reflection by construction, has unequal sides and, choosing its three angles distinct, unequal normal cosines. Disjointness and clearance persist by the strict margins just checked. This gives the asserted open family by the same persistence lemma. $\square$

For a normal alternating channel take $P = 2$, $L_0 = L_1 = g$ and $\nu_0 = \nu_1 = 1$. Then $p_b = 0$, $\chi = 2\gamma$, and (8.4) reduces to $D_{2n,b} = a_b / \sinh(2n\gamma)$. Reversal identifies the two half-line actions and amplitudes. Passing from arclength to the fixed transverse graph chart changes an amplitude by the corresponding endpoint Jacobian, normalized by its value at zero. Thus the relative law specializes to Theorem 7.2, not to a different normalization of it. The intrinsic scalar coordinate in Corollary 8.4 specializes to Theorem 10.3.

The extension above concerns the geometric forward mechanism and its physical normalization. It neither assumes nor proves that the two boundary actions of a general itinerary can be inverted by the alternating two-contact block formula. The complete signed inverse, analytic-germ results, global lattice reconstruction and finite-preparation bounds that follow keep all their original hypotheses and conclusions.

9. FROM JACOBI IDENTITIES TO A RELATIVE PHYSICAL LAW

The proof of the relative law uses three different types of information. The first is finite-dimensional algebra, the second is localization of nonlinear perturbations, and the third is stability of a physical sublevel integral. We separate these statements before deriving a nonlinear compatibility law from their combination.

9.1. The classical operator input. For a periodic discrete Lagrangian orbit, the discrete Hill formula of Bolotin and Treschev [1, Theorem 2.1] relates $\det(P - I)$ to the periodic action Hessian divided by the product of mixed edge determinants. The Hessian in our endpoint problem instead has Dirichlet boundary conditions. Its relevant scalar identity is the following elementary Schur-complement formula. We include its proof to specify the boundary convention, not as a new determinant identity.

Proposition 9.1 (Dirichlet cofactor and normalization). *Let $L(x_0, \dots, x_j) = \sum_{i=0}^{j-1} \ell_i(x_i, x_{i+1})$ have a smooth stationary interior at fixed endpoints $x_0 = u, x_j = v$. Suppose its interior Hessian H is positive definite and tridiagonal, with mixed edge derivatives $\ell_{i,12} = -J_i < 0$. For the stationary value $W(u, v)$,*

$$-W_{uv} = \frac{\prod_{i=0}^{j-1} J_i}{\det H}. \quad (9.1)$$

For a reference interior Hessian H^0 and reference edges J_i^0 , put $G^0 = (H^0)^{-1}$ and $\Delta H = H - H^0$. Then

$$\frac{-W_{uv}}{(-W_{uv})^0} = \frac{\prod_i (J_i/J_i^0)}{\det(I + G^0 \Delta H)}. \quad (9.2)$$

If $H = D(I - T)$, D is diagonal with $D_{ii} \geq d_0 > 0$, T has only its two adjacent diagonals nonzero, and $\|T\|_{\ell^\infty \rightarrow \ell^\infty} \leq q < 1$, then

$$|(H^{-1})_{ik}| \leq \frac{q^{|i-k|}}{d_0(1-q)}. \quad (9.3)$$

These estimates have constants independent of the number of sites.

Proof. For $j \geq 2$, differentiating the stationary equations eliminates the interior and gives $W_{uv} = -J_0 J_{j-1} (H^{-1})_{1,j-1}$. The corner cofactor of a tridiagonal matrix is $J_1 \cdots J_{j-2}$, so Cramer's rule proves (9.1). At $j = 1$ the interior determinant is the empty determinant, equal to one, and the same assertion is immediate. Taking the ratio at the reference point proves (9.2). Finally, $H^{-1} = \sum_{n \geq 0} T^n D^{-1}$. The (i, k) entry of T^n vanishes for $n < |i - k|$, while every entry has absolute value at most q^n . Summing the remaining series proves (9.3). $\square$

The separated inverse structure is classical: Meurant's Theorem 2.3 [5] gives inverse entries by two scalar recurrences, and his Theorem 2.4 describes their monotone decrease under strict diagonal dominance. Nabben [6] studies quantitative inverse decay for nonsymmetric tridiagonal and band matrices. Our use of a weighted Green operator belongs to this operator framework. In particular neither the corner cofactor nor exponential inverse decay is claimed here as a new mechanism.

What must additionally be proved for the present probability problem is summability of the *nonlinear perturbation evaluated on the entire stationary bridge*. A bound on each matrix entry, without its location, would not give the required determinant limit. The relevant continuity statement is recorded next.

Proposition 9.2 (A dimension-independent relative determinant bound). *For trace-class operators T, T' on a common Hilbert space with $\|T\|, \|T'\| \leq q < 1$, define*

$$\mathcal{L}(T) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \operatorname{tr}(T^n).$$

Then

$$|\mathcal{L}(T) - \mathcal{L}(T')| \leq (1 - q)^{-1} \|T - T'\|_1. \quad (9.4)$$

Suppose T, T' depend smoothly on finitely many parameters, their trace-class derivatives through order k are uniformly bounded, and the corresponding derivatives of $T - T'$ have trace norm at most ϵ . Then

$$\|\mathcal{L}(T) - \mathcal{L}(T')\|_{C^k} \leq C_k \epsilon.$$

The constants depend on q , the parameter dimension and the stated bounds, not on the dimension of a finite-dimensional truncation.

Proof. The algebraic telescoping formula for $T^n - (T')^n$ and the trace ideal inequality give $|\operatorname{tr}(T^n - (T')^n)| \leq nq^{n-1} \|T - T'\|_1$. Division by n and summation prove the first assertion. After k parameter derivatives, each product has at most k differentiated factors. In the difference of the two differentiated products, telescope the factors once more and place the difference factor in trace norm. Every other differentiated factor has a bounded operator norm. All but at most $k+1$ of the factors still have norm at most q . There are at most $C_k n^{k+1}$ terms. The finitely many products with $n \leq k+1$ are bounded separately; the other terms are bounded by $C_k \epsilon n^{k+1} q^{n-k-1}$. Their sum, including $1/n$, converges. The same estimates justify differentiating the original series by uniform convergence. No smallness of a differentiated operator has been used. $\square$

In the billiard application, Lemma 5.2 gives $|x_i| \leq C(|u|\rho^i + |v|\rho^{j-i})$, with the corresponding localized derivative bounds. Since a Hessian entry depends on only neighboring coordinates, its perturbation has a uniformly summable entrywise absolute value. Matrix units have trace norm one; hence

$$\|\Delta H_j\|_1 \leq C(|u| + |v|), \quad \|\partial^\alpha \Delta H_j\|_1 \leq C_\alpha.$$

The two-block construction in Theorem 7.2 then compares the normalized finite determinant with the direct sum of the two half-line determinants. Its trace-norm error tends to zero exponentially. Proposition 9.2 converts that specific geometric estimate into convergence of the logarithmic amplitude. Inverse decay supplies access to the blocks; the localized perturbation estimate supplies their summable determinant error.

9.2. The passage through physical integration. A second, logically separate requirement is uniformity as the offset vanishes. The following statement isolates this passage. All integrals are over a fixed neighborhood containing the indicated sublevel sets. The parameter set is compact and finite dimensional.

Proposition 9.3 (Stability of normalized sublevel integration). *Let $E, \tilde{E}$ be smooth functions of $y \in \mathbb{R}^2$ and of parameters, with value and y -gradient zero at $y = 0$. Assume their Hessians at zero are uniformly positive and their higher derivatives are uniformly bounded on a common neighborhood. Let $a, \tilde{a}$ be smooth amplitudes with uniform derivative bounds. There is a common $d_* > 0$ such that*

$$\mathcal{J}(E, a)(d) = d^{-2} \int (d - E(y))_+ a(y) \, dy$$

extends smoothly from $d > 0$ to $[0, d_]$. For each fixed k , bounds through a sufficiently high finite order $r(k)$ give*

$$\|\mathcal{J}(E, a) - \mathcal{J}(\tilde{E}, \tilde{a})\|_{C^k} \leq C_k(\|E - \tilde{E}\|_{C^{r(k)}} + \|a - \tilde{a}\|_{C^{r(k)}}). \quad (9.5)$$

The norms on the left include mixed parameter and offset derivatives.

Proof. Taylor's integral formula gives $E(y) = y^t M_E(y) y / 2$, with $M_E(y) = 2 \int_0^1 (1-s) D^2 E(sy) \, ds$. After decreasing the common neighborhood, the maps $y \mapsto M_E(y)^{1/2} y$ and their counterparts for $\tilde{E}$ have uniformly invertible differentials. Their inverses Φ_E are obtained by the contraction used in Lemma 6.1. Subtracting the inverse equations gives their Lipschitz dependence in C^0 on E ; differentiating and isolating the highest derivative on the same uniformly invertible differential proves the corresponding finite-order estimates by induction. Smoothness of the positive matrix square root is used only on a compact positive-definite set.

Set $A_E(w) = a(\Phi_E(w)) |\det D\Phi_E(w)|$. The change of variables $w = \sqrt{2d} z$ gives the fixed-domain formula

$$\mathcal{J}(E, a)(d) = 2 \int_{|z| < 1} (1 - |z|^2) A_E(\sqrt{2d} z) \, dz. \quad (9.6)$$

The inverse-map estimates bound $A_E - A_{\tilde{E}}$ in each fixed C^s norm by a sufficiently high finite norm of the two input differences. The integral in (9.6), as a function of $t = \sqrt{2d}$, is smooth and even. A smooth even function $h(t)$ has a smooth right representation in t^2 with its first k derivatives controlled by derivatives of h through order $2k$. For example $h'(t)/t = \int_0^1 h''(st) \, ds$; iteration, applied also to the even function h'' , proves the assertion and its difference estimate. Parameter derivatives commute with these integrals. Applying it to (9.6) proves the smooth extension and (9.5). All sublevel sets lie inside the common chart by the uniform quadratic lower bound, so there is no additional chart-boundary term. $\square$

Apply this proposition to (E_j, b_j) and $(S_0 \oplus S_p, B_0 \otimes B_p)$. The exact physical multiplier is $q_p / (2\pi A \sinh(j\gamma))$, by (6.1). Thus the exponentially small flux is retained *outside* a uniformly stable normalized integral. In this form the proof yields all fixed-order derivatives on a closed collar and the rare-event-preserving experiment transfer, rather than only an identity at the normal orbit.

The operator identities above, the trace estimate and the smooth integration principle are reusable elementary ingredients. The billiard-specific work is to verify their simultaneous hypotheses for physical stationary records with arbitrary collision order. The resulting law has, in addition to the experiment comparison in Part II of the complete technical manuscript, the nonlinear parity consequence established below. This consequence concerns the complete boundary functions at positive offset and does not follow from a finite list of leading amplitudes.

10. HYPERBOLIC COORDINATES AND THE INTRINSIC BOUNDARY MEASURE

The relative product law has a classical analytic model. We first calculate that model in physical endpoint coordinates, and then identify the boundary amplitudes intrinsically for the smooth billiards of Theorem 2.1. These are complementary statements: the first assumes normalizing charts; the second constructs a one-dimensional scalar density with the amplitude of the already

proved half-line law. The scalar coordinate itself is classical, as recalled in Lemma 10.2. No smooth two-dimensional normal-form theorem is used in this identification.

10.1. The analytic mixed-boundary calculation. De Simoi–Kaloshin–Leguil [2], in the normal-form discussion on printed page 10 of arXiv:1905.00890v4, recall the analytic symplectic normal form near a hyperbolic period-two orbit. The local existence statement is distinct from the symmetry assumptions in their geometric inverse theorem; see also their discussion on printed page 13. The following physical endpoint calculation was supplied in the independent memorandum [7]. We include its proof and normalizations. The proposition is conditional on the indicated charts, and is not attributed as a probability theorem to [2].

Proposition 10.1 (Relative flux in analytic hyperbolic coordinates). *Let*

$$N(s, p) = (\Delta(sp)s, \Delta(sp)^{-1}p), \quad \Delta(0) = \lambda \in (0, 1), \quad (10.1)$$

where Δ is positive and analytic near zero. Let

$$\mathcal{C}_L(s, p) = (U(s, p), \mathcal{P}(s, p)), \quad \mathcal{C}_R(q, t) = (V(q, t), \mathcal{Q}(q, t))$$

be analytic canonical charts sending zero to zero, with $U_s(0, 0)V_t(0, 0) \neq 0$. For the local branch of $\mathcal{C}_R \circ N^n \circ \mathcal{C}_L^{-1}$, prescribe the initial physical position u and terminal physical position v . There are a fixed box in (u, v) , an integer n_0 , and $\sigma \in (0, 1)$ such that this boundary problem has a unique local solution for all $n \geq n_0$. Its physical mixed flux $J_n = |\partial P / \partial v|_u$ satisfies

$$\left\| \frac{J_n(u, v)}{J_n(0, 0)} - \mathcal{B}_L(u)\mathcal{B}_R(v) \right\|_{C^k} \leq C_k \sigma^n \quad (10.2)$$

for every fixed k , where, writing $U(\alpha(u), 0) = u$ and $V(0, \beta(v)) = v$,

$$\mathcal{B}_L(u) = \frac{U_s(0, 0)}{U_s(\alpha(u), 0)}, \quad \mathcal{B}_R(v) = \frac{V_t(0, 0)}{V_t(0, \beta(v))}. \quad (10.3)$$

These functions are positive and equal one at zero. If the physical branch has a generating action W_n , normalized by $P = -W_{n,u}$ and $Q = W_{n,v}$, then there are functions $\mathcal{S}_L, \mathcal{S}_R$, vanishing with their first derivatives at zero, for which

$$\|W_n - W_n(0, 0) - \mathcal{S}_L \oplus \mathcal{S}_R\|_{C^k} \leq C_k \sigma^n. \quad (10.4)$$

All assertions hold with mixed parameter derivatives for a supplied jointly smooth family of these functions and charts on common domains, provided their fixed-order norms are uniformly bounded, their projection derivatives are uniformly nonzero, and $0 < \lambda_- \leq \lambda \leq \lambda_+ < 1$.

Proof. Temporarily prescribe $s = s_0$ and $t = p_n$ rather than u, v . The invariant $I = sp$ and the orbit endpoints obey

$$I = st\Delta(I)^n, \quad p_0 = t\Delta(I)^n, \quad q = s_n = s\Delta(I)^n. \quad (10.5)$$

For a single map put $\lambda_+ = \lambda$. Choose $\lambda_+ < \mu < \sigma < 1$ and decrease the I interval so that $\Delta \leq \mu$. On a fixed sufficiently small (s, t) box, $|st|n\mu^{n-1}\|\Delta'\|_\infty < 1/2$ for every $n \geq 1$, since $\sup_n n\mu^{n-1} < \infty$. The right side of the first identity in (10.5) maps a common small closed I interval into itself. It is a contraction there. Its solution satisfies $|I| \leq C\mu^n$. The intermediate coordinates are $(s\Delta(I)^i, t\Delta(I)^{n-i})$, so the whole orbit remains inside the chosen local chart.

Set

$$a_n = \Delta(I)^n, \quad D_n = 1 - nI \frac{\Delta'(I)}{\Delta(I)}.$$

The contraction estimate gives $|D_n| \geq 1/2$. Differentiating (10.5), without dividing by either endpoint, gives

$$T_n := \left. \frac{\partial p_0}{\partial t} \right|_s = \left. \frac{\partial q}{\partial s} \right|_t = \frac{\Delta(I)^n}{D_n}. \quad (10.6)$$

For example, $I_t = sa_n/D_n$ and $(p_0)_t = a_n + tn\Delta(I)^{n-1}\Delta'(I)I_t = a_n/D_n$. This also proves the formula on both axes.

Here are the differentiated relative estimates. At every fixed order, derivatives of the function $\Delta(I)^n$ with I treated as an independent variable are bounded by $C_k(1+n)^k\mu^n$; a finite number of small n are absorbed in C_k . Implicit differentiation of the first equation in (10.5) leaves D_n on the highest derivative of I . Its remaining terms contain the preceding derivatives of I , derivatives of st , and derivatives of Δ^n . Induction therefore gives

$$\|I\|_{C^k} \leq C_k(1+n)^{m_k}\mu^n.$$

These norms can include the supplied parameters. Since $\log(\Delta(I)/\lambda) = Ih(I)$ with a jointly smooth bounded function h , the same induction gives

$$\left\| n \log \frac{\Delta(I)}{\lambda} \right\|_{C^k} + \|D_n - 1\|_{C^k} \leq C_k(1+n)^{r_k}\mu^n.$$

The identity at $I = 0$ holds throughout the parameter family; thus parameter derivatives of λ do not leave an unsuppressed term in this normalized expression. Exponentiating and using $|D_n| \geq 1/2$ proves

$$\|\lambda^{-n}T_n - 1\|_{C^k} \leq C_k\sigma^n. \quad (10.7)$$

One fixed strict margin $\mu < \sigma$ absorbs the polynomial at each fixed order; the constants, not the margin, depend on that order.

The physical boundary map in the mixed coordinates is

$$\Phi_n(s, t) = (U(s, ta_n), V(sa_n, t)), \quad \Phi_\infty(s, t) = (U(s, 0), V(0, t)). \quad (10.8)$$

It converges to Φ_∞ in every fixed derivative norm at an exponential rate. Choose fixed inner and outer boxes on which the two limiting scalar projections are diffeomorphisms, with the closure of the inner box strictly inside the image of the outer one. Conjugating Φ_n by Φ_∞^{-1} gives the identity plus a uniformly small C^1 perturbation. The equation for its inverse is a contraction on a small ball about each point of the inner box, uniformly in that point. This gives a unique inverse on a common nonshrinking physical box for all $n \geq n_0$. Subtracting the inverse equations gives its C^0 convergence. Differentiation isolates the same uniformly invertible differential and proves the C^k convergence by induction. In particular $(s, t) = \Phi_n^{-1}(u, v)$ tends to $(\alpha(u), \beta(v))$ at the stated rate.

Put $\mathcal{D}_n = \det D\Phi_n$. Canonicity of the initial chart and (10.6) give, on the boundary surface,

$$du \wedge dP = ds \wedge dp_0 = T_n ds \wedge dt, \quad du \wedge dv = \mathcal{D}_n ds \wedge dt.$$

Consequently the exact physical, rather than canonical, flux is

$$J_n(u, v) = \left| \frac{T_n(s, t)}{\mathcal{D}_n(s, t)} \right|, \quad (s, t) = \Phi_n^{-1}(u, v). \quad (10.9)$$

At zero this denominator is exactly

$$\mathcal{D}_n(0, 0) = U_s(0, 0)V_t(0, 0) - \lambda^{2n}U_p(0, 0)V_q(0, 0). \quad (10.10)$$

On the physical box its limit is $U_s(\alpha(u), 0)V_t(0, \beta(v))$, uniformly bounded away from zero. Equations (10.7)–(10.10) therefore prove (10.2) and (10.3), including all fixed mixed derivatives. This step is necessary: canonical boundary coordinates are not physical configuration observations.

Finally define

$$\mathcal{S}_L(u) = - \int_0^u \mathcal{P}(\alpha(x), 0) \, dx, \quad \mathcal{S}_R(v) = \int_0^v \mathcal{Q}(0, \beta(y)) \, dy.$$

The physical first variations of W_n converge to the two displayed integrands, with all fixed derivatives, by (10.8). Integrate the difference of the gradients along the two coordinate segments from $(0, 0)$ to (u, v) . The values at the origin fix the additive constants and prove (10.4). $\square$

For the two-step billiard return the contracting eigenvalue in (10.1) is $\lambda = e^{-2\gamma}$, and n returns mean $j = 2n$ flights. At contact b its stable and unstable linear directions in physical position–momentum coordinates are $(1, -a_b)$ and $(1, a_b)$, by (5.1). They are nonvertical, regardless of equality or inequality of the curvatures. Use the same canonical chart at both ends. Write its first derivative as

$$DC(0) = \begin{pmatrix} A & B \\ -a_b A & a_b B \end{pmatrix}.$$

Canonicity gives $AB = (2a_b)^{-1}$. Thus (10.10) and (10.9) recover the exact physical normalization

$$J_n(0, 0) = \frac{2a_b \lambda^n}{1 - \lambda^{2n}} = \frac{a_b}{\sinh(2n\gamma)}. \quad (10.11)$$

This is an exact agreement with the Jacobi formula, not merely an agreement of leading exponential orders. If both constructions apply to the same analytic billiard, uniqueness of the normalized limit and evaluation at the other endpoint zero give $B_b = \mathcal{B}_L = \mathcal{B}_R$. Their action limits agree as well, with the stated zero normalizations. The normal-form function Δ alone does not specify the physical amplitudes: the projection charts enter (10.3) essentially.

10.2. The classical scalar comparison. Smooth linearization on a one-dimensional contracting branch is classical. It requires neither a symplectic chart on a neighborhood nor an analytic hypothesis. This is the one-dimensional case of Sternberg’s theorem; see [8] and the precise regularity statement in the introduction of Eynard–Bontemps–Navas [9]. We record the elementary product construction in the normalization needed here. The scalar comparison and its application to the width were pointed out in the memorandum [10]. Its role is to separate a classical coordinate from the identification of the physical determinant amplitude.

Lemma 10.2 (Normalized scalar linearization). *Let R_ξ be a jointly smooth family on a fixed interval about zero, with parameter ξ in a compact set, such that*

$$R_\xi(0) = 0, \quad R'_\xi(0) = \lambda_\xi, \quad 0 < \ell \leq R'_\xi(u) \leq q < 1.$$

The family is understood to be defined on an open neighborhood of the compact parameter set. On a common smaller interval the series

$$h_\xi(u) = \log \frac{R'_\xi(u)}{\lambda_\xi}, \quad (10.12)$$

$$L_\xi(u) = \sum_{i=0}^{\infty} h_\xi(R_\xi^i(u)), \quad \mathfrak{B}_\xi(u) = \exp L_\xi(u), \quad \mathfrak{z}_\xi(u) = \int_0^u \mathfrak{B}_\xi(x) \, dx$$

converges with every fixed mixed parameter and endpoint derivative. The coordinate $\mathfrak{z}_\xi$ is increasing, normalized by $\mathfrak{z}_\xi(0) = 0$ and $\mathfrak{z}'_\xi(0) = 1$, and satisfies

$$\mathfrak{B}_\xi(R_\xi(u)) R'_\xi(u) = \lambda_\xi \mathfrak{B}_\xi(u), \quad \mathfrak{z}_\xi \circ R_\xi = \lambda_\xi \mathfrak{z}_\xi. \quad (10.13)$$

It is the unique normalized differentiable local linearizer. For every fixed k and $\sigma \in (\lambda_+, 1)$, where $\lambda_+ = \max_\xi \lambda_\xi$, one has

$$\|\lambda_\xi^{-n} R_\xi^n - \mathfrak{z}_\xi\|_{C^k} + \|\lambda_\xi^{-n} (R_\xi^n)' - \mathfrak{B}_\xi\|_{C^k} \leq C_{k,\sigma} \sigma^n. \quad (10.14)$$

Here and below such norms include any fixed collection of mixed parameter and endpoint derivatives on the common domain.

Proof. Suppress ξ in compositions, but not in differentiation. Put $x_n = R_\xi^n(u)$. Since $R_\xi(0) = 0$ and $|R'_\xi| \leq q$, $|x_n| \leq q^n|u|$. At every fixed positive mixed order m , differentiation of $x_{n+1} = R_\xi(x_n)$ leaves $R'_\xi(x_n)$ multiplying the highest derivative of x_n . Every other term either contains a lower derivative of x_n , or is a pure parameter derivative of R_ξ evaluated at x_n . The latter vanishes at $x_n = 0$, because the fixed point is common to the family. Induction and the mean value theorem therefore give

$$\sup_{|\alpha|+r=m} |\partial_\xi^\alpha \partial_u^r x_n| \leq C_m(1+n)^{M_m} q^n.$$

Indeed, at the inductive step the right-hand forcing in the highest order recurrence is bounded by a polynomial in n times q^n ; iteration of that recurrence only increases the polynomial degree. Uniform bounds on the finitely many derivatives of R_ξ involved follow from compactness. This argument covers endpoint derivatives, parameter derivatives and their mixtures.

The function h_ξ is smooth, $h_\xi(0) = 0$, and all its pure parameter derivatives vanish at zero. The same chain-rule argument bounds each fixed mixed derivative of $h_\xi(x_n)$ by $C_m(1+n)^{N_m} q^n$. Summability proves convergence of (10.12) with all the asserted derivatives. In particular $L_\xi(0) = 0$ and $\mathfrak{B}_\xi(0) = 1$. Shifting the summation index gives $L_\xi(u) - L_\xi(R_\xi(u)) = h_\xi(u)$. Exponentiation proves the first identity in (10.13). Integration from zero proves the second, since both sides vanish there.

If $\tilde{\mathfrak{z}}$ is another normalized differentiable linearizer, set $H = \tilde{\mathfrak{z}} \circ \mathfrak{z}^{-1}$. Then $H(0) = 0$, $H'(0) = 1$, and $H(\lambda z) = \lambda H(z)$. For every z in a sufficiently small interval, $H(z) = \lambda^{-n} H(\lambda^n z) \rightarrow z$. This proves uniqueness.

The exact relation $R_\xi^n = \mathfrak{z}_\xi^{-1} \circ (\lambda_\xi^n \mathfrak{z}_\xi)$ gives the sharper rate based on λ_+ rather than the preliminary contraction bound q . The inverse function theorem on a common smaller interval makes $\mathfrak{z}_\xi^{-1}$ jointly smooth. Its value and first derivative at zero are 0 and 1; write it as $x + x^2 A_\xi(x)$. Thus

$$\lambda_\xi^{-n} R_\xi^n(u) - \mathfrak{z}_\xi(u) = \lambda_\xi^n \mathfrak{z}_\xi(u)^2 A_\xi(\lambda_\xi^n \mathfrak{z}_\xi(u)).$$

Every fixed mixed derivative is bounded by $C_k(1+n)^{N_k} \lambda_+^n$; the positive lower bound on λ_ξ controls derivatives of its powers. Apply the same argument with one additional endpoint derivative for the density estimate. A fixed strict margin $\lambda_+ < \sigma$ absorbs the polynomial factors. Finally, the finite product identity

$$\exp \left\{ \sum_{i=0}^{n-1} h_\xi(R_\xi^i(u)) \right\} = \lambda_\xi^{-n} (R_\xi^n)'(u) \quad (10.15)$$

follows directly from the chain rule and agrees with the density limit. $\square$

The alternating scalar identity is equally classical. Suppose that $\varphi_b(0) = 0$, $\varphi'_b(0) = r_b > 0$ and $R_b = \varphi_{1-b} \circ \varphi_b$, with $r_0 r_1 = \lambda < 1$. Let $\mathfrak{z}_b$ be the normalized scalar linearizer of R_b . The identity $\varphi_b \circ R_b = R_{1-b} \circ \varphi_b$ shows that $r_b^{-1} \mathfrak{z}_{1-b} \circ \varphi_b$ is another normalized linearizer of R_b . Uniqueness gives

$$\mathfrak{z}_{1-b} \circ \varphi_b = r_b \mathfrak{z}_b, \quad \mathfrak{B}_{1-b}(\varphi_b(u)) \varphi'_b(u) = r_b \mathfrak{B}_b(u).$$

These scalar facts by themselves do not identify the amplitude in a normalized finite physical bridge. That identification is the content of the following theorem and its exact Schur-concatenation proof.

10.3. A smooth intrinsic characterization. We now return to the full smooth setting. Write $W_{j,b}$ for the stationary action starting at type b , and let

$$\varphi_b(u) = x_1^{(b)}(u), \quad \mathfrak{r}_b = \sqrt{\frac{c_b}{c_{1-b}}}, \quad r_b = \mathfrak{r}_b \varrho, \quad R_b = \varphi_{1-b} \circ \varphi_b, \quad \varrho = e^{-\gamma}, \quad \lambda = \varrho^2.$$

Here $x^{(b)}$ is the physical half-line in Lemma 7.1. Thus φ_b is the first-hit position along the stable branch, not a freely chosen change of variables. Notice that $r_0 r_1 = \lambda$ and $0 < r_b < 1$. Indeed $\sqrt{c_b/c_{1-b}} e^{-\gamma} = c_b/(c_0 c_1 + \sqrt{c_0 c_1 (c_0 c_1 - 1)}) < 1$, since $c_{1-b} > 1$.

Theorem 10.3 (The boundary amplitude as a linearizing density). *On a fixed smaller collar, uniformly for the smooth compact families of Theorem 2.1, the half-line amplitudes satisfy*

$$B_{1-b}(\varphi_b(u)) \varphi'_b(u) = r_b B_b(u), \quad b = 0, 1. \quad (10.16)$$

The increasing smooth coordinates

$$\zeta_b(u) = \int_0^u B_b(x) \, dx \quad (10.17)$$

are normalized by $\zeta_b(0) = 0$, $\zeta'_b(0) = 1$ and obey

$$\zeta_{1-b}(\varphi_b(u)) = r_b \zeta_b(u), \quad \zeta_b(R_b(u)) = \lambda \zeta_b(u). \quad (10.18)$$

For each b , ζ_b is the unique differentiable local coordinate with that normalization which linearizes R_b . Moreover, for a fixed $\sigma \in (\lambda_+, 1)$, where λ_+ is the maximum of λ on the compact family,

$$\|\lambda^{-n} R_b^n - \zeta_b\|_{C^k} + \|\lambda^{-n} (R_b^n)' - B_b\|_{C^k} \leq C_k \sigma^n. \quad (10.19)$$

The norms include every fixed mixed parameter and endpoint derivative. No normalizing chart for the two-dimensional return map is a hypothesis.

Proof. Put $o = 1 - b$. Uniqueness of the half-line tail gives the exact relations

$$S_b(u) = \ell_b(u, \varphi_b(u)) - g + S_o(\varphi_b(u)), \quad \ell_{b,2}(u, \varphi_b(u)) + S'_o(\varphi_b(u)) = 0. \quad (10.20)$$

The second relation also follows from first variation of the middle endpoint. Its w derivative is $\ell_{b,22}(u, w) + S''_o(w)$, uniformly positive on the small box. The implicit function theorem and the already constructed half-line therefore give

$$\varphi'_b(u) = \frac{-\ell_{b,12}(u, \varphi_b(u))}{\ell_{b,22}(u, \varphi_b(u)) + S''_o(\varphi_b(u))} > 0. \quad (10.21)$$

At zero this is $(c_o + g a_o)^{-1} = \sqrt{c_b/c_o} e^{-\gamma} = r_b$. After decreasing the common interval, both first-hit maps are contractions into it. All their mixed derivatives are controlled by the same positive denominator. The graph of stable initial physical states is $P = -S'_b(u)$: differentiating the first identity in (10.20) gives $P = -\ell_{b,1}$, while the terminal momentum is $\ell_{b,2} = -S'_o$. The reversed unstable graph is $P = S'_b(u)$. This verifies the dynamical interpretation directly from the physical generating action.

We prove the amplitude identity from exact finite concatenation. For a tail of j flights starting at o , let $w_j = w_j(u, v)$ solve

$$\ell_{b,2}(u, w_j) + W_{j,o,1}(w_j, v) = 0.$$

Strict convexity identifies the resulting stationary value with $W_{j+1,b}(u, v)$. Set $H_j = \ell_{b,22}(u, w_j) + W_{j,o,11}(w_j, v)$. Eliminating the middle endpoint gives the exact Schur identity

$$-W_{j+1,b,12}(u, v) = \frac{-\ell_{b,12}(u, w_j)}{H_j} [-W_{j,o,12}(w_j, v)]. \quad (10.22)$$

The denominator is uniformly positive. Theorem 7.2 and its differentiated action limit imply $w_j \rightarrow \varphi_b(u)$ and $H_j \rightarrow \ell_{b,22}(u, \varphi_b(u)) + S''_o(\varphi_b(u))$ on the fixed box. The inverse equations, with the same positive denominator, give this convergence in every fixed mixed norm.

The terminal type of both sides of (10.22) is $c = (o + j) \bmod 2$. The exact reference constants are

$$d_{j,o}^0 = \frac{\sqrt{a_o a_c}}{\sinh(j\gamma)}, \quad d_{j+1,b}^0 = \frac{\sqrt{a_b a_c}}{\sinh((j+1)\gamma)}.$$

Their ratio tends to $\sqrt{a_o/a_b}e^\gamma = r_b^{-1}$, with all fixed parameter derivatives. Divide (10.22) by $d_{j+1,b}^0$ and let j tend to infinity along either parity. Relative factorization and (10.21) give

$$B_b(u)B_c(v) = r_b^{-1}\varphi'_b(u)B_o(\varphi_b(u))B_c(v).$$

Since $B_c > 0$, cancellation proves (10.16). Integration from zero proves (10.18). This argument uses the previously proved relative theorem to identify its amplitude; it does not claim that a scalar linearization alone proves the finite-bridge theorem.

For uniqueness, conjugate any other normalized differentiable linearizer by ζ_b^{-1} . The resulting function h has $h(0) = 0$, $h'(0) = 1$ and $h(\lambda z) = \lambda h(z)$. Hence $h(z) = \lambda^{-n}h(\lambda^n z) \rightarrow z$, so $h(z) = z$. For the quantitative assertion, the exact identity is

$$R_b^n(u) = \zeta_b^{-1}(\lambda^n \zeta_b(u)).$$

Write $\zeta_b^{-1}(x) = x + x^2 h_b(x)$, where h_b is jointly smooth on a common interval. Division by λ^n leaves $\zeta_b(u) + \lambda^n \zeta_b(u)^2 h_b(\lambda^n \zeta_b(u))$. At each fixed derivative order this last term, including parameter derivatives of λ , is bounded by $C_k(1+n)^{m_k} \lambda_+^n$. Its endpoint derivative proves the second estimate in (10.19); the fixed strict margin gives the displayed rate. This also proves all claimed uniformities without extending the scalar coordinate to a symplectic chart off the invariant branch. $\square$

Corollary 10.4 (Physical determinant and scalar product). *For the physical return $R_b = \varphi_{1-b} \circ \varphi_b$ in Theorem 10.3, put $h_b(u) = \log(R'_b(u)/\lambda)$. The physical edge product U_b , Green operator $G^{(b)}$ and Hessian perturbation ΔH_b of (7.6)–(7.7) satisfy*

$$\frac{\exp U_b(u)}{\det(I + G^{(b)} \Delta H_b(u))} = B_b(u) = \exp \left\{ \sum_{i=0}^{\infty} h_b(R_b^i(u)) \right\}. \quad (10.23)$$

All expressions have the same value 1 at zero, and the series converges in every fixed mixed norm on the common collar. More precisely, the finite scalar product has the exact remainder

$$\log B_b(u) - \sum_{i=0}^{n-1} h_b(R_b^i(u)) = \log B_b(R_b^n(u)). \quad (10.24)$$

For each fixed k and each $\sigma \in (\lambda_+, 1)$ the right side has C^k norm at most $C_{k,\sigma} \sigma^n$.

Proof. The first equality is the original relative Fredholm determinant construction, with its convergent trace series. Theorem 10.3 identifies its positive density by exact physical concatenation. Composing its two first-flight cocycles gives $B_b(R_b(u))R'_b(u) = \lambda B_b(u)$. Let $\mathfrak{B}_b$ be the scalar density in Lemma 10.2. The positive quotient $B_b/\mathfrak{B}_b$ is invariant under R_b and equals one at zero. Iteration to the contracting fixed point makes the quotient identically one. This proves (10.23). Telescoping the logarithmic cocycle proves (10.24) for every finite n . For the mixed norm bound use $R_b^n = \zeta_b^{-1}(\lambda^n \zeta_b)$ and write $\log B_b(\zeta_b^{-1}(x)) = x C_b(x)$ with C_b jointly smooth. The remainder is then $\lambda^n \zeta_b(u) C_b(\lambda^n \zeta_b(u))$. The same differentiated estimate as in (10.14) proves the stated bound. $\square$

In particular, the infinite scalar product is not introduced as a new linearization theorem. Equation (10.23) is an identification of two normalized constructions for the same physical branch. The finite-bridge trace-norm comparison, the physical reference twist, the first-hit selection and the residual-time integration still have their separate proofs. The scalar product does not replace them.

Corollary 10.5 (Stable-action width determined by the even laws). *Put $\hat{S}_b = S_b \circ \zeta_b^{-1}$. Then the limiting laws can be written*

$$\mathcal{F}_{bc}(d) = \frac{\sqrt{a_b a_c}}{\pi d^2} \int (d - \hat{S}_b(x) - \hat{S}_c(y))_+ dx dy. \quad (10.25)$$

For small $E > 0$, let $u_{b,-}(E) < 0 < u_{b,+}(E)$ solve $S_b(u) = E$. The two even laws determine the normalized widths

$$\begin{aligned} \mathcal{W}_b(E) &:= \sqrt{\frac{a_b}{2}} [\zeta_b(u_{b,+}(E)) - \zeta_b(u_{b,-}(E))] \\ &= \int_0^E x^{-1/2} \mathcal{V}_b(x) dx. \end{aligned} \quad (10.26)$$

Thus the smooth invariant is the complete normalized sublevel width of the half-line action in its intrinsic linearizing coordinate.

Proof. The change of variables $x = \zeta_b(u)$ has differential $dx = B_b(u) du$. This proves (10.25). The pushforward calculation in (11.1) gives

$$\int_{S_b(u) < E} B_b(u) du = \sqrt{\frac{2}{a_b}} \int_0^E x^{-1/2} \mathcal{V}_b(x) dx.$$

It is also exactly $\zeta_b(u_{b,+}(E)) - \zeta_b(u_{b,-}(E))$. Theorem 11.1 determines $\mathcal{V}_b$ from the even law, proving the assertion. That theorem's proof uses only the half-line formula and Volterra uniqueness, not this corollary. $\square$

Conversely, on $E > 0$,

$$\mathcal{V}_b(E) = \sqrt{E} \mathcal{W}_b'(E), \quad \mathcal{W}_b(0) = 0, \quad \mathcal{W}_b(E) = 2\sqrt{E} + O(E^{3/2}). \quad (10.27)$$

The first expression extends smoothly to $E = 0$ with value one. Thus the normalized width and the energy profile are equivalent coordinates on the same invariant, not two independent invariants. The pushforward theorem determines a sublevel measure; the width formula expresses that measure in the classical stable coordinate.

The normalization in (10.26) is useful when the separate curvatures are not supplied. The even laws determine $\mathcal{W}_b$ without knowing a_b separately. With a_b supplied, they also determine the unnormalized coordinate width. When both participating contact graphs are even, S_b and B_b are even and ζ_b is odd, so this width specifies both inverse branches of $\hat{S}_b$. For a general contact it specifies their difference. The physical contact-graph inverse in Section 12 is a further geometric argument, not an inference that a symmetrized width determines an arbitrary asymmetric boundary.

10.4. The comparison at the level of the physical theorem. Proposition 10.1 explains the analytic relative product mechanism, including its physical projection and exact reference normalization. If the limiting endpoint Hessians are positive, Proposition 9.3 also transports its estimates through the normalized residual integral. At an isolated analytic billiard channel the positivity and first-hit property are verified in Sections 4–5. An odd-flight analytic comparison can likewise compose the terminal chart with one physical flight. Thus neither the parity distinction nor the absence of global finite horizon is an obstruction to that local analytic mechanism. They

are requirements handled simultaneously in our physical theorem, not separate claims of a new normal-form effect.

Theorem 2.1 does not supply normalizing charts as data or hypotheses. It starts with smooth obstacle graphs and positive geometric margins. Its half-line and finite-bridge contractions, localized trace-ideal estimates, and common sublevel charts prove the mixed parameter and offset bounds on one closed collar, with the exact phase normalization and the physical event selection. A pointwise analytic normal form supplies none of these uniform chart estimates for an arbitrary smooth family. The conditional family clause in Proposition 10.1 states exactly what would be needed to use that route instead. No assertion that such a smooth normal-form route is impossible, or absent from all prior literature, is required here.

Theorem 10.3 makes the geometric meaning of the determinant amplitude explicit in the full smooth class: it is the derivative of the normalized scalar linearizer along the physical stable branch, with an exact first-flight cocycle linking the two contact types. Corollary 10.5 identifies what the complete invariant measures in that coordinate. These conclusions connect the analytic and classical scalar interpretations to the constructed smooth amplitude. Equation (10.23) gives their exact relation; (10.27) records that the width is a coordinate on the previously determined invariant. Neither interpretation is counted as an additional rigidity theorem.

The remaining steps concern information that the forward benchmark does not reconstruct: the Volterra inverse for the complete widths; the explicit independent-contact geometric inverse, its analytic physical image and its two-flight coordinates; and the regularized binary experiment with its charged smooth-class calibration. The preparation theorem uses certified physical smoothness bounds, not access to Δ or to a canonical chart. Its sufficient rate is not claimed to be optimal. The central result is the joint passage from smooth physical records to an intrinsic boundary measure, to its complete observable invariant, and to a finite-preparation inverse. The classical analytic product mechanism is part of that explanation, not the novelty claimed for the package.

10.5. An exact mixed-parameter benchmark. The strict exponential margin in Lemma 10.2 is part of its mixed-parameter assertion. The following exact family, recorded in the independent memorandum [11], explains why the multiplier itself cannot in general replace that margin.

Proposition 10.6 (A strict-margin example). *On a common interval about zero put*

$$R_{\lambda,a}(u) = \frac{\lambda u}{1 + a(1 - \lambda)u}, \quad z_a(u) = \frac{u}{1 + au}, \quad B_a(u) = (1 + au)^{-2}.$$

Let $0 < \lambda_- \leq \lambda \leq \lambda_+ < 1$ and let a range over a compact interval containing zero in its interior. For a sufficiently small common endpoint interval these maps are smooth contractions and

$$R_{\lambda,a}^n(u) = \frac{\lambda^n u}{1 + a(1 - \lambda^n)u}, \tag{10.28}$$

$$\lambda^{-n}(R_{\lambda,a}^n)'(u) = [1 + a(1 - \lambda^n)u]^{-2}, \tag{10.29}$$

$$\lambda^{-n}R_{\lambda,a}^n(u) - z_a(u) = \frac{a\lambda^n u^2}{[1 + a(1 - \lambda^n)u](1 + au)}. \tag{10.30}$$

The normalized errors converge with every fixed mixed derivative at rate $C_{k,\sigma}\sigma^n$ for each $\lambda_+ < \sigma < 1$. There is no such bound with $C\lambda_+^n$ for the mixed derivative $\partial_\lambda \partial_a$ of (10.30) on a nontrivial endpoint interval.

Proof. The denominators are uniformly positive after restricting the endpoint interval. Also $R_{\lambda,a}'(u) = \lambda/[1 + a(1 - \lambda)u]^2$ is then positive and uniformly less than one. Direct substitution gives $z_a \circ R_{\lambda,a} = \lambda z_a$ and $z_a^{-1}(x) = x/(1 - ax)$. Iteration and one endpoint differentiation prove (10.28)–(10.29). Subtraction proves (10.30). At each fixed order, differentiation gives

a polynomial in n times λ_+^n , multiplied by uniformly bounded rational functions; the lower bound on λ controls the finitely many inverse powers introduced. A strict margin absorbs that polynomial. The density estimate follows either directly or by one additional endpoint derivative.

At $a = 0$ the mixed derivative of the error is exactly

$$\partial_\lambda \partial_a (\lambda^{-n} R_{\lambda,a}^n(u) - z_a(u)) \Big|_{a=0} = n \lambda^{n-1} u^2.$$

At $\lambda = \lambda_+$ and any fixed $u \neq 0$, its ratio to λ_+^n is nu^2/λ_+ , which is unbounded. Finally the exact identity

$$\lambda^{-n} (R_{\lambda,a}^n)'(u) B_a(R_{\lambda,a}^n(u)) = B_a(u)$$

checks the scalar product remainder with no asymptotic expansion. $\square$

This is a benchmark for mixed estimates, not a billiard realization or an additional linearization theorem. The physical theorem retains its already strict exponential margin and its independent relative finite-bridge proof.

11. BOUNDARY ENERGY PROFILES AND NONLINEAR PARITY COMPATIBILITY

Fix one of the general channels of Theorem 2.1 and retain observations from both orientations. For an even number of flights, the first and last impacts have the same contact type; for an odd number they have different types. Write $\mathcal{F}_{00}, \mathcal{F}_{11}$ for the two even limiting functions and $\mathcal{F}_{01} = \mathcal{F}_{10}$ for the odd one. Their normalization is that of (2.6), with the same free area A and multiplier γ . Thus they are scalar functions derived from selected physical event probabilities, not postulated independent boundary densities.

11.1. An intrinsic positive factorization. For each contact type put $a_b = S_b''(0) > 0$ and introduce the signed Morse coordinate and its increasing inverse,

$$t_b(u) = \operatorname{sgn}(u) \sqrt{2S_b(u)}, \quad u = \chi_b(t).$$

The value at $u = 0$ is defined by smooth extension. Set

$$\begin{aligned} f_b(t) &= B_b(\chi_b(t)) \chi_b'(t), \\ \mathcal{V}_b(x) &= \frac{\sqrt{a_b}}{2} \{f_b(\sqrt{2x}) + f_b(-\sqrt{2x})\}, \quad x \geq 0. \end{aligned} \tag{11.1}$$

We call $\mathcal{V}_b$ the normalized boundary energy profile. It is a symmetrized weighted energy datum: no assertion that it determines a nonsymmetric contact graph is implicit in this terminology.

Theorem 11.1 (Nonlinear boundary compatibility). *On a sufficiently small common closed collar the functions $\mathcal{V}_b$ are smooth, positive, and satisfy $\mathcal{V}_b(0) = 1$. For $b, c \in \{0, 1\}$ the limiting physical laws have the representation*

$$\mathcal{F}_{bc}(d) = \frac{2}{\pi d^2} \int_{\substack{x,y>0 \\ x+y<d}} (d-x-y) \frac{\mathcal{V}_b(x) \mathcal{V}_c(y)}{\sqrt{xy}} \, dx \, dy. \tag{11.2}$$

Let $*$ denote Volterra convolution on the positive half-line, used only at arguments in this collar, and define

$$k_b(x) = x^{-1/2} \mathcal{V}_b(x), \quad K_{bc}(d) = \frac{\pi}{2} \frac{d^2}{dd^2} \{d^2 \mathcal{F}_{bc}(d)\}. \tag{11.3}$$

Then $K_{bc} = k_b * k_c$, $K_{bc}(0) = \pi$, and

$$K_{01} * K_{01} = K_{00} * K_{11}. \tag{11.4}$$

Within the class of smooth normalized profiles, $\mathcal{F}_{00}$ and $\mathcal{F}_{11}$ determine $\mathcal{V}_0, \mathcal{V}_1$ uniquely and therefore determine the complete odd law $\mathcal{F}_{01}$ on the same collar. The assertions hold without

equal-curvature, even-graph or symmetry assumptions. All the forward identities have the mixed geometric regularity of the relative law.

Proof. Strict convexity makes the signed Morse coordinate smooth and increasing near zero, with derivative $\sqrt{a_b}$ at zero. Consequently f_b is smooth, positive and has value $a_b^{-1/2}$. Its even part is a smooth function of t^2 , by the even-function argument in Proposition 9.3. This proves the asserted properties of $\mathcal{V}_b$, including its parameter regularity. Reversing the choice of transverse direction merely interchanges the two terms in (11.1).

Under $x = S_b(u) = t^2/2$, the pushforward of $B_b(u) du$ has, for $x > 0$, density

$$\frac{f_b(\sqrt{2x}) + f_b(-\sqrt{2x})}{\sqrt{2x}} = \sqrt{\frac{2}{a_b}} x^{-1/2} \mathcal{V}_b(x).$$

Insert this expression twice into (2.5), whose prefactor for types b, c is $\sqrt{a_b a_c}/(\pi d^2)$. The Hessian factors cancel and give (11.2). This calculation also proves equality of the two odd orientations. It is valid for each even orientation by applying the relative law after interchanging the contact types.

The kernels k_b are locally integrable. Fubini's theorem rewrites (11.2) as

$$\frac{\pi}{2} d^2 \mathcal{F}_{bc}(d) = \int_0^d (d-s)(k_b * k_c)(s) ds.$$

The convolution is continuous down to zero: substitution $x = ds$ gives

$$(k_b * k_c)(d) = \int_0^1 \frac{\mathcal{V}_b(ds) \mathcal{V}_c(d(1-s))}{\sqrt{s(1-s)}} ds,$$

whose value at zero is π . Differentiation proves (11.3). Associativity and commutativity of these locally integrable convolutions, again by Fubini on a bounded simplex, give (11.4). No extension of the physical law beyond the collar is used.

It remains to prove uniqueness, since a formal square root of Taylor coefficients alone would not suffice for smooth functions. Suppose $k = x^{-1/2}V(x)$ and $\ell = x^{-1/2}\tilde{V}(x)$, with smooth $V(0) = \tilde{V}(0) = 1$, have $k * k = \ell * \ell$ on the collar. Put $f = k - \ell$, $a = k + \ell$ and $k_*(x) = x^{-1/2}$. Then $a * f = 0$ and

$$a(x) = 2x^{-1/2} + x^{1/2}w(x)$$

for a smooth w . The identity $k_* * k_* = \pi$ gives

$$(k_* * a)(x) = 2\pi + R(x), \quad R(x) = x \int_0^1 \frac{s^{1/2}}{(1-s)^{1/2}} w(xs) ds.$$

Here $R(0) = 0$ and R' is bounded. Convolving $a * f = 0$ with k_* and differentiating its absolutely continuous integral yields

$$2\pi f + R' * f = 0$$

almost everywhere. Hence $|f(x)| \leq C \int_0^x |f(s)| ds$. The absolutely continuous function $\int_0^x |f(s)| ds$ has zero initial value, and the elementary Gronwall inequality makes it zero throughout the collar. Thus $k = \ell$. Applying this argument to each of K_{00} and K_{11} proves uniqueness of both profiles; (11.2) then determines the odd law. $\square$

Theorem 11.2 (Stability of the complete boundary energy profile). *Fix $D > 0$ in the common collar. Let $V, \tilde{V}$ be two smooth normalized profiles, with $V(0) = \tilde{V}(0) = 1$ and $\|V'\|_\infty, \|\tilde{V}'\|_\infty \leq B$ on $[0, D]$. Let $F, \tilde{F}$ be their same-type laws defined by (11.2), and put $K = (\pi/2)(d^2 F)''$, $\tilde{K} = (\pi/2)(d^2 \tilde{F})''$. Then*

$$\|V - \tilde{V}\|_{C^0([0, D])} \leq \frac{D}{\pi} e^{BD/2} \|K' - \tilde{K}'\|_{C^0([0, D])} \leq C_{B, D} \|F - \tilde{F}\|_{C^3([0, D])}. \quad (11.5)$$

Thus the two complete boundary profiles are locally Lipschitz functions of the two same-type laws on their physical image, when the latter are given this specified differentiated norm. The resulting odd law is Lipschitz in C^0 on uniformly bounded profile sets.

Proof. Use $k = x^{-1/2}V$, $\ell = x^{-1/2}\tilde{V}$, $f = k - \ell$, $a = k + \ell$, $k_* = x^{-1/2}$, and put $Q = K - \tilde{K} = a * f$. The convolution in the preceding proof has the more direct expression

$$(k_* * a)(x) = 2\pi + R(x), \quad R(x) = \int_0^1 \frac{V(xs) + \tilde{V}(xs) - 2}{\sqrt{s(1-s)}} ds.$$

It follows that $R(0) = 0$ and

$$\|R'\|_\infty \leq 2B \int_0^1 \frac{s}{\sqrt{s(1-s)}} ds = \pi B.$$

Since $Q(0) = 0$, differentiation under a convolution, or integration by parts before differentiation, gives

$$2\pi f + R' * f = (k_* * Q)' = k_* * Q'.$$

The right side is bounded by $2\sqrt{x} \|Q'\|_\infty$. The normalized initial values imply that f extends continuously with $f(0) = 0$. Therefore

$$|f(x)| \leq \frac{\sqrt{D}}{\pi} \|Q'\|_\infty + \frac{B}{2} \int_0^x |f(s)| ds.$$

Gronwall's inequality gives $\|f\|_\infty \leq \sqrt{D} e^{BD/2} \|Q'\|_\infty / \pi$. Multiplication by $\sqrt{x}$ proves the first estimate. For the second,

$$Q'(d) = \frac{\pi}{2} \{6(F - \tilde{F})'(d) + 6d(F - \tilde{F})''(d) + d^2(F - \tilde{F})'''(d)\},$$

which is bounded by the asserted C^3 norm. Finally the integral (11.2) is a bilinear map of the profiles with a positive kernel of normalized total mass one. Subtracting its two products proves the C^0 Lipschitz assertion for the odd law. $\square$

This is stability of normalized energy profiles, not of an arbitrary nonsymmetric contact graph. It is an inverse on the image of the forward profile map; no claim that every positive scalar function is a convolution square is needed. The derivative norm in (11.5) is part of the observation statement. The unregularized theorem does not replace it by uniform error of raw probabilities. The finite-preparation theorem in Section F.A.5 adds regularization and supplied smoothness bounds to address that different data problem. For finite jets, an explicit polynomial inverse is available without solving a function-valued equation.

11.2. Finite jets and observable identities. For a smooth right germ h , write $[d^n]h = h^{(n)}(0)/n!$. Fix $M \geq 0$ and work in the truncated polynomial ring $\mathbb{R}[z]/(z^{M+1})$. Introduce

$$\begin{aligned} \mathcal{M}_{bc}^{[M]}(z) &= \sum_{n=0}^M \frac{(n+2)!}{2} [d^n] \mathcal{F}_{bc} z^n, \\ \mathcal{A}_b^{[M]}(z) &= \sum_{n=0}^M \left(\frac{1}{2}\right)_n [x^n] \mathcal{V}_b z^n, \quad \left(\frac{1}{2}\right)_0 = 1. \end{aligned} \tag{11.6}$$

The notation $(1/2)_n$ denotes the rising factorial. Truncation is important: even for analytic input, no convergence radius of a factorially reweighted infinite series is being asserted.

Theorem 11.3 (Rank-one jets and stable finite-order inversion). *The transformed jets satisfy*

$$\mathcal{M}_{bc}^{[M]} = \mathcal{A}_b^{[M]} \mathcal{A}_c^{[M]}, \quad (\mathcal{M}_{01}^{[M]})^2 = \mathcal{M}_{00}^{[M]} \mathcal{M}_{11}^{[M]} \pmod{z^{M+1}}. \quad (11.7)$$

Both profiles' order- M jets are recovered from the two same-type laws by the following triangular recursion. If $m_{bb,n} = [z^n] \mathcal{M}_{bb}^{[M]}$ and $\alpha_{b,n} = [z^n] \mathcal{A}_b^{[M]}$, then

$$\alpha_{b,0} = 1, \quad \alpha_{b,n} = \frac{1}{2} \left(m_{bb,n} - \sum_{r=1}^{n-1} \alpha_{b,r} \alpha_{b,n-r} \right), \quad [x^n] \mathcal{V}_b = \frac{\alpha_{b,n}}{(1/2)_n}. \quad (11.8)$$

For fixed M these reconstruction maps, and the resulting prediction of the odd jet, are Lipschitz on bounded sets of input coefficients. They are smooth with respect to geometric parameters on the compact families of the relative law.

Proof. Write $v_{b,r} = [x^r] \mathcal{V}_b$. The beta integral on the simplex gives, for $r, s \geq 0$,

$$\int_{x,y>0, x+y<d} (d-x-y)x^{r-1/2}y^{s-1/2} dx dy = \frac{\pi(1/2)_r(1/2)_s}{(r+s+2)!} d^{r+s+2}.$$

This follows by first setting $x = dX, y = dY$ and integrating in Y , then in X ; both integrals are the ordinary beta integral. Taylor's integral remainders, bounded on the scaled simplex, justify coefficient extraction for smooth rather than only analytic profiles. Thus

$$[d^n] \mathcal{F}_{bc} = \frac{2}{(n+2)!} \sum_{r+s=n} (1/2)_r (1/2)_s v_{b,r} v_{c,s}. \quad (11.9)$$

Multiplying by $(n+2)!/2$ gives the first identity in (11.7); its determinant identity is an immediate polynomial consequence. All constant coefficients are one. Comparing the degree- n coefficient of $\mathcal{M}_{bb}^{[M]} = (\mathcal{A}_b^{[M]})^2$ leaves $2\alpha_{b,n}$ as the only unknown term, proving (11.8). Induction makes every output coefficient a polynomial in finitely many input coefficients, divided only by fixed positive numbers. Its derivative is bounded on each bounded coefficient box, proving the Lipschitz assertion. Finite parameter differentiation of these polynomial identities proves the last assertion without any infinite-order uniformity claim. $\square$

Corollary 11.4 (The first two nonlinear compatibility identities). *With primes denoting right offset derivatives at zero,*

$$2\mathcal{F}'_{01} = \mathcal{F}'_{00} + \mathcal{F}'_{11}, \quad (11.10)$$

$$\mathcal{F}''_{01} = \frac{\mathcal{F}''_{00} + \mathcal{F}''_{11}}{2} - \frac{3}{16}(\mathcal{F}'_{00} - \mathcal{F}'_{11})^2. \quad (11.11)$$

In particular the odd second derivative is at most the mean of the two even second derivatives, with the exact deficit displayed above.

Proof. The first two nonconstant coefficients of $\mathcal{M}_{bc}$ are $3\mathcal{F}'_{bc}$ and $6\mathcal{F}''_{bc}$. Equate degrees one and two in $\mathcal{M}_{01}^2 = \mathcal{M}_{00}\mathcal{M}_{11}$ and substitute the degree-one equality in the degree-two equality. This gives the two formulas, including the negative square coefficient. $\square$

These identities concern nonlinear offset coefficients. The zeroth-order condition merely says that every normalized law has value one. Neither the zeroth-order condition nor the multiplier alone determines the first two coefficients. The geometric variation in Proposition A.4.1 and the realization in Theorem A.4.2 exhibit nonconstant boundary data at fixed leading selected-channel invariants. The compatibility theorem organizes such data without imposing the identical-even hypothesis of the separate graph-jet inverse in the appendix.

11.3. Finite physical records and a quantitative defect. The preceding identities are exact statements about the limiting functions. The relative law also controls how they can be read from long but finite records. For $n \geq 1$ define

$$\begin{aligned} F_{bb,n}(d) &= \frac{2A \sinh(2n\gamma)}{d^2} \Pr(E_{2n,e_b}(d)), \quad e_0 = e, \quad e_1 = \bar{e}, \\ F_{01,n}(d) &= \frac{2A \sinh((2n+1)\gamma)}{d^2} \Pr(E_{2n+1,e}(d)). \end{aligned} \quad (11.12)$$

They are extended to $d = 0$ by their exact smooth right limits. These definitions use the selected patch labels and the normalization A, γ ; they are not unlabelled binary data with unknown normalizer. Estimation of these auxiliary quantities is a separate observation problem.

Corollary 11.5 (Finite-record compatibility and profile-jet recovery). *Let M and the number of geometric derivatives be fixed. Apply the transform (11.6) to the three finite laws in (11.12), obtaining $\mathcal{M}_{bc,n}^{[M]}$. With the maximum coefficient norm on truncated polynomials,*

$$\|(\mathcal{M}_{01,n}^{[M]})^2 - \mathcal{M}_{00,n}^{[M]} \mathcal{M}_{11,n}^{[M]}\| \leq C_M \tau^{2n}. \quad (11.13)$$

The same estimate holds for each fixed mixed geometric derivative. Suppose the measured order- M jets of the two same-type finite laws have coefficient error at most ϵ , with their known constant coefficients fixed to one. The recursion (11.8) reconstructs both boundary profile jets, and predicts the odd limiting jet, with error

$$C_M(\epsilon + \tau^{2n}). \quad (11.14)$$

For every fixed subcollar $[0, D] \subset [0, d_]$, the finite kernels $K_{bc,n} = (\pi/2)(d^2 F_{bc,n})''$ also satisfy*

$$\|K_{01,n} * K_{01,n} - K_{00,n} * K_{11,n}\|_{C^0([0,D])} \leq C_D \tau^{2n}.$$

Proof. Theorem 2.1, applied to each orientation and parity, bounds $F_{bc,n} - \mathcal{F}_{bc}$ in every fixed C^k norm by $C_k \tau^{2n}$. Coefficient extraction and the finite transform preserve this order. The determinant in (11.7) is a polynomial in bounded coefficients; subtract its value at the limiting jets to prove (11.13). The product rule proves the parameter derivative assertion. Apply the finite polynomial inverse of Theorem 11.3 to the two measured jets to get (11.14); this is also defined for noisy jets outside the physical image. Finally the relative C^2 estimate bounds $K_{bc,n} - K_{bc}$ uniformly on $[0, D]$. For bounded functions there, $\|f * g\|_\infty \leq D \|f\|_\infty \|g\|_\infty$. Telescoping the four products and using (11.4) proves the last claim. $\square$

The two oriented even laws suffice to reconstruct both normalized boundary energy profiles and to predict the odd law. Odd records are needed for the independent compatibility test, not as a third datum for this reconstruction. Neither conclusion recovers full collision arrays or an arbitrary billiard table. Unlike the fixed-flight parametric inverse below, the compatibility defect and its vanishing rate use the long-bridge relative theorem directly. Section F.A.5 supplies a regularized finite-preparation version of the full-profile inverse, with the required normalization and smoothness bounds stated explicitly.

12. DETERMINATION OF TWO INDEPENDENT EVEN CONTACTS

The boundary invariant has a geometric inverse on a class substantially larger than the identical-contact class. In this section each facing graph is even in the common transverse coordinate, but the two graphs are otherwise unrelated. Their curvatures may be different or equal. This is an additional hypothesis for geometric determination, not for Theorems 2.1 and 11.1. The gap and the two labelled contact curvatures are fixed. They are not inferred separately from the single multiplier.

Write, for $b = 0, 1$,

$$\psi_b(y) = \frac{\kappa_b}{2}y^2 + \sum_{m \geq 2} \frac{q_{b,2m}}{(2m)!}y^{2m}, \quad c_b = 1 + g\kappa_b, \quad c = \sqrt{c_0c_1}, \quad \gamma = \operatorname{arcosh} c, \quad \varrho = e^{-\gamma}. \quad (12.1)$$

For smooth graphs the series specifies jets only. Put

$$a_b = \frac{c \sinh \gamma}{gc_{1-b}}, \quad f_{b,k} = [d^k]\mathcal{F}_{bb}(d), \quad q_m = (q_{0,2m}, q_{1,2m})^t, \quad f_{m-1} = (f_{0,m-1}, f_{1,m-1})^t.$$

The two even laws are obtained by observing the two orientations of one selected physical channel. In particular, their indices retain contact labels even when the curvatures coincide.

Theorem 12.1 (Block-triangular two-contact inverse). *For every $m \geq 2$, the vector f_{m-1} depends only on $g, \kappa_0, \kappa_1, q_2, \dots, q_m$, and is affine in q_m . Its derivative with respect to that last vector is*

$$\frac{\partial f_{m-1}}{\partial q_m} = - \begin{pmatrix} P_m & Q_m \\ Q_m & P_m \end{pmatrix} \operatorname{diag}(K_{m,0}, K_{m,1}), \quad K_{m,b} = \frac{4}{2^m(m+1)(m!)^2 a_b^m}, \quad (12.2)$$

where

$$\begin{aligned} E_m &= \frac{1}{e^{4(m-1)\gamma} - 1} - \frac{1}{e^{4m\gamma} - 1}, \\ O_m &= \frac{1}{2 \sinh(2(m-1)\gamma)} - \frac{1}{2 \sinh(2m\gamma)}, \\ P_m &= \coth(2m\gamma) + 2mE_m, \quad Q_m = \operatorname{csch}(2m\gamma) + 2mO_m. \end{aligned} \quad (12.3)$$

Both eigenvalues of the symmetric factor are positive. More explicitly,

$$P_m + Q_m = \coth(m\gamma) + 2m \left\{ \frac{1}{e^{2(m-1)\gamma} - 1} - \frac{1}{e^{2m\gamma} - 1} \right\}, \quad (12.4)$$

$$P_m - Q_m = m \tanh((m-1)\gamma) - (m-1) \tanh(m\gamma) > 0. \quad (12.5)$$

Consequently, for every finite $M \geq 2$, the map

$$(q_2, \dots, q_M) \mapsto (f_1, \dots, f_{M-1}) \quad (12.6)$$

has a block-triangular analytic inverse on its image. The inverse is locally Lipschitz on compact finite-jet sets and positive compact g, κ_0, κ_1 boxes. These bounds require no separation of κ_0 and κ_1 , and no separation of the higher jets.

Equality of the two normalized even-law germs, for two pairs of real-analytic even graphs at the same labelled g, κ_0, κ_1 , implies equality of both graph germs. For smooth compact parameter families, replacing the limiting coefficients by their even finite-flight coefficients recovers the jet vector through order $2M$ with error at most $C_M \tau^j$. No bound uniform in M is asserted.

12.1. The two boundary contributions.

Proof. We first establish the last-jet derivative; the resulting block invertibility will then give the inverse. Set $\mathfrak{r}_b = \sqrt{c_b/c_{1-b}}$. The linear half-line orbit starting at type b is

$$x_i^{(b)}(u) = t_i^{(b)}u + O(u^3 \rho^i), \quad t_i^{(b)} = \begin{cases} \varrho^i, & i \text{ even}, \\ \mathfrak{r}_b \varrho^i, & i \text{ odd}. \end{cases} \quad (12.7)$$

The cubic remainder follows from evenness and the half-line contraction of Lemma 7.1. Notice that $a_b/a_{1-b} = \mathfrak{r}_b^2$.

For clarity, finite-jet dependence can be proved before making any analyticity assumption on the graphs. Differentiate the stationary half-line equations at zero in their weighted sequence space. At each order, the highest orbit derivative is multiplied by the same positive quadratic Jacobi operator. Its inverse is (7.1); all remaining terms contain lower orbit jets and finite graph

jets. This gives an induction in the endpoint degree. The weighted derivative estimates justify the differentiated action sums and the differentiated trace-class determinant series. The coefficient of d^{m-1} in the residual-time integral uses the action through degree $2m$ and the amplitude through degree $2m-2$. Indeed, scaling the two endpoints by $\sqrt{d}$ gives degrees $d^{n/2-1}$ for an action term of endpoint degree n and $d^{n/2}$ for an amplitude term. The common Morse coordinates justify this calculation with smooth remainders down to $d=0$. Thus only graph jets through order $2m$ can enter.

Keep all lower jets fixed and vary $q_{r,2m}$. The first varying homogeneous term of a flight length is the sum of the corresponding pure endpoint powers divided by $(2m)!$. Its mixed derivative is zero. In the stationary action, a boundary site is counted once and an interior site twice. The envelope identity therefore gives, with $s_{b,2m} = S_b^{(2m)}(0)$,

$$\begin{aligned}\partial_{q_{b,2m}} s_{b,2m} &= 1 + 2 \sum_{k \geq 1} \varrho^{4mk} = \coth(2m\gamma), \\ \partial_{q_{1-b,2m}} s_{b,2m} &= 2\mathfrak{r}_b^{2m} \sum_{k \geq 0} \varrho^{2m(2k+1)} = \mathfrak{r}_b^{2m} \operatorname{csch}(2m\gamma).\end{aligned}\tag{12.8}$$

This can equally be computed on finite stationary segments and then passed to the half-line. Every differentiated sum is absolutely convergent. Orbit corrections of degree at least three cannot change the degree- $2m$ terms displayed here.

Write $\beta_{b,2m-2} = [u^{2m-2}]B_b(u)$, using ordinary coefficients. At this degree the edge-product variation vanishes, since the varying length term is pure. The varying interior Hessian is diagonal, with its i th diagonal entry, at a site of varied type, equal to

$$\frac{2(t_i^{(b)})^{2m-2} u^{2m-2}}{(2m-2)!}.$$

The variation of the stationary orbit starts in degree $2m-1$. An unvaried Hessian perturbation starts in degree two. Hence orbit variation does not affect the degree under consideration, and only the linear term of the logarithmic determinant contributes there. The diagonal half-line Green entry is

$$G_{ii}^{(b)} = \frac{1 - \varrho^{2i}}{2a_{(b+i) \bmod 2}}.$$

It follows that

$$\begin{aligned}\partial_{q_{b,2m}} \beta_{b,2m-2} &= -\frac{E_m}{a_b(2m-2)!}, \\ \partial_{q_{1-b,2m}} \beta_{b,2m-2} &= -\frac{\mathfrak{r}_b^{2m} O_m}{a_b(2m-2)!}.\end{aligned}\tag{12.9}$$

For the first formula sum $(1 - \varrho^{4k})\varrho^{4(m-1)k}$ over $k \geq 1$. For the second sum $(1 - \varrho^{2(2k+1)})\varrho^{(2m-2)(2k+1)}$ over $k \geq 0$ and use $a_{1-b}^{-1} = \mathfrak{r}_b^2 a_b^{-1}$. The sums are respectively E_m and O_m . Passing from $\log B_b$ to B_b does not change the first varying coefficient: multiplication by a positive-degree unvaried term would raise its degree.

Residual time and the invertible block. For the same-type law both endpoint quadratic actions have coefficient a_b . Under the density proportional to $(d - a_b(u^2 + v^2)/2)_+$ one has

$$\begin{aligned}\int (d - a_b(u^2 + v^2)/2)_+ du dv &= \frac{\pi d^2}{a_b}, \\ \mathbb{E}(u^{2k}) &= \frac{2 \binom{2k}{k}}{2^k(k+1)(k+2)a_b^k} d^k.\end{aligned}\tag{12.10}$$

These identities follow from the angular average $\binom{2k}{k}/4^k$ and a one-dimensional radial integration. First variation of the action has no moving-boundary term, because the residual-time weight is zero at the boundary. Its contribution from $s_{b,2m}$ to $f_{b,m-1}$ is $-K_{m,b}s_{b,2m}$. The amplitude contribution from $\beta_{b,2m-2}$ has coefficient

$$\frac{4\binom{2m-2}{m-1}}{2^{m-1}m(m+1)a_b^{m-1}}.$$

Combining these coefficients with (12.8)–(12.9) gives

$$\partial_{q_{b,2m}} f_{b,m-1} = -K_{m,b}P_m, \quad \partial_{q_{1-b,2m}} f_{b,m-1} = -K_{m,b}\mathfrak{r}_b^{2m}Q_m.$$

But $K_{m,b}\mathfrak{r}_b^{2m} = K_{m,1-b}$. This proves (12.2), including the unequal-curvature factors. Lower nonlinear action and amplitude terms can multiply a first variation only at higher endpoint degree. Thus the computation remains valid at arbitrary fixed lower even jets. The last-jet derivatives are constant in that jet, proving the asserted affinity.

Both E_m and O_m are positive. The identities

$$\coth(2x) + \operatorname{csch}(2x) = \coth x, \quad \coth(2x) - \operatorname{csch}(2x) = \tanh x$$

and the geometric sums give (12.4). For the difference,

$$E_m - O_m = \frac{1}{2}\{\tanh((m-1)\gamma) - \tanh(m\gamma)\},$$

which proves (12.5). The function $\tanh x$ is strictly concave for $x > 0$ and has value zero at zero. Applying strict concavity at $(m-1)\gamma = ((m-1)/m)m\gamma$ proves the strict inequality in (12.5). Thus

$$\det \frac{\partial f_{m-1}}{\partial q_m} = K_{m,0}K_{m,1}(P_m + Q_m)(P_m - Q_m) > 0.$$

There is no denominator involving $\kappa_0 - \kappa_1$.

To write the inverse explicitly, suppose all lower graph jets are already known and let R_m denote their contribution to f_{m-1} . For $z_m = R_m - f_{m-1}$, solve

$$q_m = \operatorname{diag}(K_{m,0}^{-1}, K_{m,1}^{-1}) \frac{1}{P_m^2 - Q_m^2} \begin{pmatrix} P_m & -Q_m \\ -Q_m & P_m \end{pmatrix} z_m. \quad (12.11)$$

The recursion proves uniqueness at every finite order. Its coefficients are analytic in the positive geometric parameters and in lower finite jets: the operations are local length expansions, inversion of a positive quadratic operator, and convergent geometric sums. On any fixed compact set and finite collection of orders, the positive denominators have positive minima. The recursion and its derivatives therefore give the stated locally Lipschitz inverse.

For analytic graphs, equality of all recovered even jets implies equality of the two convergent graph series; their odd jets vanish by assumption. For smooth families, the differentiated relative theorem gives $|f_{j,b,k} - f_{b,k}| \leq C_M \tau^j$ through $k = M - 1$. Apply the finite-order Lipschitz inverse. For large j its inputs stay in the chosen compact neighborhood. The algebraic recursion is defined for every finite coefficient vector at fixed positive geometric data; the finitely many remaining even j are covered by increasing the constant. This completes the proof. $\square$

Corollary 12.2 (Labelled analytic boundary determination). *In the two-independent-even contact class, the gap, the two labelled contact curvatures, and the two normalized even-law germs determine both real-analytic contact graphs. If their participating obstacles have connected real-analytic closed boundaries, each boundary is determined in its contact frame. The conclusion allows equal contact curvatures and does not require congruent obstacles or identical graphs. It makes no determination of unobserved obstacles or of the lattice.*

Proof. The theorem determines each analytic graph germ. To compare two candidate closed boundaries at one contact, parametrize both by oriented arclength from that contact and its fixed tangent. Their curvatures are real-analytic functions on the real line, continued around the closed curves, and agree on an interval. The identity theorem makes them agree everywhere. Uniqueness for the planar Frenet equations with the common initial frame makes the two boundary images identical. Apply the argument at each participating contact. This is exact analytic continuation; no stability of distant boundary points from noisy local data is asserted. $\square$

At equal curvatures, restricting (12.2) to the diagonal $q_{0,2m} = q_{1,2m}$ recovers exactly Theorem F.A.3.1. The new direction is the antisymmetric one, whose nonvanishing coefficient is (12.5). A single identical-contact calculation does not see that coefficient. Without the evenness assumption, odd graph jets also enter the symmetrized energy datum. The theorem does not identify their fibres; the general relative law and full energy-profile inverse remain valid without that assumption.

13. SIGNED ENDPOINT RIGIDITY AND THE ALL-ORDER DETERMINANT-ONE INVERSE

The scalar onset laws integrate out the signs of the endpoint coordinates and therefore retain only a symmetrized profile. The physical endpoint record is strictly richer. If the signed endpoint positions are retained, the support recovers the two half-line actions before symmetrization. We prove here that those two actions determine the complete labelled contact jets. The proof is written at the level needed by the inverse theorem: the nonlinear half-line stationarity operator is inverted quantitatively in a weighted Banach space, the infinite action is differentiated through finite truncations, and the first occurrence of each new graph jet is isolated by homogeneous degree.

Fix oriented transverse coordinates at the two labelled contacts. For $b \in \{0, 1\}$ use an even number of flights and the orientation for which the first and last impacts have type b . Let $\Pi_{bb,d}^{\text{end},\infty}$ be the limiting conditional endpoint law after residual time has been integrated out. By Theorem 2.1, on a common endpoint box its interior density is proportional to

$$(d - S_b(u) - S_b(v))_+ B_b(u) B_b(v), \quad (13.1)$$

where $B_b > 0$.

Theorem 13.1 (All-order signed-endpoint rigidity). *Let the two labelled facing contacts be C^∞ and strictly convex. No reflection symmetry, equality of the two contacts, evenness, or supplied contact curvature is assumed. Suppose the selected channel and its signed endpoint coordinates are retained.*

There is $d_ > 0$ such that the two families*

$$\{\Pi_{00,d}^{\text{end},\infty} : 0 < d < d_*\}, \quad \{\Pi_{11,d}^{\text{end},\infty} : 0 < d < d_*\}$$

determine the half-line actions S_0, S_1 on a fixed contact neighborhood. If physical time is retained, the onset determines the gap g . Hence the gap, both contact curvatures, and every finite labelled graph jet are determined by the signed endpoint laws.

In the fixed oriented contact frames set $q_{b,n} = \psi_b^{(n)}(0)$, $q_{b,2} = \kappa_b$, and $q_n = (q_{0,n}, q_{1,n})^t$. For each finite $M \geq 2$ Taylor's formula gives

$$\psi_b(y) = \sum_{n=2}^M \frac{q_{b,n}}{n!} y^n + R_{b,M}(y), \quad |\partial_y^r R_{b,M}(y)| \leq C_M |y|^{M+1-r}, \quad 0 \leq r \leq M+1. \quad (13.2)$$

Here the bound uses a C^{M+1} bound on a common contact interval. For smooth contacts no convergence of an infinite Taylor series is asserted. Equal finite graph jets, including arbitrary

smooth remainders, are treated in Lemma 13.4 below. Put

$$a_b = S_b''(0), \quad s_n = (S_0^{(n)}(0), S_1^{(n)}(0))^t.$$

Then

$$\gamma = \operatorname{arsinh}(g\sqrt{a_0 a_1}), \quad c = \cosh \gamma, \quad c_0 = c\sqrt{a_0/a_1}, \quad c_1 = c\sqrt{a_1/a_0}, \quad \kappa_b = \frac{c_b - 1}{g}. \quad (13.3)$$

For every $n \geq 3$, s_n depends only on the leading geometry and $q_3, \dots, q_n$, is affine in q_n , and

$$D_{q_n} s_n = M_n := \begin{pmatrix} \coth(n\gamma) & \mathfrak{r}_0^n \operatorname{csch}(n\gamma) \\ \mathfrak{r}_1^n \operatorname{csch}(n\gamma) & \coth(n\gamma) \end{pmatrix}, \quad \mathfrak{r}_b = \sqrt{c_b/c_{1-b}}. \quad (13.4)$$

The determinant of M_n is exactly one. For every fixed M , the finite signed-jet forward map through order M is C^1 with an everywhere invertible differential; on compact positive geometric and finite-jet sets its least singular value is bounded away from zero. Its inverse is locally Lipschitz there.

For real-analytic contacts, equality of the two signed endpoint-law germs as $d \downarrow 0$ and of the physical onset determines the two labelled analytic contact germs. If the participating obstacle boundaries are connected and real analytic, their boundary images are determined in the fixed contact frames. For a smooth compact family and every fixed M , the corresponding finite even-flight support families recover both graph jets through order M with error $O_M(\tau^j)$.

13.1. Support thresholds and the leading geometry.

Recovery of the actions and the quadratic geometry. Positivity of B_b makes the interior support in (13.1) exactly

$$\{(u, v) : S_b(u) + S_b(v) < d\}.$$

Define

$$T_b(u, v) = \inf\{d > 0 : (u, v) \in \operatorname{int} \operatorname{supp} \Pi_{bb,d}^{\operatorname{end}, \infty}\}.$$

Then

$$T_b(u, v) = S_b(u) + S_b(v), \quad S_b(u) = T_b(u, 0), \quad (13.5)$$

because $S_b(0) = 0$. Thus the signed support recovers the unsymmetrized action.

For a selected j -flight itinerary the infimum of physical times at which the near-onset event has positive probability is jg , hence the time axis gives g . The half-line Hessian identity

$$a_b = \frac{c \sinh \gamma}{g c_{1-b}}, \quad c = \sqrt{c_0 c_1} = \cosh \gamma$$

gives

$$a_0 a_1 = \frac{\sinh^2 \gamma}{g^2}, \quad \frac{a_0}{a_1} = \frac{c_0}{c_1},$$

which is equivalent to (13.3). No curvature is supplied as an external parameter. $\square$

13.2. A quantitative weighted inverse for the nonlinear half-line. Let $\ell_b(y, z)$ be the local flight length from a point of type b with transverse coordinate y to the facing point of type $1 - b$ with coordinate z . For a half-line beginning at type b , set $x_0 = u$ and, for $i \geq 1$, write

$$\mathcal{F}_{b,i}(x, u; q) := \partial_2 \ell_{b+i-1}(x_{i-1}, x_i) + \partial_1 \ell_{b+i}(x_i, x_{i+1}) = 0, \quad (13.6)$$

where contact types are read modulo two. Fix a compact positive geometric set. Its hyperbolic exponent has a positive minimum $\gamma_- > 0$. Choose

$$e^{-\gamma_-} < \rho < 1, \quad (13.7)$$

and define

$$X_\rho = \left\{ x = (x_i)_{i \geq 1} : \|x\|_\rho := \sup_{i \geq 1} \rho^{-i} |x_i| < \infty \right\}.$$

Lemma 13.2 (Weighted Green bound and nonlinear inverse). *Let $H_b^0(q) = D_x \mathcal{F}_b(0, 0; q)$ be the quadratic half-line Jacobi operator. Its inverse has kernel*

$$G_{ik}^{(b)} = \frac{g\sigma_i^{(b)}\sigma_k^{(b)}}{2c \sinh \gamma} \left(e^{-\gamma|i-k|} - e^{-\gamma(i+k)} \right), \quad i, k \geq 1. \quad (13.8)$$

There is $C_0 < \infty$, uniform on the compact geometric set, such that

$$\|G^{(b)}y\|_\rho \leq C_0\|y\|_\rho. \quad (13.9)$$

After shrinking a common endpoint box, the unique stationary half-line $x^{(b)}(u; q)$ satisfies

$$\|x^{(b)}(u; q)\|_\rho \leq C_1|u|, \quad (13.10)$$

and, with $H_b(u, q) = D_x \mathcal{F}_b(x^{(b)}(u; q), u; q)$,

$$\|H_b(u, q) - H_b^0(q)\|_{X_\rho \rightarrow X_\rho} \leq C_2|u|. \quad (13.11)$$

Consequently, after choosing u_* so that $C_0 C_2 u_* \leq 1/2$,

$$\|H_b(u, q)^{-1}\|_{X_\rho \rightarrow X_\rho} \leq 2C_0, \quad |u| \leq u_*. \quad (13.12)$$

For every fixed finite collection of endpoint and geometric derivatives, the corresponding derivatives of $x^{(b)}$ are uniformly bounded in X_ρ .

Proof. The factors $\sigma_i^{(b)}$ take only two positive values on the compact geometric set, so (13.8) gives $|G_{ik}^{(b)}| \leq C e^{-\gamma|i-k|}$. If $|y_k| \leq \|y\|_\rho \rho^k$, then for $k \leq i$

$$e^{-\gamma(i-k)} \rho^k = \rho^i (e^{-\gamma_-} / \rho)^{i-k},$$

while for $k \geq i$

$$e^{-\gamma-(k-i)} \rho^k = \rho^i (\rho e^{-\gamma_-})^{k-i}.$$

Both geometric series are uniformly summable because of (13.7); the reflected term in (13.8) is smaller. This proves (13.9).

The nonlinear part of (13.6) is local and vanishes quadratically at the zero orbit. On a weighted ball $\|x\|_\rho \leq R|u|$, its derivative differs from H_b^0 by at most $C(|u| + \|x\|_\rho)$ in X_ρ operator norm. The Green bound therefore makes the standard fixed-point map a contraction on a common ball and gives (13.10). Substitution of that bound gives (13.11). Factor

$$H_b(u, q) = H_b^0(q) [I + G^{(b)}(H_b(u, q) - H_b^0(q))].$$

The bracket differs from the identity by an operator of norm at most $1/2$, so its Neumann series converges and proves (13.12).

Differentiate the fixed-point or stationarity equation. At every order the highest orbit derivative is multiplied by the same operator $H_b(u, q)$; all remaining terms are finite sums of local coefficient derivatives and lower orbit derivatives. The uniform inverse bound, the weighted product estimate, and induction give the last assertion. This is the quantitative version of the half-line construction in Lemma 7.1. $\square$

13.3. Derivative bookkeeping, the infinite action, and the envelope limit. The stationarity equations contain first derivatives of the local flight lengths. Hence differentiating them k times can expose graph derivatives of order $k + 1$. We do not infer the jet order of the action from that stationarity count. Instead the action filtration is proved directly from its first variation.

Set

$$\mathcal{A}_b(u, q) = \sum_{i \geq 0} [\ell_{b+i}(x_i^{(b)}(u; q), x_{i+1}^{(b)}(u; q)) - g]. \quad (13.13)$$

The weighted orbit bounds imply absolute and locally uniform convergence of this series and of every fixed finite collection of its mixed derivatives. Its value is $S_b(u)$.

Lemma 13.3 (Finite-truncation envelope identity). *Fix a finite jet coordinate $q_{r,m}$, realized by a smooth local graph variation which equals $y^m/m!$ in a neighborhood containing the statistical contact patch. Then, uniformly for $|u| \leq u_*$,*

$$\partial_{q_{r,m}} S_b(u) = \sum_{i \geq 0} \partial_{q_{r,m}} \ell_{b+i}(x_i^{(b)}(u), x_{i+1}^{(b)}(u)). \quad (13.14)$$

The series and every fixed finite number of u -derivatives converge absolutely. In particular no derivative of the stationary orbit appears in the right-hand side.

Proof. Let

$$A_{b,N}(u, q) = \sum_{i=0}^{N-1} [\ell_{b+i}(x_i^{(b)}, x_{i+1}^{(b)}) - g].$$

Differentiate with respect to $q_{r,m}$. The coefficients of $\partial_{q_{r,m}} x_i$ telescope into the stationarity equations for $1 \leq i \leq N - 1$. They vanish. The only orbit-variation term left is the right boundary term

$$R_N = \partial_2 \ell_{b+N-1}(x_{N-1}^{(b)}, x_N^{(b)}) \partial_{q_{r,m}} x_N^{(b)}.$$

The first factor is $O(|u|\rho^N)$ by stationarity and the local Taylor expansion at the zero orbit; the second is $O(\rho^N)$ after any prescribed finite collection of u -derivatives, by Lemma 13.2. Hence

$$|\partial_u^k R_N| \leq C_k \rho^{2N} \quad (13.15)$$

for every fixed k (after harmless adjustment at $u = 0$). Thus the boundary term tends to zero uniformly with the derivatives used below.

The direct parameter derivatives of the i th flight are bounded by a fixed multiple of a summable power of ρ^i , because both endpoints of that flight are $O(|u|\rho^i)$ up to a shift of one index. The Weierstrass test therefore permits passage to the limit $N \rightarrow \infty$ and termwise finite-order differentiation. This proves (13.14) and the asserted tail control. $\square$

13.4. Finite smooth remainders and factorization through jets. The coordinate variation in Lemma 13.3 alone does not justify independence from all smooth remainders. The next lemma treats the remainder as a function, before choosing any finite jet coordinates.

Lemma 13.4 (Smooth finite-jet factorization). *Fix $M \geq 2$. Let $\psi_0^{[0]}, \psi_1^{[0]}$ and $\psi_0^{[1]}, \psi_1^{[1]}$ be two smooth strictly convex anchored graph pairs with the same gap and the same graph jets through order M . On a common sufficiently small interval their half-line actions satisfy*

$$|S_b^{[1]}(u) - S_b^{[0]}(u)| \leq C_M |u|^{M+1} \sum_{i \geq 0} \rho^{(M+1)i} = \frac{C_M}{1 - \rho^{M+1}} |u|^{M+1}, \quad b \in \{0, 1\}. \quad (13.16)$$

Consequently $j_0^M S_b^{[1]} = j_0^M S_b^{[0]}$. The constants and contact interval can be chosen uniformly with positive gap and curvature margins and bounded C^{M+2} graph norms on a common initial interval.

For additional differentiated estimates, the corresponding higher smooth bounds are required. Bounds on a finite list of jet coefficients alone do not supply these functional remainder bounds.

Proof. Put $\Delta\psi_r = \psi_r^{[1]} - \psi_r^{[0]}$ and $\psi_{r,t} = \psi_r^{[0]} + t\Delta\psi_r$, $0 \leq t \leq 1$. Strict convexity and the anchor conditions persist along this interpolation on a common contact interval. Equal quadratic jets and the common gap fix the quadratic half-line operator. The contraction proof of Lemma 13.2, uniformly in t , gives stationary orbits $x^{(b,t)}$ with

$$|x_i^{(b,t)}(u)| \leq C|u|\rho^i, \quad i \geq 0, \quad x_0^{(b,t)} = u. \quad (13.17)$$

The same implicit-function argument gives C^M dependence on u and smooth dependence on t at the finite orders used here. In particular $\|\partial_t x^{(b,t)}\|_\rho$ is uniformly bounded. These assertions are local statements about graph pairs; no globally realized interpolation of periodic tables is required.

For a flight of starting type r , write

$$h_{r,t}(y, z) = g + \psi_{r,t}(y) + \psi_{1-r,t}(z), \quad \ell_{r,t}(y, z) = \sqrt{h_{r,t}(y, z)^2 + (z - y)^2}.$$

Taylor's remainder bound for the equal M -jets gives $|\Delta\psi_r(y)| \leq C_M|y|^{M+1}$. At fixed endpoints,

$$\partial_t \ell_{r,t}(y, z) = \frac{h_{r,t}(y, z)}{\ell_{r,t}(y, z)} \{\Delta\psi_r(y) + \Delta\psi_{1-r}(z)\}, \quad |\partial_t \ell_{r,t}(y, z)| \leq C_M(|y|^{M+1} + |z|^{M+1}). \quad (13.18)$$

The ratio is between zero and one since the separation is positive.

Apply finite-truncation differentiation to

$$A_{b,N,t}(u) = \sum_{i=0}^{N-1} [\ell_{b+i,t}(x_i^{(b,t)}, x_{i+1}^{(b,t)}) - g].$$

Interior orbit derivatives cancel by stationarity, and $\partial_t x_0 = 0$. Thus its derivative is the sum of the direct variations in (13.18), plus exactly the boundary term

$$B_{N,t} = \partial_2 \ell_{b+N-1,t}(x_{N-1}^{(b,t)}, x_N^{(b,t)}) \partial_t x_N^{(b,t)}.$$

The first factor is $O(|u|\rho^N)$ and the second is $O(\rho^N)$; hence $B_{N,t} \rightarrow 0$ uniformly in t and on the endpoint interval. By (13.17), the direct variations are bounded by $C_M|u|^{M+1}\rho^{(M+1)i}$, a summable majorant. Instead of assuming an interchange of infinite differentiation, integrate the exact finite identity in t and then let $N \rightarrow \infty$. Uniform action convergence, the vanishing boundary term and dominated convergence give

$$S_b^{[1]}(u) - S_b^{[0]}(u) = \int_0^1 \sum_{i \geq 0} \partial_t \ell_{b+i,t}(x_i^{(b,t)}, x_{i+1}^{(b,t)}) dt.$$

This proves (13.16). The actions are C^M by the weighted construction. Taylor's formula then implies equality of their derivatives at zero through order M : a nonzero Taylor coefficient of degree at most M would contradict the displayed $O(|u|^{M+1})$ bound. Every constant used is uniform under the stated functional bounds and positive margins. $\square$

In particular, the finite map from graph jets to action jets is independent of the chosen smooth representatives. One may compute it on polynomial graph representatives with smooth cutoffs on a common small interval. On a compact positive finite-jet set these representatives and intervals can be chosen with uniform finite smooth bounds. This is the finite-dimensional map used below, not an identification of a smooth function with its formal Taylor series.

A flat variation illustrates the distinction. For sufficiently small ϵ and a smooth cutoff equal to one near zero, the two graphs

$$\psi^{[0]}(y) = \frac{\kappa}{2}y^2, \quad \psi^{[1]}(y) = \frac{\kappa}{2}y^2 + \begin{cases} \epsilon e^{-1/y^2}, & y \neq 0, \\ 0, & y = 0 \end{cases}$$

are strictly convex near zero and have identical jets of every finite order, without being equal as functions. Applied for every M , the lemma shows equality of every finite action jet. It does not assert equality of the action functions or of the smooth boundary images. Analytic boundary determination below uses analyticity only after the finite-jet inverse has been established.

Lemma 13.5 (Jet filtration and homogeneous isolation). *Let $m \geq 3$. Holding the lower graph jets fixed, one has*

$$\partial_{q_r, m} S_b(u) = O(u^m). \quad (13.19)$$

Together with Lemma 13.4, this shows that $S_b^{(n)}(0)$ depends only on the graph jets through order n , including independence from smooth remainders and flat variations. For $m = n$, the coefficient of u^n in $\partial_{q_r, n} S_b(u)$ is obtained by substituting only the linear half-line orbit into the pure endpoint monomials. The error is $O(u^{n+1})$, uniformly on compact smooth classes with the required finite derivative bounds. The resulting finite-jet coefficient is uniformly controlled on compact positive finite-jet sets.

Proof. For a flight joining transverse coordinates y, z , let $h(y, z)$ denote its positive horizontal separation. In the anchored local graph representation its length is

$$\ell(y, z) = \sqrt{h(y, z)^2 + (z - y)^2}.$$

If the varied m th graph jet occurs at the y endpoint, then, on the region where the cutoff equals one,

$$\partial_{q_r, m} \ell(y, z) = \frac{h(y, z)}{\ell(y, z)} \frac{y^m}{m!} = \frac{y^m}{m!} + O((|y| + |z|)^{m+2}); \quad (13.20)$$

for a variation at the z endpoint the same formula holds with z^m . Indeed $h/\ell = 1 + O((|y| + |z|)^2)$ at the normal contact. The signs are positive for both facing graphs because increasing either inward graph height increases the horizontal separation.

Lemma 7.1 and Lemma 13.2 give

$$x_i^{(b)}(u) = t_i^{(b)}u + R_i^{(b)}(u), \quad |R_i^{(b)}(u)| \leq C|u|^2\rho^i, \quad (13.21)$$

with

$$t_i^{(b)} = \begin{cases} e^{-i\gamma}, & i \text{ even}, \\ \mathbf{r}_b e^{-i\gamma}, & i \text{ odd}. \end{cases} \quad (13.22)$$

Hence $(x_i^{(b)}(u))^m = (t_i^{(b)})^m u^m + O(u^{m+1}\rho^{mi})$. Insert this and (13.20) into the absolutely convergent envelope sum (13.14). Every summand is $O(u^m\rho^{mi})$, proving (13.19). At degree m only the first term of (13.20) and the linear orbit in (13.22) survive; every nonlinear orbit correction raises the endpoint degree by at least one. Summability is uniform, so the total remainder is $O(u^{m+1})$. $\square$

This lemma is the derivative-order bookkeeping needed by the arbitrary-order inverse. The orbit derivative of order k may depend on a $(k+1)$ st graph derivative, but the degree- n action jet cannot: a graph variation of order $m > n$ starts at endpoint degree m in the exact envelope formula.

13.5. The determinant-one block at every degree.

Proposition 13.6 (Last-jet block). *For every $n \geq 3$, after the lower graph jets are fixed, s_n is affine in q_n and its derivative is the matrix M_n in (13.4). In particular*

$$\det M_n = 1. \quad (13.23)$$

Proof. By Lemma 13.5, the degree- n first variation is the pure endpoint-power sum on the linear orbit. A boundary site is counted once and every interior site twice. For a half-line starting at type b , varying the n th jet of the starting contact gives

$$\partial_{q_b, n} S_b^{(n)}(0) = 1 + 2 \sum_{k \geq 1} e^{-2n\gamma k} = \frac{1 + e^{-2n\gamma}}{1 - e^{-2n\gamma}} = \coth(n\gamma), \quad (13.24)$$

whereas varying the opposite contact gives

$$\partial_{q_{1-b}, n} S_b^{(n)}(0) = 2\mathfrak{r}_b^n \sum_{k \geq 0} e^{-n\gamma(2k+1)} = \mathfrak{r}_b^n \operatorname{csch}(n\gamma). \quad (13.25)$$

All multiplicities are explicit and the sums converge absolutely. Interchanging the labels gives the second row of (13.4). Since $\mathfrak{r}_0 \mathfrak{r}_1 = 1$,

$$\det M_n = \coth^2(n\gamma) - (\mathfrak{r}_0 \mathfrak{r}_1)^n \operatorname{csch}^2(n\gamma) = 1.$$

The derivative of s_n with respect to q_n is independent of q_n : the linear orbit depends only on the quadratic geometry and the degree- n homogeneous coefficient above contains no higher occurrence of q_n . Therefore the dependence is affine. $\square$

The inverse block is

$$M_n^{-1} = \begin{pmatrix} \coth(n\gamma) & -\mathfrak{r}_0^n \operatorname{csch}(n\gamma) \\ -\mathfrak{r}_1^n \operatorname{csch}(n\gamma) & \coth(n\gamma) \end{pmatrix}. \quad (13.26)$$

If R_n is the contribution of the already reconstructed lower jets, then

$$q_n = M_n^{-1}(s_n - R_n). \quad (13.27)$$

For fixed maximal order M , all coefficients and derivatives in this finite recursion are uniformly bounded on compact positive geometric and finite-jet sets.

Proposition 13.7 (Quantitative tangent isomorphism). *Fix $g > 0$ and $M \geq 2$, and define*

$$\mathcal{A}_M : (\kappa_0, \kappa_1, q_3, \dots, q_M) \mapsto (s_2, s_3, \dots, s_M), \quad s_2 = (a_0, a_1)^t.$$

Then $\mathcal{A}_M$ is C^1 and $D\mathcal{A}_M$ is an isomorphism at every strictly convex labelled contact jet. On every compact positive geometric and finite-jet set there is $c_M > 0$ such that

$$\|D\mathcal{A}_M[h]\| \geq c_M \|h\|. \quad (13.28)$$

Proof. Lemma 13.4 makes $\mathcal{A}_M$ well defined independently of smooth representatives. Choose the local polynomial representatives just described. The weighted implicit-function construction and differentiated action bounds give C^1 dependence on their finite jet coordinates; the result is independent of this choice. Formula (13.3) is a smooth inverse of the quadratic map $(\kappa_0, \kappa_1) \mapsto (a_0, a_1)$ at fixed g . This gives an invertible leading 2×2 diagonal block. For $n \geq 3$ the new pair q_n enters s_n through the diagonal block M_n , while all dependence on lower jets is strictly below the block diagonal. Hence $D\mathcal{A}_M$ is block lower triangular with invertible diagonal blocks. The least singular value is a positive continuous function on a compact set, which proves (13.28). $\square$

Completion of Theorem 13.1. The support threshold and quadratic recovery were proved first. Lemma 13.4 proves independence from arbitrary smooth remainders at each finite order. Lemmas 13.2–13.5 and Proposition 13.6 give the all-order recursion (13.27); Proposition 13.7 gives the quantitative tangent inverse.

For analytic contacts, equality of all recovered labelled jets gives equality of the convergent graph germs. Parametrize two candidate connected analytic boundaries by oriented arclength from the recovered contact frame. Their analytic curvature functions agree on an interval and hence everywhere by the identity theorem; uniqueness for the planar Frenet equations gives equal boundary images. This is exact analytic continuation, not a stability claim for distant boundary points.

For the actual finite even-flight law, positivity of the physical flux makes its endpoint support threshold the exact stationary excess action $E_{j,b}(u, v)$. The differentiated relative estimate gives

$$\|E_{j,b} - (S_b \oplus S_b)\|_{C^M} \leq C_M \tau^j.$$

Taking the slice $v = 0$ and applying the finite-order locally Lipschitz inverse proves the $O_M(\tau^j)$ reconstruction statement. $\square$

Remark 13.8 (Oriented intrinsic meaning). Odd graph jets change sign when the chosen transverse orientation is reversed. Equivalently, the recovered data determine the full oriented curvature jet with respect to arclength at each labelled contact. Thus the inverse is intrinsic after the labelled contact point, tangent line, and an orientation of that line have been fixed; no reflection convention is hidden in the theorem.

13.6. Uniform control of the highest-degree interaction. The matrices in (13.4) are not arbitrary members of a determinant-one family. Their two shape factors come from positive contact curvatures. This admissibility gives the following order-uniform bounds, observed in [17]. We also record their consequences for actual fixed-lower-jet comparisons and for the block-diagonal operator on Taylor coefficients.

Proposition 13.9 (Uniform admissible blocks). *Let $g, \kappa_0, \kappa_1 > 0$, $c_b = 1 + g\kappa_b$, $c = \sqrt{c_0 c_1} = \cosh \gamma$ and $\mathbf{v}_b = \sqrt{c_b/c_{1-b}}$. Set*

$$t = e^{-\gamma}, \quad \vartheta = \frac{1+t^2}{2} < 1, \quad C_\gamma = \frac{3+t^6}{1-t^6}, \quad K_\gamma = \frac{4}{1-t^6}. \quad (13.29)$$

For the maximum absolute row-sum norm and every integer $n \geq 3$,

$$\max\{\|M_n\|_\infty, \|M_n^{-1}\|_\infty\} \leq C_\gamma, \quad (13.30)$$

$$\max\{\|M_n - I\|_\infty, \|M_n^{-1} - I\|_\infty\} \leq K_\gamma \vartheta^n. \quad (13.31)$$

The constants and the exponential rate can be chosen uniformly whenever $\gamma \geq \gamma_ > 0$, without a further upper bound on the curvature ratio.*

If two actual smooth anchored graph pairs have this gap and equal graph jets through order $n - 1$, then their degree- n graph and action differences satisfy

$$C_\gamma^{-1} \|\Delta q_n\|_\infty \leq \|\Delta s_n\|_\infty \leq C_\gamma \|\Delta q_n\|_\infty. \quad (13.32)$$

Here C_γ is independent of n ; the functional smooth bounds needed to define and differentiate each action are still imposed at the requested finite order.

For $R > 0$, define the coefficient space

$$\mathcal{A}_R = \left\{ x = (x_n)_{n \geq 3}, \ x_n \in \mathbb{C}^2 : \|x\|_R = \sum_{n \geq 3} \frac{R^n}{n!} \|x_n\|_\infty < \infty \right\}.$$

The block-diagonal operator $(\mathcal{D}x)_n = M_n x_n$ is a bounded isomorphism of $\mathcal{A}_R$, with both operator norms at most C_γ . Its corrections to the identity satisfy

$$\|(\mathcal{D} - I)x\|_{R/\vartheta} + \|(\mathcal{D}^{-1} - I)x\|_{R/\vartheta} \leq 2K_\gamma \|x\|_R. \quad (13.33)$$

These statements concern the block-diagonal part, not the complete nonlinear graph-to-action map.

Proof. Since $c_{1-b} > 1$,

$$\mathfrak{r}_b = \frac{c}{c_{1-b}} < c, \quad \lambda_b := \mathfrak{r}_b t < ct = \vartheta < 1, \quad \lambda_0 \lambda_1 = t^2. \quad (13.34)$$

Thus the linear visit to the opposite contact has contraction factor λ_b , even when $\mathfrak{r}_b > 1$. Summing the endpoint and interior visits as in (13.24)–(13.25) gives, with the sign distinguishing the matrix from its inverse,

$$M_n^{\pm 1} = \frac{1}{1 - t^{2n}} \begin{pmatrix} 1 + t^{2n} & \pm 2\lambda_0^n \\ \pm 2\lambda_1^n & 1 + t^{2n} \end{pmatrix}. \quad (13.35)$$

The inverse formula uses $\lambda_0 \lambda_1 = t^2$ and determinant one. Both matrices have the same absolute row sums. Since $n \geq 3$,

$$\|M_n^{\pm 1}\|_\infty \leq \frac{1 + t^{2n} + 2\vartheta^n}{1 - t^{2n}} \leq \frac{3 + t^6}{1 - t^6}.$$

After subtracting the identity the diagonal numerator is $2t^{2n}$. As $t^2 < \vartheta$, the row sums are at most $4\vartheta^n/(1 - t^6)$. This proves both bounds. If $\gamma \geq \gamma_*$, replace t by $e^{-\gamma_*}$ in the upper bounds and in ϑ ; all resulting constants are uniform.

For actual smooth pairs, Lemma 13.4 first makes each finite action jet a function of the corresponding finite graph jet. Proposition 13.6 then gives $\Delta s_n = M_n \Delta q_n$, since the common lower jets fix R_n . Applying (13.30) in the two directions proves (13.32). No formal-series argument replaces this smooth factorization.

For the coefficient-space assertions, multiply each of the block bounds by $R^n/n!$ and sum. The inverse is defined coefficient by coefficient and belongs to the same space by the same bound, so it is an actual bounded inverse on $\mathcal{A}_R$. Multiplying (13.31) by $(R/\vartheta)^n/n!$ proves (13.33). In particular, truncating either correction after degree N makes an operator-norm error on $\mathcal{A}_R$ of at most $K_\gamma \vartheta^{N+1}$. Only absolutely convergent coefficient sums have been used. $\square$

The location of the remaining conditioning problem is consequently precise. With the leading geometry fixed, the derivative of the full finite triangular map on degrees $3, \dots, M$, $M \geq 3$, has the form $L_M = D_M + N_M$, where $D_M = \text{diag}(M_3, \dots, M_M)$ and N_M is strictly block lower triangular. Hence

$$L_M^{-1} = \left[\sum_{k=0}^{M-3} (-D_M^{-1} N_M)^k \right] D_M^{-1}. \quad (13.36)$$

The diagonal inverse is bounded independently of M , but this assertion alone does not bound the displayed sum. It does not control the lower-jet remainders and their derivatives, the finite smooth remainder constants, the density-to-jet operation, or continuation away from the contact. The coefficient-radius gain in (13.33) therefore belongs to the diagonal correction only. The full inverse and finite-flight reconstruction retain their stated fixed-order hypotheses; no unrestricted all-order noisy inverse is being inferred.

13.7. Analytic inversion of the complete contact action. The diagonal estimate does not control the powers in (13.36). For analytic contact germs a stronger conclusion follows from a different representation of the *complete* derivative. Stationarity removes the variation of the orbit. The initial contact then contributes an invertible multiplication operator, while every later contact evaluates the boundary variation at a strictly contracted argument. This gives an inverse in an analytic function norm, including all lower-degree couplings. The norm, the local nature of the result, and the restriction on physical collars are specified below; none of the smooth or statistical norms used earlier is silently replaced by it.

Write $D_R = \{z \in \mathbb{C} : |z| < R\}$ and give $\mathcal{B}_R = H^\infty(D_R; \mathbb{C}^2)$ the norm $\|f\|_R = \max_b \sup_{D_R} |f_b|$. For $m \geq 3$ let

$$\mathcal{X}_{R,m} = \{f \in \mathcal{B}_R : f_b^{(k)}(0) = 0 \text{ for } 0 \leq k < m, b = 0, 1\}, \quad \mathcal{X}_R = \mathcal{X}_{R,3}. \quad (13.37)$$

These are closed complex Banach spaces. Real germs mean the fixed points of $f(z) \mapsto \overline{f(\bar{z})}$. Fix $g > 0$ and a real analytic anchored pair $\psi^* = (\psi_0^*, \psi_1^*)$, holomorphic near $\overline{D_{R_0}}$, with $(\psi_b^*)''(0) = \kappa_b > 0$. The affine space $\psi^* + \mathcal{X}_R$ fixes the gap and the two quadratic jets. Denote its half-line action map by

$$\mathcal{S}(\psi) = (S_{0,\psi}, S_{1,\psi}).$$

Here the action is the actual stationary sum (7.3), not a formally assigned series.

Lemma 13.10 (Holomorphic half-lines on a fixed function space). *There are $0 < R < R_0$, an open ball $\mathcal{U}$ about ψ^* in $\psi^* + \mathcal{X}_R$, and $0 < \rho < 1$ for which both half-line orbits are holomorphic in (ψ, u) , with*

$$x_{b,0}(u) = u, \quad |x_{b,i}(u)| \leq \rho^i |u| \quad (i \geq 1, u \in D_R, \psi \in \mathcal{U}). \quad (13.38)$$

The stationary sums define a holomorphic map $\mathcal{U} \rightarrow \mathcal{S}(\psi^*) + \mathcal{X}_R$. For real members of $\mathcal{U}$ these are the physical local actions on a common smaller real collar. Positivity of curvature is required on that smaller collar, not on the whole complex disc or at its boundary.

Proof. We provide the function-space details because pointwise analytic dependence of a finite family would not suffice. Put $t = e^{-\gamma}$ and $\lambda_b = \mathbf{r}_b t$. By (13.34), $\lambda_* := \max(\lambda_0, \lambda_1) < 1$. The linear orbit is $x_{b,i}^{\text{lin}}(u) = \ell_{b,i} u$, where $\ell_{b,2k} = t^{2k}$ and $\ell_{b,2k+1} = \lambda_b t^{2k}$; hence $|\ell_{b,i}| \leq \lambda_*^i$. Choose

$$\lambda_* < \beta < \rho < \sigma < 1.$$

On sequences of bounded holomorphic functions vanishing at zero use

$$\|y\|_\beta = \sup_{i \geq 1} \beta^{-i} \|y_i\|_{H^\infty(D_R)}.$$

The fixed-reference Green kernel (7.1) is bounded in this norm. Indeed its absolute entries are bounded by a constant times $t^{|i-k|}$, and $\sup_i \sum_{k \geq 1} t^{|i-k|} \beta^{k-i} < \infty$ since $t < \beta < 1$.

Consider $\psi = \psi^* + \eta$ with $\eta \in \mathcal{X}_R$ and $\|\eta\|_R < \varepsilon R^3$. The maximum principle applied to $\eta_b(z)/z^3$ gives $|\eta_b(z)| \leq \varepsilon |z|^3$ on D_R . On $D_{\sigma R}$, Cauchy's estimates give uniform third-derivative bounds, and, for $k \leq 2$,

$$|\eta_b^{(k)}(z)| \leq C_{\sigma,k} \varepsilon |z|^{3-k}. \quad (13.39)$$

For example, write $\eta_b = z^3 v_b$, use $\|v_b\|_R \leq \varepsilon$, and differentiate the product using Cauchy bounds for v_b on a slightly larger inner disc. The reference pair has the same type of remainder bounds with a fixed constant.

The stationarity equation at an interior site of type b_i is

$$\frac{h_{i-1} \psi'_{b_i}(x_i) + x_i - x_{i-1}}{\ell_{i-1}} + \frac{h_i \psi'_{b_i}(x_i) + x_i - x_{i+1}}{\ell_i} = 0, \quad (13.40)$$

where $h_i = g + \psi_{b_i}(x_i) + \psi_{1-b_i}(x_{i+1})$ and $\ell_i = \sqrt{h_i^2 + (x_{i+1} - x_i)^2}$, with the square-root branch near g . Crucially, when $x_0 = u$ approaches the boundary of D_R , this equation uses only $\psi_b(u)$, never $\psi'_b(u)$. Every differentiated graph is evaluated at an interior site, whose argument lies in $D_{\sigma R}$. Thus an unbounded differentiation operator on $H^\infty(D_R)$ is not being used.

Subtract the linear Jacobi equation, whose diagonal is $2c_{b_i}/g$ and adjacent entries are $-1/g$. On the weighted ball $\|x - x^{\text{lin}}\|_\beta \leq C_0 R^2$, the nonlinear residual has norm at most CR^2 and Lipschitz constant at most CR in the interior sequence. These estimates follow by expanding (13.40), using (13.39) at interior arguments and the cubic value remainder at u . Locality bounds neighboring weights by fixed factors $\beta^{\pm 1}$; products of two such coordinates have norm $O(R^2)$. All denominators stay in a fixed neighborhood of g . Choose C_0 larger than twice the residual constant times the Green norm, then choose R small. The Green fixed-point map preserves this ball and has contraction constant less than $1/2$.

This construction is consistent with the protected inner arguments: for every $i \geq 1$, the correction vanishes at zero, so the Schwarz bound gives

$$|x_{b,i}(u)| \leq (\lambda_*^i + C_0 R \beta^i) |u| \leq \rho^i |u|$$

after R is decreased. The last inequality follows uniformly in i from $\lambda_*/\rho + C_0 R \beta/\rho < 1$. Compositions and graph derivatives at the protected arguments are holomorphic maps of the Banach variables: their Cauchy expansions converge uniformly while these arguments remain inside $D_{\sigma R}$. The fixed-point iterates therefore converge locally uniformly as Banach-valued holomorphic maps. Their limit is holomorphic. Shrink the parameter ball once to keep these bounds and its inverse margin uniform.

Each action summand is bounded by $CR^2 \rho^{2i}$, so their sum is a locally uniformly convergent holomorphic map into $\mathcal{B}_R$. Its constant, linear and quadratic jets are respectively zero, zero and the fixed a_b of (7.4). Action differences belong to $\mathcal{X}_R$. On $|u| < R/2$, Cauchy bounds imply that real nearby graphs remain strictly convex and separated. The real stationary sequence just constructed is the unique small half-line of Lemma 7.1, hence gives its actual local action there. No assertion that every perturbed germ extends to an entire periodic obstacle is required for this local statement. $\square$

Lemma 13.11 (The complete derivative as weighted boundary visits). *On a smaller ball as in Lemma 13.10, put*

$$w_{b,i}(u) = \frac{g + \psi_{b+i}(x_{b,i}(u)) + \psi_{b+i+1}(x_{b,i+1}(u))}{\ell_{b,i}(u)}, \quad \alpha_b = w_{b,0}, \quad v_{b,i} = w_{b,i-1} + w_{b,i} \quad (i \geq 1),$$

where types are read modulo two. There are constants $W, a_* > 0$ such that $|w_{b,i}| \leq W$ and $|\alpha_b| \geq a_*$ on D_R . For every $\eta \in \mathcal{X}_R$, the full Fréchet derivative is

$$(L_\psi \eta)_b(u) := (D\mathcal{S}(\psi)\eta)_b(u) = \alpha_b(u) \eta_b(u) + \sum_{i \geq 1} v_{b,i}(u) \eta_{b+i}(x_{b,i}(u)). \quad (13.41)$$

Write $L_\psi = A_\psi + K_\psi$ for the multiplier and the sum. Both preserve every $\mathcal{X}_{R,m}$, and

$$\|A_\psi^{-1}\| \leq a_*^{-1}, \quad \|K_\psi|_{\mathcal{X}_{R,m}}\| \leq \frac{2W\rho^m}{1 - \rho^m}. \quad (13.42)$$

The operator K_ψ is compact on $\mathcal{X}_R$; specifically it has a rank at most $2(m-3)$ approximation with error at most $2W\rho^m/((1-\rho)(1-\rho^m))$.

Proof. All length factors are close to g and $w_{b,i}$ is close to one on the chosen complex domain, so the stated bounds hold. Differentiate a finite action sum along $\psi + s\eta$. The internal orbit derivatives cancel by (13.40); the derivative of x_0 is zero. The only remaining orbit term is

$$\partial_2 \ell_{b+N-1}(x_{b,N-1}, x_{b,N}) D_\psi x_{b,N}[\eta].$$

On a smaller parameter ball, the holomorphic fixed-point construction gives $\|D_{\psi}x_{b,N}[\eta]\|_R \leq C_R\beta^N\|\eta\|_R$. The first factor is at most $CR\rho^{N-1}$. The terminal term therefore tends to zero in $H^\infty(D_R)$. The direct variation of each flight is exactly $w_{b,i}\{\eta_{b+i}(x_{b,i}) + \eta_{b+i+1}(x_{b,i+1})\}$. These variations are norm summable: $\eta \in \mathcal{X}_R$ and the Schwarz estimate give a majorant $C\|\eta\|_R\rho^{3i}$. Passing to the limit in the finite identity proves (13.41). The initial visit occurs once and all internal visits occur twice, as in the finite-jet proof.

For $\eta \in \mathcal{X}_{R,m}$ the same Schwarz estimate is $|\eta_b(z)| \leq \|\eta\|_R(|z|/R)^m$. Summing it at the contracted arguments gives (13.42). Nonvanishing of α_b makes its reciprocal a bounded holomorphic multiplier. It preserves the order of vanishing.

For compactness, in every interior evaluation replace η_b by its Taylor polynomial through degree $m-1$. Each of the $2(m-3)$ scalar coefficients then multiplies one fixed pair of holomorphic functions, obtained by summing $v_{b,i}x_{b,i}^n$ over the appropriate parity. Those sums converge in norm. The resulting operator has the stated finite rank. By Cauchy's coefficient bound, the evaluation remainder at $x_{b,i}$ is at most $\|\eta\|_R\rho^{im}/(1-\rho^i)$. Summation proves the asserted error, which tends to zero. Notice that no continuation of the weights or orbits beyond D_R is used. Compactness here is approximation by finite-rank maps on the same disc; it is not a gain of holomorphic radius for the full derivative. $\square$

Theorem 13.12 (Local analytic inversion of the full action map). *For every real analytic positive-curvature anchored pair ψ^* and gap $g > 0$ as above, one can choose $R > 0$ and neighborhoods for which*

$$\mathcal{S} : \psi^* + \mathcal{X}_R \longrightarrow \mathcal{S}(\psi^*) + \mathcal{X}_R$$

is a biholomorphism between those neighborhoods. In particular, on a sufficiently small convex ball there are constants $0 < c < C < \infty$ with

$$c\|\psi - \tilde{\psi}\|_R \leq \|\mathcal{S}(\psi) - \mathcal{S}(\tilde{\psi})\|_R \leq C\|\psi - \tilde{\psi}\|_R. \quad (13.43)$$

These constants control the complete map, not only pairs with fixed lower jets. For every $N \geq 3$, its derivative and inverse induce bounded mutually inverse operators on $\mathcal{X}_R/\mathcal{X}_{R,N+1}$, with bounds independent of N in the quotient norms induced by $\|\cdot\|_R$.

For $0 < r < R$, all contact coefficients of two such pairs satisfy

$$\sum_{n \geq 3} \frac{r^n}{n!} \max_b |\psi_b^{(n)}(0) - \tilde{\psi}_b^{(n)}(0)| \leq \frac{(r/R)^3}{c(1-r/R)} \|\mathcal{S}(\psi) - \mathcal{S}(\tilde{\psi})\|_R. \quad (13.44)$$

The same local real analytic inverse allows the gap and curvatures to vary when the gap and both action Hessians are retained as finite coordinates. It concerns local contact germs and a strong norm on their holomorphic extensions, not a weak-density or full-table statistical norm.

Proof. Fix the base point and abbreviate $L = L_{\psi^*}$, $A = A_{\psi^*}$ and $K = K_{\psi^*}$. We construct a bounded inverse for L , including its lower-order couplings. Choose one finite $m \geq 4$ so large that

$$q_m := \frac{2W}{a_*} \frac{\rho^m}{1-\rho^m} < 1. \quad (13.45)$$

On $\mathcal{X}_{R,m}$, Lemma 13.11 gives

$$L_{\text{tail}}^{-1} = \sum_{j \geq 0} (-A^{-1}K)^j A^{-1}, \quad \|L_{\text{tail}}^{-1}\| \leq \frac{a_*^{-1}}{1-q_m}. \quad (13.46)$$

This norm-convergent series acts on actual functions. It is not the finite nilpotent formula with an unbounded number of unspecified factors.

Let P_m be Taylor projection onto the pairs of polynomials of degrees $3, \dots, m-1$. It is bounded on $\mathcal{X}_R$ by Cauchy's coefficient estimate. The induced finite map

$$L_{\text{low}} = P_m L|_{P_m \mathcal{X}_R}$$

is block lower triangular. Its diagonal blocks are exactly $M_3, \dots, M_{m-1}$. One way to see this is to differentiate the actual finite-jet identity proved in Lemma 13.4 and Proposition 13.6. It also follows directly from (13.41): $w_{b,i}(0) = 1$ and $x'_{b,i}(0) = \ell_{b,i}$, so the new degree is the same weighted geometric sum of visits. Analytic continuation of this finite identity, or complex linearity at the real base, gives the complex statement. All these finitely many blocks are invertible. Therefore L_{low} has a bounded inverse on its finite-dimensional space, without requiring its strictly lower part to be small.

For an arbitrary $y \in \mathcal{X}_R$, define

$$\eta_{\text{low}} = L_{\text{low}}^{-1} P_m y, \quad \eta_{\text{tail}} = L_{\text{tail}}^{-1} (y - L \eta_{\text{low}}). \quad (13.47)$$

The second argument lies in $\mathcal{X}_{R,m}$ because its projection under P_m vanishes. These formulas solve $L\eta = y$ uniquely and boundedly. For example, if $B = \|L_{\text{low}}^{-1} P_m\|$, then

$$\|L^{-1}\| \leq B + \frac{a_*^{-1}}{1 - q_m} (1 + \|L\|B). \quad (13.48)$$

Only one fixed finite low-order problem has been inverted. All remaining orders are handled together by (13.46).

For completeness the nonlinear step can be made by contraction rather than a formal inverse expansion. Set $\mathcal{N}(\eta) = \mathcal{S}(\psi^* + \eta) - \mathcal{S}(\psi^*) - L\eta$. It is holomorphic and $D\mathcal{N}(0) = 0$. On a small ball arrange $\|L^{-1} D\mathcal{N}\| < 1/2$. For y small the equation

$$\eta = L^{-1} y - L^{-1} \mathcal{N}(\eta)$$

preserves this ball and is a contraction. Its iterates converge locally uniformly and holomorphically in y . This gives the local holomorphic inverse. Shrinking a convex ball also gives $\|D\mathcal{S} - L\| < (2\|L^{-1}\|)^{-1}$ there. Integration on the segment joining two points proves (13.43), with $c = (2\|L^{-1}\|)^{-1}$ and a local bound for the upper constant. Real data give real inverses by uniqueness and conjugation symmetry.

Every tail $\mathcal{X}_{R,N+1}$ is invariant under L . It is also invariant under L^{-1} : if the output has zero jets through N , finite triangular induction with the invertible M_n makes the input jets through N zero. Thus L and L^{-1} descend to mutually inverse quotient operators. Their norms are bounded by the parent operator norms, uniformly in N . The same construction, or continuity of the derivative and its inverse on a smaller parameter neighborhood, gives local uniformity in the base pair. This statement is about analytic quotient norms, not maximum norms of unweighted derivatives. Cauchy's estimate at zero and (13.43) then give (13.44) by summing the geometric series.

Finally, let the leading variables vary near their positive base values. The protected-domain construction is holomorphic in g, κ_0, κ_1 as well. Write $\psi_b = \kappa_b u^2/2 + v_b$, with $v \in \mathcal{X}_R$, and similarly subtract $a_b u^2/2$ from each action. The leading map $(g, \kappa_0, \kappa_1) \mapsto (g, a_0, a_1)$ has the local analytic inverse already given by

$$\gamma = \operatorname{arsinh}(g\sqrt{a_0 a_1}), \quad c_b = \cosh \gamma \sqrt{a_b/a_{1-b}}, \quad \kappa_b = (c_b - 1)/g,$$

with the positive branches near the base. In the finite-variable and $\mathcal{X}_R$ splitting the derivative is block lower triangular: the finite leading block and the infinite tail block are invertible. The same local contraction proves the stated joint inverse. $\square$

The theorem supplies a local analytic inverse for the entire action, not just its diagonal response. Its additional input is concrete: the norm controls the holomorphic extensions on one fixed disc. Smallness of a real-interval C^k error or total variation does not supply that norm. The radius and the inverse constants may depend on the analytic base pair, its gap, curvature margins and extension bounds. The result does not extend a local germ to an entire obstacle, select a discrete global matching branch, or give a uniform acquisition cost. Those remain the separate geometric and statistical statements proved elsewhere. Neither K nor the full inverse is claimed to improve holomorphic radius; only the finite-rank approximation and the tail bounds stated above are used. In particular the arbitrary triangular-matrix comparison does not refute this result: an actual action derivative has the additional multiplier-and-contracted-visits structure (13.41).

13.8. Contracted evaluations and conditional stability from real observations. The complete analytic inverse has two distinct ingredients: a stationary representation of its derivative and a function-space inversion argument. We first isolate the latter. We then combine the inverse with a quantitative restriction estimate to obtain a local conditional bound from real endpoint laws. The conclusion loses holomorphic radius and retains an analytic prior. It therefore does not identify real observation accuracy with the same-disc norm of Theorem 13.12.

Proposition 13.13 (An inversion criterion for contracted evaluations). *Let $q \geq 1$, $d \geq 1$, and let X_q be the closed subspace of $H^\infty(D_R; \mathbb{C}^d)$ consisting of functions vanishing to order at least q , with the maximum component norm. Suppose*

$$(Lf)_b(z) = (A(z)f(z))_b + \sum_{i \geq 1} \sum_{c=1}^d w_{bci}(z) f_c(\phi_{bci}(z)), \quad (13.49)$$

where all coefficients and maps are holomorphic on D_R , $\sup_{D_R} \|A^{-1}\| \leq a^{-1}$, A is bounded, and

$$|\phi_{bci}(z)| \leq \rho^i |z|, \quad \sup_{b,z} \sum_c |w_{bci}(z)| \leq W, \quad 0 < \rho < 1.$$

Write $L = A + K$. Choose an integer $m > q$ with $\nu_m = a^{-1} W \rho^m / (1 - \rho^m) < 1$. Then $L : X_q \rightarrow X_q$ is boundedly invertible if and only if the finitely many matrices

$$(B_n)_{bc} = A_{bc}(0) + \sum_{i \geq 1} w_{bci}(0) (\phi'_{bci}(0))^n, \quad q \leq n < m, \quad (13.50)$$

are invertible. The operator K is compact. This criterion applies to the derivative at a point of a holomorphic map between open subsets of X_q and then yields a local holomorphic inverse of that map.

Proof. Higher-order Schwarz estimates give

$$\|K|_{X_n}\| \leq \frac{W \rho^n}{1 - \rho^n}, \quad n \geq q.$$

Here X_n denotes the corresponding order- n vanishing subspace. The series in (13.49) converges in operator norm. Both summands, and multiplication by A^{-1} , preserve every X_n . Thus $L|_{X_m}$ is invertible by the Neumann series, with inverse norm at most $a^{-1}/(1 - \nu_m)$.

Let P_m retain Taylor degrees $q, \dots, m-1$. Cauchy's coefficient bound makes P_m a bounded projection on X_q . Modulo X_m , the matrix of L is block lower triangular with diagonal (13.50): only the constant weight and the linear part of an argument contribute to the first nonzero degree of a variation. If those blocks are invertible, solve this finite quotient first and then solve the residual in X_m , exactly as in (13.47). In particular, if $C_0 = \|(P_m L|_{P_m X_q})^{-1} P_m\|$, then

$$\|L^{-1}\| \leq C_0 + \frac{a^{-1}}{1 - \nu_m} (1 + \|L\| C_0).$$

Conversely, if the finite quotient were singular, a nonzero polynomial class p would satisfy $Lp \in X_m$. Solving $Lt = Lp$ inside X_m would give a nonzero kernel vector $p - t$. Consequently invertibility of L implies the stated finite condition.

To prove compactness, replace each evaluated f_c by its Taylor polynomial of degrees $q, \dots, n-1$. The resulting operator has rank at most $d(n-q)$. Cauchy's coefficient bound gives the operator-norm remainder estimate

$$\frac{W\rho^n}{(1-\rho)(1-\rho^n)} \rightarrow 0.$$

The truncation is evaluated only at contracted arguments; convergence of Taylor partial sums on the whole open disc is not asserted. The final assertion follows from the holomorphic inverse function theorem, or from the contraction proof already given above. $\square$

For the contact action, $q = 3$, $d = 2$, the initial multiplier is diagonal, and the only nonzero weight at visit i is $v_{b,i}$ in the component of type $b + i$. Its bound is $2W$ in the notation of Lemma 13.11. The matrices B_n are the actual M_n . The protected-domain construction and the vanishing terminal term prove that this is indeed the derivative of the physical action; they are not assumptions supplied by functional analysis. The qualitative Fredholm observation suggested in [19] is now also transparent: $A^{-1}L$ is identity plus compact, and nonresonant jets kill its analytic kernel. The finite-low-order and tail construction above additionally records an inverse bound. Neither argument makes the inverse improve holomorphic radius.

Lemma 13.14 (A bounded analytic function from its real restriction). *Fix $0 < s < r < \varrho < R$ with $r + 2s < \varrho$, and put*

$$A_0 = \frac{2(r+s)}{s}, \quad b_0 = \frac{r+s}{\varrho-s} < 1, \quad \vartheta = \frac{\log(1/b_0)}{\log(A_0/b_0)} \in (0, 1). \quad (13.51)$$

If $F \in H^\infty(D_R)$, $\|F\|_R \leq M$, and $\sup_{[-s,s]} |F| \leq \epsilon \leq M$, then

$$\|F\|_r \leq \left(1 + \frac{\varrho}{\varrho-r}\right) M^{1-\vartheta} \epsilon^\vartheta. \quad (13.52)$$

The assertion includes complex-valued functions and the case $\epsilon = 0$.

Proof. For $n \geq 1$ take the n roots x_k of $T_n(x/s)$, where T_n is the Chebyshev polynomial. Their monic nodal polynomial is $P_n(z) = 2^{1-n} s^n T_n(z/s)$, and

$$|P'_n(x_k)| \geq n(s/2)^{n-1}.$$

The Lagrange interpolation polynomial $I_n F$ therefore satisfies $\|I_n F\|_r \leq \epsilon A_0^{n-1}$: the numerator of each basis polynomial is a product of $n-1$ factors of modulus at most $r+s$. The residue formula on $|\zeta| = \varrho$ gives

$$F(z) - I_n F(z) = \frac{1}{2\pi i} \int_{|\zeta|=\varrho} \frac{F(\zeta) P_n(z)}{P_n(\zeta)(\zeta-z)} d\zeta.$$

This holds also at the interpolation nodes by continuity. The bounds $|P_n(z)| \leq (r+s)^n$ and $|P_n(\zeta)| \geq (\varrho-s)^n$ imply

$$\|F\|_r \leq \epsilon A_0^{n-1} + \frac{\varrho}{\varrho-r} M b_0^n. \quad (13.53)$$

For $0 < \epsilon < M$, choose $n = \lceil \log(M/\epsilon) / \log(A_0/b_0) \rceil$. The two terms are bounded by their respective constants times $M^{1-\vartheta} \epsilon^\vartheta$. For $\epsilon = M$, take $n = 1$; for $\epsilon = 0$, let n tend to infinity. The case $M = 0$ is immediate. All contours are strictly inside the given holomorphic domain. $\square$

This is a conditional continuation estimate, not continuity of unrestricted analytic continuation. Its elementary interpolation proof is included to display the constants; the general ill-conditioning of continuation under analytic priors is discussed in [18]. We use only the local estimate just proved, not continuation around an entire obstacle.

Theorem 13.15 (Conditional contact stability from real endpoint laws). *Fix a positive-curvature real analytic anchored pair and a positive gap. There exist radii $0 < s < s_+ < r < \varrho < R$, with $r + 2s < \varrho$, and a bounded neighborhood $\mathcal{U}$ of that pair and gap in the real part of*

$$\mathbb{C} \times \{\psi \in H^\infty(D_R; \mathbb{C}^2) : \psi_b(0) = \psi'_b(0) = 0\},$$

for which the following assertions hold. The curvatures and gap may vary within $\mathcal{U}$. Choose a fixed sufficiently small positive offset d and the radii so that $Q_+ = [-s_+, s_+]^2$ is strictly inside every relevant positive-density region. Put $Q = [-s, s]^2$ and fix $a \in (-s, s) \setminus \{0\}$.

For each of two admissible members (g, ψ) and $(\tilde{g}, \tilde{\psi})$ retain a same-type limiting endpoint density for each type, possibly after separate positive recording factors. On Q_+ these densities have the form

$$f_b(u, v) = c_b U_b(u) V_b(v) \{d - S_b(u) - S_b(v)\}, \quad c_b, U_b, V_b > 0, \quad (13.54)$$

and the analogous form for $\tilde{f}_b$. Assume common bounds $m_f \leq f_b, \tilde{f}_b \leq M_f$ and $q_{f_b}, q_{\tilde{f}_b} \geq q_* > 0$, where q_f is the scalar anchor (14.3). The two sets of recording factors need not coincide, and need not have holomorphic extensions. The origins, signs, physical units and d are fixed observation conventions.

Let $e_g = |g - \tilde{g}|$ and $e_\infty = \max_b \|f_b - \tilde{f}_b\|_{C^0(Q)}$. With ϑ as in (13.51), there is a finite constant C such that, for sufficiently small errors,

$$\|\psi - \tilde{\psi}\|_r \leq C(e_g + e_\infty^\vartheta). \quad (13.55)$$

For every $0 < t < r$ this simultaneously controls all contact coefficients:

$$\sum_{n \geq 2} \frac{t^n}{n!} \max_b |\psi_b^{(n)}(0) - \tilde{\psi}_b^{(n)}(0)| \leq C \frac{(t/r)^2}{1 - t/r} (e_g + e_\infty^\vartheta). \quad (13.56)$$

If, in addition, all four densities have a common Lipschitz bound H on Q_+ , then, with $e_1 = \max_b \|f_b - \tilde{f}_b\|_{L^1(Q_+)}$, one has

$$\|\psi - \tilde{\psi}\|_r \leq C_H(e_g + e_1^{\vartheta/3}). \quad (13.57)$$

In particular e_1 may be replaced by twice the maximum total variation of the recorded laws. For a rectangular grid on Q_+ of side at most δ , with the outside as an additional category, it may instead be replaced by

$$\eta = e_{\text{cell}} + 2\sqrt{2}H|Q_+|\delta, \quad e_{\text{cell}} = \max_b \|\mathbf{p}_\delta(P_b) - \mathbf{p}_\delta(\tilde{P}_b)\|_1. \quad (13.58)$$

The constants depend on the fixed local analytic neighborhood, the radii and the stated density/anchor margins, but not on a jet truncation order.

Proof. We specify the neighborhood before using small observational error. Lemma 13.10, including its finite leading-variable extension, gives an action map on a disc D_R near the chosen base. Apply Theorem 13.12 to the same base on a smaller disc to obtain an inverse ball on D_r , with $r < R/4$. Write Y_r for the space of pairs vanishing to order two. The bounded splitting into the Hessians and the order-three tail turns the last clause of that theorem into a local inverse of

$$(g, \psi) \mapsto (g, \mathcal{S}_g(\psi)) \quad \text{in } \mathbb{C} \times Y_r.$$

Indeed $|S_b''(0)| \leq 2r^{-2}\|S_b\|_r$, and subtracting the quadratic polynomial is a bounded operation. Thus no additional measurement of the curvatures or action Hessians is required here. On a smaller convex input ball, its two-point bound has the form

$$\|\psi - \tilde{\psi}\|_r \leq C_r\{e_g + \|\mathcal{S}_g(\psi) - \mathcal{S}_g(\tilde{\psi})\|_r\}. \quad (13.59)$$

Intersect the D_R construction neighborhood with the inverse image of this D_r ball under restriction. This is an open neighborhood of the base, independent of the observed errors. Shrink it to a bounded ball on which the action norms on D_R have a common bound B . Continuity of the holomorphic forward map provides such a B ; no inverse norm is inferred from the observations. The two constructions of the action commute with restriction: on the smaller real collar both are the unique small physical half-line action, and equality extends holomorphically to D_r . Choose $\varrho = R/2$ and then $s_+ < r/2$ small enough for the common physical collar and fixed offset, followed by $s < s_+$. These choices imply $r + 2s < \varrho$. Positive contact curvature permits a fixed nonzero anchor with a uniform action-anchor margin. The recorded density bounds remain separate assumptions on the recording experiment.

For (13.54), the four-density identity cancels c_b, U_b, V_b exactly, just as in (14.6). On the stipulated density range its ratios are Lipschitz in the uniform real norm. Moreover

$$|q_f - q_{\tilde{f}}| = \frac{|R_f(a, a) - R_{\tilde{f}}(a, a)|}{q_f + q_{\tilde{f}}},$$

so the anchor division is uniformly controlled. The final map $t \mapsto dt/(1+t)$ is Lipschitz for the nonnegative real t here. Consequently

$$\max_b \sup_{[-s, s]} |S_b - \tilde{S}_b| \leq C_1 e_\infty. \quad (13.60)$$

This argument compares two realizable laws directly; it does not assume that the straight line joining their densities factorizes. Apply Lemma 13.14 componentwise to $S - \tilde{S}$, whose D_R norm is at most $2B$. Equations (13.60) and (13.59) give (13.55). Cauchy's coefficient bound on D_r and summation of a geometric series give (13.56), including odd orders.

For the weaker norm, put $w = f_b - \tilde{f}_b$ and choose a fixed nonnegative smooth kernel k supported in the unit ball of $\mathbb{R}^2$, with integral one. For $x \in Q$ and $0 < h \leq h_0 < \text{dist}(Q, \partial Q_+)$,

$$|w(x) - (k_h * w)(x)| \leq CHh, \quad |(k_h * w)(x)| \leq \|k\|_\infty h^{-2} e_1.$$

Take $h = h_0 e_1^{1/3}$ when $0 < e_1 \leq 1$; for $e_1 = 0$ let h decrease to zero. Thus $e_\infty \leq C_H e_1^{1/3}$, proving (13.57). Only the stated first-order real bound is used at this step. The restriction of an L^1 difference to Q_+ is at most twice the total variation of the full laws. Finally, the L^1 difference between one density and its cell averages is at most $\sqrt{2}H|Q_+|\delta$. The difference of the two cell averages integrates to the sum of the absolute cell-mass differences. Adding the two approximation errors proves $e_1 \leq \eta$ in (13.58). $\square$

The neighborhood and outer holomorphic bound in this theorem are a geometric regularity prior, not a consequence of small real data error. The radius loss is essential to the argument. For example, $F_N(z) = \epsilon(z/R)^N$ has norm ϵ on D_R while all its fixed-order real norms on $[-s, s]$, $s < R$, tend to zero; its norm on D_r , $r < R$, also tends to zero. Thus the estimate does not conflict with the same-disc topology distinction. The exponent is explicit but not asserted optimal. The theorem compares locally admissible candidates and controls their complete germs on an inner disc. It neither constructs a map on arbitrary corrupted densities nor identifies a global matching branch or an unknown contact frame. The histogram bound is deterministic; no uniform preparation cost over arbitrarily small recording efficiencies follows from it. Quantified

continuation to entire obstacles and the physical acquisition theorem retain their separate hypotheses and proofs.

14. AN AMPLITUDE-FREE INVERSE FROM ONE ENDPOINT LAW

An endpoint position in the plane supplies a point of the unknown boundary. A signed transverse endpoint supplies only one coordinate of that point. The result below concerns the latter datum. It recovers the two half-line actions from one positive-offset endpoint law of each type, without retaining longitudinal positions, residual times, waiting counts, or the unknown flux amplitudes. The gap remains a separate onset datum. This strengthens the support-family inverse of Theorem 13.1: a continuum of offsets is unnecessary when the density, rather than only its support, is retained.

14.1. A four-density identity. Fix a known $d > 0$. Let $I = [-r, r]$, and suppose on a neighborhood of I that $S(0) = S'(0) = 0$, $S'' > 0$, and $B > 0$. Shrink r until $\max_I S < d/4$. The endpoint law has on I^2 the positive smooth density

$$f(u, v) = Z^{-1} B(u) B(v) \{d - S(u) - S(v)\}, \quad Z > 0. \quad (14.1)$$

Its continuation outside this interior square may vanish at the moving boundary. A law in this class determines the displayed continuous density uniquely: two continuous versions equal almost everywhere on an open set are equal everywhere there. Thus values on the slices below are determined by the probability law, although a finite sample does not supply them without estimation.

Choose a fixed $a \in I \setminus \{0\}$. For a positive density f define

$$R_f(u, v) = \frac{f(u, v) f(0, 0)}{f(u, 0) f(0, v)}, \quad (14.2)$$

$$q_f = \sqrt{1 - R_f(a, a)}, \quad T_f(u) = \frac{1 - R_f(u, a)}{q_f}, \quad (14.3)$$

$$\mathcal{S}_d(f)(u) = \frac{dT_f(u)}{1 + T_f(u)}, \quad \mathcal{B}(f)(u) = \frac{f(u, 0)}{f(0, 0)} (1 + T_f(u)). \quad (14.4)$$

The square root in (14.3) is the positive scalar root at the fixed nonzero anchor, not a pointwise square root differentiated at the degenerate minimum $u = 0$.

Theorem 14.1 (Single-offset law inversion). *For (14.1), all quantities in (14.3)–(14.4) are defined and*

$$\mathcal{S}_d(f) = S, \quad \mathcal{B}(f) = B/B(0) \quad \text{on } I. \quad (14.5)$$

Consequently one same-type signed endpoint law at one known positive offset determines the complete action germ and the normalized amplitude germ. Neither an amplitude model nor its normalizing constant is supplied.

Proof. Set $t(u) = S(u)/(d - S(u))$. Direct cancellation in (14.2) gives

$$R_f(u, v) = \frac{d\{d - S(u) - S(v)\}}{(d - S(u))(d - S(v))} = 1 - t(u)t(v). \quad (14.6)$$

Strict convexity and the normalization of S imply $S(a) > 0$, hence $q_f = t(a) > 0$. Therefore $T_f(u) = t(u)$, and solving this relation for $S(u)$ gives the first identity in (14.5). Also

$$\frac{f(u, 0)}{f(0, 0)} = \frac{B(u)}{B(0)} \frac{d - S(u)}{d} = \frac{B(u)/B(0)}{1 + t(u)},$$

which proves the second. No symmetry of S or B was used. In particular values at negative and positive u remain separate and all odd derivatives of S are retained. $\square$

14.2. Fixed-order stability and finite flights.

Proposition 14.2 (Interior stability of the law inverse). *Fix an integer $M \geq 0$, the interval I , the anchor a , and $d > 0$. On a compact family of densities (14.1) with uniformly positive f , uniformly bounded C^M norms, and $q_f \geq q_* > 0$, there are $\varepsilon_*, C_M > 0$ such that*

$$\|\tilde{f} - f\|_{C^M(I^2)} < \varepsilon_* \implies \|S_d(\tilde{f}) - S\|_{C^M(I)} + \|\mathcal{B}(\tilde{f}) - B/B(0)\|_{C^M(I)} \leq C_M \|\tilde{f} - f\|_{C^M(I^2)}.$$

The perturbed function need not factorize or have total mass one. It must be defined on a neighborhood of the closed square. The constants are fixed-order constants; they are not uniform as $M \rightarrow \infty$, $a \rightarrow 0$, or $d \downarrow 0$.

Proof. Evaluation on each fixed slice is a bounded linear map on C^M . All density denominators in (14.2) have a common positive lower bound. Products and reciprocal functions are locally Lipschitz in C^M on sets with such bounds, as follows by differentiating $hh^{-1} = 1$ inductively. Thus the ratios are locally Lipschitz in C^M . The scalar $1 - R_f(a, a)$ is at least q_*^2 , so its positive square root has a bounded derivative on a common neighborhood. The denominator $q_{\tilde{f}}$ is then at least $q_*/2$. For the exact density, $1 + T_f \geq 1$; after a further uniform reduction of ε_* , $1 + T_{\tilde{f}} \geq 1/2$. Applying the same product and reciprocal bounds to (14.4) proves the claim. This proof avoids differentiation of $\sqrt{t(u)^2}$ at zero. $\square$

The hypotheses can be arranged uniformly on any compact smooth contact class with positive curvature and gap bounds. Choose one d inside its common positive-offset collar, then a common I on which $S_b(u) < d/4$. A fixed nonzero $a \in I$ has $S_b(a) \geq ca^2$, hence a common positive anchor bound. The amplitude and normalization bounds give a positive lower bound for f on I^2 .

Corollary 14.3 (Finite-flight law reconstruction). *Fix $M \geq 2$ on such a compact class. Let $f_{j,b,d}$ be the successful signed endpoint density for an even j -flight bridge, in the exact contact transverse chart, at the fixed offset d . On the selected interior square, for all sufficiently large even j ,*

$$\|S_d(f_{j,b,d}) - S_b\|_{C^M(I)} \leq C_M \tau^j. \quad (14.7)$$

More generally an estimated density $\hat{f}_{j,b,d}$ with interior C^M error at most ϵ , together with a gap estimate of error at most ϵ_g , recovers both labelled graph jets through order M with error at most

$$C_M(\tau^j + \epsilon + \epsilon_g), \quad (14.8)$$

provided the errors are sufficiently small. This is a stability statement in a density norm, not an assertion that total variation alone controls arbitrarily high derivatives.

Proof. In the transverse chart the finite endpoint density is a positive normalizing constant times $b_j(u, v)(d - E_j(u, v))_+$. The differentiated relative theorem gives $b_j = B_b \otimes B_b + O_{C^M}(\tau^j)$ and $E_j = S_b \oplus S_b + O_{C^M}(\tau^j)$. On the chosen square the cutoff argument is bounded away from zero. Over the whole common endpoint box the inequality $|x_+ - y_+| \leq |x - y|$, the uniform amplitude bounds, and its finite volume give an $O(\tau^j)$ error for the normalizing integral. That integral has a positive uniform lower bound at fixed d . Division therefore proves $\|f_{j,b,d} - \hat{f}_{j,b,d}\|_{C^M(I^2)} \leq C_M \tau^j$. Apply Proposition 14.2.

Before applying the contact-jet recursion, subtract the zeroth and first Taylor terms of the estimated action. This bounded projection fixes the true anchored action, preserves the error bound, and does not change any coefficient of order at least two. Thus a small finite-flight or density error is not interpreted as an unknown change of contact origin or tangent. For completeness, if $a_b = S_b''(0)$, the leading inversion is

$$\gamma = \operatorname{arsinh}(g\sqrt{a_0 a_1}), \quad c_b = \cosh \gamma \sqrt{a_b/a_{1-b}}, \quad \kappa_b = (c_b - 1)/g.$$

At each $3 \leq n \leq M$, use $q_n = M_n^{-1}(s_n - R_n)$, where R_n is already determined by lower-order graph jets and M_n is the determinant-one block of (13.4). Each of these finitely many operations is smooth on a neighborhood of the compact positive geometric set. Induction in n gives (14.8). No infinite-order condition bound is inferred from the determinant identity. $\square$

14.3. The intrinsic data map.

Theorem 14.4 (Periodic determination from one endpoint law of each type). *For every measured channel e , retain its gap g_e , its signed transverse coordinate conventions, and the two conditional endpoint laws $\Pi_{e,00,d_e}^{\text{end},\infty}$ and $\Pi_{e,11,d_e}^{\text{end},\infty}$ at one known sufficiently small positive offset d_e . No longitudinal coordinate, residual time, waiting count, or amplitude is part of these conditional-law data. Then:*

- (1) *For smooth strictly convex contacts the data determine every finite labelled contact jet, without reflection symmetry or supplied curvatures.*
- (2) *For connected real-analytic incident boundaries they determine the complete boundary images in their separate channel frames.*
- (3) *Suppose the data are realized by a labelled periodic dispersing billiard with connected real-analytic strictly convex obstacle boundaries and the marked lifted-channel graph is connected. Assume that this graph contains a rank-two anchoring pair of signature-rigid cycles based at one channel frame and a rooted signature-rigid spanning tree reaching every obstacle orbit, as in Theorem 17.4. Then the data determine the complete marked periodic table and the Euclidean realization of its rank-two lattice, including its Gram form and scale, up to one simultaneous element of $\text{SE}(2)$.*

On a compact class all d_e can be the same fixed $d_0 > 0$.

Proof. Theorem 14.1 recovers each action on a neighborhood of zero. The weighted half-line construction, finite-truncation envelope identity and homogeneous isolation in Theorem 13.1 then recover the labelled graph jets. The density cancellation is not a replacement for that inverse: it supplies its unsymmetrized action input. Analytic continuation gives the complete incident boundary images. The intrinsic gluing theorem matches their framed signatures.

For part (3), fix the common starting channel frame of the anchoring pair, and choose an obstacle orbit r incident to that frame. A signature-rigid incidence on an anchoring cycle gives a framed point of this obstacle whose complete oriented curvature signature occurs uniquely. Lemma 16.11 therefore permits us to root the supplied spanning tree at r . This reorients the existing tree; it introduces no channel, incidence, deck displacement, or registration datum. In particular, the analytic image at the new root is already placed in the chosen anchoring frame.

The two signature-rigid cycles determine translation holonomies $V = (v_1 \ v_2)$ in this same frame. If M has their two independent marked deck displacements as columns, then Theorem 17.3 gives

$$L = VM^{-1}, \quad L^T L = M^{-T} V^T V M^{-1}.$$

This determines the lattice, including its scale, but does not by itself place obstacle orbits outside the anchoring cycles. Those placements follow by induction along the rerooted tree. At each edge, the already placed parent boundary has a unique framed point with the prescribed oriented signature. Matching that frame fixes the adjacent channel; its recovered analytic image places the child obstacle, with the marked deck displacement interpreted through the same matrix L . The tree reaches every obstacle orbit, as required by Theorem 17.4.

To verify uniqueness in one gauge, consider two admissible realizations of the data and align their starting channel frames by one element of $\text{SE}(2)$. The local inverse and analytic continuation give the same incident boundary images. Unique signature matching along the two cycles gives the same V , hence the same L . At each successive tree edge, unique matching at the already

common parent frame gives the same child placement in both realizations. Induction identifies every obstacle orbit; the common lattice then identifies all its translates. The realizability hypothesis supplies existence and admissibility of this reconstruction, including compatibility with the remaining measured channels. Changing the starting frame applies one simultaneous Euclidean motion to the entire reconstruction, not independent motions to its obstacles. None of these steps uses ambient position samples or an externally supplied lattice metric. Uniform collar bounds give the last assertion. $\square$

This theorem concerns an intrinsic, law-valued observation map with specified signed transverse origins and axes, as does the preceding support-family inverse. It does not treat an unknown centering transformation as a common statistical kernel. Section F.47 obtains such origins and axes from a charged physical pilot, and Section F.48 implements the law inverse by bounded tests after that pilot. Exact longitudinal positions are available in that richer acquisition protocol and permit the independent direct sampling route of Section F.45. Neither route is claimed to be statistically equivalent to the other.

14.4. Finite preparations and complete analytic contact recovery. We now pass from limiting laws to finite experiments while keeping the complete analytic contact pair as the target. The order of this passage matters. A finite-bridge density does not in general have the separated form (13.54), and an empirical histogram has still less reason to have that form. We compare realizable limiting laws before applying their inverse. This avoids extending a factorization identity to an arbitrary observed density.

Fix the local analytic neighborhood, radii, squares and anchor of Theorem 13.15. Let $\mathcal{K}$ be a nonempty class of periodic realizations whose anchored contact pairs and gaps belong to this neighborhood. Require uniform positive curvature, separation, clear-channel and roof margins on the selected itinerary, and $0 < A_- \leq A(T) \leq A_+ < \infty$. We use ideal endpoint recording in this subsection. The signed frames and physical units are the observation conventions already specified for the law inverse. The sampling theorem below first treats an exactly calibrated channel; estimated calibration is treated afterwards. The class need not be finite dimensional.

All contact graphs have a common bound on D_R . Cauchy estimates on a strictly smaller physical collar give the finitely many real derivative bounds needed in the relative construction. By decreasing the analytic neighborhood and the collar, the weighted contractions, diagonal dominance and trace-norm estimates have uniform margins. Thus their endpoint estimates are uniform on $\mathcal{K}$; no differentiation with respect to an infinite-dimensional parameter is being asserted. Fix a compact positive-offset interval I_d about d , contained in the common onset collar. Decrease Q_+ once so that it lies strictly inside the positive region for every offset in I_d and every member of the class.

Lemma 14.5 (Interior density and acceptance bounds). *There are $j_0 \geq 2$, $C, H, F, c > 0$, $\omega > 0$, and $\Gamma \geq \sup_{T \in \mathcal{K}} \gamma(T) > 0$ such that, for every even $j \geq j_0$, both starting types b , and $d' \in I_d$, the actual successful endpoint density $f_{j,b,d'}^T$ and its limiting density $f_{b,d'}^T$ satisfy*

$$\|f_{j,b,d'}^T - f_{b,d'}^T\|_{C^0(Q_+)} \leq Ce^{-\omega j}, \quad (14.9)$$

$$\|f_{j,b,d'}^T\|_{\text{Lip}(Q_+)} + \|f_{b,d'}^T\|_{\text{Lip}(Q_+)} \leq H. \quad (14.10)$$

Their zero extensions are bounded by F on a common endpoint box, and are zero outside it. Their densities are uniformly Lipschitz in the offset in the uniform norm. The probability $p_{j,b,d'}^T$ of this selected success in one normalized full-phase preparation obeys

$$p_{j,b,d'}^T \geq ce^{-\Gamma j}. \quad (14.11)$$

Proof. Suppress the indices T, b and put

$$F_{j,d'} = b_j(d' - E_j)_+, \quad F_{\infty,d'} = B_b(u)B_b(v)(d' - S_b(u) - S_b(v))_+.$$

Integrate over the fixed box containing their supports, and write the normalizers as $Z_{j,d'}$ and $Z_{\infty,d'}$. The relative estimate (2.4) and the Lipschitz property of positive part give $\|F_{j,d'} - F_{\infty,d'}\|_\infty \leq Ce^{-\omega j}$ and the same bound for the normalizer difference. A fixed small square at the origin gives $Z_{j,d'}, Z_{\infty,d'} \geq z_- > 0$, after enlarging j_0 . Division proves (14.9), in fact in the uniform norm on the full box. On Q_+ the cutoff is inactive, so the first endpoint-derivative estimate gives (14.10). Bounded numerators and the lower normalizer bound give F . Also

$$\|F_{j,d'} - F_{j,d''}\|_\infty \leq C|d' - d''|, \quad |Z_{j,d'} - Z_{j,d''}| \leq C|d' - d''|.$$

These estimates, and their limiting versions, prove offset continuity without differentiating a moving boundary.

The exact full-phase formula gives

$$p_{j,b,d'}^T = \frac{d_{j,b}^0}{2\pi A(T)} Z_{j,b,d'}, \quad d_{j,b}^0 = \frac{a_b}{\sinh(j\gamma(T))}.$$

The positive lower bound for a_b , $A(T) \leq A_+$, and $1/\sinh(j\gamma) \geq 2e^{-j\gamma}$ give (14.11). The itinerary is selected throughout; no nonminimal channel is identified with the complete maximal-count event. $\square$

Theorem 14.6 (Complete contact stability from finite bridges). *For two members $T, \tilde{T} \in \mathcal{K}$, let $j, k \geq j_0$ be even, let their physical offsets both equal d , and put*

$$e_{j,k} = \max_b \|f_{j,b,d}^T - f_{k,b,d}^{\tilde{T}}\|_{C^0(Q)}.$$

For sufficiently small errors,

$$\|\psi^T - \psi^{\tilde{T}}\|_r \leq C \left\{ |g_T - g_{\tilde{T}}| + (e_{j,k} + e^{-\omega j} + e^{-\omega k})^\vartheta \right\}. \quad (14.12)$$

Every $0 < t < r$ gives the simultaneous coefficient bound

$$\sum_{n \geq 2} \frac{t^n}{n!} \max_b |(\psi_b^T)^{(n)}(0) - (\psi_b^{\tilde{T}})^{(n)}(0)| \leq \frac{(t/r)^2}{1 - t/r} \|\psi^T - \psi^{\tilde{T}}\|_r. \quad (14.13)$$

The constants and exponent do not depend on a jet truncation order.

Proof. Apply (14.9) at the two ends of the triangle inequality comparing the limiting densities. Their $C^0(Q)$ distance is at most $e_{j,k} + C(e^{-\omega j} + e^{-\omega k})$. Theorem 13.15 applies to these two limiting laws and gives (14.12). Cauchy's coefficient estimate on D_r gives (14.13). We have not applied the separated law inverse to either finite density. $\square$

A fixed preparation budget. For the next theorem assume that g and the two contact frames are exactly calibrated and fixed throughout $\mathcal{K}$. All window times and endpoint categories are then programmed without the unknown table. Choose an even j and perform N independent normalized full-phase preparations for each starting-type experiment, each of duration $jj + d$. A preparation returns failure, or the endpoint category on the prescribed selected itinerary. Partition Q_+ into J equal squares of side h and retain its complement as an additional category. In particular, endpoints outside Q_+ are not discarded from the successful-sample denominator. Let M_b be the number of successes in group b and $C_{b,i}$ the number in cell i . For $M_b > 0$ set

$$\hat{f}_{b,i} = \frac{C_{b,i}}{M_b h^2}. \quad (14.14)$$

The counters M_b are used for normalization and may subsequently be forgotten. No ancillary interpretation of the waiting counts is needed.

Theorem 14.7 (Analytic recovery with every preparation charged). *Under the preceding hypotheses let $\alpha \in (0, 1/2)$, set*

$$x = \log \frac{16(J+1)}{\alpha}, \quad m_0 = \frac{c}{2} N e^{-\Gamma j},$$

and assume $cN e^{-\Gamma j} \geq 8 \log(8/\alpha)$. For every $\xi > 0$ there is a measurable estimator with values in $\mathcal{K}$ such that, with probability at least $1 - \alpha$,

$$\|\hat{\psi} - \psi\|_r \leq C \left(e^{-\omega j} + h + \sqrt{\frac{x}{m_0 h^2}} + \frac{x}{m_0 h^2} + \xi \right)^\vartheta, \quad (14.15)$$

whenever the quantity in parentheses is sufficiently small. The experiment uses exactly $2N$ preparations and evolution time $2N(jg + d)$, including all failures. No finite-dimensional model for the contact functions is imposed.

More precisely, for $L_N = \log(C_ N/\alpha)$ and all sufficiently large N/L_N , an even choice*

$$j_N = 2 \left\lfloor \frac{\log(N/L_N)}{2(4\omega + \Gamma)} \right\rfloor \geq j_0, \quad h_N \asymp \left(\frac{L_N e^{\Gamma j_N}}{N} \right)^{1/4}, \quad 0 < \xi \leq N^{-1}, \quad (14.16)$$

with h_N rounded to an equal-square grid, gives

$$\sup_{T \in \mathcal{K}} \Pr_T \left\{ \|\hat{\psi} - \psi^T\|_r > C \left(\frac{L_N}{N} \right)^{\vartheta\omega/(4\omega+\Gamma)} \right\} \leq \alpha. \quad (14.17)$$

The same event controls all coefficients by (14.13). The rate is an upper bound, not an optimality assertion.

Proof. The binomial lower-tail estimate and (14.11) give

$$\Pr\{\min_b M_b < m_0\} \leq 2 \exp(-cN e^{-\Gamma j}/8) \leq \alpha/4. \quad (14.18)$$

For example the usual exponential Markov argument at parameter $-\log 2$ bounds $\Pr\{M \leq Np/2\}$ by $e^{-Np/8}$; the displayed bound also holds when the threshold is decreased to m_0 . Conditional on $M_b = m$, the successful marks have their actual iid conditional law. This assertion follows directly by factoring the joint density at each fixed set of m successful indices and summing over those sets. We use a fixed number of attempts, not a stopped experiment conditioned on completing its cap.

For a cell probability $p_i \leq Fh^2$, the Bernoulli exponential bound gives, simultaneously in both types and all J cells, except on an event of probability at most $4Je^{-x} \leq \alpha/4$,

$$\left| \hat{f}_{b,i} - h^{-2} \int_{C_i} f_{j,b,d} \right| \leq \sqrt{\frac{2Fx}{mh^2}} + \frac{2x}{3mh^2}. \quad (14.19)$$

For completeness, for a Bernoulli variable X of mean p and $0 < \lambda < 3$,

$$\log \mathbb{E} e^{\lambda(X-p)} \leq p(e^\lambda - 1 - \lambda) \leq \frac{p\lambda^2}{2(1-\lambda/3)}.$$

The same upper bound with $|\lambda|$ holds for the lower tail. Applying exponential Markov and optimizing gives $|\bar{X} - p| \leq \sqrt{2px/m} + 2x/(3m)$ with failure probability at most $2e^{-x}$. This proves (14.19). The bound is uniform for every $m \geq m_0$; integrating over these counts and adding (14.18) yields a simultaneous probability at least $1 - \alpha$.

Here is an explicit measurable selection construction. The image

$$\{(f_{0,d}^T|_{Q_+}, f_{1,d}^T|_{Q_+}) : T \in \mathcal{K}\} \subset C(Q_+)^2$$

is a separable metric space. Choose once a countable dense subset of this image and a realizing table T_l for each of its members. Define

$$D_l = \max_{b,i} \left| h^{-2} \int_{C_i} f_{b,d}^{T_l} - \widehat{f}_{b,i} \right|.$$

On nonempty success groups take the first l with $D_l < \inf_{q \geq 1} D_q + \xi$, and otherwise take a fixed member of the class. The countable infimum and this first-index rule are measurable. The output contact pair is measurable in the D_r norm as well, since its range is countable. Existence of this estimator is not an assertion of a finite computational search bound.

Put $\varepsilon_N = \sqrt{2Fx/(m_0h^2)} + 2x/(3m_0h^2)$. Density of the selected image, the approximation lemma and (14.19) give $\inf_l D_l \leq Ce^{-\omega j} + \varepsilon_N$. Two triangle inequalities therefore bound the cell-average difference between the fitted and true limiting densities by $2Ce^{-\omega j} + 2\varepsilon_N + \xi$. On each cell, their common Lipschitz bound supplies

$$\|f_{b,d}^{\widehat{T}} - f_{b,d}^T\|_{C^0(Q_+)} \leq 2Ce^{-\omega j} + 2\varepsilon_N + \xi + 2\sqrt{2}Hh. \quad (14.20)$$

This is a comparison of two realizable limiting laws, not a regularity claim about the empirical histogram. Apply Theorem 13.15 to obtain (14.15).

To prove the rate, put $a_N = L_N/N$ and $\zeta = \omega/(4\omega + \Gamma)$. Rounding j_N down by less than two gives

$$e^{-\omega j_N} \leq e^{2\omega} a_N^\zeta, \quad \left(\frac{L_N e^{\Gamma j_N}}{N} \right)^{1/4} \leq a_N^\zeta.$$

The rounded grid satisfies $h_N \asymp (L_N e^{\Gamma j_N}/N)^{1/4}$ and $J_N \leq C(N/L_N)^{2\zeta}$. Since $2\zeta < 1/2$, enlarging the fixed C_* ensures $x \leq CL_N$. Also $Ne^{-\Gamma j_N}/L_N \geq (N/L_N)^{4\zeta} \rightarrow \infty$, so the success-count condition holds. The square-root term in (14.15) is $O(a_N^\zeta)$ and the linear term is $O(a_N^{2\zeta})$. The selection error N^{-1} is smaller. This proves (14.17) and the asserted charges. $\square$

The exponent records three different operations: relative finite-flight error, rare-event thinning, and estimation of a two-dimensional density with a first-order real bound. The outer analytic prior and the inner radius determine ϑ . The experiment is a sequence of independent full-phase preparations, not a mixing assertion for one continuously observed orbit. The theorem does not assert a rate for global analytic continuation, lattice registration, unknown channel discovery, or arbitrarily inefficient recording. Those operations retain their own hypotheses. In particular the exactly calibrated budget in (14.17) must not be called a calibration-inclusive rate.

Estimated offset and endpoint coordinates. The finite-budget estimate also has a calibration-aware form. Consider now variable gaps in $\mathcal{K}$. Suppose an independent preliminary experiment has deterministic cap B_{pil} and, except with probability β , supplies correct signed channel labels, a gap estimate $|\widehat{g} - g| \leq v_g$, and observable endpoint projections inducing a Borel map Φ with $\|\Phi - \text{id}\|_\infty \leq r_c$ on the actual endpoint collar. Program the subsequent windows at $j\widehat{g} + d + \epsilon_t$, with $|\epsilon_t| \leq v_t$. Require $\Delta = jv_g + v_t$ small enough that all actual offsets belong to I_d , and $0 \leq r_c \leq h/2$. The observed variables after this preliminary experiment are fixed maps of the raw record; no unknown true chart is applied by the observer.

Corollary 14.8 (A charged bound with calibration errors). *Under these hypotheses perform the two groups of N fresh attempts as above. With m_0, x and the count condition of Theorem 14.7,*

there is a measurable estimator satisfying, with probability at least $1 - \alpha - \beta$,

$$\|\hat{\psi} - \psi\|_r \leq C \left[v_g + \xi + \left(e^{-\omega j} + \Delta + h + \frac{r_c}{h} + \sqrt{\frac{x}{m_0 h^2}} + \frac{x}{m_0 h^2} + \xi \right)^{\vartheta} \right]. \quad (14.21)$$

The complete deterministic preparation charge is $B_{\text{pil}} + 2N$.

Proof. Fix a true table and a good preliminary history. For a square cell C_i of side h , a change of its category under Φ can occur only in the r_c -neighborhood of its boundary. For $r_c \leq h/2$ that neighborhood has area at most

$$(h + 2r_c)^2 - (h - 2r_c)^2 = 8hr_c.$$

The global density bound in Lemma 14.5 therefore gives a cell-mass error at most $8Fhr_c$ and a cell-average error at most $8Fr_c/h$. This includes the outer edges of Q_+ ; the outside category is still included in M_b . No density or Jacobian for $\Phi_{\#}P$ is assumed. Its cell masses are at most $Fh^2 + 8Fhr_c \leq 5Fh^2$, which suffices for (14.19) with F replaced by $5F$.

Uniform offset continuity adds $C\Delta$ and the finite-flight lemma adds $Ce^{-\omega j}$. The success bound is uniform on I_d , hence the count and cell concentration estimates hold conditionally on every good preliminary history. Choose countably many representatives dense in the joint image $(g, f_0|_{Q_+}, f_1|_{Q_+})$. In the previous first-index selection allow only candidates with $|g_l - \hat{g}| < v_g + \xi$. On a good history this set is nonempty and approximates the true density image arbitrarily well: the strict slack $\xi > 0$ keeps the true image inside the permitted gap strip. The selected gap differs from the true one by at most $2v_g + \xi$. Off that event use any fixed default when necessary. The allowed indices and their residual infimum are measurable.

Repeat (14.20) with the two additional biases $C\Delta + 8Fr_c/h$, and apply the two-point analytic inverse with the selected gap error. This proves (14.21) conditionally. Integrate over good histories and add β . The two subsequent groups always use exactly $2N$ attempts; adding the preliminary cap proves the charge without conditioning on cap completion. $\square$

For the persistent position-sensor design, the existing calibration theorem F.47.2 supplies $jv_g \leq \varepsilon_{\text{pil}}$ and $r_c \leq C_X \sqrt{\varepsilon_{\text{pil}}}$ under its stated compact marked-class hypotheses, with the explicit cap (F.47.5). It is a substantive input to this application. With an exact clock, choosing $\varepsilon_{\text{pil}} \leq c_1 h^4$ makes both Δ and r_c/h $O(h)$, and $r_c \leq h/2$ for small h . The charge is then $B_{\text{pil}}(j, \varepsilon_{\text{pil}}, \beta) + 2N$, not $2N$. The pilot cost can also be estimated explicitly. This yields a slower rate in the complete preparation budget rather than treating calibration as a free oracle.

Corollary 14.9 (Complete preparation budget with the position pilot). *Assume in addition the compact marked-class hypotheses of Theorem F.47.2, persistent position recording, and an exact clock. The number of marked channels is fixed, and the channel and starting-type labels used by that theorem are retained. There are $C_0, C_1 > 0$ such that, for $0 < \alpha < 1/2$ and all sufficiently large $B/[\log(C_0 B/\alpha)]^2$, a parameter-independent experiment with at most B preparations, counting the pilot and every rejected attempt, has a measurable estimator satisfying*

$$\sup_{T \in \mathcal{K}} \Pr_T \left\{ \|\hat{\psi} - \psi^T\|_r > C_1 \left(\frac{[\log(C_0 B/\alpha)]^2}{B} \right)^{\vartheta \omega / (12\omega + \Gamma)} \right\} \leq \alpha. \quad (14.22)$$

Here Γ is enlarged, if necessary, to bound also the hyperbolicity exponent in the pilot mass estimate. The total evolution time is at most $CB \log(C_0 B/\alpha)$. All coefficient bounds (14.13) hold on the same event.

Proof. Clip the preliminary gap estimate to the known interval $[g_-, g_+]$ (and use a fixed value there on a failed pilot). This cannot enlarge the good-history error and bounds every programmed time on all histories. Use preliminary failure probability $\alpha/2$, and apply Corollary 14.8 with

subsequent failure probability $\alpha/2$. The pilot has success lower bound $p_*(j, \varepsilon) = c_* \varepsilon^2 e^{-\Gamma j}$ at a grid point within $[\varepsilon, 2\varepsilon]$ of onset. Its grid contains at most $C(j+1)/\varepsilon$ points. Thus the exact integer cap (F.47.5) satisfies, for $j \geq j_0$ and $0 < \varepsilon < \varepsilon_0$,

$$B_{\text{pil}}(j, \varepsilon, \alpha/2) \leq C(j+1)e^{\Gamma j} \varepsilon^{-3} \log(C/\alpha). \quad (14.23)$$

The ceiling terms are absorbed since the logarithm is bounded below and ε is small.

Choose an equal-square grid with $h \asymp e^{-\omega j}$, put $\varepsilon = c_1 h^4$ with fixed sufficiently small $c_1 > 0$, and take

$$N = \left\lceil C_2 e^{\Gamma j} h^{-4} \log \frac{C_2}{\alpha h} \right\rceil, \quad 0 < \xi \leq h^2.$$

For sufficiently large fixed C_2 , the count condition holds and both concentration terms are $O(h)$ (the linear term is $O(h^2)$). The biases $e^{-\omega j}$, Δ , and r_c/h are $O(h)$; the additive gap error is $O(h^4/j)$. Hence (14.21) gives error Ch^ϑ . The complete cap satisfies

$$\begin{aligned} B_{\text{pil}} + 2N &\leq C(j+1)e^{\Gamma j} h^{-12} \log \frac{C}{\alpha h} \\ &\leq C(j+1)e^{(12\omega+\Gamma)j} (\log(C/\alpha) + \omega j). \end{aligned}$$

The grid-rounding constants are fixed.

Let $L = \log(C_0 B/\alpha)$ and choose the even integer

$$j = 2 \left\lfloor \frac{\log\{B/(C_3 L^2)\}}{2(12\omega + \Gamma)} \right\rfloor \geq j_0.$$

Then $j+1 \leq CL$ and $\log(C/\alpha) + \omega j \leq CL$. Taking the fixed C_3 sufficiently large makes the last cap at most B , including all ceilings. Rounding down by less than two gives $h \leq C(L^2/B)^{\omega/(12\omega+\Gamma)}$, which proves (14.22). To bound evolution time on every history, retain the ceiling in the pilot grid. With $q = j(g_+ - g_-)/\varepsilon$ and $L_{\text{grid}} = \lceil q \rceil + 2$, its final time satisfies

$$t_{L_{\text{grid}}} - jg_+ = (\lceil q \rceil - q + 2)\varepsilon \in [2\varepsilon, 3\varepsilon]. \quad (14.24)$$

Thus every programmed pilot time is strictly less than $jg_+ + 3\varepsilon$. On the successful stopping event used in the pilot proof one has the sharper bound $T_{e,b} \leq jg_e + 2\varepsilon$; that event-specific estimate is not an all-history bound on the prescribed grid. A failed scan may reach $t_{L_{\text{grid}}}$. The cap already counts all $L_{\text{grid}} + 1$ grid points, and is unchanged. Both the pilot times and the programmed later times are bounded by $C(j+1)$. Multiplying by the deterministic cap gives the stated evolution-time bound, including failed histories. The labels, grid, tolerance, stopping caps and subsequent window times are either fixed in advance or measurable functions of the preliminary record, as required. $\square$

The power 12 in this bound comes from the concrete pilot rather than from an asserted intrinsic information barrier: the onset scan costs ε^{-3} , while displacement control for histogram cells uses $\varepsilon \asymp h^4$. Neither preparation-budget exponent is claimed optimal. This result complements the retained full-table acquisition result F.41.10. It recovers complete local analytic germs at a quantitative rate; global continuation, matching and lattice recovery retain their separate hypotheses and stability moduli.

14.5. Equivariant reconstruction of perturbed densities. The inverse on the exact factorized family is independent of its anchor. An extension to nearby densities is not. We now specify an extension which commutes with a change of the common transverse orientation, including away from the factorized family. The magnitude of the anchor is fixed once; its sign is part of the coordinate convention, not an additional observation.

Write $\varsigma(u) = -u$ and $f^r(u, v) = f(-u, -v)$. For each nonzero $\alpha \in I$, make the anchor in (14.3)–(14.4) explicit:

$$q_\alpha(f) = \sqrt{1 - R_f(\alpha, \alpha)}, \quad T_\alpha(f)(u) = \frac{1 - R_f(u, \alpha)}{q_\alpha(f)}, \quad (14.25)$$

$$S_{d,\alpha}(f) = \frac{dT_\alpha(f)}{1 + T_\alpha(f)}, \quad B_\alpha(f)(u) = \frac{f(u, 0)}{f(0, 0)}\{1 + T_\alpha(f)(u)\}. \quad (14.26)$$

For $M \geq 0$, let $\mathcal{U}_\alpha^M$ be the set of real C^M functions on a neighborhood of I^2 satisfying

$$\inf_{I^2} f > 0, \quad 1 - R_f(\alpha, \alpha) > 0, \quad \inf_I \{1 + T_\alpha(f)\} > 0. \quad (14.27)$$

All these conditions are open in the restricted C^M norm. Neither factorization nor a mass normalization is required. Fix $0 < a < r$. In orientation sector $\epsilon \in \{+1, -1\}$ use $\alpha = \epsilon a$ and set

$$\mathcal{F}_{d,a}(\epsilon, f) = (\epsilon, S_{d,\epsilon a}(f), B_{\epsilon a}(f)).$$

The common reversal acts by $(\epsilon, f) \mapsto (-\epsilon, f^r)$, and on the output by changing the tag and precomposing both functions with ς . For a channel collection the same tag is used for every contact and edge; fixed positive anchor magnitudes may differ between edges.

Proposition 14.10 (Transported-anchor equivariance and stability). *Reflection is an isometric bijection $\mathcal{U}_\alpha^M \rightarrow \mathcal{U}_{-\alpha}^M$, and on this entire domain*

$$\begin{aligned} T_{-\alpha}(f^r) &= T_\alpha(f) \circ \varsigma, \\ S_{d,-\alpha}(f^r) &= S_{d,\alpha}(f) \circ \varsigma, \quad B_{-\alpha}(f^r) = B_\alpha(f) \circ \varsigma. \end{aligned} \quad (14.28)$$

In particular $\mathcal{F}_{d,a}$ is equivariant, also for nonfactorized perturbations. On sets on which

$$\inf f \geq m > 0, \quad \|f\|_{C^M} \leq K, \quad q_{\epsilon a}(f) \geq q_* > 0, \quad \inf(1 + T_{\epsilon a}(f)) \geq \nu > 0, \quad (14.29)$$

it is Lipschitz in each sector with one constant $C_M = C(M, I, a, d, m, K, q_*, \nu)$, the same in the two sectors. The estimate holds for every pair of functions satisfying these bounds, not only for a perturbation compared with an exact model member.

Every compact exact-model family satisfying the hypotheses of Proposition 14.2, together with its reflected copy, has a neighborhood of this form when the anchor is transported. On the exact family the output is $(\epsilon, S, B/B(0))$, independently of the admissible anchor magnitude.

Proof. Substitution in the four-density ratio gives

$$R_{f^r}(u, v) = R_f(-u, -v), \quad q_{-\alpha}(f^r) = q_\alpha(f).$$

It follows that $T_{-\alpha}(f^r)(u) = T_\alpha(f)(-u)$. Substituting in (14.26), including the density ratio on the axis, proves all of (14.28). Each strict inequality in (14.27) is preserved. Reflection is an involution and preserves the supremum of every partial derivative of order at most M , proving the domain and isometry assertions.

For completeness, all uniform estimates can be proved between the two endpoints, without requiring a segment to remain in the formula domain. Restriction to the axis or to a fixed slice has operator norm at most one for the usual maximum-derivative norm. Leibniz's rule bounds products in C^M . If $h \geq m$, differentiating $hh^{-1} = 1$ bounds the C^M norm of h^{-1} in terms of m^{-1} and $\|h\|_{C^M}$. For two such functions, the identity $h^{-1} - k^{-1} = (k - h)h^{-1}k^{-1}$ gives the corresponding difference bound. Applying these estimates to R_f gives

$$\|R_f - R_g\|_{C^M(I^2)} \leq C\|f - g\|_{C^M(I^2)}.$$

At the chosen anchor,

$$|q_\alpha(f) - q_\alpha(g)| = \frac{|R_f(\alpha, \alpha) - R_g(\alpha, \alpha)|}{q_\alpha(f) + q_\alpha(g)} \leq \frac{C}{2q_*}\|f - g\|_{C^M(I^2)}.$$

The lower bound on q_α now gives a difference bound for T_α . Its C^M norm is uniformly bounded by the same estimates. The reciprocal bound using ν , followed by the product bound, proves the claims for $S_{d,\alpha}$ and B_α . Reflection preserves every bound used, so the constants agree in the two sectors.

For the last assertion, on an exact model $1 + T_\alpha \geq 1$ and $q_\alpha = S(\alpha)/(d - S(\alpha)) > 0$. The compact lower bounds give a common neighborhood with positive density denominators, anchor radical and $1 + T_\alpha$. Its reflected copy has exactly the same bounds. Exact recovery and anchor independence follow from (14.6). No differentiated square root at the action minimum is taken. $\square$

Corollary 14.11 (Orbit stability and finite-order contact reconstruction). *The transported-anchor action and amplitude reconstruction descends to the common-orientation quotient with the same fixed-order Lipschitz constant. On a compact positive contact class, the finite-jet reconstruction of Corollary 14.3, with this anchor convention, is also equivariant. Its quotient error bound through any fixed order $M \geq 2$ remains $C_M(\tau^j + \varepsilon + \varepsilon_g)$. Here ε is the interior C^M density error and ε_g is the gap error, as in that corollary.*

Proof. For tagged representatives (ϵ, f) , (δ, g) , let $\eta \in \{0, 1\}$ be the unique value with $\epsilon = (-1)^\eta \delta$. Define their orbit distance by

$$\bar{d}_M([\epsilon, f], [\delta, g]) = \|f - g^{r^\eta}\|_{C^M(I^2)}.$$

It is independent of the chosen representatives because reflection is an isometry. Define the output distance by the sum of the two corresponding function norms, using the same tag alignment. Equivariance and the same-sector estimate in Proposition 14.10 prove the quotient estimate with the same constant. For finitely many contact laws use the corresponding sum or maximum product norm throughout.

The anchored projection $Ph(u) = h(u) - h(0) - uh'(0)$ is bounded on $C^M(I)$ and satisfies $P(h \circ \varsigma) = (Ph) \circ \varsigma$. Reflection fixes the gap and both quadratic action coefficients. It therefore fixes the leading curvatures and γ in (13.3). At degree n , it multiplies both graph and action coefficients by $(-1)^n$. The finite smooth-jet factorization and reflection of the half-line action show that the lower-jet remainder obeys

$$R_n(((-1)^n q_m)_{2 \leq m < n}) = (-1)^n R_n((q_m)_{2 \leq m < n}).$$

One may verify this identity on smooth polynomial representatives; finite smooth-jet factorization makes it independent of their cutoffs and remainders. The block M_n depends only on the reflection-invariant quadratic geometry. Thus the recursion $q_n = M_n^{-1}(s_n - R_n)$ commutes with reflection, by induction in n . Its finitely many operations are smooth on the compact positive set and on a sufficiently small neighborhood of it. Combining their Lipschitz bounds with the density estimate and the finite-flight estimate proves the stated error bound. The alignment argument above then gives the same bound on orientation orbits. $\square$

The extension just constructed is an action, amplitude and finite-jet reconstruction on its specified neighborhood. An arbitrary perturbed density is not asserted to be the exact law of a periodic billiard, and its output is not asserted to admit analytic continuation or global incidence gluing. The exact global classification still uses the exact law image. All constants remain fixed-order constants.

Remark 14.12 (Why a fixed signed anchor cannot be used in both sectors). For a fixed signed anchor the identity is instead $S_{d,a}(f^r) = S_{d,-a}(f) \circ \varsigma$. This cannot in general be replaced by $S_{d,a}(f) \circ \varsigma$. For example, take $d = 1$ and

$$f_\zeta(u, v) = \frac{2}{\pi}(1 - u^2 - v^2)_+ \{1 + \zeta uv(u + v)\}, \quad 0 < \zeta \leq 1/100.$$

This is a probability density on the unit disk: its perturbation integrates to zero under simultaneous reversal, and its multiplier is positive. On $[-1/3, 1/3]^2$ it is a positive smooth perturbation of the exact density with $S(u) = u^2$. Set $a = 1/4$, $D = 1/225$, and $h = 2\zeta a^3(224/225)$. Direct substitution yields

$$T_a(f_\zeta^r)(a)^2 = D + h, \quad T_a(f_\zeta)(-a)^2 = \frac{D^2}{D - h}, \quad (D + h) - \frac{D^2}{D - h} = -\frac{h^2}{D - h} \neq 0.$$

Both unsquared quantities are positive. Hence the action outputs differ as well. At $\zeta = 1/100$ the last difference is $-49/2092500$. This is a witness concerning an off-model formula, not physical-table nonidentifiability. Transporting the anchor restores the identity exactly, without removing the perturbation neighborhood.

14.6. What one density adds to one support. The family of support thresholds determines the action, but the support at one offset need not do so. The following functional comparison isolates the additional information used by the four-density identity.

Proposition 14.13 (A support-preserving action change). *On $[-1, 1]$ let $d = 1$, $S(u) = u^2$, and, for $0 < \epsilon \leq 1/10$, set*

$$\phi_\epsilon(s) = s + \epsilon s(1 - s)(s - 1/2), \quad \tilde{S}(u) = \phi_\epsilon(S(u)).$$

Both actions vanish to second order at zero, are strictly convex, and are different. Nevertheless

$$S(u) + S(v) < 1 \iff \tilde{S}(u) + \tilde{S}(v) < 1. \quad (14.30)$$

Their endpoint densities with any positive separable amplitudes cannot have the same four-density invariant on an interior neighborhood of zero.

Proof. For $0 \leq s \leq 1$,

$$\phi'_\epsilon(s) = 1 + \epsilon(-3s^2 + 3s - 1/2) \geq 1 - \epsilon/2 > 0, \quad \phi_\epsilon(1 - s) = 1 - \phi_\epsilon(s).$$

Consequently, for $s, t \in [0, 1]$, $s + t < 1$ is equivalent to $\phi_\epsilon(t) < \phi_\epsilon(1 - s) = 1 - \phi_\epsilon(s)$. This proves (14.30). Also

$$\tilde{S}''(u) = 2\{\phi'_\epsilon(s) + 2s\phi''_\epsilon(s)\}_{s=u^2}, \quad \phi'_\epsilon(s) + 2s\phi''_\epsilon(s) = 1 + \epsilon(-15s^2 + 9s - 1/2) \geq 1 - 13\epsilon/2 > 0.$$

The zero value and derivative at the origin follow by composition. For $0 < S(u) < 1/2$ the correction is nonzero, so the actions differ on every neighborhood of zero. On a sufficiently small interior square, (14.6) and the fixed-anchor inverse recover S from its four-density invariant, irrespective of the positive amplitude. Equality of the invariants would therefore make the two actions equal, a contradiction. $\square$

This example belongs to the functional action–density class. It is not an assertion that both actions have been realized by physical billiard tables satisfying the global hypotheses. The physical theorem uses the relative-law construction for that realization. The comparison shows precisely why retaining density, rather than just one support, enters Theorem 14.1.

14.7. Analytic continuation of a registered boundary germ.

Lemma 14.14 (A registered analytic germ determines the connected boundary). *Let $C, C' \subset \mathbb{R}^2$ be connected embedded real-analytic strictly convex closed curves. Suppose there are points $p \in C$, $p' \in C'$ and an orientation-preserving Euclidean motion A carrying the oriented point–tangent frame at p to that at p' , such that $A(C)$ and C' agree on a nonempty boundary arc containing p' . Then*

$$A(C) = C'.$$

Equivalently, equality of a nonempty registered analytic boundary germ implies equality of the complete connected boundary image.

Proof. Parametrize C and C' by positively oriented arclength from p and p' , with lengths L and L' . Their positive signed curvature functions κ and κ' are real analytic. Equality of the registered arcs gives equality of the curvature germs at the origins. Analytic continuation on the universal arclength covers therefore gives one real-analytic positive function $\tilde{\kappa} : \mathbb{R} \rightarrow (0, \infty)$ which is both L -periodic and L' -periodic and restricts to the two curvature functions.

Strict convexity gives total turning

$$\int_0^L \tilde{\kappa}(s) ds = 2\pi = \int_0^{L'} \tilde{\kappa}(s) ds.$$

If, say, $L' < L$, positivity would give $\int_{L'}^L \tilde{\kappa} > 0$, contradicting these two equalities; the opposite inequality is identical. Hence $L = L'$.

Finally the planar Frenet system

$$X' = T, \quad T' = \tilde{\kappa}N, \quad N' = -\tilde{\kappa}T$$

has unique solutions for a prescribed oriented point–tangent frame. The motion A matches those initial data, so the two arclength parametrizations agree on $[0, L]$. Their complete boundary images are therefore equal. $\square$

15. INTERACTION DATA WITH UNKNOWN TRANSVERSE ORIGINS

The geometric inverse uses the interaction of the two endpoint coordinates, not the individual flux amplitudes. This distinction also permits the transverse contact origins to be recovered from the law. We prove this for an observation in which the two numerical origins are unknown and successful records are subject to unknown, separate endpoint efficiencies. The signed units and directions of the transverse coordinates, the gap and the excess time remain specified. In particular, an unknown origin is not an unknown nonlinear change of the physical coordinate.

15.1. The recorded law and its interaction. Let I be a bounded interval containing zero. Let S be smooth on a neighborhood of $\bar{I}$, with

$$S(0) = S'(0) = 0, \quad S'' > 0.$$

Fix $d > 0$ so small that $\{u \in I : S(u) \leq d\}$ is compactly contained in I . For unknown $\xi, \eta \in \mathbb{R}$ and positive smooth functions A and C on neighborhoods of $\xi + \bar{I}$ and $\eta + \bar{I}$, respectively, consider the probability density

$$\tilde{f}(x, y) = \frac{A(x)C(y)}{Z} [d - S(x - \xi) - S(y - \eta)]_+, \quad (x, y) \in (\xi + I) \times (\eta + I), \quad (15.1)$$

extended by zero. Here Z is the positive integral of the numerator. The positive set

$$D = \{(x, y) : S(x - \xi) + S(y - \eta) < d\} \quad (15.2)$$

is understood in the displayed coordinate box. It is bounded, connected, and has interval fibers. Its closure is inside that box. Strict positivity of A, C is important: no part of the cap is hidden by a zero recording efficiency. The law determines D and its smooth interior density; a continuous density is determined everywhere in its open positive set by its almost-everywhere class.

For a same-type bridge, the physical limiting density of Theorem 7.3 has this form with $A(\xi + u) = B(u)$ and $C(\eta + v) = B(v)$. Unknown recording probabilities $e_1(x), e_2(y) > 0$ multiply these factors before normalization. Conversely, since B is positive on the compact collar, the positive smooth ratios $A(\xi + u)/B(u)$ and $C(\eta + v)/B(v)$ can each be rescaled to recording probabilities at most one without changing the normalized law. Thus every pair of effective factors in this class is allowed, and A, C need not agree. For fixed smooth efficiencies, the corresponding reweighted finite-flight laws have this limit: bounded continuous reweighting

preserves weak convergence and the limiting normalizing mass is positive. No uniform bound over efficiencies tending to zero is asserted.

On D put

$$\mathcal{K}_{\tilde{f}}(x, y) = \partial_x \partial_y \log \tilde{f}(x, y) = \frac{\tilde{f} \tilde{f}_{xy} - \tilde{f}_x \tilde{f}_y}{\tilde{f}^2}. \quad (15.3)$$

This mixed logarithmic derivative is the classical local dependence function of Holland and Wang [14]; the related odds-ratio description of association is developed in [15]. We use those invariants in physical transverse coordinates, not after a transformation to marginal quantiles. The facts needed below are proved directly. No sampling-equivalence or efficiency theorem from association models is imported into the billiard experiment.

Lemma 15.1 (Recovery of the two transverse origins). *The law (15.1) determines ξ and η , independently of A, C, Z . More precisely, with $H = d - S(x - \xi) - S(y - \eta)$,*

$$\mathcal{K}_{\tilde{f}}(x, y) = -\frac{S'(x - \xi)S'(y - \eta)}{H^2}. \quad (15.4)$$

Among vertical fibers of D , exactly the fiber at $x = \xi$ has identically zero interaction. Among horizontal fibers, exactly the one at $y = \eta$ has that property. If $(\xi, y_) \in D$ with $y_* \neq \eta$, then*

$$\partial_x \mathcal{K}_{\tilde{f}}(\xi, y_*) = -\frac{S''(0)S'(y_* - \eta)}{[d - S(y_* - \eta)]^2} \neq 0. \quad (15.5)$$

The analogous horizontal root is simple as well.

Proof. On D , $\log \tilde{f} = \log A + \log C + \log H - \log Z$. The mixed derivative of the first, second and fourth terms vanishes. Since $H_{xy} = 0$, differentiating $\log H$ gives (15.4). Strict convexity implies that S' vanishes exactly at zero. Every nonempty vertical fiber is an open interval of positive length, and so contains a point different from η . If $x \neq \xi$, the interaction is nonzero at such a point. For $x = \xi$ it vanishes on the entire fiber. This proves the vertical characterization; interchanging the coordinates proves the horizontal one. Differentiating (15.4) at $x = \xi$ eliminates the term containing $S'(0)$ and yields (15.5). All derivatives are taken in the positive interior, not across the boundary of the cap. $\square$

For the complete positive cap, there is also a support-only centering method: its fiber widths attain their unique maxima at the two origins. Lemma 15.5 gives that comparison. The interaction proof above is useful because its simple roots, and the finite-smooth estimate below, are confined to a positive interior rectangle. Theorem 15.6 and Corollary 15.7 state what survives when only such an interior window is recorded.

15.2. Recovery of the action, with separate endpoint weights. After determining the two origins, set $f^c(u, v) = \tilde{f}(\xi + u, \eta + v)$. Choose $r > 0$ with $[-r, r] \subset I$ and $\max_{[-r, r]} S < d/4$, and let $a \in (-r, r) \setminus \{0\}$. Define

$$R(u, v) = \frac{f^c(u, v)f^c(0, 0)}{f^c(u, 0)f^c(0, v)}, \quad (15.6)$$

$$q = \sqrt{1 - R(a, a)}, \quad T(u) = \frac{1 - R(u, a)}{q}, \quad \mathcal{S}(\tilde{f})(u) = \frac{dT(u)}{1 + T(u)}. \quad (15.7)$$

Theorem 15.2 (The action as an invariant of the recorded interaction). *For every law (15.1), the construction above gives $q > 0$ and $\mathcal{S}(\tilde{f}) = S$ on $[-r, r]$. Hence the law recovers the signed*

action germ without supplied transverse origins or endpoint weights. Once S is known, the effective factors have the locally determined ratios

$$\frac{A(\xi + u)}{A(\xi)} = \frac{f^c(u, 0)}{f^c(0, 0)}(1 + T(u)), \quad \frac{C(\eta + v)}{C(\eta)} = \frac{f^c(0, v)}{f^c(0, 0)}(1 + T(v)). \quad (15.8)$$

For fixed d , two centered positive densities in this class have the same action germ if and only if their ratio is a product of a function of u and a function of v on some common square about zero. Thus the germ is a complete local invariant for separate endpoint reweighting after the origins are recovered. This does not separate a physical flux factor from an unknown recording factor multiplying it.

Proof. Write $A_0(u) = A(\xi + u)$ and $C_0(v) = C(\eta + v)$. Direct substitution, without assuming $A_0 = C_0$, gives

$$R(u, v) = \frac{d[d - S(u) - S(v)]}{[d - S(u)][d - S(v)]}, \quad 1 - R(u, v) = \frac{S(u)}{d - S(u)} \frac{S(v)}{d - S(v)}.$$

Since $a \neq 0$ and $S'' > 0$, the positive scalar root in (15.7) equals $S(a)/(d - S(a))$. It follows that $T(u) = S(u)/(d - S(u))$ for both signs of u . Solving this identity proves the action formula. In particular, no pointwise square root at the minimum is involved. Substitution in the two axis densities proves (15.8).

Equal actions imply separate factorization of the ratio of the densities by their defining formulas. Conversely, multiplying by a positive function of u and a positive function of v leaves R unchanged. The same nonzero anchor and (15.7) then recover the same action. Equivalently, equality of the two ratios R_f, R_h on a centered square gives the explicit factorization

$$\frac{f(u, v)}{h(u, v)} = \frac{f(u, 0)}{h(u, 0)} \frac{f(0, v)}{h(0, v)} \frac{h(0, 0)}{f(0, 0)}.$$

This assertion is local in the common positive square. It does not claim that arbitrary prescribed marginal laws are compatible with a given nonrectangular cap, nor that a joint recording factor depending on both coordinates would cancel. $\square$

15.3. Finite-order regularity of the extraction. The identification of a zero fiber is an exact operation on a law. For a local error estimate it is preferable to use simple roots on two fixed slices. This also makes the derivative with respect to the observation explicit and avoids differentiating a set-valued zero-fiber operation.

Proposition 15.3 (A local differentiable extraction with finite loss). *Fix a law $\tilde{f}_0$ of the preceding class. For every integer $m \geq 0$, there are a closed rectangle Q compactly contained in its positive set, a number $r > 0$, and an open neighborhood $\mathcal{V}$ of $\tilde{f}_0|_Q$ in $C^{m+3}(Q)$ on which maps*

$$f \mapsto (\Xi(f), \Upsilon(f), \mathcal{S}_*(f)) \in \mathbb{R}^2 \times C^m([-r, r])$$

are C^1 . They equal (ξ, η, S) for laws of the recorded class whose origins and action data are in the indicated local branch. On a sufficiently small convex ball in $\mathcal{V}$,

$$|\Xi(f) - \Xi(h)| + |\Upsilon(f) - \Upsilon(h)| + \|\mathcal{S}_*(f) - \mathcal{S}_*(h)\|_{C^m([-r, r])} \leq C_m \|f - h\|_{C^{m+3}(Q)}. \quad (15.9)$$

The constants are local and depend on positive density, simple-root and anchor margins. For a differentiable recorded family, at its base law,

$$\dot{\xi} = -\frac{\dot{\mathcal{K}}(\xi, y_*)}{\partial_x \mathcal{K}(\xi, y_*)}, \quad \dot{\eta} = -\frac{\dot{\mathcal{K}}(x_*, \eta)}{\partial_y \mathcal{K}(x_*, \eta)}. \quad (15.10)$$

In particular a stationary interior density has stationary origins and stationary action at every separately fixed smooth order, even when its separate endpoint weights vary.

Proof. Choose noncentral x_*, y_* close enough to ξ_0, η_0 that a closed rectangle $Q \Subset D_0$ contains (ξ_0, η_0) , both slice anchors, and a small centered square, with positive room to its boundary. Its density has a positive lower bound. The map

$$f \mapsto \mathcal{K}_f = f_{xy}/f - f_x f_y / f^2$$

is C^1 from the open positive part of $C^{m+3}(Q)$ into $C^{m+1}(Q)$. Evaluating at (x, y_*) or (x_*, y) gives a C^1 map in f and the remaining scalar variable. The nonzero derivatives in (15.5) permit the scalar implicit function theorem. It produces unique roots $\Xi(f)$ near ξ_0 and $\Upsilon(f)$ near η_0 , and both root maps are C^1 . On the model these roots are the actual origins whenever the latter are in those neighborhoods. The exact global identification remains Lemma 15.1; no global root selection for arbitrary nonmodel perturbations is needed.

Shrink the neighborhood and choose r so that $(\Xi(f) + [-r, r]) \times (\Upsilon(f) + [-r, r])$ is inside Q for every f under consideration. Translation followed by restriction defines

$$f \mapsto f(\Xi(f) + u, \Upsilon(f) + v)$$

as a C^1 map into $C^{m+1}([-r, r]^2)$, with the surplus derivatives in the hypothesis. Its differential in a direction h is the translated h plus the translated $f_x D\Xi[h] + f_y D\Upsilon[h]$. Choose a fixed $a \neq 0$ in this square. The base value $1 - R_0(a, a)$ is strictly positive. On a smaller neighborhood every axis denominator, the scalar anchor root, and $1 + T$ retain positive lower bounds. Products, quotients, slice evaluations and a scalar square root are C^1 in these finite smooth norms. Composing them gives $\mathcal{S}_*$ by (15.7).

Continuity of the derivative at f_0 bounds its operator norm on a sufficiently small ball. Integration of that derivative on the segment from f to h proves (15.9). This uses a finite smooth observation norm; it is not a total-variation estimate. Differentiating the two root equations on their fixed slices proves (15.10). If $\dot{f} = 0$ in these smooth norms, then $\dot{\mathcal{K}} = 0$ and both root derivatives vanish. The translated density is then stationary, and differentiation of the scalar anchor and action formulas makes $\dot{S} = 0$. Unknown derivatives of A, C, Z have disappeared through the interaction and ratio identities, not through an assumption that they were known or constant. $\square$

For a small nonmodel perturbation the extracted action need not have zero first derivative. Subtracting its constant and linear terms fixes that anchoring and changes no coefficient of degree at least two. With a pair of recorded laws and the gap, the finite smooth inverse of Proposition 14.2 therefore gives, for each fixed contact order M , a local bound for the recovered graph coefficients by the gap error plus the two C^{M+3} density errors in (15.9). Its constants also depend on the inherited positive geometric bounds. No infinite-order condition bound or uniform estimate near a vanishing recording efficiency follows from this statement.

15.4. The complete geometric inverse with these nuisances. For a selected channel e and each same-type law $b = 0, 1$, let $\tilde{P}_{e,b}$ be its complete translated and separately weighted law as in (15.1). The gaps g_e , positive offsets d_e , obstacle and deck marks, and signed transverse length conventions are retained. The two origins and both recording profiles in each law are unknown, and may differ between endpoint types and channels.

Theorem 15.4 (Geometric determination from uncentered weighted laws). *These recorded laws and the gaps determine every finite labelled contact jet of smooth strictly convex channels. For analytic noncircular tables with a selected skeleton as in Theorem A, the set of compatible tables, after allowing the stated nuisance origins and positive separate recording profiles, is exactly the finite table fiber of the original ideal-law datum. In particular the complete marked lattice is recovered on every branch, and the fiber is a singleton on $\mathcal{A}_N$.*

For a C^1 common-strip analytic-support table family with a persistent selected design and fixed positive offsets, allow also C^1 origins and positive recording profiles with C^1 dependence in each finite smooth norm on the common recorded collars. If the gaps and all recorded interior densities are stationary, the table variation is a common infinitesimal proper Euclidean motion. Conversely, every such table variation admits stationary recorded laws by keeping its intrinsic origins and recording profiles fixed. Thus the projection of the joint data-derivative kernel onto table variations is exactly the common Euclidean tangent space.

Proof. Lemma 15.1 identifies the origins in each record, and Theorem 15.2 recovers both signed action germs. The gap and their quadratic coefficients give

$$\gamma = \operatorname{arsinh}(g\sqrt{a_0a_1}), \quad c_b = \cosh \gamma \sqrt{a_b/a_{1-b}}, \quad \kappa_b = (c_b - 1)/g, \quad a_b = S_b''(0).$$

The finite-order identities of Lemma 13.4 and Theorem 13.1 then give $q_n = M_n^{-1}(s_n - R_n)$ successively. The smooth remainder result is still necessary; cancellation of recording factors does not replace it. Every finite signed jet, including its odd terms, is recovered.

For analytic boundaries the resulting contact germs determine the complete ordered channel-frame images. Finite matching and the displacement cochain of Theorem 21.3 enumerate the compatible tables and recover $L = (v_1 \ v_2)M^{-1}$ on each accepted branch. Every realization of the recorded laws yields these same jets and images, and so belongs to that list. Conversely, each accepted branch places every recovered ordered channel pair by a proper motion. Its local half-line actions and normalized physical amplitudes therefore agree with the original ones; any global phase-volume constant cancels in the conditional law. Reusing the original numerical origins and endpoint recording profiles realizes the complete recorded laws on that branch. This proves both inclusions of the table fibers, not merely finiteness of a list containing them. Asymmetric uniqueness follows from the same matching theorem.

For the differential assertion, work on common positive rectangles about the base contacts. The inherited forward differentiability and the assumed regularity of the profiles give the finite smooth dependence required in Proposition 15.3. The normalizer is differentiable by the positive-part calculation in Lemma 22.1; translations can first be removed in that integral, so the same regular level set applies. Stationarity of the recorded densities gives $\dot{\xi} = \dot{\eta} = 0$ and $\dot{S}_b = 0$. Differentiate the displayed leading recovery and then

$$\dot{q}_n = M_n^{-1}(\dot{s}_n - \dot{R}_n - \dot{M}_n q_n).$$

Finite induction makes every graph-jet variation zero. Analyticity of the support variation on a smaller common strip and the identity theorem give stationary complete channel-frame images, by Lemma 22.3. The actual local congruence branch of Lemma 21.2 makes their relative angular and translation velocities zero. The same finite cochain then makes the lattice derivative zero in one common gauge. Undoing that gauge yields exactly the common infinitesimal Euclidean motion. This argument never infers an injective derivative from exact injectivity.

For the converse choose a common Euclidean motion of the table and keep all nuisance profiles fixed as functions of their intrinsic numerical coordinates. Gaps and the ideal laws are invariant, hence the recorded laws are stationary. This is an existence assertion about a joint kernel vector. Arbitrary changes of the recording profiles need not preserve a law even when the table itself is stationary. $\square$

15.5. Relation to the other observations. The extraction recovers numerical origins, not an unknown clock, channel direction, transverse scale or deck marking. The axes remain the specified oriented transverse axes of each channel; this is not an arbitrary fixed laboratory projection while the channel direction varies. It requires the positive cap not to be cropped before either origin is visible. An inseparable joint recording factor $w(x, y)$ adds $\partial_x \partial_y \log w$ to (15.3) and is a different

model. Similarly, an arbitrary nonlinear coordinate change alters the physical action; it is not an unobserved translation.

Theorem 15.4 concerns exact laws, with the local finite smooth bound (15.9). It is not a parameter-independent Markov kernel transforming individual uncentered samples into ideal samples, a uniform finite-histogram theorem, or a semiparametric efficiency assertion. The unknown recording factors may drastically reduce the chance of accepting a successful bridge. The preparation budgets and density margins in Part III of the complete technical manuscript cannot be transferred uniformly without additional quantitative bounds on those factors. The physical pilot there also determines quantities other than transverse origins, and its proved sensor and clock assumptions remain in force. In this way the same boundary law has distinct roles: its interaction determines geometry, while its amplitude and retained auxiliary records govern acquisition and local statistical information.

15.6. What the complete support already determines. There are two different reasons for using an interior interaction. It eliminates separate recording factors, and its local regularity does not require control of a support boundary. The second point should not be confused with nonidentifiability of the origins from a complete cap. The following observation, communicated in [16], identifies exactly what that support supplies.

Lemma 15.5 (Centering from complete fiber widths). *In the setting of (15.1), the positive set D alone determines both origins. Its vertical fiber length has a unique maximum at $x = \xi$, and its horizontal fiber length has a unique maximum at $y = \eta$. Both maxima are nondegenerate. No symmetry of S is required.*

Proof. For $0 < q \leq d$, strict convexity and the compact-sublevel assumption give exactly two inverse branches $v_-(q) < 0 < v_+(q)$ of $S(v) = q$, lying in I . The inverse function theorem applies to both branches, including in a neighborhood of $q = d$, since their derivatives $S'(v_\pm(q))$ are nonzero and the sublevel has positive room inside I . Put

$$W(q) = v_+(q) - v_-(q), \quad W'(q) = \frac{1}{S'(v_+(q))} - \frac{1}{S'(v_-(q))} > 0.$$

For every x with a nonempty vertical fiber, its length is

$$w(x) = W(d - S(x - \xi)). \tag{15.11}$$

This expression has its unique maximum at ξ , since S has its unique minimum at zero. Differentiation in a neighborhood of that maximum gives

$$w'(\xi) = 0, \quad w''(\xi) = -W'(d)S''(0) < 0.$$

The horizontal calculation is identical, with η in place of ξ . Strict positivity of the recording factors makes D the interior of the support of the recorded law, so these are observable fiber lengths. The width need not be symmetric about its maximum. $\square$

This comparison concerns centering, not recovery of the entire action from one support: Proposition 14.13 preserves a whole cap while changing its action germ. It is also not a replacement for Proposition 15.3, whose observation norm controls only a fixed rectangle strictly inside the positive set. The support boundary is outside that norm. The four-density cancellation, including unequal separate factors, is already the algebra of Theorem 14.1; it is not an additional independent high-order inverse. The geometric work is done by the relative law and the actual-smooth contact recursion to which the invariant is applied.

15.7. Recording only a positive interior window. The extraction can be formulated without an observed cap boundary. Let J_x, J_y be bounded open intervals with known endpoints and let $\mathcal{W} = J_x \times J_y$ be a known recording window. The numerical origins $\xi \in J_x$ and $\eta \in J_y$ are unknown. On neighborhoods of the translated closed intervals let S be smooth with $S(0) = S'(0) = 0$ and $S'' > 0$, as above. Assume

$$H(x, y) = d - S(x - \xi) - S(y - \eta) > 0 \quad \text{on } \overline{\mathcal{W}}. \quad (15.12)$$

Only records in $\mathcal{W}$ are retained. Their law has density

$$p_{\mathcal{W}}(x, y) = \frac{A(x)C(y)}{Z_{\mathcal{W}}} H(x, y) \mathbf{1}_{\mathcal{W}}(x, y), \quad Z_{\mathcal{W}} = \int_{\mathcal{W}} A(x)C(y)H(x, y) \, dx \, dy, \quad (15.13)$$

where A, C are positive and smooth near the respective closed intervals. They need not agree. Neither the density nor its support outside $\mathcal{W}$, nor the discarded mass, is part of this datum. In particular its positive support is just the fixed recording window.

The visibility assumption $\xi \in J_x, \eta \in J_y$ retains both critical coordinate lines. It specifies a class of observations, not an algorithm for locating a suitable window without prior information. The known signed transverse units, directions and excess time remain those of Section 15. We do not replace them by unknown scales or arbitrary coordinate transformations.

Theorem 15.6 (Interior-window extraction). *The law (15.13) determines ξ, η and the signed germ of S . A complete cap and an observation of its boundary are not needed. The two effective factor ratios in (15.8) are determined on a small interval about the recovered origins.*

For each fixed integer $m \geq 0$, the extraction has a local C^1 extension from an open neighborhood in $C^{m+3}(Q)$ to $\mathbb{R}^2 \times C^m([-r, r])$, where $Q \Subset \mathcal{W}$ contains the two slice anchors and the translated square. On a sufficiently small convex neighborhood it satisfies (15.9). The constants depend on the chosen positive density, root, anchor and window margins. They do not require bounds on a density outside Q or on a discarded cap boundary.

Proof. On the open window the mixed logarithmic derivative is

$$\mathcal{K}_p(x, y) = -\frac{S'(x - \xi)S'(y - \eta)}{H(x, y)^2}.$$

Every vertical fiber is the nonempty interval J_y . Since S' has only one zero and J_y has positive length, exactly the vertical line $x = \xi$ has identically zero interaction; similarly the horizontal line is $y = \eta$. This recovers the origins using no boundary value. The smooth positive representative of a density on the window is uniquely determined by its almost-everywhere class.

Choose a small $r > 0$ so that $(\xi + [-r, r]) \times (\eta + [-r, r]) \Subset \mathcal{W}$ and $\max_{[-r, r]} S < d/4$. On that square put $p^c(u, v) = p_{\mathcal{W}}(\xi + u, \eta + v)$ and choose one fixed $a \in (-r, r) \setminus \{0\}$. The window normalization is a positive scalar, and hence cancels together with both separate factors:

$$R_p(u, v) = \frac{p^c(u, v)p^c(0, 0)}{p^c(u, 0)p^c(0, v)}, \quad 1 - R_p(u, v) = \frac{S(u)}{d - S(u)} \frac{S(v)}{d - S(v)}.$$

The scalar $q = \sqrt{1 - R_p(a, a)}$ is positive and equals $S(a)/(d - S(a))$. Thus $T(u) = (1 - R_p(u, a))/q$ and $S(u) = dT(u)/(1 + T(u))$, for both signs of u . Substitution on the axes gives the asserted effective-factor ratios. Nothing in these identities involves $Z_{\mathcal{W}}$ or the unobserved mass separately.

For the local regularity assertion choose noncentral slices x_*, y_* inside the window and then a closed $Q \Subset \mathcal{W}$ containing these slices near their roots and a small centered square. Compactness and (15.12) give a positive density minimum there. The two root slopes are exactly (15.5) and its horizontal counterpart, so they are nonzero. The proof of Proposition 15.3 is local to this Q : the map $f \mapsto f_{xy}/f - f_x f_y/f^2$ is C^1 from positive $C^{m+3}(Q)$ to $C^{m+1}(Q)$; two scalar implicit function theorems give the origin maps; and restriction after their translations, followed by the slice

products and scalar anchor inverse, gives a C^1 action map into $C^m([-r, r])$. The derivative of the translation contains both $f_x D\Xi$ and $f_y D\Upsilon$, and the scalar anchor is also differentiated. None of these operations evaluates the edge of $\mathcal{W}$. A smaller convex ball has bounded derivative, which gives the stated estimate by integration on line segments. This argument applies to arbitrary nearby positive nonmodel inputs as a local extension; exact identification is claimed on the model class. $\square$

The fixed support of (15.13) cannot perform the centering in Lemma 15.5. For example, begin with an actual channel law and a window strictly inside its cap, with both origins strictly inside the coordinate intervals. Small changes of the numerical origins preserve these strict conditions. The cropped laws then have the same positive support $\mathcal{W}$ and different origins, while their interactions locate those origins by the theorem. This is a change of the recorded numerical origins of an existing channel, not a claim that an arbitrary functional action is realized by a billiard. The window theorem records the boundary-independent content of the existing interior proof; it does not introduce a new association invariant.

15.8. The geometric fiber after interior cropping.

Corollary 15.7 (The same table fiber from interior-window laws). *For every selected channel and both same-type laws, retain a known window satisfying (15.12) and containing the respective unknown origins, together with the gaps, marks, signed units, directions and fixed positive offsets. Allow independent positive smooth separate recording factors for every law. Then these window laws determine every finite labelled contact jet of smooth strictly convex channels. On the analytic noncircular class of Theorem A, their compatible table fiber is exactly the ideal-law fiber, including its unknown marked lattices. It is therefore finite, and is a singleton on $\mathcal{A}_N$.*

In the differentiable-family setting of Theorem 15.4, keep the numerical windows fixed and the visibility and positivity conditions locally strict. The projection of the joint window-data derivative kernel onto table variations is exactly the common infinitesimal proper Euclidean tangent space.

Proof. Apply Theorem 15.6 to each record. The gap and the two recovered quadratic action coefficients give the leading geometry by the formulas in Theorem 15.4. The finite-smooth factorization of Lemma 13.4 then makes each higher coefficient an actual graph-jet invariant before the triangular inverse is applied. Thus all finite signed contact jets are fixed. For analytic boundaries these jets give complete ordered channel-frame images by analytic continuation. Every table realizing the window laws therefore gives one of the compatible incidence assignments in Theorem 21.3, with its lattice $L = (v_1 \ v_2)M^{-1}$. This is one inclusion in the claimed fiber.

Conversely, each accepted ideal branch places every recovered ordered channel pair by a proper Euclidean motion. Its local physical actions and amplitudes agree with those of the original pair, in the same intrinsic transverse coordinates. Reuse the original numerical origins, windows and recording factors on each such pair. The resulting normalized window density is exactly (15.13). A global phase-volume factor cancels under conditioning. This gives the reverse inclusion; no unobserved portion of a recorded density is being assumed as extra data. The uniqueness conclusion follows on the original properly asymmetric subclass, without altering that class.

For a family, the normalizing integral on the fixed window is particularly simple. In the notation of (15.13), with $w_t = A_t C_t$,

$$\dot{Z}_{\mathcal{W}} = \int_{\mathcal{W}} (\dot{w}H + w\dot{H}), \quad \dot{H} = -\dot{S}(x - \xi) - \dot{S}(y - \eta) + S'(x - \xi)\dot{\xi} + S'(y - \eta)\dot{\eta}.$$

Local smooth bounds on the compact window justify differentiation under the integral and give continuity of the derivative. There is no moving integration boundary. Together with the inherited forward estimates this gives the finite smooth density dependence needed above.

Stationary window densities have stationary origins and stationary action jets. Differentiating the leading recovery and then $q_n = M_n^{-1}(s_n - R_n)$ at each finite order gives zero graph-jet variation, retaining the derivatives of both R_n and M_n . The variation of a common-strip analytic-support family is analytic on a smaller strip, so it vanishes on each complete channel-frame image. The actual continuous alignment branches and the displacement cochain then make the table variation zero in one common Euclidean gauge, exactly as in Theorem 15.4. Undoing the gauge gives the asserted tangent space. Conversely, a common proper Euclidean motion with fixed intrinsic numerical origins and recording profiles leaves every window law stationary. This is a statement about the projection of a joint kernel, not identification of all profile variations. $\square$

The effective factors in this window model are compatible with separate recording probabilities. On the two compact coordinate intervals the physical amplitudes are positive. Extend the logarithms of the smooth positive ratios A/B and C/B to the physical collars by smooth cutoffs, and exponentiate. The extensions stay positive and bounded; multiplying each by a sufficiently small positive constant makes it at most one. The two constants cancel in the normalized window law. The additional known crop is itself a product of the two interval indicators.

For fixed recording factors these window laws are also limits of the corresponding recorded finite-flight laws. The ideal endpoint laws converge weakly by Theorem 7.3. Multiplication by the bounded recording weight and $\mathbf{1}_W$ preserves the limit of integrals against bounded continuous tests: its discontinuities are confined to the four edges, which have zero mass for the absolutely continuous limiting law. The limiting denominator is positive, so normalizing gives (15.13). This is a fixed-observation limit, not a uniform bound over efficiencies or windows with vanishing accepted mass.

In particular, neither interior-window rigidity nor complete-cap rigidity transfers the preparation budgets of Part III of the complete technical manuscript to an unconstrained recording class. Multiplying both recording probabilities by ε leaves the normalized law unchanged and multiplies the recorded-success probability by ε^2 . A charged uniform budget therefore still requires a separate positive success bound. The interior smooth estimate, the exact finite-dimensional coordinates, and the finite-histogram theorem retain their different observation norms and hypotheses.

16. INTRINSIC LIFTED-CHANNEL GRAPHS AND ANALYTIC TABLE RIGIDITY

The signed one-channel inverse is formulated in a channel-adapted oriented frame. In a periodic billiard a measured channel also carries a deck translation. A global inverse theorem must therefore reconstruct both the relative placement of distinct channel frames and their placement relative to the periodic lattice.

Let $\Lambda_{\text{abs}} \simeq \mathbb{Z}^2$ be a marked abstract lattice with its fixed Euclidean Gram form and the orientation induced by its marked basis. An oriented Euclidean realization is an orientation-preserving linear isometry

$$\iota : \Lambda_{\text{abs}} \otimes \mathbb{R} \longrightarrow \mathbb{R}^2.$$

The quotient obstacle orbits are labelled $1, \dots, N$. An oriented measured channel is indexed exactly as in Section 2:

$$e = (a_e, b_e, \ell_e), \quad \ell_e \in \Lambda_{\text{abs}},$$

and joins a lift of obstacle a_e to the lift of obstacle b_e translated by ℓ_e . Its two incidences are denoted $(e, -)$ and $(e, +)$, with

$$r(e, -) = a_e, \quad \nu(e, -) = 0, \quad r(e, +) = b_e, \quad \nu(e, +) = \ell_e. \quad (16.1)$$

Reversing the channel replaces (a, b, ℓ) by $(b, a, -\ell)$.

Definition 16.1 (Intrinsic signed lifted-channel datum). For an oriented channel $e = (a, b, \ell)$, its intrinsic datum consists of

- (1) the lifted channel label (a, b, ℓ) in the marked abstract lattice;
- (2) the physical onset g_e ;
- (3) the two same-type signed endpoint-law germs

$$\{\Pi_{00,d}^{e,\text{end},\infty} : 0 < d < d_e\}, \quad \{\Pi_{11,d}^{e,\text{end},\infty} : 0 < d < d_e\};$$

- (4) the sign convention for the transverse endpoint coordinate.

The law is recorded in the channel-adapted frame in which the first contact is $(0, 0)$, the second is $(g_e, 0)$, and the oriented transverse axes are those used by the signed endpoint coordinates. No Euclidean placement of this frame relative to the lattice or to any other measured channel is supplied.

Theorem 13.1 and Lemma 14.14 recover two complete oriented analytic boundary images in the edge frame,

$$C_{e,-}, \quad C_{e,+},$$

representing the lifted boundaries at deck offsets 0 and ℓ_e . The remaining inverse problem is a periodic Euclidean gluing problem.

16.1. Analytic signatures and local matching. Let C be a connected oriented real-analytic strictly convex closed curve, parametrized by oriented arclength $s \in \mathbb{R}/L\mathbb{Z}$, with signed curvature $\kappa_C(s) > 0$. At $p = C(s_0)$ define

$$\mathbf{j}_C(p) = (\kappa_C(s_0), \kappa'_C(s_0), \kappa''_C(s_0), \dots).$$

Let $\text{Sym}^+(C) \subset \text{SE}(2)$ be the orientation-preserving Euclidean symmetry group of the boundary image.

Lemma 16.2 (Coincident analytic signatures are exactly symmetry orbits). *For framed points $p, q \in C$,*

$$\mathbf{j}_C(p) = \mathbf{j}_C(q)$$

if and only if an element of $\text{Sym}^+(C)$ carries the oriented framed point p to the oriented framed point q .

Proof. The reverse implication is immediate. For the forward implication choose arclength so that $p = C(0)$ and $q = C(a)$. Equality of the complete curvature jets gives equality of the real-analytic germs $\kappa_C(s)$ and $\kappa_C(s + a)$, hence equality for every s by the identity theorem. Let $A \in \text{SE}(2)$ send the oriented point-tangent frame at 0 to that at a . Uniqueness for the planar Frenet system gives $AC(s) = C(s + a)$ for all s , so $A \in \text{Sym}^+(C)$. $\square$

Thus a circle has a genuine continuous phase ambiguity and a cyclically symmetric analytic obstacle may have a finite one. These are actual inverse ambiguities, not registration conventions.

16.2. Admissible periodic gluings. Let $\tau_v(x) = x + v$. An *algebraic periodic gluing* of the intrinsic data is a pair

$$(\iota, (A_e)_{e \in E}), \quad A_e \in \text{SE}(2), \tag{16.2}$$

where ι is an oriented Euclidean realization of Λ_{abs} , such that whenever two incidences $(e, \sigma), (f, \sigma')$ carry the same quotient obstacle label,

$$\tau_{-\iota(\nu(e, \sigma))} A_e C_{e, \sigma} = \tau_{-\iota(\nu(f, \sigma'))} A_f C_{f, \sigma'} \tag{16.3}$$

as oriented labelled boundary images. Thus every recovered lift is translated back by its deck vector before copies of the same quotient obstacle are identified.

The common values in (16.3) define quotient obstacle images $C_1, \dots, C_N$. Extend them by all deck translations $\iota(\Lambda_{\text{abs}})$. We call the gluing *admissible* if this periodic family is a dispersing billiard table: distinct translated obstacle closures are disjoint and every measured edge has the prescribed facing contacts, onset and selected closest-pair channel geometry. These are conditions on the reconstructed curves and their candidate placements.

The group $\text{SE}(2)$ acts by

$$B = (R, t) : \quad \iota \mapsto R\iota, \quad A_e \mapsto B \circ A_e.$$

Both (16.3) and admissibility are invariant. Let $\mathfrak{G}_{\text{per}}(\mathcal{D})$ be the quotient of the admissible periodic gluings by this action.

Theorem 16.3 (Intrinsic periodic reconstruction and holonomy classification). *Let two labelled periodic dispersing billiard tables have the same marked abstract Euclidean lattice, the same obstacle-orbit labels, and the same connected measured lifted-channel graph. Assume every obstacle boundary is connected, real analytic and strictly convex, and that the two tables have identical intrinsic signed lifted-channel data for every measured edge. Then:*

- (1) *every incident lifted obstacle boundary is recovered as a complete analytic image in its own channel frame;*
- (2) *the observations determine the admissible periodic gluing space $\mathfrak{G}_{\text{per}}(\mathcal{D})$ without a common laboratory registration;*
- (3) *labelled periodic dispersing tables realizing the observations, modulo one simultaneous orientation-preserving Euclidean motion of the lattice and all obstacle images, are in bijection with $\mathfrak{G}_{\text{per}}(\mathcal{D})$.*

In particular, if $\mathfrak{G}_{\text{per}}(\mathcal{D})$ is a singleton, the complete labelled periodic table is intrinsically determined up to one global $\text{SE}(2)$ motion.

For the common-orientation orbit of the entire marked datum, as defined in Definition 16.12, the corresponding classification is modulo $\text{E}(2)$, by Theorem 16.14. This assertion concerns one common orientation choice, not pointwise deletion of endpoint signs.

Proof. The first assertion follows from Theorem 13.1 and Lemma 14.14. Hence every curve entering (16.3) is determined by the observations.

A table realization supplies its lattice realization ι and the actual placement A_e of every channel frame. The incidence (e, σ) lies on the lift $D_{r(e, \sigma)} + \iota(\nu(e, \sigma))$, so translating it back by the deck vector gives the same quotient obstacle image at every incidence with that label. The resulting gluing is admissible, and simultaneous global Euclidean motion changes only its quotient representative.

Conversely an admissible gluing defines

$$C_r = \tau_{-\iota(\nu(e, \sigma))} A_e C_{e, \sigma}$$

for any incidence with $r(e, \sigma) = r$. The gluing constraint makes this independent of the choice. The periodic lifts are $C_r + \iota(\lambda)$, $\lambda \in \Lambda_{\text{abs}}$. For a measured edge $e = (a, b, \ell)$, its placed endpoint curves are exactly C_a and $C_b + \iota(\ell)$, with the observed onset and oriented signed contact convention. Admissibility makes the full periodic family a dispersing billiard table with the declared measured channels. These constructions are inverse modulo the common $\text{SE}(2)$ action. $\square$

16.3. Cycle holonomy and an intrinsic uniqueness criterion. If two incidences of the same obstacle orbit have contact signatures occurring at unique framed points, Lemma 16.2 gives a unique orientation-preserving Euclidean congruence between their recovered curve copies. We call such an incidence transition *signature-rigid*.

Consider a closed oriented channel cycle

$$c = e_1^{\epsilon_1} \cdots e_m^{\epsilon_m}, \quad \epsilon_i \in \{\pm 1\},$$

for which every required incidence transition is signature-rigid. Its marked deck displacement is

$$\eta_c = \sum_{i=1}^m \epsilon_i \ell_{e_i} \in \Lambda_{\text{abs}}. \quad (16.4)$$

Compose the unique data-determined incidence congruences around the cycle in the starting channel frame and denote the resulting Euclidean motion by H_c .

Lemma 16.4 (Periodic cycle-holonomy identity). *For realizable intrinsic data, H_c has identity rotational part. Hence $H_c = \tau_{v_c}$ for a data-determined vector v_c . For every admissible periodic gluing and every placement $A_{e_0}(x) = R_{e_0}x + t_{e_0}$ of the starting channel frame,*

$$R_{e_0}v_c = \iota(\eta_c). \quad (16.5)$$

Equivalently, after using the global gauge to set the starting frame rotation to the identity, $\iota(\eta_c) = v_c$.

Proof. For two consecutive incidences of the same quotient obstacle, the gluing equation identifies their placed curve copies after the corresponding deck translations. The unique signature-rigid congruence between the two recovered edge-frame copies is therefore conjugate, by their placements, to translation by the difference of their deck offsets. Compose these identities around the cycle. All intermediate edge placements telescope. The signed deck differences sum to η_c , giving

$$A_{e_0}H_cA_{e_0}^{-1} = \tau_{\iota(\eta_c)}.$$

Conjugation by A_{e_0} preserves the rotational part, so H_c is a translation. Comparing translation vectors gives (16.5). $\square$

Definition 16.5 (Lattice-anchoring cycle). A signature-rigid cycle is *lattice-anchoring* if $\eta_c \neq 0$. Because the marked basis fixes the orientation and Gram form of the abstract lattice, the image of one nonzero vector determines the orientation-preserving realization ι relative to the starting channel frame through Lemma 16.4.

Definition 16.6 (Rooted signature-rigid spanning tree). Fix a quotient obstacle orbit r_* . A spanning tree rooted at r_* is *signature-rigid* if its edges can be ordered so that each edge joins an already reached obstacle orbit to one new orbit, and its contact on the already reached orbit has a curvature signature occurring at a unique framed point of that obstacle. For a one-vertex quotient graph the empty tree is allowed.

Corollary 16.7 (Fully intrinsic periodic table rigidity). *Under the hypotheses of Theorem 16.3, assume the measured network contains a lattice-anchoring cycle and, rooted at an obstacle orbit met by that cycle, a signature-rigid spanning tree. Then $\mathfrak{S}_{\text{per}}(\mathcal{D})$ is a singleton. Hence the intrinsic signed lifted-channel data determine the complete labelled periodic billiard table up to one simultaneous global element of $\text{SE}(2)$. No absolute contact point, absolute tangent frame, inter-channel placement, or lattice orientation relative to a channel frame is supplied in advance.*

Proof. Use the global Euclidean gauge to place the starting frame of the anchoring cycle. The cycle places at least one recovered obstacle lift and therefore a root obstacle image. Since $\eta_c \neq 0$, Lemma 16.4 fixes the orientation-preserving lattice realization ι relative to that frame. Traverse the rooted spanning tree. At each edge, the unique signature on the already placed obstacle fixes the edge-frame transform, and the known deck correction $\iota(\ell_e)$ fixes the new quotient obstacle image. Induction places all obstacle orbits. Thus there is at most one algebraic gluing modulo

the initial global gauge. Since the hypotheses concern data realized by a dispersing table, that unique gluing is admissible. Hence the admissible gluing space is a singleton. $\square$

Corollary 16.8 (One-root registration is sufficient once the lattice is known). *Suppose an oriented Euclidean realization of the periodic lattice and the Euclidean placement of one root channel frame relative to that lattice are supplied as ambient calibration. Then a rooted signature-rigid spanning tree determines every obstacle image and every remaining measured channel frame. Thus one root registration suffices; a common laboratory registration of all channels is not required.*

Remark 16.9 (Symmetry, connectivity, holonomy and admissibility). The sufficient conditions above are not necessary. Finite obstacle symmetries produce finitely many possible incidence matches, and several cycles can eliminate those choices. The exact object controlling uniqueness is the admissible periodic gluing space $\mathfrak{G}_{\text{per}}(\mathcal{D})$. It separates four genuine issues: obstacle symmetry, channel-network connectivity, periodic lattice holonomy and global geometric admissibility.

Corollary 16.10 (Common laboratory registration as a special case). *If the Euclidean placement of the lattice and of every measured channel frame is supplied externally, then the periodic gluing is fixed a priori. A family of measured channels meeting every obstacle orbit therefore determines the whole analytic table in that registration, recovering the registered v22 statement as a special case.*

16.4. Transporting the root to an anchoring cycle. The root of a signature-rigid spanning tree need not be prescribed at the starting frame of an anchoring cycle. The next lemma shows that these two choices are compatible under the existing signature hypotheses. Thus the propagation of obstacle placements requires no additional registration or root-incidence assumption.

Lemma 16.11 (Rerooting a signature-rigid spanning tree). *Let a rooted signature-rigid spanning tree join a finite collection of connected real-analytic strictly convex obstacle boundaries. Suppose one vertex r has a framed boundary point at which the complete oriented curvature signature occurs uniquely. Then the same undirected tree, rooted instead at r , is signature-rigid. In particular, an obstacle orbit met by a signature-rigid anchoring cycle may be chosen as the root.*

Proof. First, uniqueness of the complete signature at one framed point of a boundary C implies

$$\text{Sym}^+(C) = \{\text{id}\}.$$

Indeed, every symmetry carries that framed point to one with the same signature, by Lemma 16.2. Uniqueness makes it fix the point and its oriented unit tangent. An orientation-preserving Euclidean motion fixing such a frame is the identity. The same lemma then shows that the complete signature occurs uniquely at every framed point of C . Thus uniqueness is a property of this boundary, rather than of a specially chosen incidence on it.

Let r_0 be the original root. Every vertex which has a child in the original rooted tree has a unique-signature contact on an outgoing edge, by Definition 16.6. The preceding argument makes all its framed boundary signatures unique. Write the unique tree path from r_0 to r as

$$r_0 = v_0, v_1, \dots, v_k = r.$$

Rerooting reverses precisely the edges of this path. On the reversed edge from v_i to v_{i-1} , with $1 \leq i < k$, the new parent v_i was the old parent of v_{i+1} and therefore has unique signatures at every framed point. For $i = k$, the same conclusion follows from the hypothesis on r . Every reversed edge consequently has a unique-signature contact on its new parent. Off the path, the parent-child relations are unchanged and retain their original property. This proves the assertion; when $k = 0$, including the one-vertex empty tree, nothing is changed.

Every required incidence transition on a signature-rigid cycle has a unique framed signature by its definition. A vertex met by that cycle therefore satisfies the hypothesis on r , proving the last assertion. $\square$

For the rank-two recovery of Theorem 17.4, fix the common starting channel frame of the anchoring pair and choose a root obstacle met there. Lemma 16.11 transports any supplied rooted signature-rigid spanning tree to that obstacle. The two cycle holonomies determine the lattice in this frame, and induction along the rerooted tree then places every obstacle orbit. The same argument applies to the single-offset reconstruction in Theorem 14.4. The tree is only reoriented; no edge, obstacle label, deck displacement, or observation is added or removed.

16.5. The common-orientation quotient of the marked datum. There are two different operations on signed observations: changing one orientation convention for the entire experiment, and discarding signs from individual observations. Only the former is involved in passing from oriented to unoriented Euclidean classification. We define it at the level of the complete law-valued datum.

Let $J(x, y) = (x, -y)$ and $r(u, v) = (-u, -v)$. For a signed channel-law collection $\mathcal{D}$, write $\mathcal{R}_0\mathcal{D}$ for the collection obtained by replacing, for every measured edge, both contact types and every retained offset,

$$\Pi_{bb,d}^{e,\text{end},\infty} \quad \text{by} \quad r_*\Pi_{bb,d}^{e,\text{end},\infty}. \quad (16.6)$$

The same involution is used throughout the collection. The obstacle labels, the directed edge labels, the marked deck elements and the onset gaps are unchanged. In particular this operation does not reverse the directed channel (a, b, ℓ) . For a density it reads $f_{e,b,d}^r(u, v) = f_{e,b,d}(-u, -v)$. When residual time is present its coordinate is left fixed, and when a full transcript is present its designs, counts, failure symbols and stopping rule are transported together with the same change of endpoint coordinates.

The orientation of a marked lattice requires some care. In the preceding signed formulation ι is required to preserve the orientation of the marked basis. A reflection of the physical table does not preserve this sector. To describe the common reflection action, let $\epsilon \in \{+1, -1\}$ denote the orientation character of ι . Write $\mathcal{D}_\epsilon = (\epsilon, \mathcal{D})$, and let $\mathfrak{G}_{\text{per}}^\epsilon(\mathcal{D})$ be the gluing space defined by (16.2)–(16.3), except that the isometry ι has orientation character ϵ . The placements A_e are still proper Euclidean motions of the consistently oriented channel frames. The original signed statement is the $\epsilon = +1$ sector. The proof of its classification applies in either sector: it uses the incidence equations and Euclidean isometry of ι , not the sign of its determinant.

The character ϵ records the common orientation convention relative to the marked lattice; it is not a supplied lattice angle or a laboratory registration. In an unoriented representative it must be transported with the transverse convention, rather than erased independently.

Definition 16.12 (Common-orientation datum). On the complete marked signed data define

$$\mathcal{R}(\epsilon, \mathcal{D}) = (-\epsilon, \mathcal{R}_0\mathcal{D}), \quad \mathcal{R}^2 = \text{id}. \quad (16.7)$$

The common-orientation datum is the two-element orbit

$$\overline{\mathcal{D}} = \{(\epsilon, \mathcal{D}), (-\epsilon, \mathcal{R}_0\mathcal{D})\}.$$

Thus it retains all relative sign choices, including their relation to the marked lattice convention, but not a choice of the one common orientation. It is an orbit of the entire collection of probability laws, not a mixture of those laws and not a pushforward under pointwise absolute values.

Lemma 16.13 (Equivariance of recovery and gluing). *Reflection induces a well-defined involution between the two signed gluing spaces belonging to $\overline{\mathcal{D}}$. On representatives it is*

$$(\iota, (A_e), (C_{e,\sigma})) \longmapsto (J\iota, (JA_eJ), (JC_{e,\sigma})). \quad (16.8)$$

Reflected boundary images are given their canonical positive boundary orientation. The recovered quotient obstacle images are JC_r , and all labelled deck translations are transported by the same J .

Proof. A reflected contact graph is $\psi_b^r(y) = \psi_b(-y)$. Reflection preserves flight lengths, the reflection law, arclength and the collision flux measure. Sending every endpoint coordinate to its negative therefore gives (16.6); the gap and residual-time coordinate do not change. Consequently the signed inverse and analytic globalization recover exactly the reflected boundary images. Equivalently the half-line actions and their graph jets satisfy

$$S_b^r(u) = S_b(-u), \quad (q_{b,n})^r = (-1)^n q_{b,n}. \quad (16.9)$$

These identities also follow directly by reflecting each finite stationary bridge and then taking the half-line limit. They do not require convergence of an infinite Taylor series for a smooth graph.

Since $\det J = -1$, the realization $J\iota$ lies in the opposite lattice sector while $JA_e J \in \text{SE}(2)$. The identity $J\tau_v = \tau_{Jv}J$ gives, incidence by incidence,

$$\tau_{-J(\nu)}(JA_e J)(JC_{e,\sigma}) = J(\tau_{-\iota(\nu)}A_e C_{e,\sigma}). \quad (16.10)$$

Thus every gluing equation is preserved. Reflection also preserves disjointness, strict convexity, facing contacts, flight lengths and the selected channel geometry; hence it preserves admissibility. Applying the construction twice gives the original gluing.

Finally a different proper representative $B \circ A_e$, with lattice realization $R_B \iota$, changes the reflected representative by the single proper motion $JB J$. Formula (16.8) is therefore well defined after the proper Euclidean quotient. $\square$

Theorem 16.14 (Unoriented marked periodic classification). *Under the analytic and measured-network hypotheses of Theorem 16.3, let $\overline{\mathcal{D}}$ be the common-orientation datum of Definition 16.12. Define*

$$\overline{\mathfrak{G}}_{\text{per}}(\overline{\mathcal{D}}) = \left(\mathfrak{G}_{\text{per}}^\epsilon(\mathcal{D}) \sqcup \mathfrak{G}_{\text{per}}^{-\epsilon}(\mathcal{R}_0 \mathcal{D}) \right) / \langle \mathcal{R} \rangle. \quad (16.11)$$

The labelled periodic tables realizing this datum, modulo one common element of $E(2)$ acting on the lattice and every obstacle image, are in bijection with $\overline{\mathfrak{G}}_{\text{per}}(\overline{\mathcal{D}})$. In particular, a singleton signed gluing space gives a singleton unoriented gluing space and determines the table up to one common Euclidean isometry. The conclusion applies to a single positive offset at each contact type whenever the signed reconstruction is made using the single-offset inverse.

Proof. Choose either representative of $\overline{\mathcal{D}}$. The signed classification, in its indicated lattice sector, identifies proper-motion classes of realizing tables with the corresponding signed gluing space. Lemma 16.13 identifies the classifications for the two representatives by the same reflection of the entire table.

More explicitly, if two realizing tables are related by a proper Euclidean motion, the signed classification identifies their signed gluing classes. If they are related by an improper motion, write that motion as BJ with $B \in \text{SE}(2)$. Reflection first interchanges the two signed data representatives by $\mathcal{R}$, and B then identifies their proper classes. Conversely, equality in (16.11) is represented either by equality in one signed gluing space or by equality after its common reflected identification. The inverse signed classification yields respectively a proper or an improper global Euclidean motion of the realizing tables. This proves both directions of the bijection. Reflection maps the two signed gluing spaces bijectively, so the singleton assertion follows. The argument uses only the signed inverse, not the number of offsets used to recover its actions. $\square$

Corollary 16.15 (Holonomy and fixed-order stability under the quotient). *The signature-rigid gluing criteria are invariant under the common reflection. If two independent marked cycle*

columns form M , their recovered holonomy columns form V , and $L = VM^{-1}$, then

$$V^r = JV, \quad L^r = JL, \quad (L^r)^\top L^r = L^\top L. \quad (16.12)$$

Thus the common-orientation quotient retains the recovered Gram form and scale. Any fixed-order stability estimate for the equivariant signed inverse on its exact law image, valid for the representative pairs used in reflection-invariant orbit metrics, descends with the same constant to those metrics. On arbitrary perturbed densities the corresponding action, amplitude and finite-jet assertion uses the transported-anchor extension of Proposition 14.10 and Corollary 14.11. It does not use a fixed signed anchor in both orientation sectors.

Proof. Canonical positive arclength reverses under reflection. Hence the curvature signature transforms as $(\kappa, \kappa', \kappa'', \dots) \mapsto (\kappa, -\kappa', \kappa'', \dots)$. Equality and uniqueness of signature matches are preserved. Every incidence congruence is conjugated by J , as is its product around a cycle. Since $J\tau_v J = \tau_{Jv}$, the holonomy vector becomes Jv ; marked cycle columns are unchanged. Formula (16.12) follows from $J^\top J = I$. A rooted rigid spanning tree is carried to the same rooted rigid combinatorial tree with all its matches reflected.

For the stability assertion on exact laws, let F be the equivariant signed inverse between the relevant fixed-order data and reconstruction spaces, with reflection-invariant metrics d and d' , and suppose $d'(F(D), F(E)) \leq C_M d(D, E)$ for every representative pair entering the orbit comparison. If the estimate is formulated sectorwise, use only same-sector representative comparisons; their reflected estimates have the same constant. Define the orbit distances by

$$\bar{d}([D], [E]) = \min_{\eta \in \{0,1\}} d(D, \mathcal{R}^\eta E), \quad \bar{d}'([X], [Y]) = \min_{\eta \in \{0,1\}} d'(X, \mathcal{R}^\eta Y).$$

For a sectorwise metric the minima are restricted to the nonempty set of same-sector comparisons. Choose a minimizing η and use equivariance of F . Then

$$\bar{d}'([F(D)], [F(E)]) \leq d'(F(D), F(\mathcal{R}^\eta E)) \leq C_M \bar{d}([D], [E]).$$

For contact jets and densities on symmetric compact boxes, reflection invariance follows from (16.9) and the usual fixed-order supremum norms. This statement retains the order-dependent constants and hypotheses of the signed estimates; it asserts no order-independent stability for analytic continuation. For perturbed densities the same argument applies to the explicitly equivariant extension in (14.28), on its stated domain. The fixed-anchor formula alone has no such off-model equivariance; see Remark 14.12. No arbitrary perturbed density is thereby declared globally realizable. $\square$

Remark 16.16 (Law-valued orientation orbits are not folded records). For a continuously distributed endpoint pair, the pushforward under $(u, v) \mapsto (|u|, |v|)$ has density, on the open positive quadrant,

$$f_{\text{abs}}(a, b) = f(a, b) + f(-a, b) + f(a, -b) + f(-a, -b). \quad (16.13)$$

The coordinate axes have zero mass in this setting. This sum is not the law-valued orbit $\{f, f \circ r\}$. Even pushing each observed pair to its orbit under simultaneous reversal is a sample-space coarsening, not an orbit of the complete probability law. Randomly symmetrizing a law is a further operation, and independently reversing different channel collections gives an action of $\{\pm 1\}^E$, rather than the single diagonal group used above.

For example, on $[-1, 1]^2$, the positive normalized symmetric densities

$$f_1(u, v) = \frac{1}{4}(1 + \alpha(u + v)), \quad f_2(u, v) = \frac{1}{4}(1 + 2\alpha(u + v)), \quad 0 < \alpha < \frac{1}{4},$$

have the same folded density, identically one on $(0, 1)^2$, but their law-valued common-reversal orbits are different: the coefficient of $u + v$ can only change sign under r , not double. This example distinguishes the observation maps in a class of densities. It is not a claim that these two densities

are realized by physical billiard tables, nor a physical nonidentifiability theorem for folded billiard observations. All rigidity statements above use the specified law-valued common-orientation datum.

17. RANK-TWO HOLONOMY AND RECOVERY OF THE PERIODIC LATTICE

The preceding intrinsic reconstruction theorem was stated relative to a marked abstract lattice carrying a prescribed Euclidean Gram form. The holonomy identity in fact contains more information. With two independent deck cycles based in one channel frame, the Euclidean realization of the marked lattice, including its metric and absolute scale, is determined by the channel data. We record that stronger form here and use it in the global statistical theorem below.

Let $\Lambda_{\text{top}} \simeq \mathbb{Z}^2$ be only an oriented marked free abelian group of rank two. No inner product on $\Lambda_{\text{top}} \otimes \mathbb{R}$ is supplied. A periodic Euclidean realization is an orientation-preserving linear isomorphism

$$L : \Lambda_{\text{top}} \otimes \mathbb{R} \longrightarrow \mathbb{R}^2.$$

It induces the Gram form

$$G_L(\eta, \zeta) = \langle L\eta, L\zeta \rangle_{\mathbb{R}^2}. \quad (17.1)$$

The lifted channel label $e = (a, b, \ell)$, with $\ell \in \Lambda_{\text{top}}$, remains part of the marked acquisition protocol. Thus the deck class is known combinatorially, but neither its Euclidean length nor its direction is supplied.

An algebraic gluing is now defined exactly as in Section 16, except that the unknown lattice variable is $L \in \text{GL}^+(2, \mathbb{R})$, rather than an isometry from a pre-metrized lattice. The incidence relation is

$$\tau_{-L\nu(e, \sigma)} A_e C_{e, \sigma} = \tau_{-L\nu(f, \sigma')} A_f C_{f, \sigma'} \quad (17.2)$$

whenever the two incidences carry the same quotient-obstacle label. Admissibility again means that the periodic translates form a dispersing billiard and that the declared measured edges are the prescribed facing channels. Simultaneous Euclidean motion acts on the whole realization in the usual way.

Definition 17.1 (Rank-two anchoring pair). Two signature-rigid closed channel cycles c_1, c_2 , based at the same oriented channel frame, form a *rank-two anchoring pair* if their marked deck displacements

$$\eta_i = \sum_{e \in c_i} \epsilon_e \ell_e \in \Lambda_{\text{top}}, \quad i = 1, 2,$$

are linearly independent in $\Lambda_{\text{top}} \otimes \mathbb{R}$.

The incidence congruences are determined entirely by the recovered analytic curve copies. Hence, as in Lemma 16.4, their composition around c_i is a data-determined Euclidean motion $H_i = \tau_{v_i}$. The next proposition is the metric-free version of the cycle-holonomy identity.

Proposition 17.2 (Metric-free cycle holonomy). *For realizable intrinsic data and every signature-rigid cycle c based at an edge frame whose placement has rotational part R_0 ,*

$$R_0 v_c = L \eta_c. \quad (17.3)$$

In particular, no Euclidean structure on the abstract deck group is used in deriving the identity.

Proof. The proof of Lemma 16.4 uses only the deck-addition law and Euclidean translations of the realized table. Replacing $\iota(\nu)$ there by $L\nu$ leaves the telescoping argument unchanged:

$$A_{e_0} H_c A_{e_0}^{-1} = \tau_{L\eta_c}.$$

The rotational part of H_c is therefore the identity, and comparison of translation vectors gives (17.3). $\square$

Theorem 17.3 (Recovery of lattice shape, scale and Gram form). *Assume the intrinsic signed channel data contain a rank-two anchoring pair c_1, c_2 . Then the data determine the full Euclidean realization of the marked lattice relative to the common starting channel frame, up to the one global rotation already contained in the global SE(2) gauge. More precisely, after fixing that gauge so that the starting-frame rotation is the identity, there is a unique linear map*

$$L : \Lambda_{\text{top}} \otimes \mathbb{R} \longrightarrow \mathbb{R}^2, \quad L\eta_i = v_i \quad (i = 1, 2). \quad (17.4)$$

Consequently the full marked Gram matrix, including absolute scale, is the data-determined matrix

$$[G_L]_{\mathcal{B}} = M_{\mathcal{B}}^{-T} V^T V M_{\mathcal{B}}^{-1}, \quad (17.5)$$

where $M_{\mathcal{B}}$ has as columns the coordinates of η_1, η_2 in the marked basis $\mathcal{B}$, and $V = (v_1 \ v_2)$.

Proof. The vectors η_1, η_2 form a basis of $\Lambda_{\text{top}} \otimes \mathbb{R}$. Equation (17.3) therefore prescribes a unique linear map L once the common starting-frame rotation has been fixed. Realizability implies that L is invertible and orientation preserving, so v_1, v_2 are linearly independent with the compatible orientation. Formula (17.5) is simply $G_L = L^T L$ written in the marked basis. A change of the global Euclidean gauge left-multiplies L by a rotation and hence leaves G_L unchanged. $\square$

Theorem 17.4 (Uncalibrated intrinsic periodic rigidity). *Let two labelled periodic dispersing billiard tables have the same oriented marked deck group $\Lambda_{\text{top}} \simeq \mathbb{Z}^2$, the same obstacle-orbit labels and the same connected measured lifted-channel graph. No lattice Gram form, lattice scale, lattice direction, common channel registration or absolute contact frame is supplied. Assume every obstacle boundary is connected, real analytic and strictly convex, the intrinsic signed channel data agree, the measured graph contains a rank-two anchoring pair, and a rooted signature-rigid spanning tree reaches every obstacle orbit. Then the complete marked periodic table and the Euclidean Gram form of its lattice are determined by the observations up to one simultaneous global element of SE(2).*

Proof. The signed one-channel inverse and analytic continuation recover every incident obstacle curve in its own channel frame. Fix the common starting frame of the rank-two anchoring pair, and choose an incident obstacle orbit r . A signature-rigid incidence on an anchoring cycle gives a framed point of this obstacle with a unique complete oriented curvature signature. Lemma 16.11 therefore roots the supplied spanning tree at r , without changing an edge, a deck label or the observation map. The recovered image at this root is already placed in the chosen anchoring frame.

Theorem 17.3 fixes the complete lattice realization in this same frame. The rerooted signature-rigid spanning tree then propagates the placements of all quotient obstacle images exactly as in Corollary 16.7, with every deck correction interpreted through that one lattice realization. Thus there is at most one algebraic gluing. Existence is not hidden in the word “admissible”: it is provided by the standing hypothesis that the data are generated by an actual periodic dispersing table. Hence that algebraic gluing is admissible and is the unique realization modulo the initial global Euclidean gauge. $\square$

Remark 17.5 (What is marked and what is recovered). The deck label ℓ is a combinatorial lift label in the marked covering and is part of the acquisition design; its Euclidean vector is not. Unknown or corrupted deck-class labels constitute a different inverse problem and are not silently inferred here. The classification of all admissible gluings is a bookkeeping theorem; the rigidity content is the rank-two holonomy recovery plus signature-rigid propagation. Theorem 17.4 therefore removes the metric calibration singled out in the preceding formulation while keeping the measurement protocol explicit.

18. FINITE ANALYTIC SIGNATURES AND STABILITY OF INTRINSIC GLUING

Exact analytic uniqueness, finite-signature matching, and a compact inverse modulus are three different assertions. We prove them in that order. An outside-arc gap alone does not prove uniqueness inside the correct arc. The local assertion will instead follow from a finite jet embedding, whose order is selected uniformly from the complete analytic signatures.

Fix the marked measured graph, two signature-rigid anchoring cycles with independent marked deck displacements, and a rooted signature-rigid spanning tree. Let $\mathfrak{K}$ be a compact class of their realizations, modulo one simultaneous global element of $\mathrm{SE}(2)$. The designation *signature-rigid* has its meaning in Section 16: each transition used by these cycles and the tree has a unique matching oriented framed point, not merely a unique point after arbitrarily choosing a symmetry phase. Obstacles not used for such transitions may have exact symmetries; the classification theorem for general gluing spaces remains unchanged.

Assume uniform positive curvature, gap, separation, diameter, free-area and persistent-channel collar bounds. For some $\rho, B > 0$, the oriented arclength curvature functions extend holomorphically to strips of width ρ , with absolute value at most B . The class is closed in a fixed root-frame gauge and compact in every fixed labelled C^q topology. A channel margin $\chi > 0$ excludes switching to a competing closest pair. The data-determined holonomy vectors satisfy

$$\sigma_{\min}(v_1(T) \ v_2(T)) \geq \sigma_{\mathrm{hol}} > 0. \quad (18.1)$$

The compactness assumption includes the lattice and marked placements; it is not inferred from a bound on the curvature of a single obstacle alone. These are the uniform geometric assumptions used below.

A transition obstacle has trivial orientation-preserving symmetry group. Indeed any such nonidentity symmetry carries each boundary point to a different point with the same complete oriented signature, contradicting the designated unique framed match. A nonidentity rigid motion preserving a compact strictly convex body is a rotation about an interior point and has no boundary fixed point. In particular, transition obstacles are not circles. This observation uses the exact signature-rigidity assumption already required for unique global gluing; it is not a new reflection or genericity assumption.

18.1. Uniform finite embeddings. Parametrize each transition obstacle by normalized oriented arclength $x \in \mathbb{R}/\mathbb{Z}$, with its origin at a designated incidence. The physical arclength derivative is denoted by ∂_s . Define

$$J_{T,i}^{[m]}(x) = (\kappa_{T,i}(x), \partial_s \kappa_{T,i}(x), \dots, \partial_s^m \kappa_{T,i}(x)).$$

All derivatives with respect to x below are of this vector-valued map. Uniform upper and lower perimeter bounds make normalized and physical arclength distances equivalent. Fixed positive weights could be placed on the coordinates without changing any assertion.

Theorem 18.1 (Uniformly selected finite signature embedding). *There are an integer m , numbers $v, r, \sigma > 0$, and $U, B_2 < \infty$, uniform on $\mathfrak{K}$ and its finite list of transition obstacles, such that, writing $J = J_{T,i}^{[m]}$,*

$$|J'(x)| \geq v, \quad |J'(x)| \leq U, \quad |J''(x)| \leq B_2, \quad (18.2)$$

$$|J(x) - J(y)| \geq (v/2) \operatorname{dist}(x, y), \quad \operatorname{dist}(x, y) \leq r, \quad (18.3)$$

$$|J(x) - J(y)| \geq \sigma, \quad \operatorname{dist}(x, y) \geq r. \quad (18.4)$$

Thus a sufficiently rich finite signature, not necessarily an initially chosen scalar curvature signature, is a uniform embedding of every transition boundary.

Proof. If every derivative of positive order of an analytic curvature function vanished at one point, it would be constant by the identity theorem. Since transition obstacles are not circles, at each (T, i, x) some component of the derivative of a finite signature is nonzero. The parameter space of these triples is compact: use normalized arclength and the finite obstacle list. Each nonvanishing condition persists on a neighborhood. A finite subcover gives an order m_1 for which $|(J^{[m_1]})'|$ is bounded below by a positive constant v . Holomorphic strip bounds give a uniform second derivative bound for this finite vector. Taylor's integral formula then gives

$$|J^{[m_1]}(y) - J^{[m_1]}(x)| \geq v|y - x| - \frac{1}{2}\|(J^{[m_1]})''\|_\infty|y - x|^2 \geq (v/2)|y - x|$$

on a uniform short arc. Choose its radius $r < 1/4$.

For pairs at normalized arclength distance at least r , equality of all signature coordinates would, by Lemma 16.2, give a nonidentity orientation-preserving symmetry. No such symmetry exists on a transition obstacle. Each pair is therefore separated by some finite coordinate. The set of (T, i, x, y) with distance at least r is compact. A second finite subcover supplies an order $m \geq m_1$ and a positive minimum separation σ . Adding coordinates preserves the local lower bound already proved for the m_1 -block. The strip bounds supply finite uniform U, B_2 for the enlarged vector. This proves (18.2)–(18.4). $\square$

The argument also separates a prescribed target arc from its complement: shrink the radius below r , using (18.3) on the intermediate annulus and (18.4) outside. Therefore an existing outside-arc margin may be retained, but it is not used to infer (18.3). The latter is an independently proved consequence of analytic nonconstancy and compactness.

Lemma 18.2 (Unique matching for noisy finite signatures). *There are $\epsilon_0, C > 0$, uniform on the class, with the following property. Fix a true target $J(x_0)$. Suppose a recovered signature curve $\tilde{J}$, expressed in the same normalized arclength parameter, and a target vector y satisfy*

$$\|\tilde{J} - J\|_{C^2} \leq \epsilon, \quad |y - J(x_0)| \leq \epsilon, \quad 0 < \epsilon < \epsilon_0.$$

Then the minimum of $|\tilde{J}(x) - y|^2$ on the parameter circle is attained at one point $\hat{x}$, and

$$\text{dist}(\hat{x}, x_0) \leq C\epsilon. \tag{18.5}$$

Locally this match is Lipschitz in the target vector and in the recovered curve in C^1 . A mere C^0 perturbation of an arbitrary signature curve is not asserted to have a unique nearest point.

Proof. Choose a radius $a < r/4$ so small that $2B_2Ua < v^2/16$, decreasing it if necessary to handle bounded perturbations of the derivatives. Theorem 18.1 gives a positive uniform separation between $J(x_0)$ and every $J(x)$ outside this arc, namely at least $\min(va/2, \sigma)$. For sufficiently small ϵ , no minimizer can lie outside the arc: the value of the residual norm at x_0 is at most 2ϵ , while outside it is larger.

On the arc put $f(x) = \frac{1}{2}|\tilde{J}(x) - y|^2$. Its second derivative is

$$f''(x) = |\tilde{J}'(x)|^2 + (\tilde{J}(x) - y) \cdot \tilde{J}''(x).$$

The derivative lower bound, the choice of a , and then a sufficiently small ϵ_0 give $f'' \geq v^2/8$ throughout the arc. Hence the minimum is unique and interior. At the minimum the residual norm is at most 2ϵ , so

$$|J(\hat{x}) - J(x_0)| \leq 4\epsilon.$$

The local embedding bound proves (18.5), for example with $C = 8/v$. Finally compare the equations $f'(\hat{x}) = 0$ for two nearby data sets. The derivative of f' is at least $v^2/8$, and its change under a perturbation of y and $\tilde{J}$ is bounded by a constant times $|\Delta y| + \|\Delta \tilde{J}\|_{C^1}$. The mean value theorem gives the stated local Lipschitz dependence. $\square$

For example, a scalar curvature with a strict minimum can take each nearby larger value twice inside the correct arc. Adding a finite number of its oriented derivatives as above resolves this failure; declaring the original scalar match unique would not. Exact finite symmetries can instead be handled on a fixed symmetry quotient for a classification problem, but they must not be assigned an arbitrary phase and called a unique Euclidean gluing. The unique-reconstruction theorem uses the signature-rigid transitions specified at the start of the section.

18.2. Propagation through the marked graph.

Lemma 18.3 (Persistence of the prescribed gluing). *On $\mathfrak{R}$, the prescribed incidence matches are unique and continuous. More precisely, if their recovered finite signature curves and targets satisfy the hypotheses of Lemma 18.2, and the recovered point–tangent frames are close in C^1 , then the incidence congruences and their finite graph compositions depend locally Lipschitz-continuously on these quantities. The lattice map obeys*

$$L = VM^{-1}, \quad \|\tilde{L} - L\| \leq \|\tilde{V} - V\| \|M^{-1}\|, \quad (18.6)$$

where the fixed invertible matrix M has the two marked deck vectors as columns and $V = (v_1 \ v_2)$. Rank two persists for $\|\tilde{V} - V\| < \sigma_{\text{hol}}/2$.

Proof. The matching lemma supplies the unique in-arc point; its global separation excludes other arcs. Point and oriented tangent determine the required orientation-preserving rigid motion between the two recovered curve frames. The elementary formula for that motion is smooth in the point and unit tangent data on a fixed orientation chart. Products of a fixed finite number of such motions are locally Lipschitz on bounded point data. Consequently the tree placements and the two cycle holonomies inherit a uniform local bound, with a constant depending on the finite graph. The channel margin prevents a different closest pair from replacing the selected one. Formula (18.6) is direct matrix multiplication by the fixed M^{-1} , not inversion of a noisy matrix V . The singular-value perturbation inequality proves rank persistence. Also

$$\|\tilde{L}^t \tilde{L} - L^t L\| \leq 2\|L\| \|\tilde{L} - L\| + \|\tilde{L} - L\|^2,$$

so the Gram form and subsequent deck corrections are stable as well. $\square$

This lemma concerns already recovered signature curves in the stated norms. It does not turn noisy contact Taylor coefficients into a well-conditioned unrestricted analytic continuation algorithm. The next argument supplies the distinct, non-effective continuity route needed for global statistical consistency.

18.3. The compact inverse modulus. Normalize the action coordinates by

$$x_{e,b,m}(T) = \rho^m S_{e,b}^{(m)}(0; T)/m!, \quad m \geq 2,$$

using any fixed positive scale ρ , and the gaps by g_* . For a fixed enumeration $|e|$ of the finite channel set define

$$\begin{aligned} d_{\mathcal{D}}(T, T') &= \sum_{e,b} \sum_{m \geq 2} 2^{-(m+|e|+3)} \min\{1, |x_{e,b,m}(T) - x_{e,b,m}(T')|\} \\ &\quad + \sum_e 2^{-(|e|+2)} \min\{1, |g_e(T) - g_e(T')|/g_*\}. \end{aligned} \quad (18.7)$$

The truncation by one and the summable weights make this metric compatible with coordinatewise convergence of the intrinsic data. This conclusion requires no assertion that the action holomorphic radius equals the curvature holomorphic radius.

Theorem 18.4 (Uniform inverse modulus). *For every fixed $q \geq 0$,*

$$\omega_q(t) = \sup\{d_q(T, T') : T, T' \in \mathfrak{K}, d_{\mathcal{D}}(T, T') \leq t\} \quad (18.8)$$

is finite, nondecreasing, and tends to zero as $t \downarrow 0$. If $\mathcal{D}_M$ consists of the gaps and action jets through order M , then

$$d_q(T, T') \leq \omega_q(C\|\mathcal{D}_M(T) - \mathcal{D}_M(T')\|_M + C2^{-M}). \quad (18.9)$$

The finite norm is the maximum norm of the normalized retained coordinates; C depends only on the finite graph and the chosen weights.

Proof. Equality of the complete action data and gaps gives the full analytic contact germs by Theorem 13.1. Analytic continuation, exact signature-rigid gluing, and Theorem 17.3 determine the marked periodic table in the root-frame gauge. This exact injectivity uses the complete signatures; it does not depend on a possibly noisy finite matching rule. The complete data map is continuous coordinate by coordinate, by the half-line inverse estimates, and hence continuous in (18.7). A continuous injection from compact $\mathfrak{K}$ into this metric data space has a uniformly continuous inverse on its image. This proves the assertion about ω_q . The terms with $m > M$ in the data metric sum to at most $C2^{-M}$; the retained terms are bounded by the normalized finite maximum norm times C . Substitution gives (18.9). $\square$

Remark 18.5 (Two distinct stability statements). The local finite-signature estimates are proved by immersion, separation and strict convexity of the matching criterion. The global modulus is proved by exact injectivity and compactness. Neither proof substitutes for the other. The tail $C2^{-M}$ is a product-metric truncation bound, not an exponential analytic-continuation error rate. No effective formula for ω_q or a uniform-in-order jet condition number is asserted.

19. A NONSYMMETRIC PERIODIC REALIZATION WITH FOUR MEASURED CHANNELS

We give a family in which the contact geometry, the two deck cycles and intrinsic registration can all be evaluated in one construction. The parameters used to construct the table are not supplied to its observer. In particular, the coordinates in the following construction verify realizability; they are not additional observations. The example uses selected channels, not a decomposition of the whole collision-count event.

Put $n(\theta) = (\cos \theta, \sin \theta)$ and $t(\theta) = (-\sin \theta, \cos \theta)$. If $h + h'' > 0$, the curve

$$x_h(\theta) = h(\theta)n(\theta) + h'(\theta)t(\theta) \quad (19.1)$$

has derivative $(h + h'')t$, outward normal n , support function h , and curvature $(h + h'')^{-1}$. To see the support assertion directly, for fixed θ_0 the derivative of $x_h(\theta) \cdot n(\theta_0)$ is $(h + h'')(\theta) \sin(\theta_0 - \theta)$. Its unique maximum occurs at $\theta = \theta_0$. Thus (19.1) is the boundary of a strictly convex body, denoted by K_h .

Consider the two support functions

$$\begin{aligned} h_1(\theta) &= 1 + \epsilon \left(\cos 2\theta + \frac{1}{4} \cos(3\theta + \pi/4) \right), \\ h_2(\theta) &= \frac{3}{4} + \delta \left(\cos 2\theta + \frac{1}{4} \cos(3\theta + \pi/6) \right), \quad \epsilon, \delta \in [1/200, 1/100]. \end{aligned} \quad (19.2)$$

Let T be a real 2×2 matrix with $\|T - I\| \leq 1/100$, and let $|w| \leq 1/100$. Define

$$L = T \begin{pmatrix} 8 & 0 \\ 0 & 10 \end{pmatrix}, \quad v = T(4, 5)^t + w, \quad D_1 = K_{h_1}, \quad D_2 = v + K_{h_2}. \quad (19.3)$$

The periodic obstacles are $D_a + Lm$, $a \in \{1, 2\}$, $m \in \mathbb{Z}^2$. The marked basis of the abstract deck group is denoted by $\mathbf{e}_1, \mathbf{e}_2$; neither column of L is given as a datum.

Theorem 19.1 (A uniformly admissible nonsymmetric family). *Every member of (19.2)–(19.3) is a periodic dispersing table with connected analytic boundaries. Every two distinct obstacle closures have distance greater than $43/10$. The curvature ranges on the two obstacle types are disjoint:*

$$\frac{20}{21} \leq \kappa_1 \leq \frac{20}{19}, \quad \frac{5}{4} \leq \kappa_2 \leq \frac{10}{7}. \quad (19.4)$$

Neither obstacle has a nonidentity Euclidean symmetry. Modulo its period translations, the complete table has no Euclidean symmetry either.

The four labelled channels

$$e_{ij} = (1, 2, -ie_1 - je_2), \quad i, j \in \{0, 1\}, \quad (19.5)$$

are unobstructed closest-pair channels for their specified obstacle pairs. Every such closest segment has distance at least $383/200$ from every third obstacle. Their contacts and gaps depend real analytically on the parameters and have common contact and positive-offset collars. All their incidence transitions are signature-rigid. The cycles

$$c_1 = e_{00}e_{10}^{-1}, \quad c_2 = e_{00}e_{01}^{-1} \quad (19.6)$$

form a rank-two anchoring pair based at e_{00} ; the edge e_{00} is a signature-rigid spanning tree rooted at obstacle 1.

Proof. For $h = r + z(\cos 2\theta + \frac{1}{4}\cos(3\theta + \phi))$,

$$h + h'' = r - 3z \cos 2\theta - 2z \cos(3\theta + \phi).$$

The curvature radii lie in $[19/20, 21/20]$ and $[7/10, 4/5]$, respectively. This proves strict convexity and (19.4). The support bound gives $K_{h_a} \subset \overline{B}(0, R)$ with $R = 81/80$ for both types. At $T = I, w = 0$, the set of obstacle centers is

$$\{(8m, 10n) : m, n \in \mathbb{Z}\} \cup \{(4 + 8m, 5 + 10n) : m, n \in \mathbb{Z}\}.$$

Its minimum separation is $\sqrt{41}$. Since $|Tz| \geq (99/100)|z|$, all distinct perturbed centers have separation at least $(99/100)\sqrt{41} - 1/100$. Subtracting $2R$ gives

$$\text{dist}(D_a + Lm, D_b + Ln) \geq \frac{99}{100}\sqrt{41} - \frac{407}{200} > \frac{43}{10}$$

for distinct obstacles; the last strict inequality follows already from $\sqrt{41} > 32/5$. The smallest singular value of T is positive and $\det T > 0$, so the quotient is a genuine flat torus. Its free area is positive. More explicitly, integration of $\frac{1}{2}h(h + h'')$ gives

$$A = 80 \det T - \frac{25\pi}{16} + \frac{7\pi}{4}(\epsilon^2 + \delta^2) > 0. \quad (19.7)$$

We next exclude symmetry without imposing any contact registration. The first Fourier coefficients of both supports vanish. The vector

$$s(K) = \frac{1}{\pi} \int_0^{2\pi} h_K(\theta) n(\theta) d\theta \quad (19.8)$$

is covariant under translations and orthogonal transformations, so any symmetry of either body fixes the origin. A rotational symmetry through α would preserve its nonzero modes 2 and 3, and hence satisfy $e^{2i\alpha} = e^{3i\alpha} = 1$. Thus it is the identity. Reflection in a line at angle α would require $2\alpha \in \pi\mathbb{Z}$ and $3\alpha + \phi \in \pi\mathbb{Z}$. Neither $\phi = \pi/4$ nor $\phi = \pi/6$ permits this: substituting $\alpha = k\pi/2$ gives, respectively, $(6k + 1)/4$ and $(9k + 1)/6$ as nonintegral multiples of π . The two perimeters, 2π and $3\pi/2$, differ. A symmetry of the periodic table must therefore preserve obstacle types. Its orthogonal part would be a symmetry of K_{h_1} and is the identity; its translation belongs to $L\mathbb{Z}^2$.

For clearance, first consider the center segment from $(0, 0)$ to $(4, 5)$. Every third reference center has distance at least 4 from the rectangle $[0, 4] \times [0, 5]$: this follows directly by separating

the integer coordinates of the two displayed center sets. The other three rectangles are reflections of this one. After applying T , the distance of a third center from the corresponding center segment is at least $4(99/100)$. Moving the second-type centers by w changes a segment by at most $|w|$ in Hausdorff distance and a third center by at most $|w|$. Thus the lower bound becomes $4(99/100) - 2/100$. A segment joining points of the two radius- R bodies is contained in the radius- R neighborhood of the center segment, by convex interpolation of its endpoints. Subtracting a further R for the third body proves the claimed clearance:

$$4\frac{99}{100} - \frac{2}{100} - 2\frac{81}{80} = \frac{383}{200}. \quad (19.9)$$

This argument covers every lattice translate, not a finite numerical list.

For completeness, the contacts can be found from a scalar equation. Put

$$v_{ij} = T(4 - 8i, 5 - 10j)^t + w, \quad d_{ij} = |v_{ij}|, \quad \vartheta_{ij} = \arg v_{ij},$$

and maximize

$$F_{ij}(\theta) = v_{ij} \cdot n(\theta) - h_1(\theta) - h_2(\theta + \pi). \quad (19.10)$$

The variable d_{ij} here is a center distance, not the positive time offset d used elsewhere. The perturbation in $h_1 + h_2$ from $7/4$ is at most $1/40$. Comparison with $\theta = \vartheta_{ij}$ excludes $|\theta - \vartheta_{ij}| \geq \pi/3$ from the set of maxima. On the remaining arc,

$$F''_{ij} \leq -d_{ij}/2 + 1/8 < 0,$$

because $\|h''_1\| + \|h''_2\| \leq 1/8$. There is a unique maximum θ_{ij} and it depends analytically on the parameters. With

$$p_{ij} = x_{h_1}(\theta_{ij}), \quad q_{ij} = v_{ij} + x_{h_2}(\theta_{ij} + \pi), \quad g_{ij} = F_{ij}(\theta_{ij}), \quad (19.11)$$

the equation $F'_{ij} = 0$ gives $q_{ij} - p_{ij} = g_{ij}n(\theta_{ij})$. The two supporting half-planes show that all cross-body distances are at least g_{ij} , so these points realize the closest pair. The clearance bound excludes a third impact on this segment. Positive curvature, positive gap and the strict clearance margins give its alternating contact collar. Compactness of the parameter set and the finite number of channels give a common collar and, by the relative theorem, a common positive-offset collar. No finite-horizon assertion is used.

One can also choose gates without using an unknown contact point as an observation. Let $\vartheta_{ij}^0 = \arg(4 - 8i, 5 - 10j)$. The center perturbation is less than $3/40$ and $d_{ij} > 6.32$, so $|\vartheta_{ij} - \vartheta_{ij}^0| < 0.012$. The equation $F'_{ij} = 0$ and $\|h'_1\| + \|h'_2\| \leq 11/200$ give $|\theta_{ij} - \vartheta_{ij}| < 0.009$. Hence the first contact is within 0.062 of $n(\vartheta_{ij}^0)$ and the second within 0.131 of $(4 - 8i, 5 - 10j)^t - \frac{3}{4}n(\vartheta_{ij}^0)$. Fixed planar disks of radius $1/4$ about these reference points therefore contain all the appropriate contacts with a uniform interior margin. On each obstacle these gates are disjoint. Shrinking the contact collars inside them preserves the strict third-obstacle clearance. These are prescribed selected gates; their common reference coordinates in this proof are not passed to the intrinsic observer as an inter-channel registration.

Finally, Lemma 16.2 and absence of proper obstacle symmetries make every required incidence transition rigid. The two deck sums in (19.6) are e_1 and e_2 , and both cycles start with e_{00} . The stated tree reaches the only other obstacle orbit. This checks each part of the anchoring and spanning hypotheses in a single realization. $\square$

19.1. Recovering the channel frames from the recovered curves. For a complete convex image C , let

$$z_k(C) = \frac{1}{2\pi} \int_0^{2\pi} h_C(\theta) e^{-ik\theta} d\theta, \quad k \geq 2.$$

These coefficients do not change under translation. They are derived from the boundary image recovered by analytic continuation, not added to the endpoint-law observations. Identify the oriented plane with $\mathbb{C}$, so a unit complex number denotes a rotation.

Proposition 19.2 (Two-harmonic registration in the realized family). *Let $C_{e,1}$ and $C_{e,2}$ be the two complete boundary images recovered in the frame of channel e , whose origin is its first contact. Set $e_* = e_{00}$ and $c_{e,a} = s(C_{e,a})$. The Euclidean congruence placing the first-type copy of edge e onto that of e_* is*

$$U_e = \frac{z_3(C_{e,1})z_2(C_{e_*,1})}{z_3(C_{e_*,1})z_2(C_{e,1})}, \quad A_e(y) = U_e y + c_{e_*,1} - U_e c_{e,1}. \quad (19.12)$$

In exact data $|U_e| = 1$. If $b_e = A_e(c_{e,2})$, then the lattice columns and Gram matrix in this one common frame are

$$V = (b_{00} - b_{10} \quad b_{00} - b_{01}), \quad G = V^T V. \quad (19.13)$$

The redundant fourth channel satisfies

$$b_{00} - b_{11} = (b_{00} - b_{10}) + (b_{00} - b_{01}). \quad (19.14)$$

On this compact family, registration and the displayed lattice recovery are locally Lipschitz in the support functions of the recovered complete curve images, in the uniform norm. This statement does not assert a Lipschitz analytic-continuation map from contact jets or endpoint laws.

Proof. Write the actual edge placement as $y \mapsto R_e y + p_e$, with rotation angle θ_e . Its first-type image is $C_{e,1} = R_e^T(K_{h_1} - p_e)$. Therefore

$$z_k(C_{e,1}) = e^{ik\theta_e} z_k(K_{h_1}), \quad z_2(K_{h_1}) = \epsilon/2, \quad z_3(K_{h_1}) = (\epsilon/8)e^{i\pi/4}.$$

Taking the ratio for mode 3 and dividing by that for mode 2 yields $e^{i(\theta_e - \theta_{e_*})}$. This proves the rotation formula without selecting a branch of an argument. Translation covariance of (19.8) proves the formula for A_e . Indeed $c_{e,1} = R_e^T(-p_e)$, and hence

$$A_e(y) = R_{e_*}^T R_e y + R_{e_*}^T(p_e - p_{e_*}), \quad b_{ij} = R_{e_*}^T(v_{ij} - p_{e_*}).$$

Taking differences proves (19.13) and (19.14). These differences are precisely the translation holonomies of the two cycles, since their marked displacement matrix is I . The copy $C_{e_*,1}$ and the placed copy $C_{e_*,2}$, together with $V\mathbb{Z}^2$, recover every obstacle translate.

For stability, the maps $h \mapsto z_k$ and $h \mapsto s$ have uniform-norm operator bounds 1 and 2. On the stated family $|z_2| \geq 1/400$ and $|z_3| \geq 1/1600$. The quotients in (19.12) are therefore locally Lipschitz under small uniform perturbations; normalize a perturbed U_e by its modulus if it is not exactly a rotation. This normalization is smooth near the unit circle. The centers lie in a common bounded set, so the translation formula, the differences in V , and $V^T V$ inherit local Lipschitz bounds. This proof concerns the registration stage after the complete curve images have been obtained. $\square$

19.2. The observation-to-table reconstruction in this family.

Corollary 19.3 (Eight signed laws and four onsets). *There is $d_0 > 0$, common to the family, such that the four marked gaps and the two same-type signed limiting endpoint laws at offset d_0 for each of the four channels determine the whole marked table and its Euclidean lattice up to one simultaneous proper Euclidean motion. The parameters T, w, ϵ, δ , the support coefficients, the contact curvatures, and the placements of different channel frames are not supplied. The two actions at a channel are not both even. In particular, replacing signed endpoint laws by their pointwise sign-erased versions is not the observation used in this corollary.*

Proof. Choose d_0 inside the uniform positive-offset collar in Theorem 19.1. At each edge and each contact type, form the four-density ratio of Theorem 14.1 on a common positive interior square. A nonzero anchor recovers $S_{e,b}$ without knowing $B_{e,b}$ or the normalization. In particular, if $a_{e,b} = S''_{e,b}(0)$, then

$$\gamma_e = \operatorname{arsinh}(g_e \sqrt{a_{e,0} a_{e,1}}), \quad \kappa_{e,b} = \frac{\cosh \gamma_e \sqrt{a_{e,b}/a_{e,1} - b} - 1}{g_e}.$$

For every $n \geq 3$ the smooth finite-remainder theorem and the homogeneous isolation in Theorem 13.1 give $q_{e,n} = M_{e,n}^{-1}(s_{e,n} - R_{e,n})$. The geometric bounds above do not replace that nonlinear inverse; they ensure its hypotheses hold simultaneously at all eight contacts. The recovered analytic germs determine the complete images $C_{e,a}$, by Lemma 14.14. Now use Proposition 19.2 to place all edges in the e_{00} frame and recover V and both obstacle images. This is also the concrete common-frame implementation of Theorem 14.4.

If both actions in a channel were even, reflecting both transverse graphs would leave the two actions unchanged. The uniqueness of the contact-jet inverse would make both graph jets even. Analyticity would then make each whole obstacle invariant under reflection in its contact normal, contradicting Theorem 19.1. Thus the construction is not reduced to the reflection-symmetric local case. $\square$

For a fully specified member, take

$$T = \begin{pmatrix} 1 & 1/200 \\ 0 & 1 \end{pmatrix}, \quad w = (1/200, -1/200)^t, \quad \epsilon = 1/125, \quad \delta = 3/400. \quad (19.15)$$

Its four center vectors are $(4.030, 4.995)$, $(-3.970, 4.995)$, $(3.980, -5.005)$ and $(-4.020, -5.005)$ in the construction frame. Solving the strictly concave scalar problems (19.10) gives the actual contacts and onsets; these are not assumed to lie on the center directions. The exact recovered Gram matrix, which is independent of the unknown starting-frame rotation, is

$$G = \begin{pmatrix} 64 & 2/5 \\ 2/5 & 40001/400 \end{pmatrix}. \quad (19.16)$$

The finite diagnostic supplied with the source solves all four contact equations and computes (19.12) from the edge-frame support functions. It does not fit the matrix G directly or give the reconstruction routine the generating matrix T . The numerical comparison with (19.16) checks this finite geometric computation, not the all-order boundary-law theorem.

19.3. Persistence beyond a finite-dimensional model.

Proposition 19.4 (Analytic neighborhoods of the realization). *The family above has a neighborhood under real-analytic support-function perturbations in which the four selected channels, strict clearance, absence of obstacle symmetry, signature-rigid spanning structure and rank-two anchoring pair persist. One may independently add to h_1, h_2 any real-analytic r_1, r_2 with $\|r_a\|_{C^2} \leq 10^{-6}$. Consequently the intrinsic determination applies on an infinite-dimensional analytic neighborhood, not only to tables with the finite-dimensional shape and placement description in (19.2)–(19.3). On each compact analytic subfamily satisfying the regularity conventions of the global physical theorem, its charged long-bridge reconstruction conclusion applies with the same four channel labels.*

Proof. Here $\|r\|_{C^2} = \max_{0 \leq k \leq 2} \|r^{(k)}\|_\infty$. The perturbation in curvature radius is at most $2 \cdot 10^{-6}$ and that in the enclosing radius at most 10^{-6} . All separation, curvature and clearance inequalities retain strict margins. The scalar contact maximum remains nondegenerate, so the channels and their gate margins persist. Fourier coefficients change by at most 10^{-6} . In particular the nonzero

modes 2 and 3, of minimum moduli $1/400$ and $1/1600$, remain nonzero and exclude every proper rotation.

For reflections, write the arguments of these two coefficients as β_2, β_3 . A necessary condition is $2\beta_3 - 3\beta_2 \in \pi\mathbb{Z}$. In the unperturbed family this quantity is $\pi/2$ or $\pi/3$. Each argument changes by at most $2 \cdot 10^{-6}/(1/1600) \leq 0.0032$, so the required congruence remains impossible. The vector (19.8), which may now move, removes the translation part of any candidate symmetry. The two perimeters remain different. Thus signature uniqueness persists, and the same combinatorial cycles and tree prove the intrinsic determination. The two-harmonic formula continues to hold because the same unknown coefficients cancel between copies; it does not require the absence of higher Fourier modes.

For example, fixing any strip width $\sigma > 0$ permits infinitely many independent sufficiently small perturbations $a_n \cos n\theta$ and $b_n \sin n\theta$, with exponentially summable coefficients. Hence this is genuinely an analytic neighborhood rather than a finite-dimensional reparametrization. On a compact analytic subfamily the positive margins and the hypotheses of the observable calibration and global physical reconstruction theorems give a common collar, the finite separating tests, and the charged preparation caps. Running their budget-indexed stages with the present four labels proves the last assertion. Compactness is used at this last step; the whole bounded analytic neighborhood is not silently asserted to be compact in an unspecified topology. $\square$

The example thus implements the entire intrinsic route: nonlinear long-bridge laws, amplitude-free action inversion, both contact jets, analytic images, unique registration, and an uncalibrated rank-two lattice. Its explicit registration bounds do not supply an effective rate for the preceding analytic continuation. In the richer planar-position protocol, the direct-position benchmark remains available as well. The two routes continue to use different retained information, even on this same nonsymmetric periodic family.

20. FINITE CHANNEL SKELETONS AND GENERIC PERIODIC RIGIDITY

The incidence hypotheses of the intrinsic inverse are not restricted to specially arranged families. We prove that a finite clear channel skeleton exists in every positively separated periodic table, and that its signature-rigidity requirements hold on an open dense set. The resulting inverse uses $N + 1$ selected channels for N labelled obstacle orbits. The selection is a geometric existence statement, not an algorithm for discovering unmarked channels from a trajectory.

Write $K_a = \overline{D}_a$, $1 \leq a \leq N$, and fix representatives in the plane. Let $L \in \mathrm{GL}^+(2, \mathbb{R})$ realize the marked deck group $\mathbb{Z}^2$. Throughout this section distinct members of

$$\mathcal{B} = \{K_a + Lm : a \in \{1, \dots, N\}, m \in \mathbb{Z}^2\}$$

are disjoint compact strictly convex bodies with smooth boundary and positive curvature. Analyticity will be imposed only when signatures and complete image recovery are used. An edge is counted without its reverse; loops and multiple edges in the quotient graph are allowed.

20.1. Clear closest segments and descent through an obstruction. For two distinct bodies $A, B \in \mathcal{B}$, let $d(A, B) > 0$ be their distance and let $[p_{AB}, q_{AB}]$ be their closest segment. The closest points are unique. Indeed, strict convexity of the squared norm makes the minimizing vector in $B - A$ unique, and the two supporting faces perpendicular to this vector consist of single points. We call the pair *clear* if its closed closest segment meets no member of $\mathcal{B} \setminus \{A, B\}$. Thus a third-body tangency is an obstruction, not a clear pair.

Lemma 20.1 (Uniform separation and finite distance sublevels). *There is $\sigma > 0$ such that $d(A, B) \geq \sigma$ for all distinct bodies in $\mathcal{B}$. For every $D < \infty$, only finitely many pairs at distance*

at most D occur modulo common deck translation. A clear closest segment has positive distance from the union of all third bodies.

Proof. Choose R with $K_a \subset \overline{B}(0, R)$ for every a , and write $s = s_{\min}(L) > 0$. Then

$$d(K_a, K_b + L\ell) \geq s|\ell| - 2R. \quad (20.1)$$

For any fixed finite upper distance, the possible ℓ therefore belong to a finite set. Select one pair and use its finite positive distance as an upper bound when taking the minimum over all pairs. The minimum is attained among finitely many disjoint pairs and is positive. This proves the first two assertions. For the last, bodies intersecting the closed unit neighborhood of a fixed compact segment form a finite set, by the same estimate. The distance from the segment to each of these third bodies is positive when the pair is clear; all remaining bodies have distance greater than one. Taking a finite minimum proves the assertion. $\square$

Lemma 20.2 (Obstruction descent). *For every pair $A \neq B$ there is a finite path of clear pairs*

$$A = A_0, A_1, \dots, A_k = B$$

such that

$$\sum_{i=1}^k d(A_{i-1}, A_i) \leq d(A, B), \quad k \leq \left\lfloor \frac{d(A, B)}{\sigma} \right\rfloor. \quad (20.2)$$

In particular, the graph of clear pairs on all lifted obstacles is connected.

Proof. If the pair is not clear, choose a third body C and $z \in C \cap [p_{AB}, q_{AB}]$. Separation implies that z is in the open segment. With $d = d(A, B)$, the two shorter distances obey

$$\begin{aligned} d(A, C) &\leq |p_{AB} - z|, & d(C, B) &\leq |z - q_{AB}|, \\ d(A, C) + d(C, B) &\leq d, & \max\{d(A, C), d(C, B)\} &\leq d - \sigma. \end{aligned} \quad (20.3)$$

The last inequality uses $|p_{AB} - z|, |z - q_{AB}| \geq \sigma$. Induct on $\lfloor d/\sigma \rfloor$. A blocked pair cannot have $d < 2\sigma$; each pair on the right of (20.3) has a strictly smaller induction index. Concatenate their clear paths. Their total edge distance is at most $d(A, C) + d(C, B) \leq d$. Every terminal edge has distance at least σ , proving the bound on k . Erasing any repeated-vertex subwalk gives a path with the same bounds. This proof also treats tangential obstructions. No length inside an intermediate body is charged: the assertion concerns a graph path, not a physical billiard itinerary or a bound on its travel time. $\square$

20.2. A finite skeleton with two independent deck cycles. An oriented quotient edge is $e = (a, b, \ell)$, joining K_a to $K_b + L\ell$. A path lifts uniquely once its starting lift is fixed, and its deck gain is the signed sum of its edge labels. For a closed quotient walk this gain belongs to $\mathbb{Z}^2$.

Theorem 20.3 (Finite clear skeleton). *Every table in the setting above has a set E of $N + 1$ clear selected channels whose quotient multigraph is connected and contains two closed walks with linearly independent deck gains. It is the union of a spanning tree and two further edges. All selected contacts have positive third-obstacle clearance and admit fixed gates and alternating contact collars. The same finite labels, tree, gains and gates persist for sufficiently small C^2 perturbations of the supports and small perturbations of L that preserve the stated setting.*

More precisely, set

$$D_0 = \max(\{d(K_1, K_a) : 2 \leq a \leq N\} \cup \{d(K_1, K_1 + Le_i) : i = 1, 2\}). \quad (20.4)$$

The channels may be selected among the finite set of clear pairs with gap at most D_0 . With R, s as in Lemma 20.1, their labels satisfy $|\ell| \leq (D_0 + 2R)/s$.

Proof. Apply Lemma 20.2 to the $N - 1$ pairs (K_1, K_a) and to the two pairs $(K_1, K_1 + Le_i)$. Every edge on the resulting finite paths has gap at most D_0 . Project their union to a finite quotient graph Γ . It is connected: the first group of paths reaches every obstacle type. The last two paths project to closed walks based at type 1 whose deck gains are exactly e_1 and e_2 . In particular the gains of all closed walks in Γ generate the whole deck group, not merely a real one-dimensional subspace. Finiteness and the label bound follow from (20.1).

Choose a spanning tree $\mathcal{T}$ of Γ , rooted at type 1. For its root-to- a path let $m_a \in \mathbb{Z}^2$ be the signed deck sum, with $m_1 = 0$. For an oriented edge $e = (a, b, \ell_e)$ define

$$\eta_e = m_a + \ell_e - m_b. \quad (20.5)$$

This is zero for tree edges. The gain of any closed walk is the signed sum of its η_e , since the m_a terms telescope. Conversely, for a non-tree edge, η_e is the gain of the walk from the root to a along the tree, then across e , and back along the tree from b to the root. The non-tree gains consequently generate the deck-gain group of Γ . Two of them, say η_{f_1}, η_{f_2} , are linearly independent over $\mathbb{R}$. Take

$$E = \mathcal{T} \cup \{f_1, f_2\}.$$

It has $N - 1 + 2 = N + 1$ edges, including when $N = 1$ and the tree is empty. The two gains need not form an integer basis. Their real independence is exactly the hypothesis needed to recover L from $L\eta_{f_i}$; one must not replace their displacement matrix by the identity.

We verify the geometric assertions rather than identifying a graph edge with an observation without a collar. At a closest pair, use transverse coordinates whose positive directions agree in the plane. The Hessian of the flight length at its normal contact is

$$\begin{pmatrix} \kappa_a + g^{-1} & -g^{-1} \\ -g^{-1} & \kappa_b + g^{-1} \end{pmatrix}, \quad \det = \kappa_a \kappa_b + (\kappa_a + \kappa_b)/g > 0. \quad (20.6)$$

Thus the closest point is nondegenerate. The local implicit construction and strict convexity make the closest contacts continuous under C^2 perturbation and smooth, respectively analytic, along smooth, respectively analytic, finite-dimensional families. Their connecting segments inherit the positive clearance of Lemma 20.1.

Distinct clear incidences on a given lifted obstacle have distinct contact points. Otherwise their closest segments issue from the same point along the same outward normal ray. The nearer endpoint lies on the farther segment and on a third body, contradicting clearness; equality of the two endpoints contradicts disjointness. Since E is finite, choose disjoint small planar gates around its incident contacts on each representative obstacle, and choose fixed reference gates also at the finitely many lifted endpoints used by E . Periodic copies at the reference table are assigned their marks before observation. Shrink these gates so that their connecting segments stay in the clear tubes. During a perturbation the finitely many laboratory gates stay fixed; the unknown lattice is not used to relocate them. They are not exact contact observations. After decreasing the perturbation neighborhood they still contain the corresponding contacts with positive interior margins.

There is no unexamined infinity of third obstacles in this persistence argument. On a small parameter neighborhood all bodies have a common radius bound and $s_{\min}(L)$ has a positive lower bound. Only finitely many translates can approach any of the finitely many selected segments; their positive margins persist. The same argument preserves all-pair separation. Formula (20.6), positive curvature, and the fixed gates give the alternating contact collars. The labels and the two integer gain vectors do not change under this perturbation. $\square$

Lemma 20.4 (Locality of a selected clear channel). *The selected-channel conclusions of Theorem 2.1, the signed contact inverse, and the single-offset density inverse apply to every channel of Theorem 20.3. These channels need not attain the minimum gap of the whole table. Each*

fixed table has a common positive offset collar for this finite channel set. On compact smooth or analytic subfamilies with the uniform higher-norm assumptions of the relative theorem, the collar and every fixed-order bound may be chosen uniformly.

Proof. The event under consideration specifies the same two facing patches and alternation as the selected event of Section 2. Every one of its flights has length at least this pair's gap g_e . In a window $jj_g + d$, the total excess of the j flights above jj_g , and both unused end intervals, is at most d . On fixed patches strict convexity and (20.6) give a positive quadratic lower bound on one-flight excess near the closest points, and a positive lower bound away from that neighborhood. Thus a sufficiently small d localizes all the selected impacts in the contact collars. Short unused end intervals remain in their clear tubes after the offset collar is shrunk once more.

The weighted Jacobi construction and its determinant comparison use the local positive gap, positive curvatures and twist, not a comparison with another pair's gap. Their resulting bridges remain within the same clear tubes. The half-line construction and the fixed-domain sublevel integration therefore give precisely the selected relative law. Its single-offset density and smooth contact inverse depend on that law and the selected gap, so they apply unchanged. Taking a minimum over finitely many channels proves the fixed-table collar assertion; taking the uniform constants in the stated compact-family hypotheses proves the last one. A C^2 geometric neighborhood alone is not asserted to bound arbitrarily high derivatives or to provide a compact analytic class. $\square$

Proposition 20.5 (The channel count for two-cycle reconstruction). *A connected quotient multigraph on N obstacle types that admits two linearly independent deck-cycle gains has at least $N + 1$ edges. Consequently Theorem 20.3 attains the smallest channel count for this connected two-cycle reconstruction mechanism.*

Proof. If the graph has m edges, a spanning tree leaves $m - N + 1$ non-tree edges. As in (20.5), their gains generate every closed walk gain. Real rank two requires $m - N + 1 \geq 2$. $\square$

This is a graph-theoretic count of selected channels, not a lower bound on the number of preparations, the dimension of a probability law, or the information needed by every conceivable inverse observation map.

20.3. Generic signature rigidity. Let $\mathcal{P}_N^\omega$ denote the space of such labelled periodic tables with real-analytic boundaries. A parameter consists of L and the support functions h_a of the representatives K_a ; translations are retained in the first harmonics. Give this space the relative C^2 topology on supports and the ordinary topology on L . The assertions below hold also with any fixed stronger C^r support topology. This definition of genericity makes no compactness or common complex-extension assertion.

For a convex body K , put

$$s(K) = \frac{1}{\pi} \int_0^{2\pi} h_K(\theta) n(\theta) d\theta, \quad z_k(K) = \frac{1}{2\pi} \int_0^{2\pi} h_K(\theta) e^{-ik\theta} d\theta.$$

The vector $s(K)$ is translation and orthogonal covariant. The coefficients z_k for $k \geq 2$ are translation invariant; the centered body $K - s(K)$ has zero first harmonic.

Proposition 20.6 (Open dense obstacle asymmetry). *The subset $\mathcal{A}_N \subset \mathcal{P}_N^\omega$ for which every obstacle has trivial orientation-preserving Euclidean symmetry group is open and dense. It contains the open dense subset $\mathcal{H}_N$ defined by, for every a ,*

$$z_2(K_a)z_3(K_a) \neq 0, \quad \text{Im}(z_3(K_a)^2 \overline{z_2(K_a)})^3 \neq 0. \quad (20.7)$$

Every individual obstacle of a table in $\mathcal{H}_N$ has no nonidentity Euclidean symmetry, including reflections. No restriction is placed on its other analytic harmonics.

Proof. A symmetry of a bounded body fixes its covariant center. For the centered support, rotation by α is a symmetry exactly when $e^{-ik\alpha}z_k = z_k$ for all $k \geq 2$. A body has trivial proper symmetry exactly when the indices of its nonzero coefficients have greatest common divisor one. In that case finitely many such indices already have greatest common divisor one: successive greatest common divisors are positive integers and stabilize. Those finitely many nonzero coefficients remain nonzero after a sufficiently small support perturbation, proving openness of $\mathcal{A}_N$.

Nonzero coefficients of orders two and three force a rotational symmetry to be the identity. A reflection across a line at angle α through the center transforms z_k to $e^{-2ik\alpha}\overline{z_k}$. Invariance of both coefficients would imply

$$2\alpha + \arg z_2 \in \pi\mathbb{Z}, \quad 3\alpha + \arg z_3 \in \pi\mathbb{Z}.$$

Eliminating α makes $z_3^2\overline{z_2}^3$ real. Thus (20.7) excludes reflections as well. The conditions are open because Fourier coefficients are continuous in the uniform norm.

To prove density, perturb each support by

$$u_{a,2} \cos 2\theta + v_{a,2} \sin 2\theta + u_{a,3} \cos 3\theta + v_{a,3} \sin 3\theta.$$

These four real parameters vary its second and third complex coefficients independently in a neighborhood. Their zero sets and the zero set of the nonzero real polynomial $\text{Im}(z_3^2\overline{z_2}^3)$ have empty interior. An arbitrarily small choice therefore satisfies (20.7). Since $h_a + h_a'' > 0$ and separation are open conditions, the perturbations preserve positive curvature and all-pair disjointness. They are trigonometric polynomials, so they preserve analyticity and can be arbitrarily small in any fixed C^r norm. The lattice and every other harmonic may be left unchanged. This proves density of $\mathcal{H}_N$, hence of $\mathcal{A}_N$. $\square$

Theorem 20.7 (Generic finite-channel determination). *For every $T \in \mathcal{A}_N$ there are $N + 1$ marked selected channels and a positive offset d_0 such that their $N + 1$ onsets and their $2N + 2$ same-type signed limiting endpoint laws at d_0 determine the complete labelled periodic table and its Euclidean lattice up to one simultaneous orientation-preserving Euclidean motion. No inter-channel placements, contact curvatures, support coefficients or lattice Gram form are given. The competing realizations may be any tables in $\mathcal{P}_N^\omega$ realizing these selected data in the contact-law observation class of Theorem 14.4, not only members of $\mathcal{A}_N$.*

The channel labels, gates, spanning tree and two independent cycle gains can be kept fixed in a geometric neighborhood of T . In particular, this finite-channel conclusion holds on an open dense subset of $\mathcal{P}_N^\omega$. For the common-orientation orbit of the entire signed datum the conclusion is modulo one element of $E(2)$, not under independent sign erasures in different channels.

Proof. Select E by Theorem 20.3 and choose a common d_0 by Lemma 20.4. The four-density inverse of Theorem 14.1 recovers both unsymmetrized half-line actions at each channel from its two laws. The smooth finite-remainder inverse of Theorem 13.1 then recovers both complete contact jets. Lemma 14.14 determines the two connected analytic boundary images in that channel's own frame.

Every boundary of T has trivial proper symmetry. Lemma 16.2 implies that every one of its complete oriented contact signatures occurs at exactly one framed point. Hence all incidence transitions in E are signature-rigid. Any competing realization has the same complete incident images and inherits this property; asymmetry is not imposed separately on the competitor.

The two fundamental closed walks may be based at the same root obstacle. Choose one incident measured channel as the common starting frame. Unique congruences between root-boundary copies transport each walk's initial and final incidence to that frame. Equivalently, prepend a zero-net-gain backtrack ee^{-1} when a common initial edge is required. These operations add neither a

measured channel nor a deck displacement. They concern compositions of incidence congruences, not realization of a multi-edge closed walk as a specular trajectory. The resulting walks are a rank-two anchoring pair in the sense of Definition 17.1. The selected tree is signature-rigid and can be rooted at this same obstacle.

Theorem 14.4, or equivalently its common-frame construction, now gives the conclusion. Explicitly, align the starting frames in two realizations. Their unique incidence congruences give the same cycle translations v_1, v_2 . With $M = (\eta_{f_1} \ \eta_{f_2})$ and $V = (v_1 \ v_2)$, they have the same $L = VM^{-1}$, including its Gram form and scale. Unique matching along the tree then places every remaining obstacle image in that frame, with deck corrections interpreted through this one L . Realizability supplies existence and admissibility; the argument proves uniqueness. Persistence follows from the clear-skeleton theorem and openness of $\mathcal{A}_N$. The final assertion is the equivariance of Theorem 16.14 applied to the entire marked datum. $\square$

20.4. An explicit recovered-image reconstruction on the harmonic locus. On $\mathcal{H}_N$ the incidence stage has a finite formula, independent of how many higher harmonics the unknown obstacles possess. This also gives a quantitative statement in a clearly specified topology.

Proposition 20.8 (Harmonic registration and the displacement cochain). *Suppose $E = \mathcal{T} \cup \{f_1, f_2\}$ is the skeleton above and every obstacle has nonzero second and third support coefficients. From the complete boundary images already recovered in the separate channel frames, all channel rotations, both lattice columns and all obstacle placements can be computed by the following finite operations.*

For two copies C_s, C_t of the same labelled obstacle, the unique proper congruence taking C_s to C_t has rotation

$$U_{s \rightarrow t} = \frac{z_3(C_s)z_2(C_t)}{z_3(C_t)z_2(C_s)}, \quad A_{s \rightarrow t}(y) = U_{s \rightarrow t}y + s(C_t) - U_{s \rightarrow t}s(C_s), \quad (20.8)$$

where the plane is identified with $\mathbb{C}$ and $|U_{s \rightarrow t}| = 1$ in exact data. Propagate these rotations along the tree, starting with one root incidence in a fixed channel frame. Let R_e be the resulting rotation of edge e into that common frame, also obtained for the two non-tree edges by matching their initial obstacle copies. Set

$$d_e = R_e\{s(C_{e,+}) - s(C_{e,-})\}, \quad p_1 = 0, \quad p_a = \sum_{e \in [1,a]_{\mathcal{T}}} \epsilon_e d_e. \quad (20.9)$$

For $f_i = (a_i, b_i, \ell_i)$ put

$$v_i = p_{a_i} + d_{f_i} - p_{b_i}, \quad L = (v_1 \ v_2)M^{-1}, \quad c_a = p_a - Lm_a, \quad M = (\eta_{f_1} \ \eta_{f_2}). \quad (20.10)$$

Then c_a are the representative support centers, with $c_1 = 0$, and centering and rotating one recovered copy of each type gives its complete boundary at c_a . Formula (20.10) also recovers $G = L^t L$.

For fixed labels and tree, this inverse stage is locally Lipschitz in the uniform norms of the support functions of the recovered complete images, uniformly when their radii and centers are bounded, their second and third coefficient moduli have positive lower bounds, and $\|M^{-1}\|$ is bounded. It is not a Lipschitz assertion from endpoint laws or finite contact jets to entire analytic boundaries.

Proof. For a rotation $U = e^{i\alpha}$ and translation t ,

$$z_k(UC + t) = e^{-ik\alpha} z_k(C) \quad (k \geq 2), \quad s(UC + t) = Us(C) + t.$$

If $C_t = UC_s + t$, substituting the coefficient relation in (20.8) recovers U , with no choice of a branch of an argument. The center identity gives the translation. Nonzero modes two and three

exclude any competing proper rotation. Starting with the root image, apply this rule to the parent copy in each tree edge; rotating that edge also provides the child's image in the common orientation. Induction provides all obstacle orientations. The parent type of either non-tree edge is now available, giving its rotation as well. All templates used in this induction are recovered images, not supplied supports of the unknown table.

Let c_a and L temporarily denote the true representative centers and lattice in the resulting common orientation, translating so that $c_1 = 0$. The two endpoint centers of $e = (a, b, \ell_e)$ have displacement

$$d_e = c_b + L\ell_e - c_a.$$

Summing on the tree gives $p_a = c_a + Lm_a$. Substitution in (20.10) gives $v_i = L\eta_{f_i}$ and all claimed identities. Thus the formulas use the unknown realization only to prove correctness; their inputs do not contain c_a , L , or an inter-channel registration.

For stability, $h \mapsto z_k$ has uniform-norm operator bound one and $h \mapsto s$ has bound two. Quotients in (20.8) are locally Lipschitz away from their stated zero denominators. For perturbed noncongruent images, replace a computed U by $U/|U|$; this is smooth near the unit circle. Finite tree propagation preserves a local Lipschitz bound. Support functions of bodies in a common radius bound are uniformly Lipschitz in angle, so the small rotation errors have the same bound in the uniform support norm. The remaining operations are finite sums, multiplication by the fixed M^{-1} and translations by bounded marked deck vectors. They preserve the asserted local bound. Finally

$$\|\tilde{L}^t \tilde{L} - L^t L\| \leq 2\|L\| \|\tilde{L} - L\| + \|\tilde{L} - L\|^2.$$

This controls the recovered Gram form in exactly the stated image topology. $\square$

Additional acquisition input. The following application uses Theorem F.48.3 of the complete technical manuscript, including its uniform-family assumptions and its physical observation space. In particular its calibration stage records planar endpoint positions in a common frame for the two types of each channel, uses the physical clock, and charges all independent preparations, including failures. These are additional acquisition assumptions, not conclusions of the exact law-valued inverse. The present corollary is not used in the proofs of the geometric reconstruction or its differential.

Corollary 20.9 (Persistent finite-design physical reconstruction). *Let $\mathfrak{K}$ be a compact analytic subfamily contained in one of the persistent skeleton neighborhoods of Theorem 20.7. Assume the uniform analytic, chart and physical-record hypotheses of Theorem F.48.3, with the marked $N + 1$ -channel design supplied by that neighborhood. Then the acquisition and uniform reconstruction conclusions of that theorem hold for this design, with one common positive offset and all preparation failures charged. In particular its finite capped policies recover the complete labelled table, including the unknown marked lattice, modulo one common proper Euclidean motion. The minimum even flight number may tend to infinity as in its prescribed-budget protocol.*

Proof. We apply Theorem F.48.3; the task here is to verify its geometric design hypotheses, not to reconstruct its acquisition estimates from exact injectivity. The persistent skeleton fixes connected incidence, a signature-rigid spanning tree, and two independent marked cycle gains. The transition matches are unique on this properly asymmetric neighborhood. The finite list of channels has fixed gates and positive clearance; the stipulated compact-family smooth bounds give common local laws by Lemma 20.4. The integer gain matrix M is fixed. Since L is nonsingular throughout the compact class, the smallest singular value of the recovered holonomy matrix LM has a positive minimum. No Euclidean lattice vector is supplied by this argument.

Theorem 20.7 gives injectivity on this class. The finite analytic signature and compact-inverse results of Section 18 provide the geometric inverse hypotheses used by the cited acquisition

theorem. Its remaining uniform analytic, chart and physical-record hypotheses are precisely those assumed in the statement. Theorem F.48.3 therefore supplies the charged same-final-flight pilot, capped fresh-stage estimator, and increasing-order prescribed-budget construction. Applying those conclusions gives the assertion. The design is fixed before acquisition; no new sensor, uncharged selection stage, or common design across unrelated skeleton neighborhoods is introduced. $\square$

The existence of a skeleton at each table does not select one common design throughout an arbitrary compact set crossing several such neighborhoods. Nor do $N + 1$ channels mean finitely many scalar observations: an exact endpoint law is infinite-dimensional. The generic theorem removes the need to posit a special incidence architecture within this marked observation model. It does not replace that model by an ordinary marked length spectrum, erase the endpoint signs, or equate it with the richer position experiment. The local nonlinear relative law and the smooth finite-remainder inverse remain the analytic inputs to this broader consequence.

21. FINITE SYMMETRY AND THE PERIODIC RECONSTRUCTION BRANCHES

The local signed inverse and analytic image recovery do not require asymmetry. We separate the finite alternatives caused by rotational symmetry from the continuous ambiguity present for a circle. The same distinction will sharpen the differential theorem in the next section. Throughout this section the selected design is fixed and realizable. All competing realizations use the same signed contact-law model on an admissible collar at the prescribed offsets.

21.1. Congruence choices and a local angular lift. For a convex body K let

$$s(K) = \frac{1}{\pi} \int_0^{2\pi} h_K(\theta) n(\theta) d\theta, \quad z_k(K) = \frac{1}{2\pi} \int_0^{2\pi} h_K(\theta) e^{-ik\theta} d\theta.$$

The center s is proper-Euclidean equivariant, and z_k , $k \geq 2$, is invariant under translations. Write $\text{Sym}^+(K)$ for the group of proper Euclidean isometries preserving K .

Lemma 21.1 (Finite congruence alternatives). *If K is noncircular, $\text{Sym}^+(K)$ is a finite cyclic group of order*

$$m(K) = \gcd\{k \geq 2 : z_k(K) \neq 0\}.$$

For two properly congruent copies C_s, C_t of K , the set of proper congruences between them has exactly $m(K)$ members. Their translations are determined by their rotations and the two centers.

Proof. Every symmetry fixes $s(K)$. After centering it is a rotation. A centered body is a disk precisely when all harmonics of orders $k \geq 2$ vanish: Fourier uniqueness then makes its support function constant. Otherwise choose one nonzero z_k . An allowed rotation $e^{i\alpha}$ must satisfy $e^{-ik\alpha} = 1$, so there are finitely many such rotations. A rotation preserves the support function precisely when this equality holds for every nonzero harmonic. The common solutions are the $m(K)$ th roots of unity. The gcd of the infinite nonempty set is already the gcd of a finite subset, since successive positive integer gcds eventually stabilize. Given one congruence, all others differ by a symmetry of K . For each selected rotation U , its translation is $s(C_t) - Us(C_s)$. $\square$

Lemma 21.2 (Local alignment without a global phase choice). *Let $C_s(t), C_t(t)$ be C^1 families of properly congruent support images, with noncircular base images. Suppose their actual proper congruences form a continuous family through the base alignment $U_0 = e^{i\alpha_0}$. Then one nonzero base harmonic z_k , $k \geq 2$, determines a C^1 local branch of their rotation and translation through that alignment. If both image derivatives vanish, the derivatives of that rotation and translation vanish. The assertion does not require the symmetry group to be constant along the family.*

Proof. Choose k with $z_k(C_s(0)) \neq 0$; the corresponding coefficients in both copies remain nonzero. Put $r(t) = z_k(C_t(t))/z_k(C_s(t))$ and $r_0 = e^{-ik\alpha_0}$. In a small disk about 1 use the logarithm with $\text{Log } 1 = 0$. Along the actual congruent family,

$$U(t) = U_0 \exp\{-k^{-1} \text{Log}(r(t)/r_0)\}, \quad a(t) = s(C_t(t)) - U(t)s(C_s(t)). \quad (21.1)$$

Indeed, choose the continuous actual angular lift through α_0 and restrict it by $|\alpha(t) - \alpha_0| < \pi/k$. Then $\text{Log}(r(t)/r_0) = -ik(\alpha(t) - \alpha_0)$, proving (21.1). The displayed formula is C^1 , and

$$\frac{\dot{r}}{r} = -ik\dot{\alpha}, \quad \dot{\alpha} = \frac{i}{k} \frac{\dot{r}}{r}. \quad (21.2)$$

The last expression is real along the congruent family. Stationary support images have zero center and harmonic derivatives, so $\dot{\alpha} = 0$ and $\dot{a} = 0$. Other congruence branches can exist, but are separated from the selected angular lift. The branch followed by the actual family is the one required here; a base symmetry which is broken need not extend to nearby pairs. Symmetry breaking does not affect the nonzero harmonic or this argument. No phase is selected globally from unlabelled noisy copies by (21.1). $\square$

21.2. A finite reconstruction of the exact fiber. Let $E = \mathcal{T} \cup \{f_1, f_2\}$ be a clear skeleton as in Theorem 20.3. Its graph on obstacle labels is connected, and its two independent cycle gains form a fixed invertible integer matrix M . In addition, form a graph whose vertices are the channels in E , connecting two channels whenever they contain an occurrence of the same obstacle label. This graph is connected: a path in the obstacle graph gives a chain of incident channels. Choose a rooted spanning tree $\mathcal{U}$ in this channel graph. For each link ν of $\mathcal{U}$ fix the particular two occurrences of its common obstacle label, denoted $a(\nu)$. These are combinatorial choices, including for loops and multiple edges, not supplied poses.

Theorem 21.3 (Finite fibers for noncircular obstacles). *Suppose every obstacle of a realizable analytic table T_0 is noncircular. Its gaps and same-type signed laws at one known sufficiently small positive offset per channel determine all realizations of the same marked datum, modulo a common proper Euclidean motion, as a finite set. With the tree $\mathcal{U}$ just chosen its cardinality is at most*

$$\prod_{\nu \in \mathcal{U}} m(K_{a(\nu)}). \quad (21.3)$$

The bound counts congruence assignments and need not be attained. When every $m(K_a) = 1$, the fiber consists of a single table.

Proof. Theorem 14.1, the smooth signed jet inverse, and analytic continuation recover the entire pair of obstacle images $C_{e,-}, C_{e,+}$ in every channel frame. This step uses no proper asymmetry. It also determines their centers and finite symmetry orders. Every realization of the datum must have these same images.

Fix the rotation of the root channel as the identity. Propagate rotations R_e along $\mathcal{U}$. At a link matching occurrences C_s and C_t , Lemma 21.1 supplies exactly $m(K_{a(\nu)})$ proper congruences $C_t = UC_s + a$. For each choice set $R_t = R_s U^{-1}$. Thus $R_t(C_t - s(C_t)) = R_s(C_s - s(C_s))$. All assignments are finite and their number is bounded by (21.3). Their translations are not propagated as unknown lattice displacements; only the rotations have been fixed so far.

For a complete assignment, require all centered, rotated occurrences of each obstacle to agree, not merely those on $\mathcal{U}$. Assignments failing this test are discarded. Denote the resulting centered

representative images by $\widehat{K}_a$. Orient the obstacle tree $\mathcal{T}$ from a root obstacle and define

$$\begin{aligned} D_e &= R_e\{s(C_{e,+}) - s(C_{e,-})\}, & p_1 &= 0, \\ p_a &= \sum_{e \in [1,a]_{\mathcal{T}}} \epsilon_e D_e, & m_a &= \sum_{e \in [1,a]_{\mathcal{T}}} \epsilon_e \ell_e, \\ v_i &= p_{a_i} + D_{f_i} - p_{b_i}, & \eta_i &= m_{a_i} + \ell_{f_i} - m_{b_i} \quad (i = 1, 2), \\ L &= (v_1 \ v_2)M^{-1}, & M &= (\eta_1 \ \eta_2), \\ c_a &= p_a - Lm_a, & K_a &= \widehat{K}_a + c_a. \end{aligned} \tag{21.4}$$

This uses the actual M^{-1} , without assuming $\det M = 1$. The cochain identities give, for every tree edge and both remaining edges,

$$c_b + L\ell_e - c_a = D_e. \tag{21.5}$$

For tree edges this follows directly by telescoping; for f_i it is $L\eta_i = v_i$. Each assignment therefore produces at most one marked candidate in the root-center and root-direction gauge.

Retain a candidate only when $\det L > 0$, the lifted obstacles are pairwise disjoint, and the selected pairs have the stipulated clear contacts and collars. These conditions are part of realization, not consequences of the integer gain matrix. They require only finitely many lattice checks once a nonsingular candidate has been produced: if all representative bodies and selected segments lie in a ball of radius R , a translated object can meet one of them only for marks bounded by a constant times $R/s_{\min}(L)$. Exact shape equality is an equality of the already known analytic images. This description is a finite-branch inverse of exact function-valued data, not a numerical decision algorithm for arbitrary finite precision inputs.

Every actual realization gives one of the congruence assignments and satisfies (21.4), hence occurs among the retained candidates. Conversely, for a retained candidate, centered shape agreement and (21.5) place each entire ordered pair exactly as its recovered channel pair, by one proper motion. Its closest segment, oriented transverse convention and local actions and amplitudes therefore agree with those recovered from the datum. The relative law theorem yields exactly the same normalized conditional laws. Global phase-volume normalization cancels under conditioning; it is not extra input here. All selected design conditions have been tested. Thus every retained candidate is a realization. Duplicate candidates can be identified in the common gauge. Finally, realizability makes the set nonempty, and when all symmetry orders are one the upper bound is one. $\square$

The theorem removes no discrete ambiguity by an arbitrary phase rule. Nor does it assert global uniqueness for all noncircular tables. Its differential counterpart uses the local branch through the actual base alignment and therefore needs only the absence of a continuous rotational stabilizer.

21.3. A circular comparison for the same selected design.

Proposition 21.4 (A circle and a variable marked Gram form). *For one disk orbit of radius $1/10$, let*

$$L_t = \begin{pmatrix} 1 & -\sin t \\ 0 & \cos t \end{pmatrix}, \quad |t| < 1/10.$$

The two selected channels with gains e_1, e_2 are clear on a common neighborhood and have constant gaps and same-type signed limiting conditional laws at a common sufficiently small positive offset. Nevertheless the marked lattice Gram form has a nonzero derivative at zero, and the variation is not a common Euclidean motion.

Proof. Both columns of L_t have length one. Thus the two disk pairs have gap $4/5$ and each ordered pair is constant up to its own proper channel motion. The same-type local actions and amplitudes, and hence their conditional laws, are constant. Their common offset and collars can be chosen after clearance is verified.

Put $\epsilon = \|L_t - I\| = 2|\sin(t/2)| < 1/10$. For every nonzero integer mark m , $|L_t m| \geq (1 - \epsilon)|m|$, so distinct disk centers are separated by at least $9/10$. Consider either unit center segment from 0 to $L_t e_i$. If a third center $L_t m$ had distance at most $1/2$ from that segment, then $|L_t m| \leq 3/2$, whence $|m| \leq 5/3$. Apart from the two incident centers, m is therefore one of the eight nearest nonzero integer neighbors. At $t = 0$ its distance from $[0, e_i]$ is at least one. Perturbing that center and segment decreases the distance by at most $(|m| + 1)\epsilon \leq (\sqrt{2} + 1)/10 < 1/4$. This contradicts the assumed distance at most $1/2$. Consequently every third disk has distance greater than $2/5$ from the center segment, and also from its shorter boundary-to-boundary closest segment. Uniformly small gates are admissible.

Finally,

$$L_t^T L_t = \begin{pmatrix} 1 & -\sin t \\ -\sin t & 1 \end{pmatrix}, \quad \dot{L}_0 = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}, \quad \left. \frac{d}{dt} \right|_0 L_t^T L_t = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

A common rotation would preserve the Gram form. Thus this is not such a motion. The example concerns the two specified channels; no finite-horizon condition is asserted. $\square$

This comparison is outside the noncircular class and does not contradict the exact or differential theorem. It exhibits the geometric role of the hypothesis for this design, not a universal failure of reconstruction from other designs or richer observations.

21.4. A noncircular fiber with exactly two realizations. Finite ambiguity in Theorem 21.3 occurs inside its noncircular class. The following example was suggested in the referee-style memorandum [13]. We give the all-lattice clearance and completeness arguments, and then show that one additional marked gap distinguishes the two branches. This last observation is specific to the example; it is not a uniform branch-selection assertion for arbitrary noncircular tables.

Proposition 21.5 (An exact two-table fiber). *Let K have support function*

$$h(\theta) = \frac{1}{10} + \frac{1}{1000} \cos(4\theta), \quad a = 2^{-1/2}, \quad L_+ = \begin{pmatrix} 1 & a \\ 0 & a \end{pmatrix}, \quad L_- = \begin{pmatrix} 1 & -a \\ 0 & a \end{pmatrix}. \quad (21.6)$$

The one-obstacle tables $T_{\pm} = (K, L_{\pm})$ are admissible analytic periodic dispersing tables. For the two channels with fixed deck marks e_1, e_2 , they have the same gaps and the same two same-type signed conditional endpoint laws at every common sufficiently small positive offset. The fiber of this marked datum, modulo one common proper Euclidean motion, consists of exactly these two tables. They have different marked Gram forms, although the intrinsic data derivative at either table is injective modulo common motion on the family class of Theorem 22.5.

Proof. The support is entire and

$$h + h'' = \frac{1}{10} - \frac{15}{1000} \cos(4\theta) \geq \frac{17}{200} > 0. \quad (21.7)$$

It therefore defines a strictly convex analytic body with positive curvature. Its support center is zero, its proper rotation group is exactly C_4 , and $K \subset \overline{B}(0, R)$ with $R = 101/1000$. Both lattice determinants equal $a > 0$. Their Gram eigenvalues are $1 - a, 1 + a$, so, for any nonzero integer mark m ,

$$|L_{\pm} m| \geq \sqrt{1 - a} |m| \geq \sqrt{1 - a} > 2R. \quad (21.8)$$

This proves disjointness of every pair of distinct lifted closures, not just of the selected pairs.

The two selected center displacements have length one. In the first channel their direction is 0, while in the second it is $\pi/4$ for T_+ and $3\pi/4$ for T_- . At all these normals $h' = 0$. The supporting points therefore lie on the corresponding center segment, and the parallel supporting lines give the normal closest pair. Strict convexity makes this pair unique. Since $h(\theta + \pi) = h(\theta)$, the gaps are

$$g_1 = 1 - 2 \left(\frac{1}{10} + \frac{1}{1000} \right) = \frac{399}{500}, \quad g_2 = 1 - 2 \left(\frac{1}{10} - \frac{1}{1000} \right) = \frac{401}{500}. \quad (21.9)$$

We check clearance against the entire lattice. Let $v_i = L_{\pm} e_i$. For a lattice center not on the line $\mathbb{R}v_i$, its distance from that line is $|\det(v_i, L_{\pm} m)|/|v_i|$, a nonzero integer multiple of a . A lattice center on that line is an integer multiple of v_i ; unless it is one of the two incident centers, its distance from $[0, v_i]$ is at least one. Thus every third obstacle has distance at least $a - R > 3/5$ from the unit center segment. The actual closest segment is a subset of this segment. Uniformly small contact gates and a common positive-offset interval are consequently available in both tables.

The first ordered obstacle pair agrees in the two tables. Rotation by $\pi/2$ preserves K and takes $L_+ e_2 = (a, a)$ to $L_- e_2 = (-a, a)$. Hence it identifies their second ordered pairs as well. These are proper motions, preserving the oriented signed contact convention. The local stationary actions and normalized amplitudes, and therefore the two same-type conditional laws, agree in each channel. The overall phase-volume constant cancels under conditioning; equality of the conditional laws does not assert equality of the successful-preparation frequencies.

To prove that the fiber has no additional members, use the signed smooth inverse and analytic continuation in Theorem 21.3. They fix the complete ordered images in both channel frames for every realization of the datum. Fix the first channel direction and the support center. Its center displacement is then e_1 and its centered body is K . The four possible relative C_4 rotations of the second pair give exactly

$$v_2 \in \{(a, a), (-a, a), (-a, -a), (a, -a)\}. \quad (21.10)$$

Here the marked gain matrix is the identity, so the cochain uniquely gives $L = (e_1 \ v_2)$; no continuous placement variable is left. The determinants of these four candidates are respectively $a, a, -a, -a$. Precisely the first two satisfy the positive-orientation condition, and both have already passed separation and clearance. This proves exact two-element completeness of the fiber. The product bound in (21.3) is four in this example and is strictly larger than the number of realizations.

Finally,

$$L_+^t L_+ = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}, \quad L_-^t L_- = \begin{pmatrix} 1 & -a \\ -a & 1 \end{pmatrix}. \quad (21.11)$$

A common Euclidean motion preserves a marked Gram form. Changing integer basis is not permitted when the deck marks are fixed, so the two candidates are distinct in the announced quotient. The body is noncircular and the selected skeleton persists. The differential assertion follows from Theorem 22.5 at each candidate, with its actual local alignment branch. It does not identify the two distinct global branches. $\square$

21.5. A further marked channel separates the two branches.

Proposition 21.6 (Separation by one additional gap). *For the two tables in Proposition 21.5, the channel with deck mark $e_1 + e_2$ is clear in both tables. Its closest gaps satisfy*

$$g_{+, e_1 + e_2} > 1 > g_{-, e_1 + e_2}. \quad (21.12)$$

Thus the original two-channel datum together with this one additional marked gap determines the table uniquely within its original exact fiber. Conditional endpoint laws on the additional channel are not needed for this distinction.

Proof. Put $w_{\pm} = L_{\pm}(e_1 + e_2)$ and

$$\ell_+ = |w_+| = \sqrt{2 + \sqrt{2}}, \quad \ell_- = |w_-| = \sqrt{2 - \sqrt{2}}.$$

Since K and $K + w_{\pm}$ lie in radius- R disks about their centers, their distance obeys

$$\ell_{\pm} - 2R \leq g_{\pm, e_1 + e_2} \leq \ell_{\pm} + 2R. \quad (21.13)$$

The upper estimate needs no assumption that the closest points lie on the center line. Here $\ell_+ - 101/500 > 1$ and $\ell_- + 101/500 < 1$, which prove (21.12).

It remains to show that these closest segments are genuine clear channels, rather than merely distances between two bodies. For a lattice center $L_{\pm}(m, n)$ off the line $\mathbb{R}w_{\pm}$,

$$\text{dist}(L_{\pm}(m, n), \mathbb{R}w_{\pm}) = \frac{|\det(w_{\pm}, L_{\pm}(m, n))|}{\ell_{\pm}} = \frac{a|n - m|}{\ell_{\pm}} \geq \frac{a}{\ell_{\pm}}.$$

A center on the line has $m = n$. Excluding $0, w_{\pm}$, its distance from $[0, w_{\pm}]$ is at least $\ell_{\pm}$. Hence every third center is separated from that center segment by at least

$$c_* := \min\{\ell_-, \ell_+, a/\ell_-, a/\ell_+\} = a/\ell_+ > 2R.$$

For endpoints $x \in K$ and $y \in K + w_{\pm}$, each point of $[x, y]$ is within R of $[0, w_{\pm}]$: subtract the corresponding convex combination of $0, w_{\pm}$ and use the radius bound at both ends. This applies in particular to the closest segment, which need not be a subset of the center segment. A third obstacle lies in the radius- R disk about its center. Its distance from the closest segment is therefore at least $c_* - 2R > 0$. Positive curvature and strict separation give the unique normal closest pair and allow uniformly small gates. The added channels are clear, completing the proof. $\square$

The two propositions separate three different statements. The two original function-valued channel observations have a genuinely nontrivial finite exact fiber. At each member the intrinsic derivative has no non-Euclidean kernel. A third, differently marked distance observation resolves this particular discrete ambiguity. Neither the circular comparison nor these propositions assert a failure of the noncircular finite-fiber theorem, a universal minimal channel count, or a uniform statistical selector on the entire noncircular class.

22. INFINITESIMAL RIGIDITY AND FINITE OBSERVABLE COORDINATES

The exact single-offset inverse has a differential consequence which does not follow from injectivity alone. We prove it for variations of entire analytic tables, allowing the contacts, obstacle shapes and marked lattice to vary. The analytic hypothesis is imposed on the parameter derivative as well as on each table. At the end of the section, a finite-dimensional model is introduced only for the scalar-coordinate conclusion.

22.1. Analytic variations and fixed intrinsic coordinates. For $\rho > 0$, let $\mathcal{H}_{\rho}$ be the real Banach space of bounded 2π -periodic holomorphic functions on $|\text{Im } z| < \rho$ which are real on the real axis, with the supremum norm. A *common-strip analytic-support family* is a map

$$t \longmapsto (h_{1,t}, \dots, h_{N,t}, L_t) \in \mathcal{H}_{\rho}^N \times \text{GL}^+(2, \mathbb{R}) \quad (22.1)$$

which is C^1 on an open parameter interval and represents admissible tables. Families with a parameter in an open subset of $\mathbb{R}^p$ are defined in the same way. Positive curvature, separation, and the chosen clear-channel design persist near the base parameter. A dot means differentiation

at that parameter. Cauchy's estimates imply C^1 dependence in each real C^m support norm and holomorphy of $\dot{h}_i$ on the strip. Changes to moving contact frames preserve these properties on every smaller strip. No common bound for all orders of differentiation is asserted.

Fix a table in $\mathcal{C}_N$ and a skeleton $E = \mathcal{T} \cup \{f_1, f_2\}$ from Theorem 20.3. Freeze all obstacle, channel and deck marks, the transverse signs, and one sufficiently small positive offset d_e for each edge. The same offset may be used on all edges. In the smoothly moving intrinsic frame of each channel, its contacts have coordinates $(0, 0)$ and $(g_e, 0)$ and the transverse variable is u . Fix contact-centered squares $Q_{e,b} = (-r_{e,b}, r_{e,b})^2$ whose closures lie in the respective positive-density regions, and nonzero scalar anchors $a_{e,b} \in (-r_{e,b}, r_{e,b})$. These choices persist on a parameter neighborhood. Write

$$f_{e,b,t} = \frac{dP_{e,b,t}}{du dv}$$

for the limiting conditional density in this frame. The frame is a coordinate convention for the intrinsic experiment, not an observed inter-channel pose.

A common Euclidean gauge can be fixed by putting the support center of a root obstacle at the origin and the oriented axis of a root channel along the first coordinate axis. The support center is the vector $s(K) = \pi^{-1} \int_0^{2\pi} h_K(\theta) n(\theta) d\theta$. The root channel has positive length; its direction is smooth. This gives a local C^1 slice for the three-dimensional proper Euclidean action. When needed, restrict the strip after this change of gauge. All conclusions about zero tangent vectors in this slice mean equality of the support-function and lattice derivatives, not a particular parametrization of the boundary curves.

Lemma 22.1 (Differentiability on an interior square). *For a family (22.1), each gap, intrinsic contact frame and finite contact jet is C^1 . On every fixed positive-density square, $t \mapsto f_{e,b,t}$ is C^1 in each prescribed finite C^m norm, after restricting the parameter neighborhood as necessary. Consequently*

$$\left. \frac{d}{dt} \right|_0 \int \phi dP_{e,b,t} = \int_{Q_{e,b}} \phi(u, v) \dot{f}_{e,b}(u, v) du dv, \quad \phi \in C_c^\infty(Q_{e,b}). \quad (22.2)$$

The derivatives in this statement are at fixed excess time d_e in the moving intrinsic transverse coordinates.

Proof. For a facing pair, the two endpoint stationarity equations have an invertible Hessian. In contact transverse coordinates the quadratic part of the distance has Hessian

$$\begin{pmatrix} \kappa_0 + g^{-1} & -g^{-1} \\ -g^{-1} & \kappa_1 + g^{-1} \end{pmatrix}, \quad \det = \kappa_0 \kappa_1 + g^{-1}(\kappa_0 + \kappa_1) > 0.$$

The support parametrization and its finite derivatives are C^1 by Cauchy's estimates. The implicit function theorem therefore gives C^1 contact points, gap and frame; it also gives C^1 local graph functions at every finite smooth order. Clearance fixes the selected facing pair and its gates on this neighborhood.

Apply the weighted half-line construction and the differentiated relative law to this family. The contraction constants are uniform on a sufficiently small neighborhood with positive gap and curvature margins. Differentiating the fixed-point equation solves for the parameter derivative with the same bounded inverse; finite endpoint derivatives use the higher smooth bounds already supplied by (22.1). The summable endpoint and terminal estimates of Lemmas 13.2 and 13.3 justify the corresponding action derivatives. For the amplitude, we make the functional-family derivative explicit. Suppress the endpoint type and write

$$H_t^0 = H_t(0), \quad G_t = (H_t^0)^{-1}, \quad \Delta H_t(u) = H_t(u) - H_t^0, \quad \mathcal{T}_t(u) = G_t \Delta H_t(u).$$

The uniformly positive tridiagonal reference operator is C^1 in operator norm. The resolvent identity gives

$$\dot{G} = -G\dot{H}^0G, \quad \Delta\dot{H} = \dot{H}(u) - \dot{H}^0, \quad \dot{\mathcal{T}} = \dot{G}\Delta H + G\Delta\dot{H}. \quad (22.3)$$

In particular the derivative of the moving reference is not omitted. For each fixed endpoint order m , locality and the weighted half-line estimates imply, on a uniform small neighborhood,

$$\sum_{i,j \geq 1} |\partial_u^r \partial_t^a (\Delta H_t(u))_{ij}| \leq C_m \sum_{i \geq 1} \rho^i < \infty, \quad 0 \leq r \leq m, \quad a = 0, 1.$$

Only neighboring entries occur. The terms without endpoint decay in $\dot{H}(u)$ cancel those in $\dot{H}^0$; the remaining terms contain an orbit coordinate or its parameter derivative. Fixed index powers are absorbed in the strict exponential margin. Finite entry sums therefore converge with their derivatives uniformly in trace norm. Thus $\mathcal{T}_t$ is C^1 in trace norm, in each fixed endpoint norm, and $\|\mathcal{T}_t\| \leq q < 1$ after the collar is shrunk once.

The edge sum U_t of (7.6) has the same summability: $g_t[-\ell_{t,uv}(0,0)] = 1$ identically in t , so its reference derivative cancels before the edges are summed. In the absolutely convergent trace series (7.7), cyclicity and the trace-ideal inequality give

$$\frac{d}{dt} \operatorname{tr}(\mathcal{T}_t^n) = n \operatorname{tr}(\mathcal{T}_t^{n-1} \dot{\mathcal{T}}_t), \quad |\operatorname{tr}(\mathcal{T}_t^{n-1} \dot{\mathcal{T}}_t)| \leq q^{n-1} \|\dot{\mathcal{T}}_t\|_1.$$

Summing this geometric majorant proves

$$\frac{\dot{B}}{B} = \dot{U} - \operatorname{tr}((I + \mathcal{T})^{-1} \dot{\mathcal{T}}). \quad (22.4)$$

Fixed endpoint differentiation introduces only fixed polynomial factors in the series index and bounded resolvent factors, which remain summable. This proves the amplitude's C^1 dependence at each finite smooth order, directly from the same relative construction. The half-line actions have the C^1 dependence already established above.

It remains to check the normalization, since its integral meets a moving boundary. In a fixed box U containing the selected supports, put $A_t(u, v) = S_t(u) + S_t(v)$ and $w_t(u, v) = B_t(u)B_t(v)$. The normalizer is

$$Z_t = \int_U w_t(d_e - A_t)_+.$$

The level set $A_t = d_e$ has two-dimensional measure zero. Indeed S'_t vanishes only at zero, whereas $d_e > 0$; the gradient of A_t is nonzero at every point of this level set. The positive-part map is Lipschitz. Its difference quotients are bounded by a common bound for the parameter derivative of A_t . Dominated convergence gives

$$\dot{Z} = \int_U \{\dot{w}(d_e - A)_+ - w\dot{A} \mathbf{1}_{\{A < d_e\}}\}.$$

The same argument as the parameter varies gives continuity of this derivative: the indicators converge almost everywhere off the limiting level set. The positive lower bound for Z_t persists. On a positive interior square the cutoff is inactive, so the product and quotient rules prove the asserted $C^1(C^m)$ statement. Integrating against the fixed compactly supported test function proves (22.2). In particular no derivative of a Dirac mass on the cap boundary is introduced into this argument. $\square$

22.2. Linearization of the signed contact inverse.

Lemma 22.2 (A zero law derivative has zero contact-jet derivative). *In a common-strip analytic-support family of one channel, suppose $\dot{g} = 0$ and $\dot{f}_b = 0$ on the chosen interior square for both same-type laws at their fixed known positive offset. Then every labelled anchored graph coefficient has zero derivative. This holds for odd as well as even orders and does not assume a known amplitude.*

Proof. We display the differential of the amplitude-free inverse. For one type omit b , put $q = \sqrt{1 - R_f(a, a)} > 0$, and write $t(u) = (1 - R_f(u, a))/q$. At a fixed positive density,

$$\dot{R}_f(u, v) = R_f(u, v) \left\{ \frac{\dot{f}(u, v)}{f(u, v)} + \frac{\dot{f}(0, 0)}{f(0, 0)} - \frac{\dot{f}(u, 0)}{f(u, 0)} - \frac{\dot{f}(0, v)}{f(0, v)} \right\}, \quad (22.5)$$

$$\dot{t}(u) = -\frac{\dot{R}_f(u, a)}{q} + \frac{t(u)\dot{R}_f(a, a)}{2q^2}, \quad \dot{S}(u) = \frac{d_e \dot{t}(u)}{(1 + t(u))^2}. \quad (22.6)$$

All these operations are C^1 in any prescribed finite C^m norm: evaluation on a fixed slice is bounded linear, the denominators are positive, and the sole square root is a scalar evaluated at the nonzero anchor. Lemma 22.1 and Theorem 14.1 allow the ordinary chain rule. Thus $\dot{f} = 0$ implies $\dot{S} = 0$, including every finite derivative at zero. The two signed coordinates have not been identified.

Write $a_b = S_b''(0)$. The explicit leading recovery is

$$\gamma = \operatorname{arsinh}(g\sqrt{a_0 a_1}), \quad c_b = \cosh \gamma \sqrt{a_b/a_{1-b}}, \quad \kappa_b = (c_b - 1)/g.$$

It follows that $\kappa_b = 0$. At order $n \geq 3$, use the actual finite-jet identity

$$s_n = M_n q_n + R_n,$$

where R_n depends only on lower graph jets and the gap, and M_n is the invertible block in (13.4). Its validity for smooth representatives, including dependence on finite rather than formal jets, is supplied by Lemma 13.4. Each finite map is smooth, as also follows from the finite Bellman recursion. Differentiating gives

$$\dot{q}_n = M_n^{-1} \{ \dot{s}_n - \dot{R}_n - \dot{M}_n q_n \}. \quad (22.7)$$

Induction, beginning with the leading recovery, makes every term on the right zero. The orders zero and one are fixed by the contact anchors. The induction is carried out separately at each finite n ; it neither differentiates an infinite inverse series nor uses an order-independent bound on M_n^{-1} . $\square$

Lemma 22.3 (Analytic propagation of a stationary contact variation). *If all anchored contact-graph coefficients of a common-strip analytic-support family have zero derivative and $\dot{g} = 0$, the entire incident obstacle images are stationary to first order in that channel frame. Their support functions therefore have zero derivative in every real C^m norm.*

Proof. Express the two bodies in the same moving channel frame. The contact origins are 0 and $(g, 0)$, whose derivatives vanish under the hypothesis. The normal directions at the contacts are fixed, and any reversal needed to use the positively oriented tangent is fixed as well. The local graph-to-support conversion is a smooth triangular map at each finite order, with positive curvature as its nondegeneracy condition; see (A.1.1). Hence every derivative in normal angle of the support variation vanishes at the contact normal.

For clarity, the analyticity needed at this step concerns the variation itself. If $A_t x = R_{\alpha_t} x + a_t$ is the placement of a channel frame, the support of a body in that frame is obtained from

its global support by an angular translation and a linear first-harmonic term. Its parameter derivative is a sum of the translated $\dot{h}$, a multiple of the translated h' , and first harmonics. Each is holomorphic on any fixed smaller strip. This follows from (22.1), Cauchy's estimates and the C^1 frame dependence, not from pointwise analyticity of h_t alone. The identity theorem now makes the support variation zero on the connected strip, hence on the entire real normal circle. Its real C^m norm is zero for each m . No norm-continuous inverse for analytic continuation is used or concluded. $\square$

22.3. Differentiable incidence matching and the lattice.

Lemma 22.4 (Finite harmonic registration on the asymmetric locus). *For each obstacle with trivial proper Euclidean symmetry, choose a finite set $F \subset \{2, 3, \dots\}$ of nonzero support harmonics whose indices have greatest common divisor one. These choices persist locally. Incidence congruences, the rotations of all selected channel frames into one common frame, the lattice L , and representative obstacle placements are C^1 functions of the complete channel-frame images along common-strip analytic-support families. If all these images are stationary to first order, the reconstructed gauge-fixed table is stationary to first order. For noncircular obstacles, the same conclusions hold on a local alignment branch through the actual base realization, without asserting globally single-valued registration.*

Proof. The finite witness exists by Proposition 20.6: otherwise the nonzero centered support harmonics have a common divisor larger than one, or there are none, giving respectively a nontrivial rotation or a circle. The greatest common divisors of successive finite subsets stabilize, so a finite witness suffices when the full divisor is one. Nonzero coefficients remain nonzero on a neighborhood.

Choose integers a_k with $\sum_{k \in F} a_k k = 1$. For two copies C_s, C_t of this obstacle, with $C_t = UC_s + a$ and $U = e^{i\alpha}$, define

$$r_k = \frac{z_k(C_t)}{z_k(C_s)}, \quad U = \prod_{k \in F} r_k^{-a_k}, \quad a = s(C_t) - Us(C_s), \quad z_k(C) = \frac{1}{2\pi} \int_0^{2\pi} h_C(\theta) e^{-ik\theta} d\theta. \quad (22.8)$$

Indeed $r_k = e^{-ika}$, so the product is $e^{i\alpha}$. Only integer powers of nonzero scalars occur; no argument or root branch is chosen. Centers and coefficients are bounded linear functionals of the real support function. Formula (22.8) is consequently C^1 on its nonvanishing domain. It recovers the unique proper incidence congruence. For $F = \{2, 3\}$ it reduces to (20.8). Thus the present argument does not restrict the asymmetric locus to the second-and-third-harmonic subclass.

Propagate these congruences along the finite spanning tree from a fixed root incidence. Match the initial obstacle of either remaining edge to get its rotation as well. This is a finite composition of C^1 operations. In the resulting orientation let R_e be the rotation of edge e and set

$$d_e^{\text{cen}} = R_e \{s(C_{e,+}) - s(C_{e,-})\}, \quad p_1 = 0, \quad p_a = \sum_{e \in [1,a]_{\mathcal{T}}} \epsilon_e d_e^{\text{cen}}.$$

Here m_a denotes the sum of signed deck marks along the same tree path. For $f_i = (a_i, b_i, \ell_i)$ the marked gain is $\eta_i = m_{a_i} + \ell_i - m_{b_i}$ and

$$v_i = p_{a_i} + d_{f_i}^{\text{cen}} - p_{b_i}, \quad L = (v_1 \ v_2) M^{-1}, \quad c_a = p_a - L m_a, \quad M = (\eta_1 \ \eta_2). \quad (22.9)$$

These are the displacement-cochain identities of Proposition 20.8, whose telescoping proof depends on the recovered rotations, not on using harmonics two and three in particular. The matrix M is fixed and invertible; it need not be unimodular. Centering and rotating one recovered copy of each obstacle, then translating it to c_a , completes the reconstruction in the root-center gauge.

All operations are C^1 along the family. In particular

$$\dot{L} = (\dot{v}_1 \ \dot{v}_2)M^{-1}.$$

If every input support variation is zero, every coefficient and center derivative, every relative-rotation derivative, and then every quantity in (22.9) has zero derivative. The output support variations and lattice variation are zero. This proves the last assertion without assuming a differentiable inverse to analytic continuation.

Local registration at a noncircular table. When the base obstacle has a nontrivial finite symmetry group, the Bezout formula in Lemma 22.4 need not exist. Use instead the local angular lift of Lemma 21.2, separately at each link of a fixed channel-incidence spanning tree. Such branches exist because the actual continuously moving channel frames give congruences between each fixed pair of labelled occurrences. Every noncircular base obstacle has one nonzero harmonic, so these local choices persist along a C^1 family, whether or not its finite symmetry is preserved. Stationary channel-frame support images give stationary relative rotations and translations on these branches. All subsequent center and cochain operations in (22.9) are the same finite C^1 operations. Hence stationary images again give $\dot{L} = 0$ and stationary representative supports in the common gauge. The actual base placements are used only to specify the local branch for this derivative argument, not supplied as extra observations. The finite alternatives in Theorem 21.3 are not removed globally by this local choice. $\square$

Theorem 22.5 (Infinitesimal single-offset periodic rigidity). *Let T_t be a common-strip analytic-support family through $T_0 \in \mathcal{C}_N$, with a persistent selected skeleton and fixed positive offsets as above. The following are equivalent:*

- (1) $\dot{g}_e = 0$ and $\dot{f}_{e,b} = 0$ on $Q_{e,b}$ for every e, b ;
- (2) $\dot{g}_e = 0$ and $\left. \frac{d}{dt} \right|_0 \int \phi dP_{e,b,t} = 0$ for every e, b and every real $\phi \in C_c^\infty(Q_{e,b})$;
- (3) the table derivative is generated by one common infinitesimal proper Euclidean motion.

In particular the data derivative is injective in the root-center and root-direction gauge, including variations of the marked lattice.

Proof. The first two assertions are equivalent by Lemma 22.1. For the nontrivial direction, a continuous nonzero $\dot{f}$ has constant strict sign on some small open ball. A nonnegative smooth function supported in that ball has nonzero integral against $\dot{f}$, contradicting the second assertion.

Assume the first assertion. Lemma 22.2 makes all contact-jet variations zero. Lemma 22.3 makes all entire channel-frame support variations zero. Lemma 22.4 then makes the gauge-fixed table variation zero. The C^1 gauge change therefore expresses the original derivative as a common infinitesimal translation and rotation. Explicitly, for constants $a \in \mathbb{R}^2$ and $\omega \in \mathbb{R}$, it has the support and lattice form

$$\dot{h}_i(\theta) = a \cdot n(\theta) - \omega h'_i(\theta), \quad \dot{L} = \omega JL, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (22.10)$$

The converse follows from invariance of gaps and signed intrinsic conditional laws under a simultaneous proper Euclidean motion. All parameter derivatives are ordinary derivatives along the stipulated family. Exact injectivity of the undifferentiated map is not used as a substitute for any step of the kernel argument. $\square$

22.4. Finite scalar coordinates on immersed models.

Theorem 22.6 (Local coordinates from finitely many gap and law tests). *Let $\Theta \subset \mathbb{R}^p$ be open, $p \geq 1$, and let $\theta \mapsto T_\theta$ be a C^1 common-strip analytic-support family in the fixed Euclidean gauge. Suppose its table derivative at θ_0 is injective and $T_{\theta_0} \in \mathcal{C}_N$. Use a persistent skeleton and fixed*

positive offsets. Then there are p scalar observables $\lambda_1, \dots, \lambda_p$, each either g_e or $\int \phi dP_{e,b}$ for a real $\phi \in C_c^\infty(Q_{e,b})$, such that

$$F(\theta) = (\lambda_1(T_\theta), \dots, \lambda_p(T_\theta)) \quad (22.11)$$

is a C^1 diffeomorphism between neighborhoods of θ_0 and $F(\theta_0)$. On a sufficiently small convex parameter ball there are constants $c, C > 0$ such that

$$c|\theta - \tilde{\theta}| \leq |F(\theta) - F(\tilde{\theta})| \leq C|\theta - \tilde{\theta}|. \quad (22.12)$$

For each fixed m , after restricting this ball if necessary, the C^m differences of the gauge-fixed support functions and the matrix difference of L are bounded by a constant times $|F(\theta) - F(\tilde{\theta})|$.

Proof. Consider on $\mathbb{R}^p$ the linear functionals

$$v \mapsto Dg_e(\theta_0)v, \quad v \mapsto \int_{Q_{e,b}} \phi Df_{e,b}(\theta_0)v, \quad \phi \in C_c^\infty(Q_{e,b}).$$

Their common kernel is zero. Indeed a vector in that kernel gives a table variation with vanishing intrinsic derivative by Theorem 22.5; the gauge and the immersion hypothesis then make the vector zero. A set of linear functionals with zero common kernel spans the dual of a finite-dimensional vector space: a proper span would have a nonzero annihilator. Select a basis of p members from this set. Each selected member is the derivative of an allowed scalar observable, not a derivative evaluation of an unknown density. Lemma 22.1 makes the resulting map F continuously differentiable. Its derivative $A = DF(\theta_0)$ is invertible, so the inverse function theorem proves the local coordinate assertion.

There is also a direct bound avoiding any assertion that pointwise nonsingularity alone implies a lower Lipschitz estimate. On a small convex ball, continuity gives $\|DF(\theta) - A\| \leq (2\|A^{-1}\|)^{-1}$. For $z = \theta - \tilde{\theta}$, integration along the joining segment therefore yields

$$|F(\theta) - F(\tilde{\theta})| \geq |Az| - \frac{|z|}{2\|A^{-1}\|} \geq \frac{|z|}{2\|A^{-1}\|}.$$

The upper bound follows from the local bound for DF . Finally the $C^1(\mathcal{H}_\rho)$ family has locally bounded first parameter derivatives in every fixed real C^m support norm, by Cauchy's estimates on a smaller strip, and bounded lattice derivatives. Integrating these derivatives on the same ball and using the lower bound proves the last assertion. $\square$

Theorem 22.6 is a model-local coordinate result for exact expectations. It does not claim a fixed finite observation vector for all analytic tables, a dimension bound for an arbitrary compact analytic class, or a statistical efficiency theorem. The full-class finite-histogram reconstruction in the complete technical manuscript uses increasing resolution and its separately stated conditioning hypotheses. The fixed-contact variation bundle and uniform positive designs in Section F.49 concern a different question: realizing compatible finite jets and detecting boundary velocities with a finite collection of offsets. Here the gaps, contacts and lattice may move, and a single offset per channel suffices for the differential law inverse. The analytic identity theorem removes the shape kernel; finite harmonic registration removes the relative-frame and lattice kernel; finite-dimensionality is used only in the final selection of scalar tests.

Acknowledgments. The author-requested AI-assisted memoranda cited here are not commissioned journal reports. The finite-versus-infinitesimal symmetry distinction and the circular comparison were communicated in [12]; the noncircular two-table example was communicated in [13]. The support-width centering comparison was communicated in [16]. The proofs and their attributions are retained separately from the reports' nonbinding placement assessments. The admissible highest-degree block estimates were communicated in [17]; their fixed-lower-jet and coefficient-space consequences are proved with them below.

Part 2. Local coordinate conversion and comparison proofs

APPENDIX A.1. GRAPH AND SUPPORT COORDINATES

The differential image argument uses a local geometric coordinate change, not the quantitative statistical hypotheses of the full technical manuscript. We record it here so that its proof is part of the principal article.

Lemma A.1.1 (Finite graph-to-support conversion). *In an outward-normal contact frame let the boundary be $(-\psi(y), y)$, where $\psi(0) = \psi'(0) = 0$ and $\psi''(0) = \kappa > 0$. On a small normal-angle interval there is a unique solution $\psi'(y(\theta)) = \tan \theta$. Its support function is*

$$h(\theta) = -\psi(y(\theta)) \cos \theta + y(\theta) \sin \theta. \quad (\text{A.1.1})$$

For each $m \geq 2$, the support jet through degree m is a smooth function of the graph jet through degree m , on positive curvature domains. The same assertion holds for parameter derivatives in the corresponding finite smooth norms. In particular, zero variations of all anchored graph coefficients imply zero variations of all anchored support coefficients.

Proof. The scalar implicit function theorem applies to $\psi'(y) = p$ at $(y, p) = (0, 0)$ since its derivative in y is κ . Strict convexity identifies this point with the unique local support point for normal $(\cos \theta, \sin \theta)$. Put $H(p) = -\psi(y(p)) + py(p)$. Stationarity gives $H'(p) = y(p)$, so $h(\theta) = \cos \theta H(\tan \theta)$. The coefficient of p^k in $y(p)$, for $k \leq m - 1$, is obtained by successively solving the scalar equation $\psi'(y(p)) = p$; it depends only on graph coefficients through order $k + 1$ and divides by the positive number κ . Since $H(0) = 0$, integration of $H' = y$ shows that the jet of H through order m needs no higher graph jet. The compositions by $\tan \theta$ and $\cos \theta$ preserve this finite-order property. The implicit function theorem and the finite chain rule justify this computation for actual smooth functions, and also justify parameter differentiation. For example $h(0) = h'(0) = 0$ and $h''(0) = \kappa^{-1}$.

When the second contact's outward normal is reversed, its positively oriented tangent must also be reversed. This known proper frame change multiplies its degree- n graph coefficient by $(-1)^n$ before the same formula is used. It introduces no unknown relative pose. Thus vanishing of all graph-coefficient variations in the common signed channel frame implies vanishing of all support-coefficient variations on both sides. $\square$

APPENDIX A.2. MARKED ONSETS AND SOURCE DOMAINS

Let $x_{e,0}, x_{e,1}$ be the normal outgoing states and f_ξ a smooth collision mark near them. Put $a_{e,b} = f_\xi(x_{e,b})$, $\bar{f}_e = (a_{e,0} + a_{e,1})/2$, and $w_{j,e} = \sum_{i=0}^j a_{e,i \bmod 2}$.

Theorem A.2.1 (Marked response and the exact axial source domain). *At the selected channel's own threshold,*

$$Z_{j,e}(\xi, q, d) = \frac{d^2 e^{qw_{j,e}(\xi)}}{2A(\xi) \sinh(j\gamma_e(\xi))} [1 + d\mathcal{R}_{j,e}^f(\xi, q, d)]. \quad (\text{A.2.1})$$

Every fixed mixed derivative of the remainder is bounded uniformly in j on compact parameter and source sets. The complete marked event in the ground collar is the sum at its positive channel offsets. For fixed ξ , real q , and sufficiently small fixed $d \geq 0$, the series $\sum_{j,e} Z_{j,e}(\xi, q, d)/d^2$ converges exactly when

$$\max_{e \in \mathcal{E}} (q\bar{f}_e - \gamma_e) < 0. \quad (\text{A.2.2})$$

Right limits are used at $d = 0$. Every fixed mixed derivative is summable on compact subsets of this domain. Finitely many real sources are allowed, with the scalar product in place of $q\bar{f}_e$.

This series ranges over different onset windows, not over all itineraries in one unrestricted long window. The source formula, interface jumps and record response are consequences of the same uniform construction, not separate claims of a general limit theory. The exact equal-gap inverse theorem is retained in Section F.A.9. It uses central symmetry to remove the two-endpoint ambiguity and equal gaps to retain all channels at every collision order. The recovered area is a symmetric function of the three recovered curvatures in that chosen family, not a fourth independent unknown. Section F.A.9.1 proves this identity and the loss of first-order sensitivity at coalescence.

APPENDIX A.3. CONDITIONAL ENDPOINT METRICS AND CURVATURE OBSERVABILITY

The scalar coefficient in (3.1) contains only the ratio in (5.2). We now retain the metric which appears before that ratio is formed. All expectations in this section are for the actual full-phase preparation, conditioned on $E_{j,e}(d)$.

Proposition A.3.1 (Quadratic endpoint statistics). *In the common transverse endpoint coordinates $Y = (u, v)$,*

$$\mathbb{E}Y = d m_{j,e}(\xi, d), \quad \frac{1}{d} \text{Cov}(Y) = \frac{1}{3} H_{j,e}^{-1} + d B_{j,e}(\xi, d). \quad (\text{A.3.1})$$

The vector and matrix remainders are smooth for $d \geq 0$ with uniform fixed-order derivative bounds. For physical positions the scalar variance in (3.2) has the corresponding leading diagonal entry of $H_{j,e}^{-1}/3$, with the same $O(d)$ remainder. In particular the leading metric is determined without a phase-space normalizer or a chosen transverse initial law.

Proof. Integrating the first residual time in (6.2) gives the conditional endpoint density, in $w = \sqrt{2d}z$ coordinates,

$$\frac{(1 - |z|^2) a_{j,e}(\xi, \sqrt{2d}z)}{\int_{|z| < 1} (1 - |z|^2) a_{j,e}(\xi, \sqrt{2d}z) dz} dz, \quad |z| < 1.$$

Its denominator is uniformly positive. The area and the exponentially small flux coefficient cancel in this conditional expression, but were included in the event probability. At $d = 0$ the density is $2(1 - |z|^2)/\pi$. Direct radial integration gives

$$\mathbb{E}_0 z = 0, \quad \mathbb{E}_0(z z^t) = I_2/6. \quad (\text{A.3.2})$$

For example $\mathbb{E}_0|z|^2 = 1/3$ and rotational symmetry divides this between the two coordinates.

The endpoint map is $Y = \Phi_{j,e}(\xi, w)$, with linear part $H_{j,e}^{-1/2}w$. For any smooth amplitude, integration on this radial domain annihilates its odd Taylor terms. Applied to each component of Ya and $YY^t a$, the smooth radial lemma gives

$$\mathbb{E}Y = d m_{j,e}, \quad \mathbb{E}(YY^t) = \frac{d}{3} H_{j,e}^{-1} + d^2 \tilde{B}_{j,e}.$$

All statements include smooth right extensions and uniform derivatives: the endpoint maps and amplitudes have the bounds in Lemma 6.1, and each division by d uses the vanishing constant term of a smooth even integral. Subtracting $(\mathbb{E}Y)(\mathbb{E}Y)^t$ proves (A.3.1).

The physical endpoint graph has the form $Q_b = q_b + \mathbf{t}y_b \pm \mathbf{n}\psi_b(y_b)$, with $|\mathbf{t}| = |\mathbf{n}| = 1$ and $\psi_b = O(y_b^2)$. Its mean is $q_b + d\tilde{m}_b$. The leading term in $|Q_b - q_b|^2$ is y_b^2 , since the two frame vectors are orthogonal. The remaining squared term starts at degree four. Apply the same radial argument and subtract the squared mean displacement. This proves the physical scalar variance formula. In particular no unknown q_b occurs in the observable itself: it is eliminated by centering the measured positions. $\square$

Proof of Theorem 3.2. The determinant-one middle matrix in (5.1) has inverse obtained by changing the signs of its off-diagonal entries. Thus

$$\mathbf{H}_{j,e}^{-1} = \frac{g}{c \sinh \gamma} D_j \begin{pmatrix} \coth(j\gamma) & \operatorname{csch}(j\gamma) \\ \operatorname{csch}(j\gamma) & \coth(j\gamma) \end{pmatrix} D_j. \quad (\text{A.3.3})$$

For odd j , $\sigma_0^2 = c_1$ and $\sigma_j^2 = c_0$. Proposition A.3.1 gives (3.3), with its differentiated error estimate. Taking the geometric mean and ratio gives (3.4). The function

$$F_j(\gamma) = \frac{\coth(j\gamma)}{\sinh \gamma}$$

is strictly decreasing from infinity to zero on $(0, \infty)$, since

$$-F_j'(\gamma) = \frac{j \operatorname{csch}^2(j\gamma)}{\sinh \gamma} + \frac{\coth(j\gamma) \cosh \gamma}{\sinh^2 \gamma} > 0. \quad (\text{A.3.4})$$

This proves unique recovery of γ , then of the two curvatures. At $j = 1$, the inversion is even explicit: $F_1(\gamma) = c/(c^2 - 1)$, so if $P = 3\sqrt{v_0 v_1}/g$, then $c = (1 + \sqrt{1 + 4P^2})/(2P)$.

We verify uniform conditioning, not just nonvanishing of a fixed-length Jacobian. On a compact positive gap-curvature box, $0 < \gamma_- \leq \gamma \leq \gamma_+ < \infty$. The second term of (A.3.4) gives a lower bound for $-F_j'$ independent of j . Its upper bound is also uniform because $j \operatorname{csch}^2(j\gamma_-) \leq C$ and $\coth(j\gamma_-) \leq \coth \gamma_-$. Both variances are bounded above and bounded away from zero. Taking their ratio and geometric mean is consequently Lipschitz; monotone inversion of F_j , then the last line of (3.4), is uniformly Lipschitz. Forward differentiation of (3.3) gives the reverse bound with the same type of constants. More explicitly, for data from two points of this box,

$$C^{-1} \|(g, \kappa_0, \kappa_1) - (\tilde{g}, \tilde{\kappa}_0, \tilde{\kappa}_1)\|_\infty \leq \|(g, v_0, v_1) - (\tilde{g}, \tilde{v}_0, \tilde{v}_1)\|_\infty \leq C \|(g, \kappa_0, \kappa_1) - (\tilde{g}, \tilde{\kappa}_0, \tilde{\kappa}_1)\|_\infty. \quad (\text{A.3.5})$$

No factor $(\kappa_0 - \kappa_1)^{-1}$ appears. Nor do we divide by the exponentially small off-diagonal entry in (A.3.3).

For data outside the exact image, minimize its distance to that image on the compact curvature box at the known g ; a minimizer exists. The true pair is an admissible comparison, so the selected image point is within twice the data error of the true image. The error is bounded by $\delta + Cd$ by Proposition A.3.1. Apply the inverse bound in (A.3.5). This proves the last assertion and also specifies a reconstruction for noisy or finite-offset data. A small error in the supplied g adds its size to the same bound, provided the conditional variances are still those of the stated physical window; we do not replace an onset-location error by a cost-free timing change. $\square$

The distinction between odd and even lengths is informational. At an even length the first and last impacts are on the same obstacle, and the two diagonal leading variances coincide. At an odd length they sample the two endpoints. The theorem therefore uses the two diagonal entries at an odd length rather than an exponentially small covariance or an assumed equality of endpoint curvatures.

Proposition A.3.2 (Independent-preparation estimation). *Suppose that $2K$ independent full-phase preparations are retained upon success of $E_{j,e}(d)$, for a fixed odd j . The two variances have estimators $\hat{V}_0, \hat{V}_1$, using only their endpoint positions and the known offset d , such that*

$$\Pr \left(\max_b |\hat{V}_b - V_{j,b}(d)| > \delta \right) \leq 4 \exp(-cK\delta^2).$$

A reconstruction as in Theorem 3.2 consequently has error at most $C[d + \sqrt{\log(4/\eta)/K}]$ with probability at least $1 - \eta$. All constants are independent of odd j . The expected number of

independent unconditioned preparations needed for these $2K$ successes is comparable to

$$Kd^{-2}e^{j\gamma_e}. \quad (\text{A.3.6})$$

No independence of successive impacts in one trajectory is assumed.

Proof. Pair the independent successful records and use, at each endpoint,

$$\widehat{V}_b = \frac{1}{K} \sum_{k=1}^K \frac{|Q_b^{(2k)} - Q_b^{(2k-1)}|^2}{2d}.$$

Independence gives its expectation as the variance in (3.2). Coercivity confines each endpoint to a set of diameter $C\sqrt{d}$, independently of j . The summands are therefore in one bounded interval $[0, C_0]$. For completeness, a random variable in an interval of length C_0 satisfies $\mathbb{E}e^{t(X-\mathbb{E}X)} \leq e^{t^2 C_0^2/8}$: the second derivative of the log moment generating function is a variance under a tilted law, bounded by $C_0^2/4$, and its value and first derivative vanish at zero. Multiplication over the independent pairs, the exponential Markov inequality, and optimization over t yield $2e^{-2K\delta^2/C_0^2}$ at each endpoint. A union bound proves the displayed estimate. The deterministic inverse estimate gives the curvature bound. Rejection from independent preparations gives independent conditional successes, and the expected waiting time is $2K/\Pr(E_{j,e}(d))$. Theorem 3.1 and $\sinh(j\gamma) \asymp e^{j\gamma}$ prove (A.3.6). Uniform conditional conditioning is therefore not a uniform unconditioned sampling cost. $\square$

A.3.1. Realized scalar fibers and their separation by records. At known g, A , all leading selected-channel count coefficients depend on c_0, c_1 only through $c_0 c_1$. Conversely, their first two values recover this product, since their ratio is $2 \cosh \gamma$. The following construction shows that this scalar fiber occurs for actual periodic billiards, not just for algebraic local jets.

Theorem A.3.3 (A geometric fiber of the full leading count hierarchy). *On the rectangular lattice $3\mathbb{Z} \times 4\mathbb{Z}$ there is a real-analytic family of strictly convex scatterers D_s , $|s| < s_0$, with D_0 the unit disk, for which the only shortest unoriented channel is horizontal,*

$$g(s) = 1, \quad A(s) = 12 - \pi, \quad c_0(s) = 2e^s, \quad c_1(s) = 2e^{-s}.$$

All maximal-count threshold locations and every leading coefficient $\lim_{d \downarrow 0} d^{-2} \Pr_s(N_{j+d} = j+1)$ are independent of s . The ordered contact curvatures vary. The two endpoint variances at any odd order distinguish s , with constants uniform in that order.

Proof. Use the support function of the disk, $h_0 \equiv 1$. Put

$$\chi_0 = \frac{1}{4}(1 + \cos \theta) \sin^2 \theta, \quad \chi_\pi = \frac{1}{4}(1 - \cos \theta) \sin^2 \theta, \quad \chi_* = \sin^4 \theta.$$

At $0, \pi$ all three functions and their first derivatives vanish. The second derivatives of χ_0 at these two points are $(1, 0)$, those of χ_π are $(0, 1)$, and those of χ_* are $(0, 0)$. These explicit trigonometric polynomials permit independent contact jets and a compensating change of area. Set

$$r_0(s) = (2e^s - 1)^{-1}, \quad r_1(s) = (2e^{-s} - 1)^{-1},$$

and

$$h_{s,a} = 1 + (r_0(s) - 1)\chi_0 + (r_1(s) - 1)\chi_\pi + a\chi_*.$$

For small s, a , $h + h'' > 0$, so this is a smooth strictly convex body. At normals $0, \pi$, its support values stay one and its first support derivatives stay zero. Its two curvature radii are exactly $r_0(s), r_1(s)$, since the compensator has zero second jet at both normals. The supporting lines between neighboring horizontal translates therefore stay a distance one apart, and their support points attain that distance. The closest horizontal segment is unchanged.

The obstacle area is $\mathcal{A}(h) = \frac{1}{2} \int_0^{2\pi} [h^2 - (h')^2] d\theta$. Its derivative in a at $(s, a) = (0, 0)$ is $\int \chi_* d\theta > 0$. Since $\int \chi_* d\theta = 3\pi/4$, the analytic implicit function theorem supplies a real-analytic $a = a(s)$ with $a(0) = 0$ and $\mathcal{A}(h_{s,a(s)}) = \pi$ exactly. This variation has not altered either contact jet. At the disk, the vertical gap is two and every other nonhorizontal pair has still larger gap. The positive gap to the horizontal minimum, disjointness of translates, and clearance persist for small s . Hence the two horizontal orientations are the entire ground-onset event. Their sum in (3.1) has leading coefficient

$$\frac{1}{(12 - \pi) \sinh(j \operatorname{arccosh} 2)},$$

independent of s . This assertion concerns leading coefficients, not equality of the finite-offset remainders or all other billiard data. Finally (3.3) gives $v_{j,1}/v_{j,0} = e^{2s}$ at every odd j . It separates the family; uniform conditioning follows from Theorem 3.2. $\square$

The fixed contact patches can be chosen once for this entire family. Even when several minimizing pairs compete, a recorded first-impact patch and its outgoing direction select the oriented channel. On a nonminimal channel one must also use its own window and the itinerary selection specified in Section 2. Counts in the ground collar alone eventually suppress a fixed gap difference $\Delta > 0$, since activation requires $j\Delta < \varepsilon_0$. The marked theorem neither changes that condition nor implicitly provides unobserved channel labels.

APPENDIX A.4. INFORMATION BEYOND THE LEADING HIERARCHY

The nonlinear law of the preceding section is not determined by the contact Hessian. We prove this by an analytic deformation with fixed area, gap and contact curvatures. Its higher coefficient can be computed from the half-line action and determinant, without extrapolating an uncontrolled finite-length error.

A.4.1. The quartic variation of a boundary law. Assume first that the two facing graphs are identical and even, and write

$$\psi(y) = \frac{\kappa}{2}y^2 + \frac{q}{24}y^4 + O(y^6), \quad c = 1 + g\kappa, \quad \gamma = \operatorname{arccosh} c, \quad a = \frac{\sinh \gamma}{g}, \quad \lambda = e^{-\gamma}. \quad (\text{A.4.1})$$

Here q is a fourth graph derivative, not a source variable. Fix g and κ and vary q in a smooth family. Higher graph jets may vary as well. The two boundary actions and amplitudes now agree; write them as S and B . Both limiting parities give one function $\mathcal{F}$.

Proposition A.4.1 (A nonlinear coefficient and its quartic derivative). *Let $\mathcal{R}_\infty(0) = \partial_d \mathcal{F}(0)$. At fixed g, κ , its derivative with respect to q is*

$$\partial_q \mathcal{R}_\infty(0) = -\frac{\cosh(2\gamma) + 2}{12a^2 \sinh(2\gamma)}. \quad (\text{A.4.2})$$

This derivative does not depend on the graph derivatives of order six or higher. More precisely, if $S(u) = au^2/2 + tu^4/24 + O(u^6)$ and $B(u) = 1 + \beta u^2 + O(u^4)$, then

$$\partial_q t = \coth(2\gamma), \quad \partial_q \beta = -\frac{1}{4a \sinh(2\gamma)}, \quad \mathcal{R}_\infty(0) = \frac{2\beta}{3a} - \frac{t}{12a^2}. \quad (\text{A.4.3})$$

Proof. The linear half-line segment is $x_i = \lambda^i u$. Because the quadratic action is stationary with its first endpoint fixed, its change upon substituting the cubic bridge correction begins at degree six. The quartic part of the stationary action is therefore the quartic part of the broken action

evaluated on that linear segment. In one flight, the q -dependent quartic term of the length is $q(x_i^4 + x_{i+1}^4)/24$. The endpoint is counted once and every interior site twice. It follows that

$$\partial_q t = 1 + 2 \sum_{i \geq 1} \lambda^{4i} = \frac{1 + \lambda^4}{1 - \lambda^4} = \coth(2\gamma).$$

This use of stationarity may equivalently be obtained by differentiating finite stationary actions and passing to the convergent half-line sum.

To degree two in u , the q variation of an edge's mixed derivative vanishes: the quartic term just displayed is a sum of two pure fourth powers. Thus the product term U in (7.6) has no q -dependent quadratic coefficient. The interior Hessian variation at that degree is diagonal, with entries $q\lambda^{2i}u^2$. Changes in the cubic bridge do not enter degree two, and products of two Hessian perturbations enter only degree four. Consequently the quadratic coefficient of the logarithmic determinant has derivative $\sum_{i \geq 1} G_{ii}\lambda^{2i}$. At equal curvatures, (7.1) gives $G_{ii} = (1 - \lambda^{2i})/(2a)$. Hence

$$\partial_q \beta = -\frac{1}{2a} \left(\frac{\lambda^2}{1 - \lambda^2} - \frac{\lambda^4}{1 - \lambda^4} \right) = -\frac{1}{4a \sinh(2\gamma)}.$$

The quadratic coefficient of B equals that of $\log B$ because both have vanishing linear coefficient by evenness.

Finally use (7.12). The density correction is $\beta(u^2 + v^2)$, whose expectation under the quadratic residual-time weight is $2\beta d/(3a)$. For the quartic action correction, first variation of the positive-part integral gives the negative integral of $t(u^4 + v^4)/24$ over $a(u^2 + v^2)/2 < d$; there is no boundary term because the residual weight vanishes there. The two elementary integrals are

$$\int (d - ar^2/2)_+ \, du \, dv = \frac{\pi d^2}{a}, \quad \int_{r^2 < 2d/a} (u^4 + v^4) \, du \, dv = \frac{2\pi d^3}{a^3}.$$

Taylor's formula in the uniform Morse coordinates justifies the remaining $O(d^2)$ relative error. This gives the last identity in (A.4.3), and substitution proves (A.4.2). Terms of degree six or higher cannot contribute to this first offset coefficient. $\square$

For an additional finite-length identity, put $L_i = (\sinh((j-i)\gamma), \sinh(i\gamma))/\sinh(j\gamma)$, $M_j = H_j^{-1}$ and $v_i = L_i M_j L_i^t$ in the equal-curvature case. Thus $L_0 = (1, 0)$ and $L_j = (0, 1)$. The same calculation before taking the limit yields

$$\partial_q \mathcal{R}_j(0) = -\frac{1}{3} \sum_{i=1}^{j-1} (G_j)_{ii} v_i - \frac{1}{24} \left(v_0^2 + v_j^2 + 2 \sum_{i=1}^{j-1} v_i^2 \right). \quad (\text{A.4.4})$$

Indeed, the logarithmic amplitude variation is $-\sum (G_j)_{ii} (L_i Y)^2$, and its quadratic conditional expectation is $-d \sum (G_j)_{ii} v_i/3$. The unweighted fourth moment in the quadratic sublevel, divided by its residual-time integral, is dv_i^2 . These observations prove (A.4.4), including $j = 1$ with an empty interior sum. Its limit is (A.4.2). This identity provides a finite check of the normalization but is not needed to define the infinite boundary determinant.

A.4.2. An analytic family invisible to all leading data.

Theorem A.4.2 (Identical leading metrics and distinct nonlinear laws). *On the lattice $3\mathbb{Z} \times 4\mathbb{Z}$ there is a real-analytic family of strictly convex scatterers D_s , $|s| < s_*$, with D_0 the unit disk, for which*

$$g = 1, \quad A = 12 - \pi, \quad \kappa_0 = \kappa_1 = 1$$

at the two horizontal contacts. The horizontal orientations exhaust the ground-onset event. All leading count amplitudes and all leading endpoint covariance matrices, at every flight number, are independent of s . Nevertheless their nonlinear boundary coefficient satisfies

$$\left. \frac{d}{ds} \mathcal{R}_\infty^{(s)}(0) \right|_{s=0} = \frac{\sqrt{3}}{2}. \quad (\text{A.4.5})$$

It therefore determines s locally with a bounded inverse derivative. For every sufficiently small fixed positive d , the limiting normalized probability $\mathcal{F}^{(s)}(d)$ also varies with s .

Proof. Consider the support function

$$h_{s,z}(\theta) = 1 + s \sin^4 \theta + z \sin^6 \theta. \quad (\text{A.4.6})$$

Its obstacle area is $\mathcal{A}(h) = \frac{1}{2} \int_0^{2\pi} (h^2 - (h')^2) d\theta$. At $(s, z) = (0, 0)$ its z derivative is $\int_0^{2\pi} \sin^6 \theta d\theta = 5\pi/8 > 0$. The analytic implicit-function theorem gives $z = z(s)$, $z(0) = 0$, with $\mathcal{A}(h_{s,z(s)}) = \pi$ exactly. Since the s derivative is $3\pi/4$, one also has $z'(0) = -6/5$.

At $\theta = 0, \pi$, both perturbations and their first two derivatives vanish. Hence $h = 1$, $h' = 0$ and the curvature radius $h + h'' = 1$. The support points remain $(1, 0)$ and $(-1, 0)$, their curvatures remain one, and the supporting lines of horizontal neighbors have separation one. These points attain that separation. At the disk the vertical gap is two; all other nonhorizontal gaps are larger. A small analytic parameter interval preserves this strict gap, disjointness, positive curvature, and clearance. Thus only the horizontal pair contributes to the ground onset. Reflection and central symmetry of (A.4.6) make the two facing graphs identical and even. Area and the two contact Hessians are fixed exactly, not merely to first order. Equations (5.2) and (A.3.1) give the assertion about every leading datum.

We compute the first higher graph jet. Let $r(\theta) = h(\theta) + h''(\theta)$ be the curvature radius. Near the right contact the boundary parametrization has derivatives $x'(\theta) = -r(\theta) \sin \theta$ and $y'(\theta) = r(\theta) \cos \theta$. Since $r'(0) = 0$, integrating these expansions and eliminating θ gives for the facing graph

$$\psi(y) = \frac{y^2}{2r(0)} + \frac{3r(0) - r''(0)}{24r(0)^4} y^4 + O(y^6).$$

In our family $r(0) = 1$ and $r''(0) = h''(0) + h^{(4)}(0) = 24s$; $\sin^6 \theta$ has zero fourth jet and does not affect this identity. Thus $q_s = 3 - 24s$. Here $\gamma = \text{arcosh } 2$ and $a = \sqrt{3}$, so $\cosh(2\gamma) = 7$, $\sinh(2\gamma) = 4\sqrt{3}$. Multiplying (A.4.2) by $q'_s = -24$ yields (A.4.5). The inverse-function theorem proves the local scalar reconstruction. Finally the joint analytic expansion $\mathcal{F}^{(s)}(d) = 1 + d\mathcal{R}_\infty^{(s)}(0) + O(d^2)$ has uniform differentiated remainder. Its s derivative at zero is $d\sqrt{3}/2 + O(d^2)$, nonzero for all sufficiently small positive d . $\square$

The family in Theorem A.3.3 varies the two contact curvatures while preserving only their scalar product. The family in Theorem A.4.2 instead keeps both contact curvatures fixed, so even the entire leading conditional metric is unchanged. It is separated by the nonlinear law, not by a relabelling of the leading inverse. Neither construction claims equality of finite-offset probabilities or of marked length spectra. Higher-jet information from other billiard observations has its own inverse theory; the assertion here is its explicit realization in a full-phase statistical threshold law with a collision-order-uniform limiting construction.

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